The left subclavian artery and vein arch over and are related to the superior lobe of the left lung as they pass over the dome of the cervical pleura and into the axilla.

Bronchial tree

The **trachea** is a flexible tube that extends from vertebral level CVI in the lower neck to vertebral level TIV/V in the mediastinum where it bifurcates into a right and a left main bronchus (Fig. 3.43). The trachea is held open by C-shaped transverse cartilage rings embedded in its wall—the open part of the C facing posteriorly. The lowest tracheal ring has a hook-shaped structure, the carina, that projects backwards in the midline between the origins of the two main bronchi. The posterior wall of the trachea is composed mainly of smooth muscle.

Each main bronchus enters the root of a lung and passes through the hilum into the lung itself. The right main bronchus is wider and takes a more vertical course

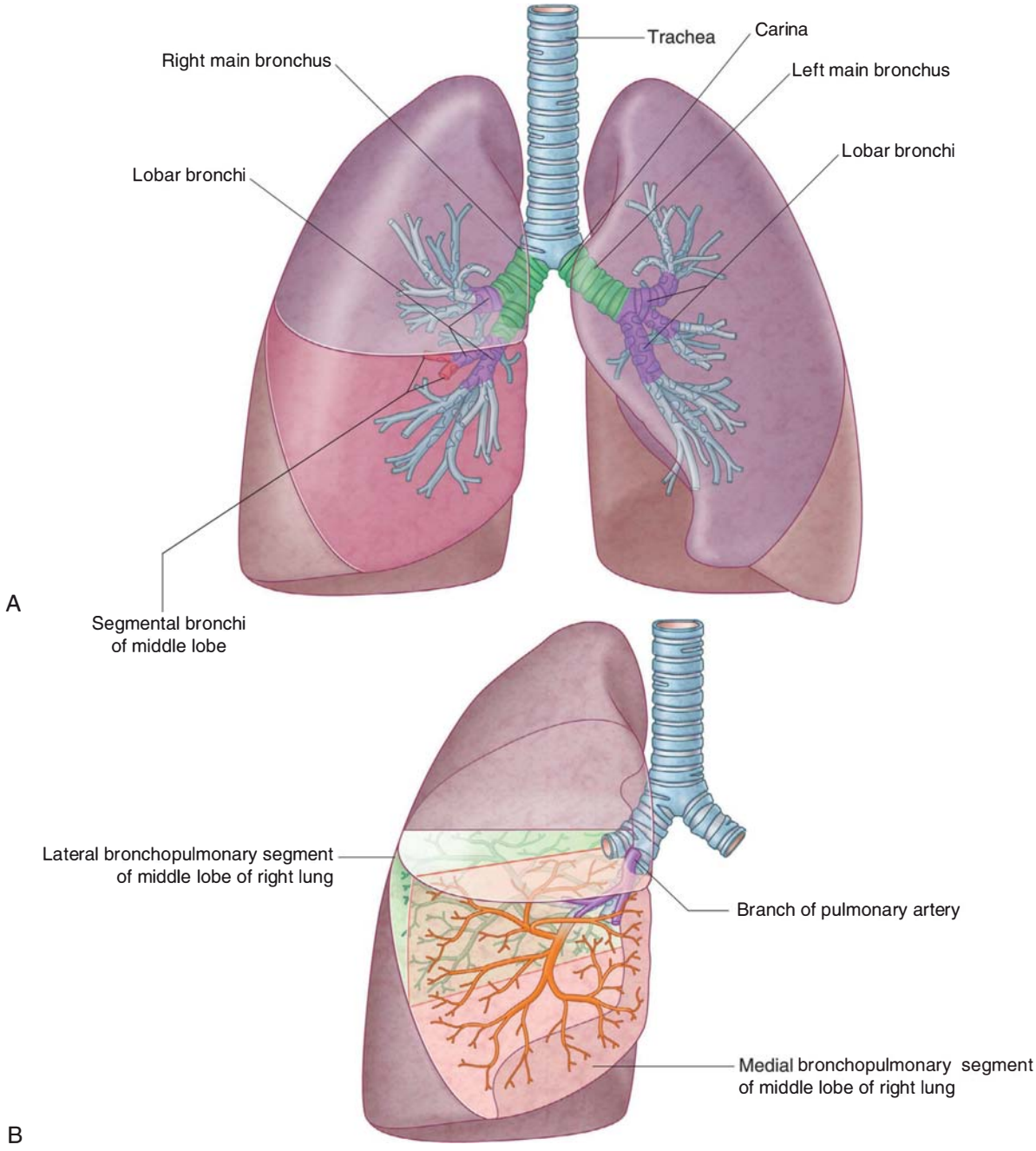


Fig. 3.43 A. Bronchial tree. **B.** Bronchopulmonary segments.

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through the root and hilum than the left main bronchus (Fig. 3.43A). Therefore, inhaled foreign bodies tend to lodge more frequently on the right side than on the left.

The main bronchus divides within the lung into **lobar bronchi** (secondary bronchi), each of which supplies a lobe. On the right side, the lobar bronchus to the superior lobe originates within the root of the lung.

The lobar bronchi further divide into **segmental bronchi** (tertiary bronchi), which supply bronchopulmonary segments (Fig. 3.43B).

Within each bronchopulmonary segment, the segmental bronchi give rise to multiple generations of divisions and, ultimately, to bronchioles, which further subdivide and supply the respiratory surfaces. The walls of the bronchi are held open by discontinuous elongated plates of cartilage, but these are not present in bronchioles.

Bronchopulmonary segments

A **bronchopulmonary segment** is the area of lung supplied by a segmental bronchus and its accompanying pulmonary artery branch.

Tributaries of the pulmonary vein tend to pass intersegmentally between and around the margins of segments.

Each bronchopulmonary segment is shaped like an irregular cone, with the apex at the origin of the segmental bronchus and the base projected peripherally onto the surface of the lung.

A bronchopulmonary segment is the smallest functionally independent region of a lung and the smallest area of lung that can be isolated and removed without affecting adjacent regions.

There are ten bronchopulmonary segments in each lung (Fig. 3.44); some of them fuse in the left lung.

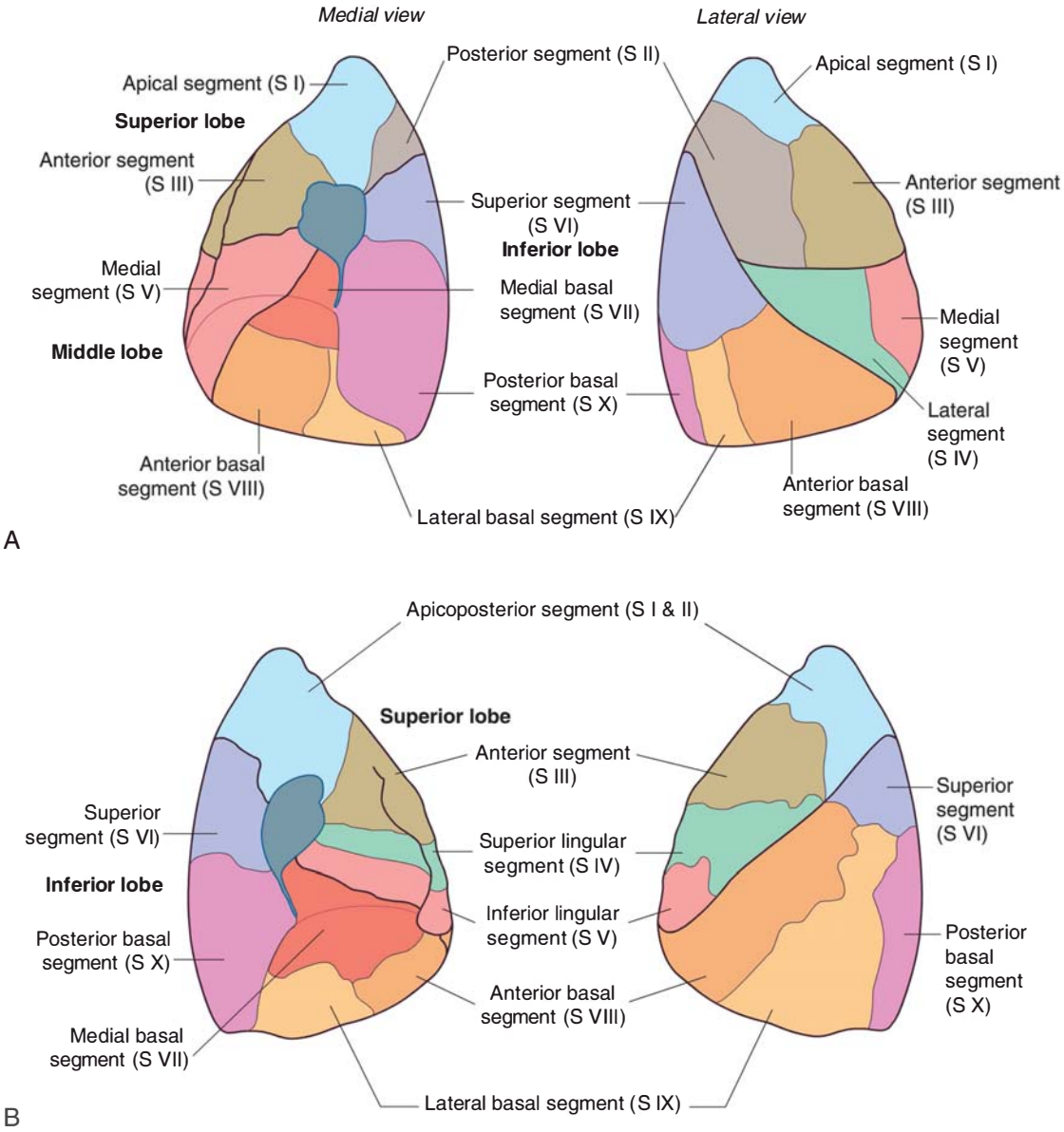


Fig. 3.44 Bronchopulmonary segments. **A.** Right lung. **B.** Left lung. (Bronchopulmonary segments are numbered and named.)

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Pulmonary arteries

The right and left pulmonary arteries originate from the **pulmonary trunk** and carry deoxygenated blood to the lungs from the right ventricle of the heart (Fig. 3.45).

The bifurcation of the pulmonary trunk occurs to the left of the midline just inferior to vertebral level TIV/V, and anteroinferiorly to the left of the bifurcation of the trachea.

Right pulmonary artery

The **right pulmonary artery** is longer than the left and passes horizontally across the mediastinum (Fig. 3.45). It passes:

- anteriorly and slightly inferiorly to the tracheal bifurcation and anteriorly to the right main bronchus, and
- posteriorly to the ascending aorta, superior vena cava, and upper right pulmonary vein.

The right pulmonary artery enters the root of the lung and gives off a large branch to the superior lobe of the lung. The main vessel continues through the hilum of the lung, gives off a second (recurrent) branch to the superior lobe, and then divides to supply the middle and inferior lobes.

Left pulmonary artery

The **left pulmonary artery** is shorter than the right and lies anterior to the descending aorta and posterior to the superior pulmonary vein (Fig. 3.45). It passes through the root and hilum and branches within the lung.

Pulmonary veins

On each side a **superior pulmonary vein** and an **inferior pulmonary vein** carry oxygenated blood from the lungs back to the heart (Fig. 3.45). The veins begin at the hilum of the lung, pass through the root of the lung, and immediately drain into the left atrium.

Bronchial arteries and veins

The bronchial arteries (Fig. 3.45) and veins constitute the "nutritive" vascular system of the pulmonary tissues (bronchial walls and glands, walls of large vessels, and visceral pleura). They interconnect within the lung with branches of the pulmonary arteries and veins.

The bronchial arteries originate from the thoracic aorta or one of its branches:

- A single **right bronchial artery** normally arises from the third posterior intercostal artery (but occasionally, it originates from the **upper left bronchial artery**).
- Two **left bronchial arteries** arise directly from the anterior surface of the thoracic aorta—the **superior left bronchial artery** arises at vertebral level TV, and the inferior one inferior to the left bronchus.

The bronchial arteries run on the posterior surfaces of the bronchi and ramify in the lungs to supply pulmonary tissues.

The **bronchial veins** drain into:

- either the pulmonary veins or the left atrium, and
- into the azygos vein on the right or into the superior intercostal vein or hemiazygos vein on the left.

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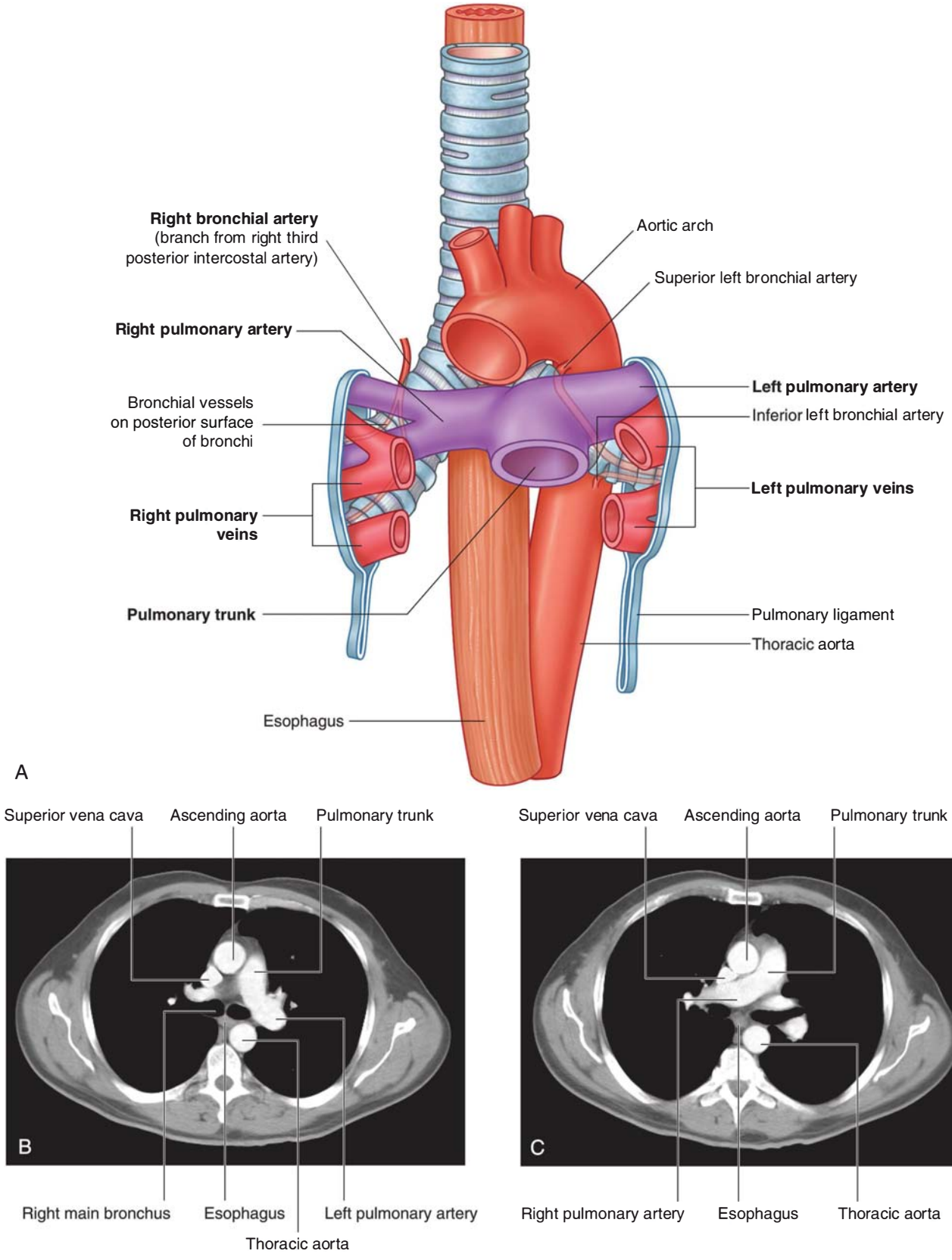


Fig. 3.45 Pulmonary vessels. **A.** Diagram of an anterior view. **B.** Axial computed tomography image showing the left pulmonary artery branching from the pulmonary trunk. **C.** Axial computed tomography image (just inferior to the image in **B**) showing the right pulmonary artery branching from the pulmonary trunk.

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Innervation

Structures of the lung, and the visceral pleura, are supplied by visceral afferents and efferents distributed through the anterior pulmonary plexus and posterior pulmonary plexus (Fig. 3.46). These interconnected plexuses lie anteriorly and posteriorly to the tracheal bifurcation and main bronchi. The anterior plexus is much smaller than the posterior plexus.

Branches of these plexuses, which ultimately originate from the sympathetic trunks and vagus nerves, are distributed along branches of the airway and vessels.

Visceral efferents from:

- the vagus nerves constrict the bronchioles;
- the sympathetic system dilates the bronchioles.

Lymphatic drainage

Superficial, or subpleural, and deep lymphatics of the lung drain into lymph nodes called **tracheobronchial nodes** around the roots of lobar and main bronchi and along the sides of the trachea (Fig. 3.47). As a group, these lymph nodes extend from within the lung, through the hilum and root, and into the posterior mediastinum.

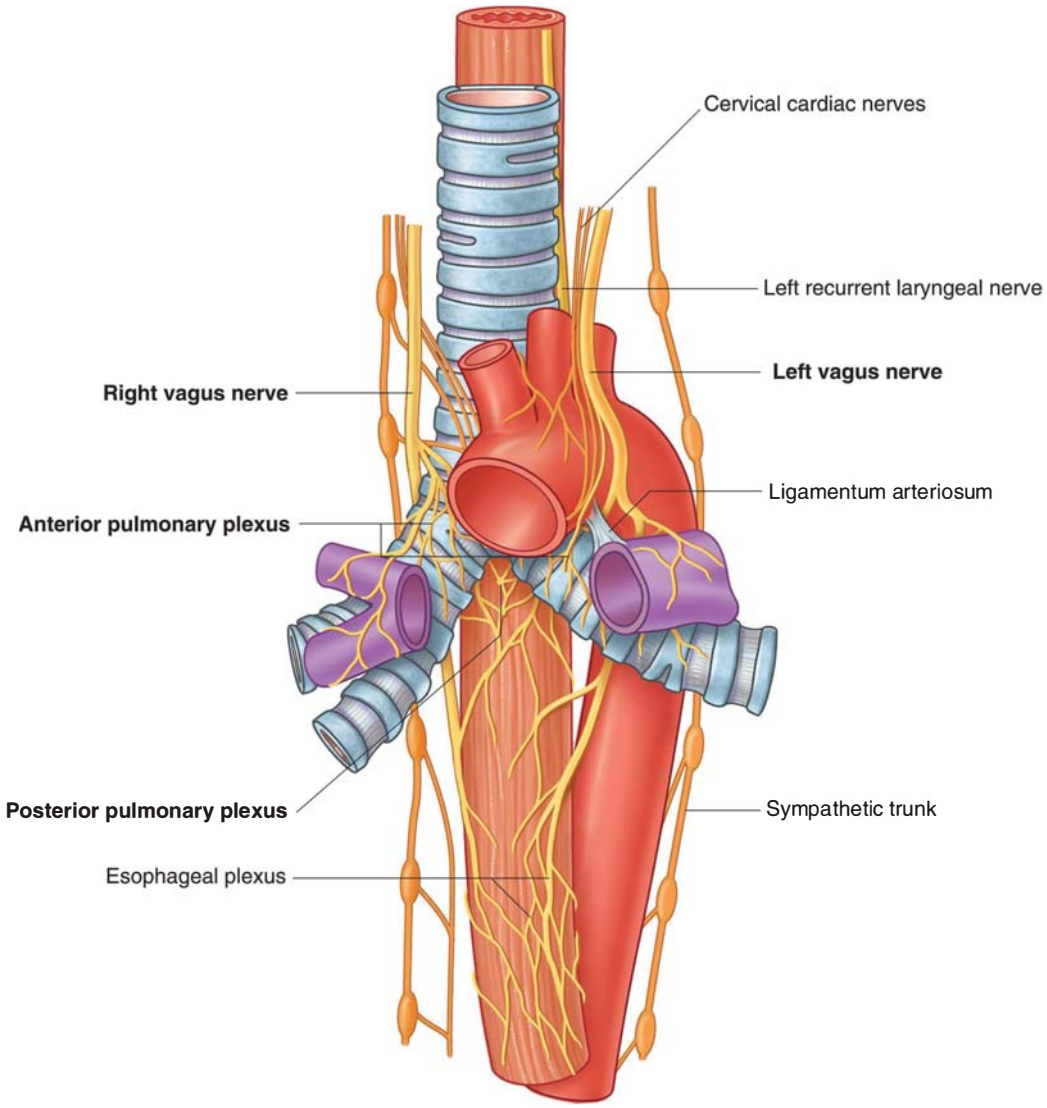


Fig. 3.46 Pulmonary innervation.

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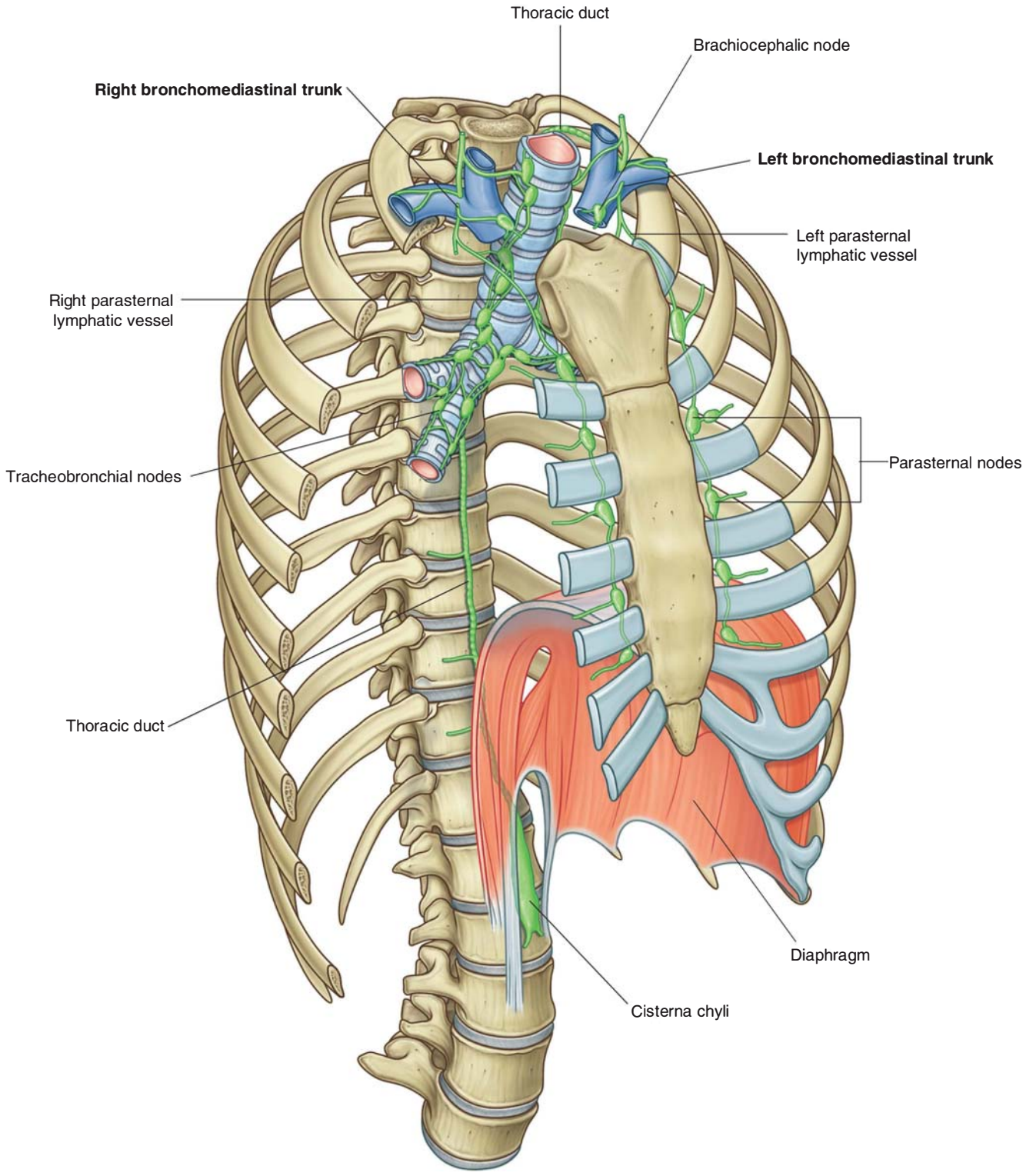


Fig. 3.47 Lymphatic drainage of lungs.

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Efferent vessels from these nodes pass superiorly along the trachea to unite with similar vessels from parasternal nodes and brachiocephalic nodes, which are anterior to brachiocephalic veins in the superior mediastinum, to form the **right** and **left bronchomediastinal trunks**. These trunks drain directly into deep veins at the base of the neck, or may drain into the right lymphatic trunk or thoracic duct.

In the clinic

Imaging the lungs

Medical imaging of the lungs is important because they are one of the commonest sites for disease in the body. While the body is at rest, the lungs exchange up to 5 L of air per minute, and this may contain pathogens and other potentially harmful elements (e.g., allergens). Techniques to visualize the lung range from plain chest radiographs to high-resolution computed tomography (CT), which enables precise localization of a lesion within the lung.

In the clinic

High-resolution lung CT

High-resolution computed tomography (HRCT) is a diagnostic method for assessing the lungs but more specifically the interstitium of the lungs. The technique involves obtaining narrow cross-sectional slices of 1 to 2 mm. These scans enable the physician and radiologist to view the patterns of disease and their distribution. Diseases that may be easily demonstrated using this procedure include emphysema, pneumoconiosis (coal worker's pneumoconiosis), and asbestosis.

In the clinic

Bronchoscopy

Patients who have an endobronchial lesion (i.e., a lesion within a bronchus) may undergo bronchoscopic evaluation of the trachea and its main branches (Fig. 3.48). The bronchoscope is passed through the nose into the oropharynx and is then directed by a control system past the vocal cords into the trachea. The bronchi are inspected and, if necessary, small biopsies are obtained.

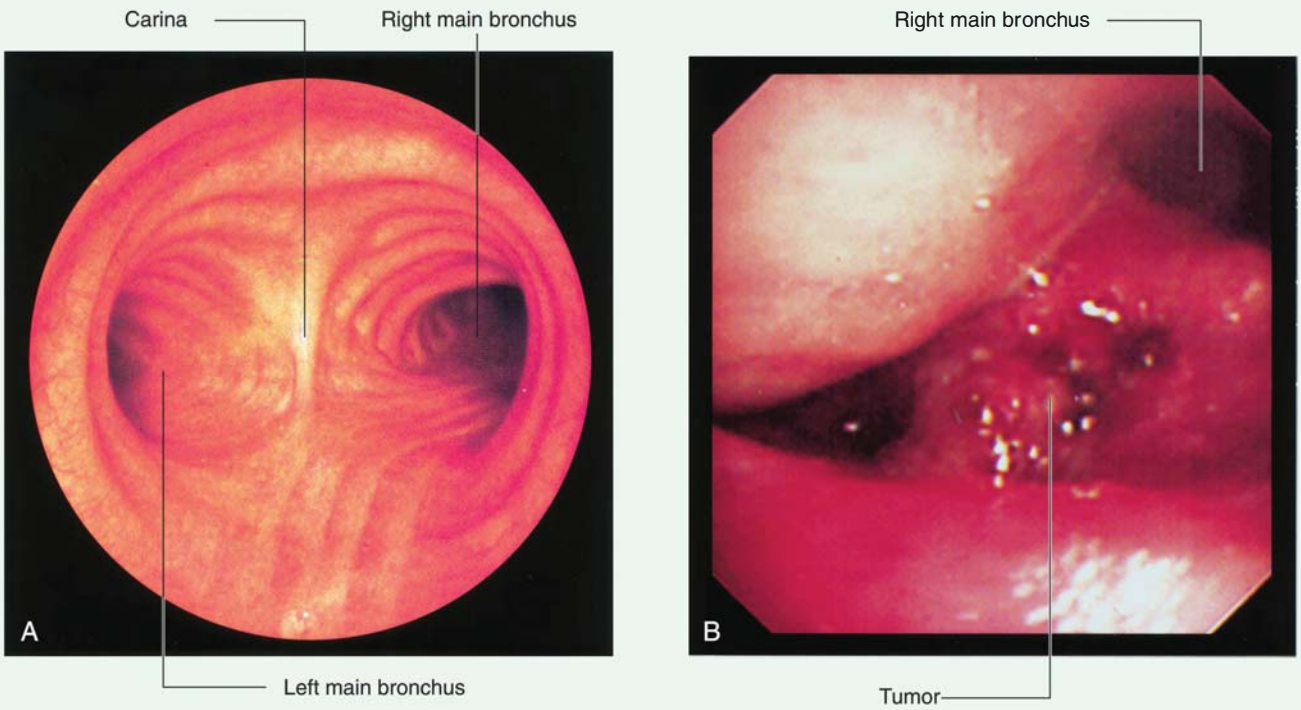


Fig. 3.48 Bronchoscopic evaluation. **A.** Of the lower end of the trachea and its main branches. **B.** Of tracheal bifurcation showing a tumor at the carina.

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In the clinic

Lung cancer

It is important to stage lung cancer because the treatment depends on its stage.

If a small malignant nodule is found within the lung, it can sometimes be excised and the prognosis is excellent. Unfortunately, many patients present with a tumor mass that has invaded structures in the mediastinum or the pleurae or has metastasized. The tumor may then be inoperable and is treated with radiotherapy and chemotherapy.

Spread of the tumor is by lymphatics to lymph nodes within the hila, mediastinum, and root of the neck.

A key factor affecting the prognosis and ability to cure the disease is the distant spread of metastases. Imaging methods to assess spread include plain radiography (Fig. 3.49A), computed tomography (CT; Fig. 3.49B), and magnetic resonance imaging (MRI). Increasingly, radionuclide studies using fluorodeoxyglucose positron emission tomography (FDG PET; Fig. 3.49C) are being used.

In FDG PET a gamma radiation emitter is attached to a glucose molecule. In areas of high metabolic activity (i.e., the tumor), excessive uptake occurs and is recorded by a gamma camera.

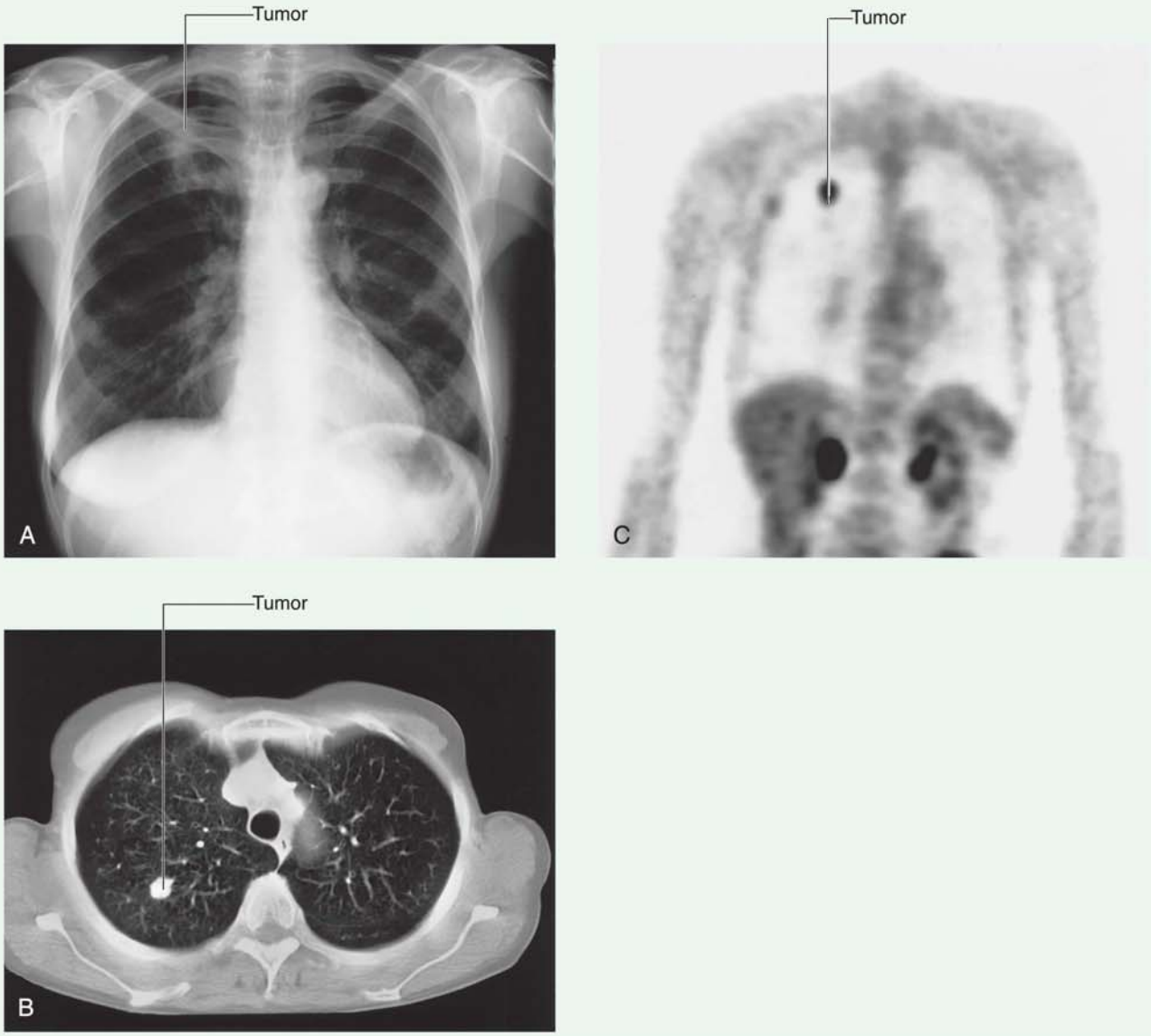


Fig. 3.49 Imaging of the lungs. **A.** Standard posteroanterior view of the chest showing tumor in upper right lung. **B.** Axial CT image of lungs showing tumor in right lung. **C.** Radionuclide study using FDG PET showing a tumor in the right lung.

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MEDIASTINUM

The **mediastinum** is a broad central partition that separates the two laterally placed pleural cavities (Fig. 3.50). It extends:

- from the sternum to the bodies of the vertebrae, and
- from the superior thoracic aperture to the diaphragm (Fig. 3.51).

The mediastinum contains the thymus gland, the pericardial sac, the heart, the trachea, and the major arteries and veins.

Additionally, the mediastinum serves as a passageway for structures such as the esophagus, thoracic duct, and various components of the nervous system as they traverse the thorax on their way to the abdomen.

For organizational purposes, the mediastinum is subdivided into several smaller regions. A transverse plane extending from the sternal angle (the junction between the manubrium and the body of the sternum) to the intervertebral disc between vertebrae TIV and TV separates the mediastinum into the:

- **superior mediastinum**, and
- **inferior mediastinum**, which is further partitioned into the **anterior**, **middle**, and **posterior mediastinum** by the pericardial sac.

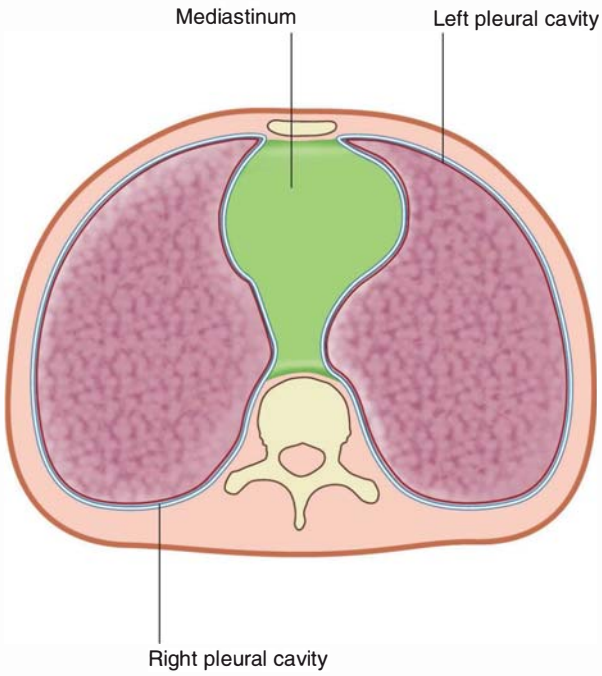


Fig. 3.50 Cross-section of the thorax showing the position of the mediastinum.

The area anterior to the pericardial sac and posterior to the body of the sternum is the anterior mediastinum. The region posterior to the pericardial sac and the diaphragm and anterior to the bodies of the vertebrae is the posterior mediastinum. The area in the middle, which includes the pericardial sac and its contents, is the middle mediastinum (Fig. 3.52).

Middle mediastinum

The **middle mediastinum** is centrally located in the thoracic cavity. It contains the pericardium, heart, origins of the great vessels, various nerves, and smaller vessels.

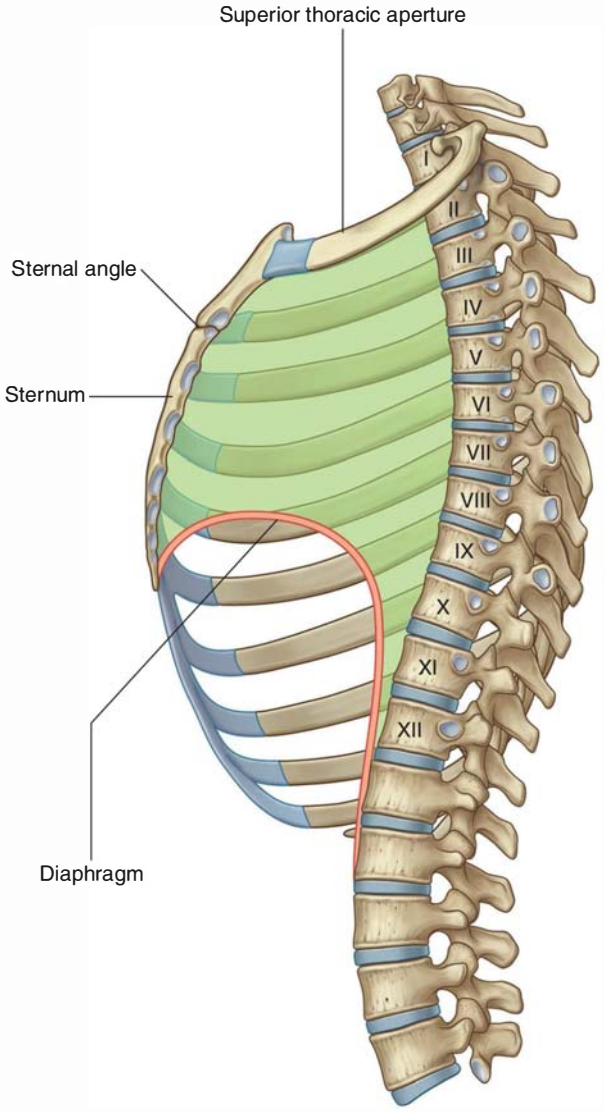


Fig. 3.51 Lateral view of the mediastinum.

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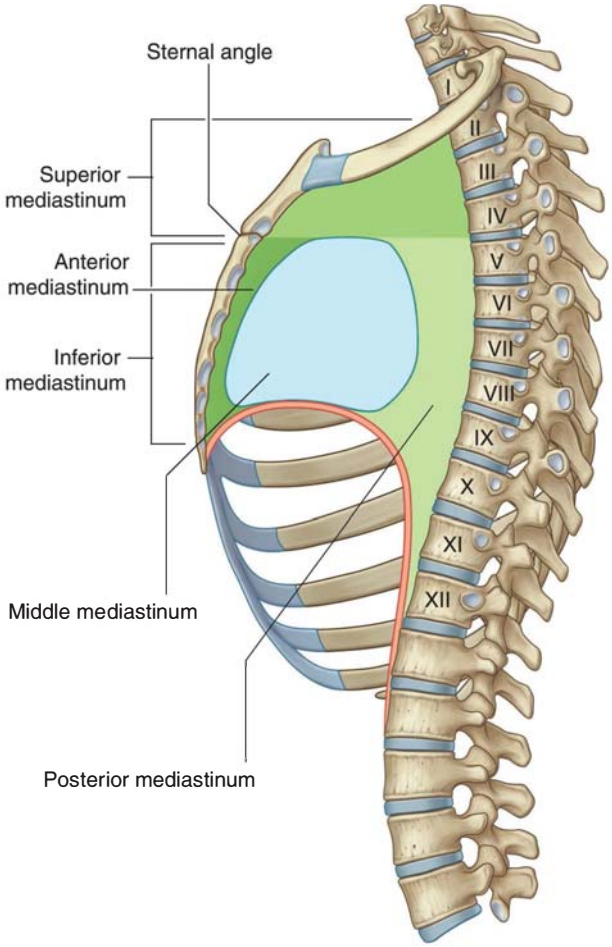


Fig. 3.52 Subdivisions of the mediastinum.

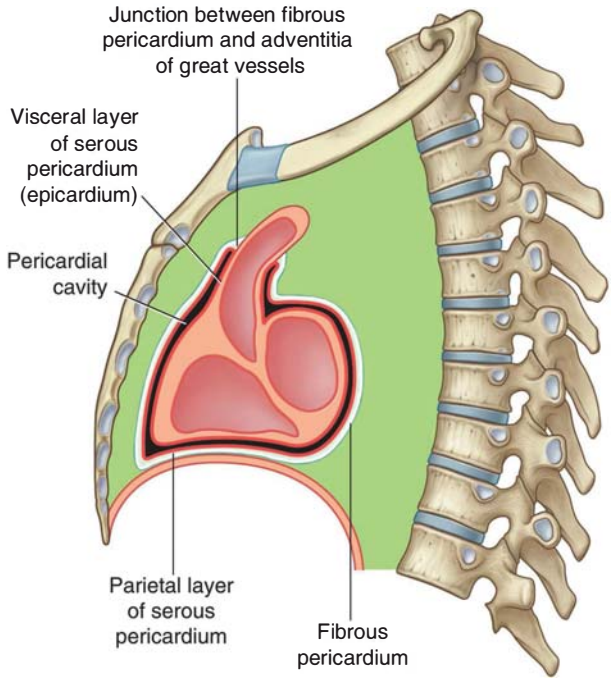


Fig. 3.53 Sagittal section of the pericardium.

Pericardium

The **pericardium** is a fibroserous sac surrounding the heart and the roots of the great vessels. It consists of two components, the fibrous pericardium and the serous pericardium (Fig. 3.53).

The **fibrous pericardium** is a tough connective tissue outer layer that defines the boundaries of the middle mediastinum. The **serous pericardium** is thin and consists of two parts:

- The **parietal layer** of serous pericardium lines the inner surface of the fibrous pericardium.
- The **visceral layer** (**epicardium**) of serous pericardium adheres to the heart and forms its outer covering.

The parietal and visceral layers of serous pericardium are continuous at the roots of the great vessels. The narrow space created between the two layers of serous pericardium, containing a small amount of fluid, is the **pericardial cavity**. This potential space allows for the relatively uninhibited movement of the heart.

Fibrous pericardium

The **fibrous pericardium** is a cone-shaped bag with its base on the diaphragm and its apex continuous with the **adventitia** of the great vessels (Fig. 3.53). The base is attached to the **central tendon of the diaphragm** and to a small muscular area of the diaphragm on the left side. Anteriorly, it is attached to the posterior surface of the sternum by **sternopericardial ligaments**. These attachments help to retain the heart in its position in the thoracic cavity. The sac also limits cardiac distention.

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The phrenic nerves, which innervate the diaphragm and originate from spinal cord levels C3 to C5, pass through the fibrous pericardium and innervate the fibrous pericardium as they travel from their point of origin to their final destination (Fig. 3.54). Their location, within the fibrous pericardium, is directly related to the embryological origin of the diaphragm and the changes that occur during the formation of the pericardial cavity. Similarly, the **pericardiacophrenic vessels** are also located within and supply the fibrous pericardium as they pass through the thoracic cavity.

Serous pericardium

The parietal layer of serous pericardium is continuous with the visceral layer of serous pericardium around the roots

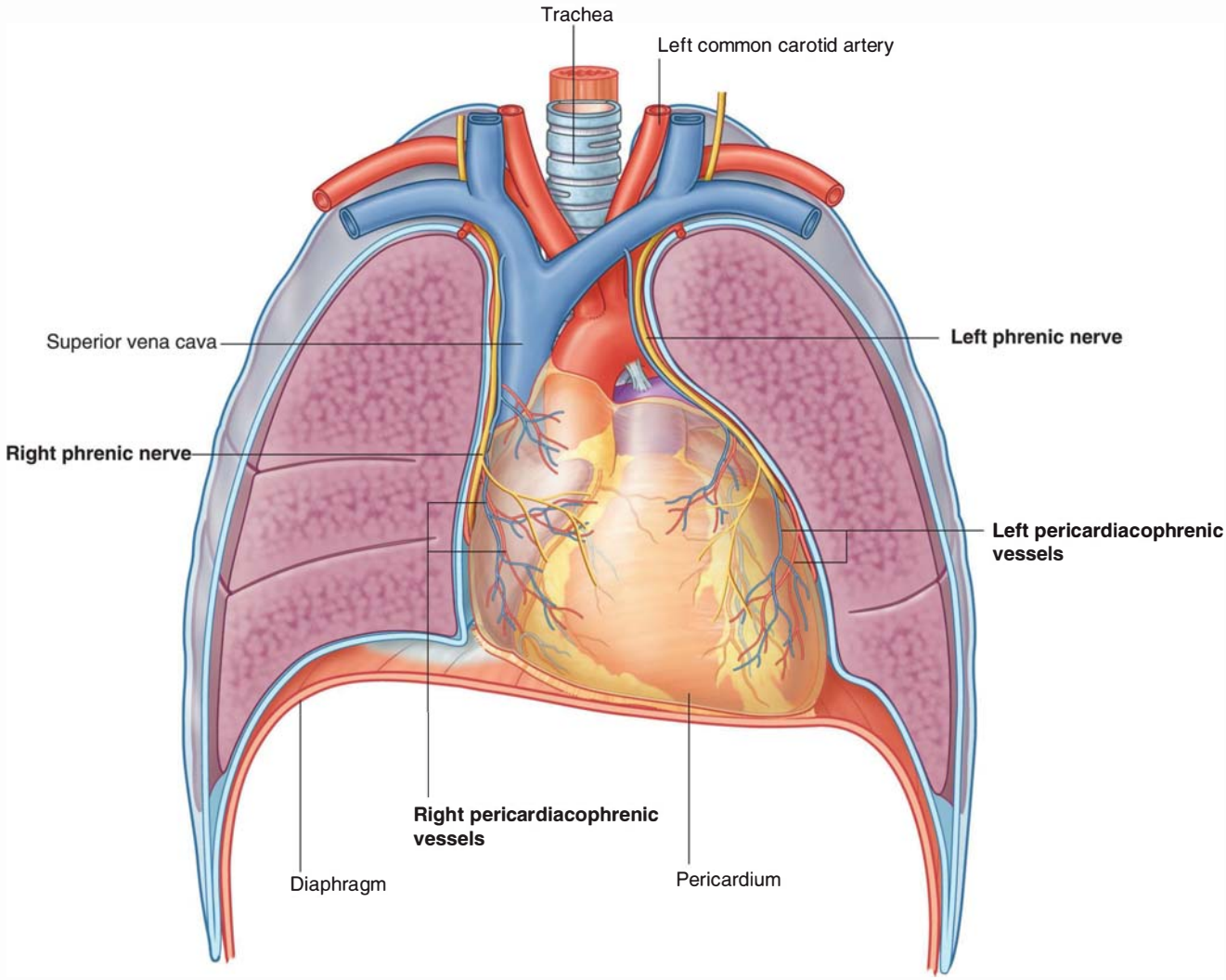


Fig. 3.54 Phrenic nerves and pericardiacophrenic vessels.

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of the great vessels. These reflections of serous pericardium (Fig. 3.55) occur in two locations:

- one superiorly, surrounding the arteries, the aorta, and the pulmonary trunk;
- the second more posteriorly, surrounding the veins, the superior and inferior vena cava, and the pulmonary veins.

The zone of reflection surrounding the veins is J-shaped, and the cul-de-sac formed within the J, posterior to the left atrium, is the **oblique pericardial sinus**.

A passage between the two sites of reflected serous pericardium is the **transverse pericardial sinus**. This sinus lies posterior to the ascending aorta and the pulmonary trunk, anterior to the superior vena cava, and superior to the left atrium.

When the pericardium is opened anteriorly during surgery, a finger placed in the transverse sinus separates arteries from veins. A hand placed under the apex of the heart and moved superiorly slips into the oblique sinus.

Vessels and nerves

The pericardium is supplied by branches from the internal thoracic, pericardiacophrenic, musculophrenic, and inferior phrenic arteries, and the thoracic aorta.

Veins from the pericardium enter the azygos system of veins and the internal thoracic and superior phrenic veins.

Nerves supplying the pericardium arise from the vagus nerve [X], the sympathetic trunks, and the phrenic nerves.

It is important to note that the source of somatic sensation (pain) from the parietal pericardium is carried by somatic afferent fibers in the phrenic nerves. For this reason, "pain" related to a pericardial problem may be referred to the supraclavicular region of the shoulder or lateral neck area dermatomes for spinal cord segments C3, C4, and C5.

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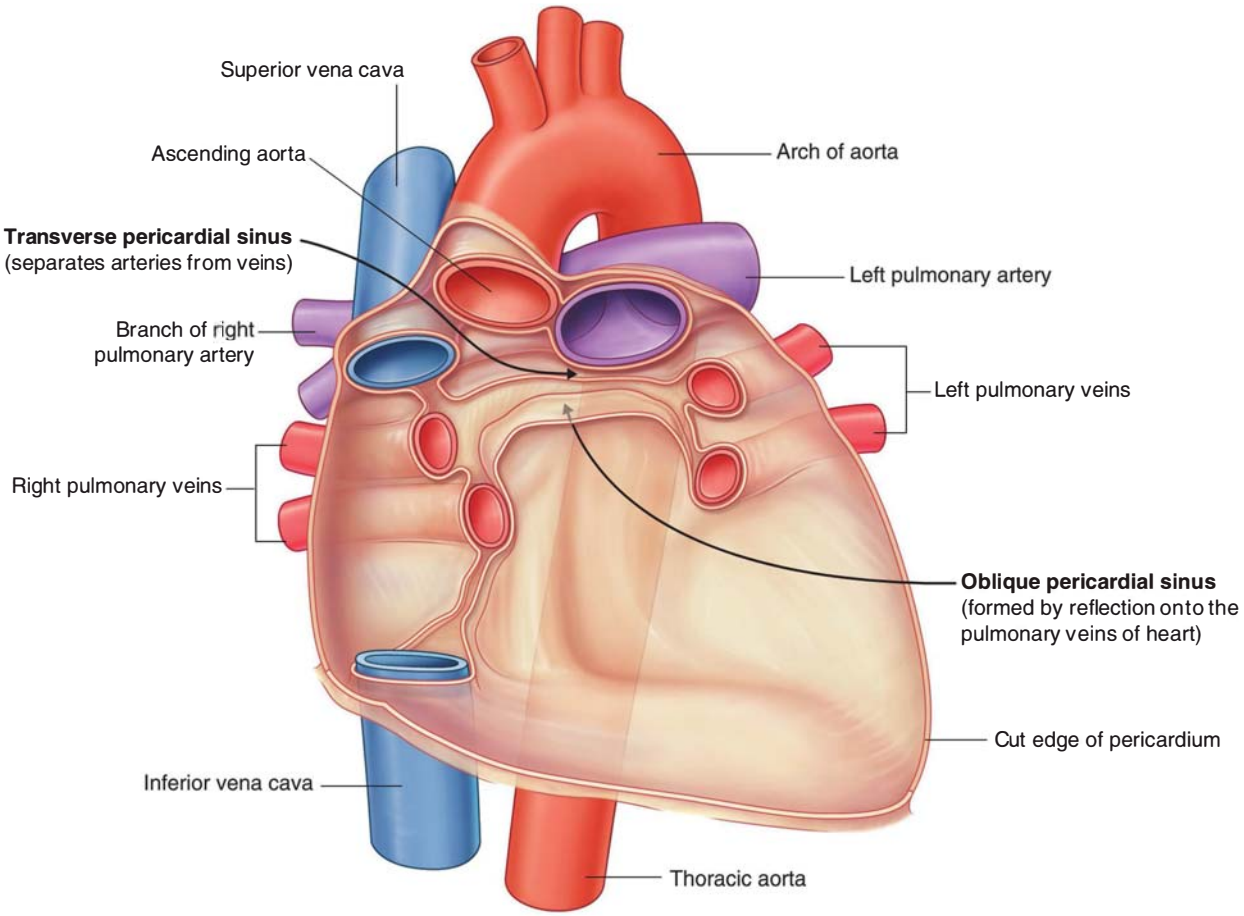


Fig. 3.55 Posterior portion of pericardial sac showing reflections of serous pericardium.

In the clinic

Pericarditis

Pericarditis is an inflammatory condition of the pericardium. Common causes are viral and bacterial infections, systemic illnesses (e.g., chronic renal failure), and after myocardial infarction.

Pericarditis must be distinguished from myocardial infarction because the treatment and prognosis are quite different. As in patients with myocardial infarction, patients with pericarditis complain of continuous central chest pain that may radiate to one or both arms. Unlike myocardial infarction, however, the pain from pericarditis may be relieved by sitting forward. An electrocardiogram (ECG) is used to help differentiate between the two conditions.

In the clinic

Pericardial effusion

Normally, only a tiny amount of fluid is present between the visceral and parietal layers of the serous pericardium. In certain situations, this space can be filled with excess fluid (pericardial effusion).

Because the fibrous pericardium is a "relatively fixed" structure that cannot expand easily, a rapid accumulation of excess fluid within the pericardial sac compresses the heart (cardiac tamponade), resulting in biventricular failure. Removing the fluid with a needle inserted into the pericardial sac can relieve the symptoms.

In the clinic

Constrictive pericarditis

Abnormal thickening of the pericardial sac (constrictive pericarditis) can compress the heart, impairing heart function and resulting in heart failure. The diagnosis is made by inspecting the jugular venous pulse in the neck. In normal individuals, the jugular venous pulse drops on inspiration. In patients with constrictive pericarditis, the reverse happens and this is called Kussmaul's sign. Treatment often involves surgical opening of the pericardial sac.

Heart

Cardiac orientation

The general shape and orientation of the heart are that of a pyramid that has fallen over and is resting on one of its sides. Placed in the thoracic cavity, the apex of this pyramid projects forward, downward, and to the left, whereas the base is opposite the apex and faces in a posterior direction (Fig. 3.56). The sides of the pyramid consist of:

- a diaphragmatic (inferior) surface on which the pyramid rests,
- an anterior (sternocostal) surface oriented anteriorly,
- a right pulmonary surface, and
- a left pulmonary surface.

Base (posterior surface) and apex

The **base of the heart** is quadrilateral and directed posteriorly. It consists of:

- the left atrium,
- a small portion of the right atrium, and
- the proximal parts of the great veins (superior and inferior venae cavae and the pulmonary veins) (Fig. 3.57).

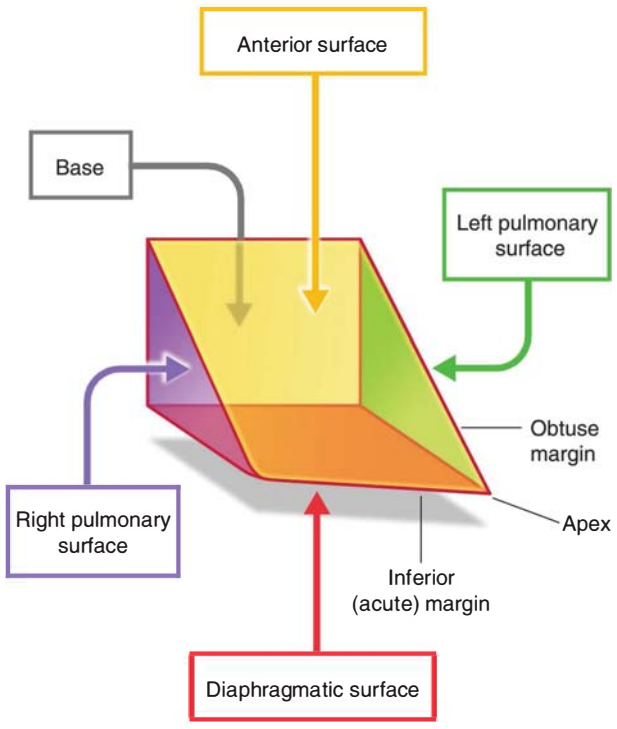


Fig. 3.56 Schematic illustration of the heart showing orientation, surfaces, and margins.

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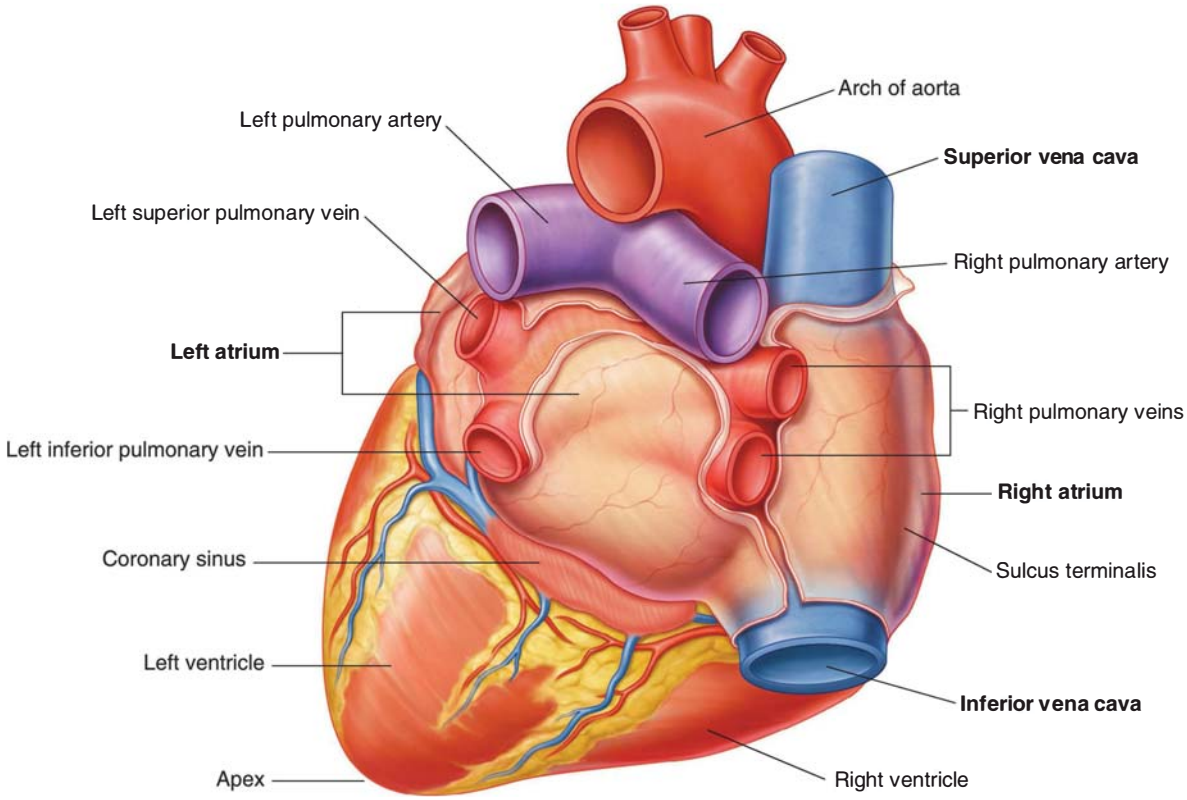


Fig. 3.57 Base of the heart.

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Because the great veins enter the base of the heart, with the pulmonary veins entering the right and left sides of the left atrium and the superior and inferior venae cavae at the upper and lower ends of the right atrium, the base of the heart is fixed posteriorly to the pericardial wall, opposite the bodies of vertebrae TV to TVIII (TVI to TIX when standing). The esophagus lies immediately posterior to the base.

From the base the heart projects forward, downward, and to the left, ending in the apex. The **apex of the heart** is formed by the inferolateral part of the left ventricle (Fig. 3.58) and is positioned deep to the left fifth intercostal space, 8–9 cm from the midsternal line.

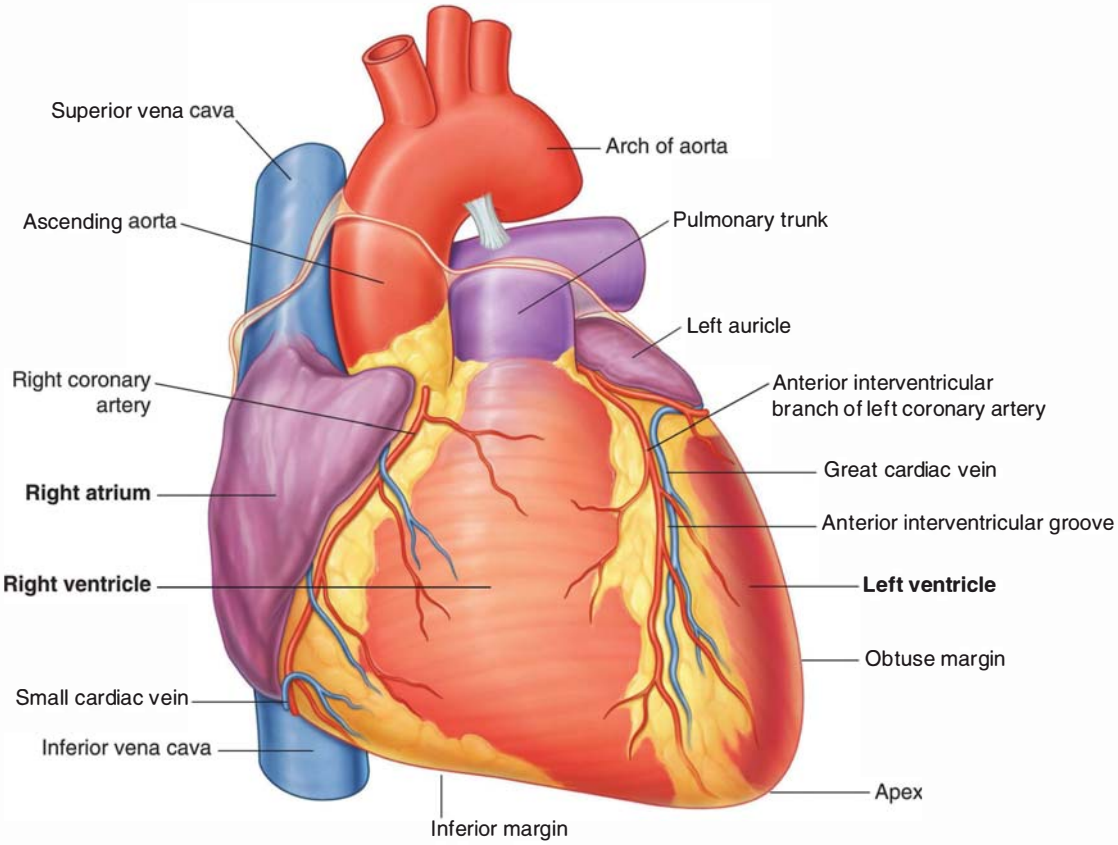


Fig. 3.58 Anterior surface of the heart.

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Surfaces of the heart

The **anterior surface** faces anteriorly and consists mostly of the right ventricle, with some of the right atrium on the right and some of the left ventricle on the left (Fig. 3.58).

The heart in the anatomical position rests on the **diaphragmatic surface**, which consists of the left ventricle and a small portion of the right ventricle separated by the posterior interventricular groove (Fig. 3.59). This surface faces inferiorly, rests on the diaphragm, is separated from the base of the heart by the coronary sinus, and extends from the base to the apex of the heart.

The **left pulmonary surface** faces the left lung, is broad and convex, and consists of the left ventricle and a portion of the left atrium (Fig. 3.59).

The **right pulmonary surface** faces the right lung, is broad and convex, and consists of the right atrium (Fig. 3.59).

Margins and borders

Some general descriptions of cardiac orientation refer to right, left, inferior (acute), and obtuse margins:

- The **right** and **left margins** are the same as the right and left pulmonary surfaces of the heart.
- The **inferior margin** is defined as the sharp edge between the anterior and diaphragmatic surfaces of the heart (Figs 3.56 and 3.58)—it is formed mostly by the right ventricle and a small portion of the left ventricle near the apex.

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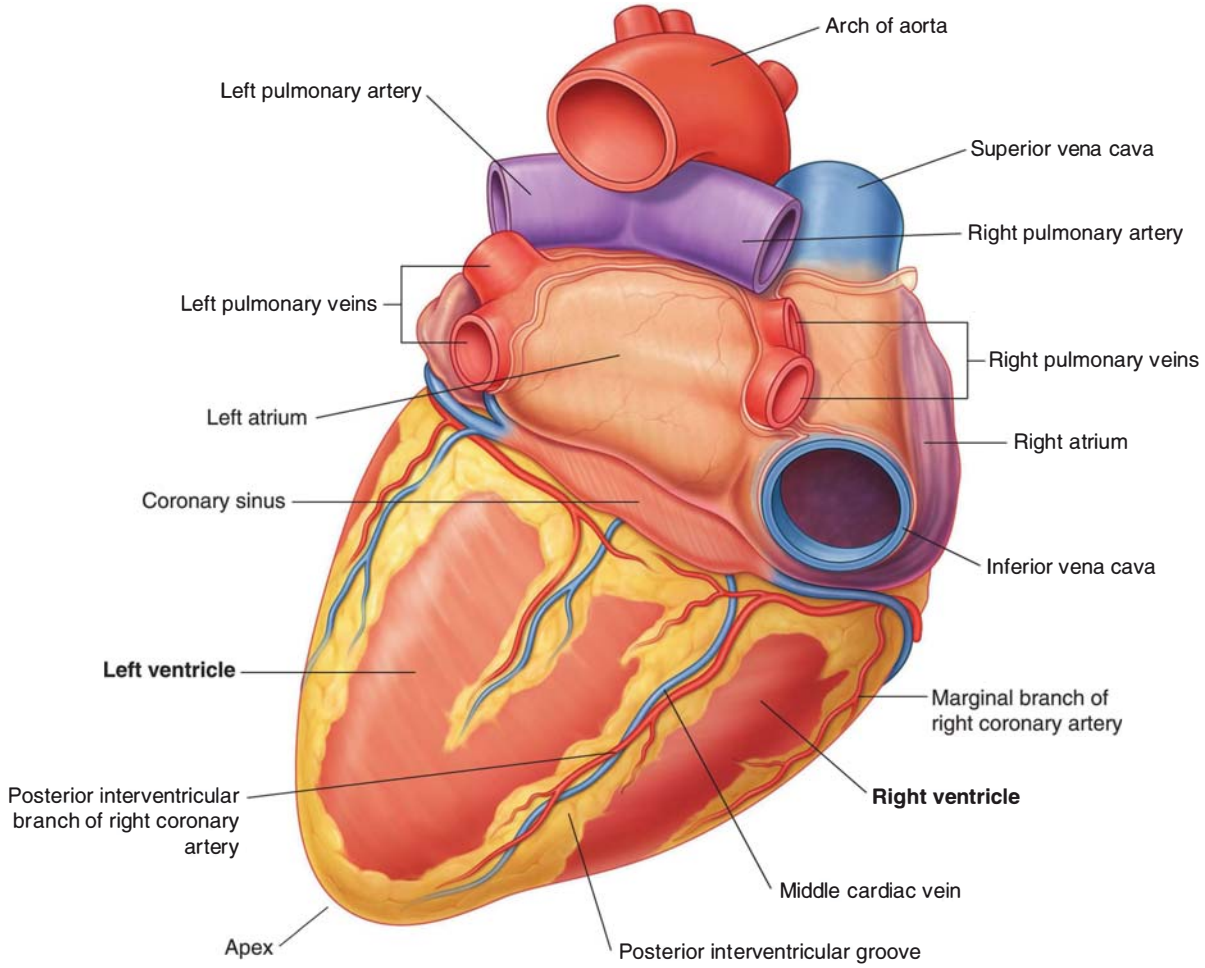


Fig. 3.59 Diaphragmatic surface of the heart.

- The **obtuse margin** separates the anterior and left pulmonary surfaces (Fig. 3.56)—it is round and extends from the left auricle to the cardiac apex (Fig. 3.58), and is formed mostly by the left ventricle and superiorly by a small portion of the left auricle.

For radiological evaluations, a thorough understanding of the structures defining the cardiac borders is critical. The right border in a standard posteroanterior view consists of the superior vena cava, the right atrium, and the inferior vena cava (Fig. 3.60A). The left border in a similar view consists of the arch of the aorta, the pulmonary trunk, and the left ventricle. The inferior border in this radiological study consists of the right ventricle and the left ventricle at the apex. In lateral views, the right ventricle is seen anteriorly, and the left atrium is visualized posteriorly (Fig. 3.60B).

External sulci

Internal partitions divide the heart into four chambers (i.e., two atria and two ventricles) and produce surface or external grooves referred to as sulci.

- The **coronary sulcus** circles the heart, separating the atria from the ventricles (Fig. 3.61). As it circles the heart, it contains the right coronary artery, the small cardiac vein, the coronary sinus, and the circumflex branch of the left coronary artery.
- The **anterior** and **posterior interventricular sulci** separate the two ventricles—the anterior interventricular sulcus is on the anterior surface of the heart and contains the anterior interventricular artery and the great cardiac vein, and the posterior interventricular sulcus is on the diaphragmatic surface of the heart and contains the posterior interventricular artery and the middle cardiac vein.

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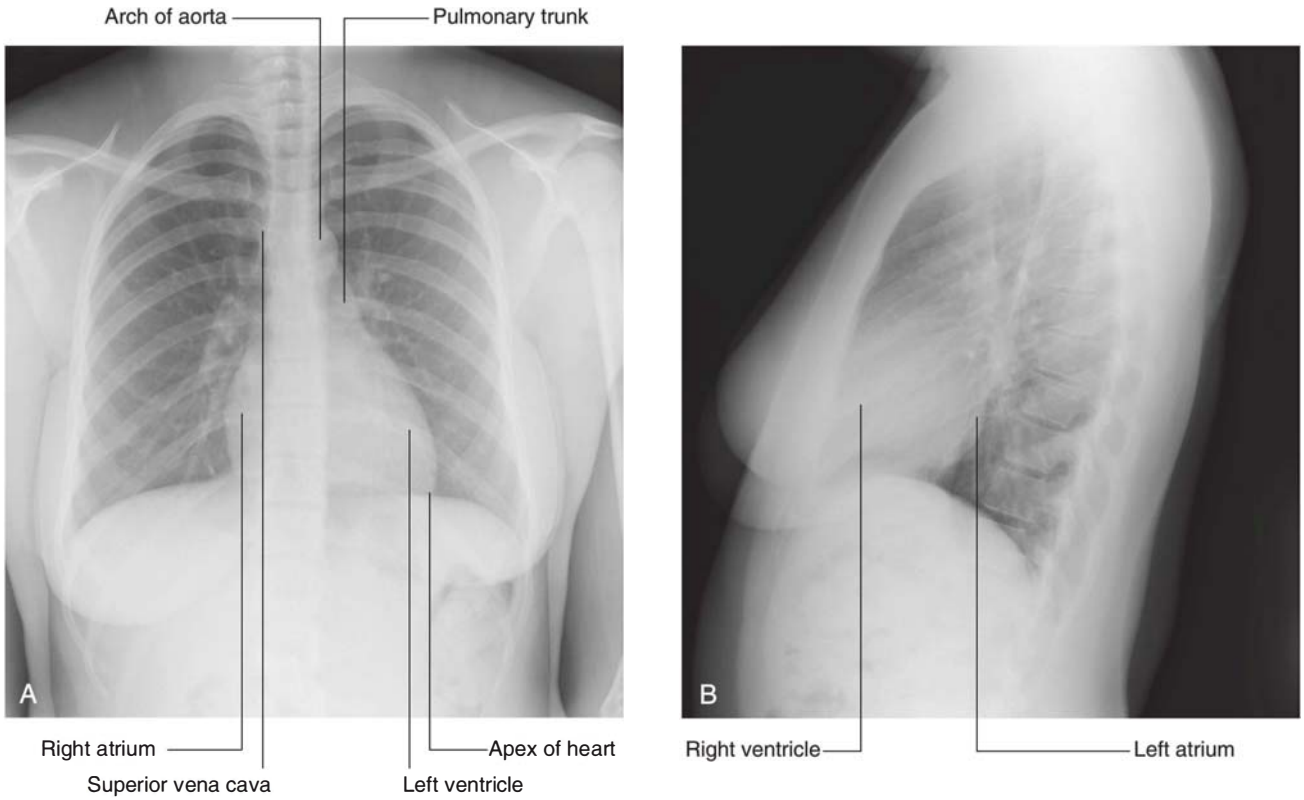


Fig. 3.60 Chest radiographs. **A.** Standard posteroanterior view of the chest. **B.** Standard lateral view of the heart.

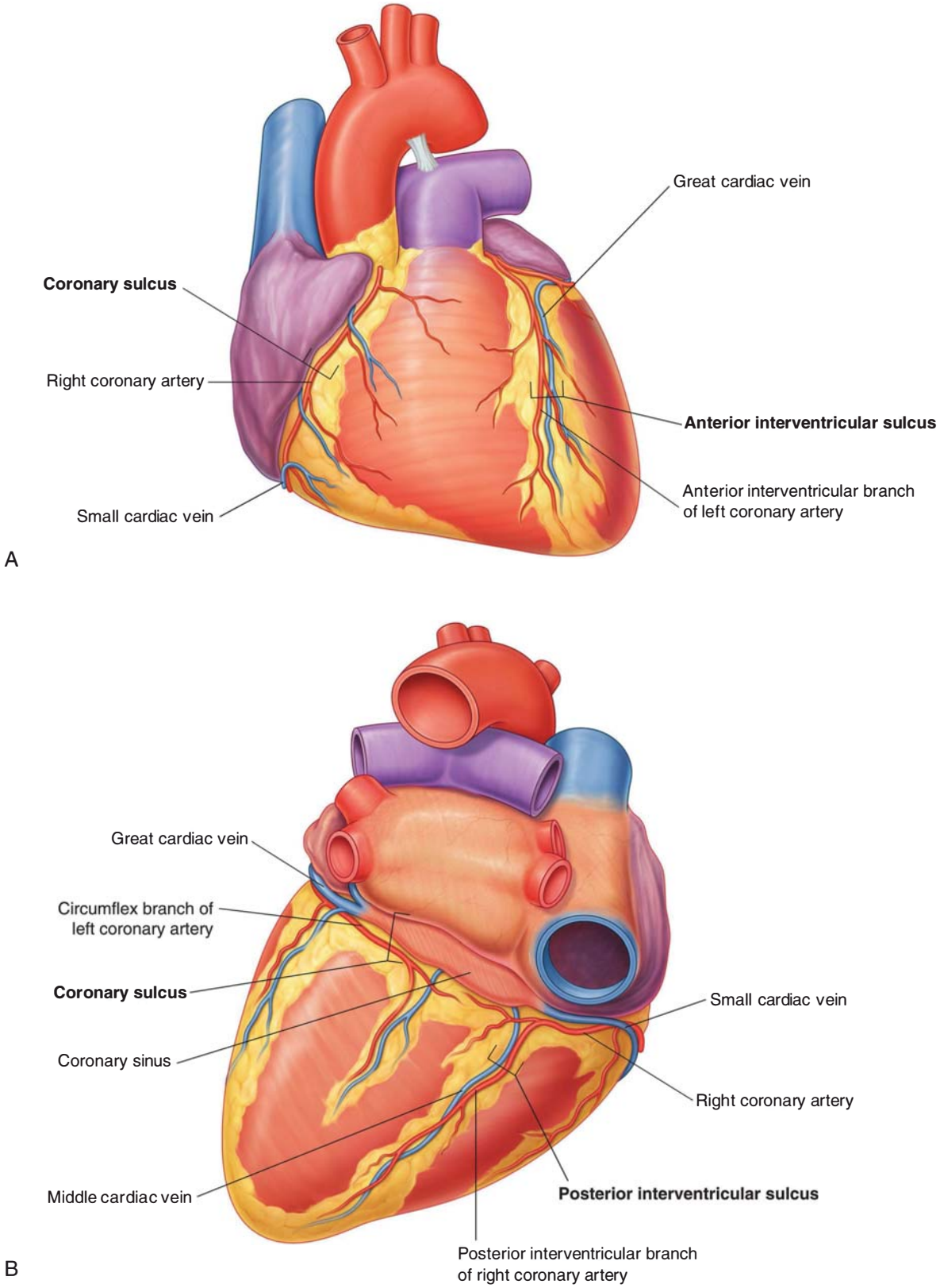


Fig. 3.61 Sulci of the heart. **A.** Anterior surface of the heart. **B.** Diaphragmatic surface and base of the heart.

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These sulci are continuous inferiorly, just to the right of the apex of the heart.

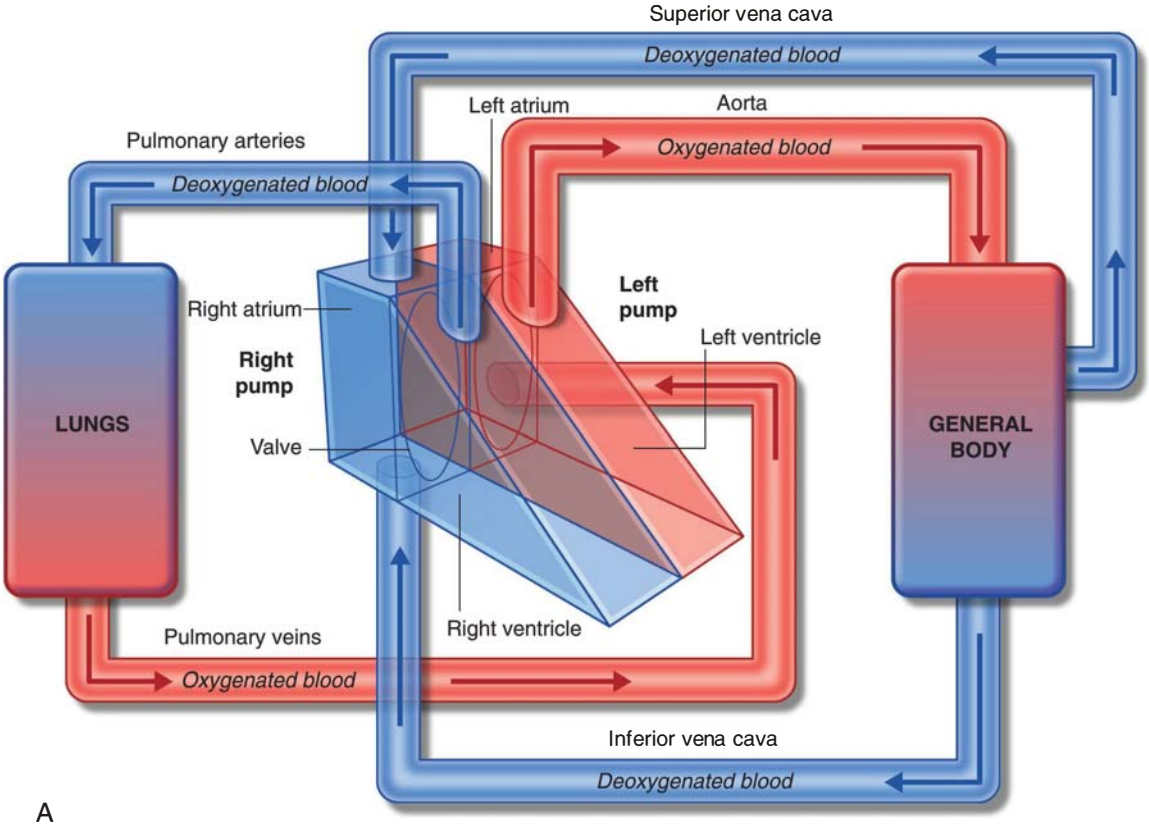
Cardiac chambers

The heart functionally consists of two pumps separated by a partition (Fig. 3.62A). The right pump receives deoxygenated blood from the body and sends it to the lungs. The left pump receives oxygenated blood from the lungs and sends it to the body. Each pump consists of an atrium and a ventricle separated by a valve.

The thin-walled atria receive blood coming into the heart, whereas the relatively thick-walled ventricles pump blood out of the heart.

More force is required to pump blood through the body than through the lungs, so the muscular wall of the left ventricle is thicker than the right.

Interatrial, interventricular, and atrioventricular septa separate the four chambers of the heart (Fig. 3.62B). The internal anatomy of each chamber is critical to its function.



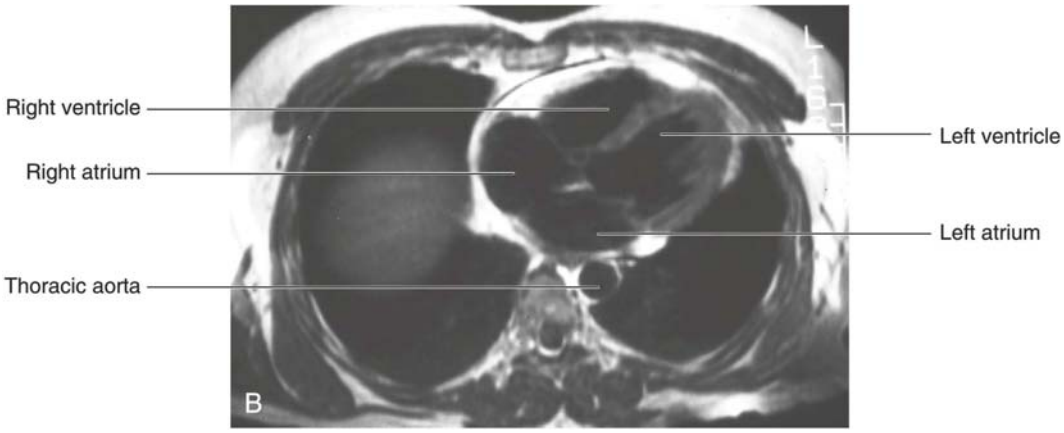


Fig. 3.62 A. The heart has two pumps. **B.** Magnetic resonance image of midthorax showing all four chambers and septa.

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Right atrium

In the anatomical position, the right border of the heart is formed by the **right atrium**. This chamber also contributes to the right portion of the heart's anterior surface.

Blood returning to the right atrium enters through one of three vessels. These are:

- the superior and inferior venae cavae, which together deliver blood to the heart from the body; and
- the coronary sinus, which returns blood from the walls of the heart itself.

The superior vena cava enters the upper posterior portion of the right atrium, and the inferior vena cava and coronary sinus enter the lower posterior portion of the right atrium.

From the right atrium, blood passes into the right ventricle through the **right atrioventricular orifice**. This opening faces forward and medially and is closed during ventricular contraction by the tricuspid valve.

The interior of the right atrium is divided into two continuous spaces. Externally, this separation is indicated by a shallow, vertical groove (the **sulcus terminalis cordis**), which extends from the right side of the opening of the superior vena cava to the right side of the opening of the inferior vena cava. Internally, this division is indicated by the **crista terminalis** (Fig. 3.63), which is a smooth, muscular ridge that begins on the roof of the atrium just in front of the opening of the superior vena cava and extends down the lateral wall to the anterior lip of the inferior vena cava.

The space posterior to the crista is the **sinus of venae cavae** and is derived embryologically from the right horn of the sinus venosus. This component of the right atrium has smooth, thin walls, and both venae cavae empty into this space.

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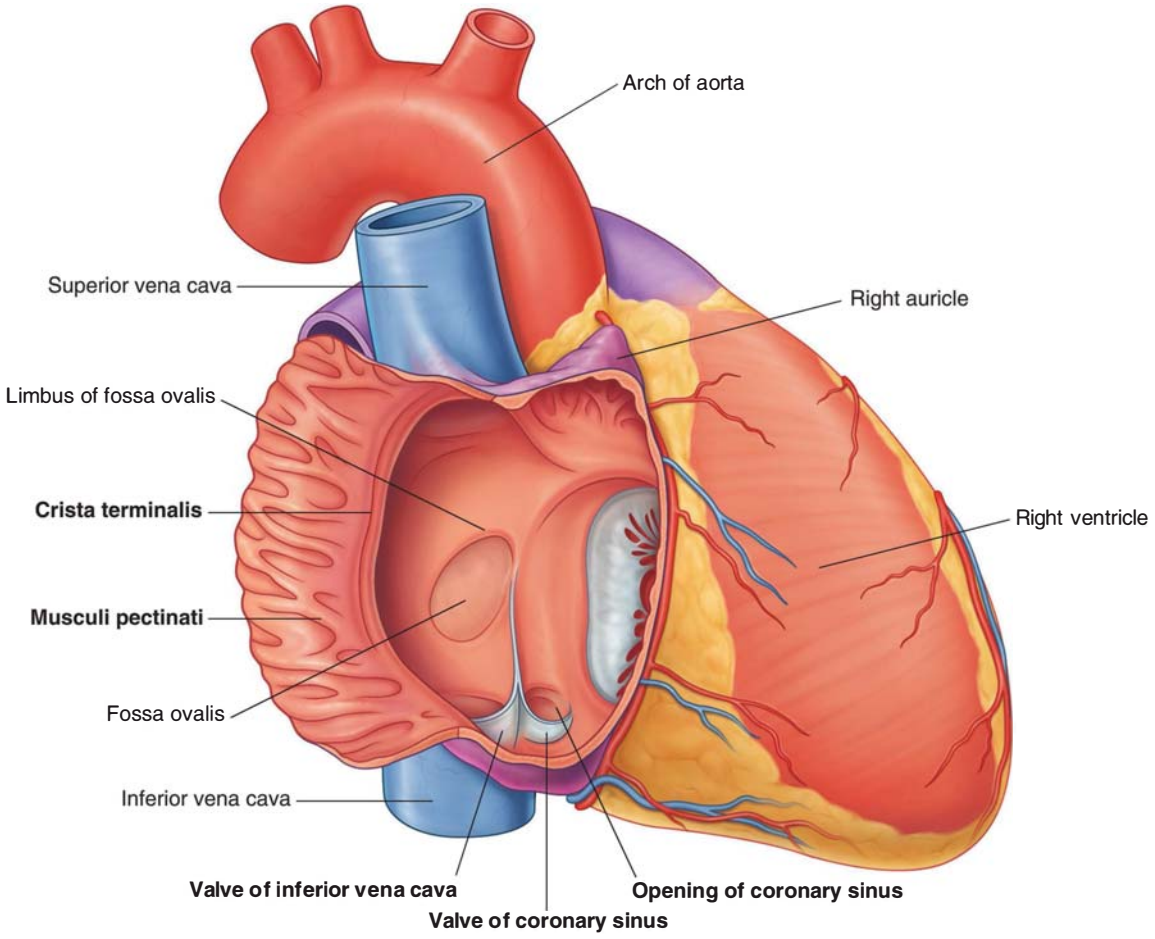


Fig. 3.63 Internal view of right atrium.

The space anterior to the crista, including the **right auricle**, is sometimes referred to as the **atrium proper**. This terminology is based on its origin from the embryonic primitive atrium. Its walls are covered by ridges called the **musculi pectinati** (**pectinate muscles**), which fan out from the crista like the "teeth of a comb." These ridges are also found in the right auricle, which is an ear-like, conical, muscular pouch that externally overlaps the ascending aorta.

An additional structure in the right atrium is the **opening of the coronary sinus**, which receives blood from most of the cardiac veins and opens medially to the **opening of the inferior vena cava**. Associated with these openings are small folds of tissue derived from the valve of the embryonic sinus venosus (the **valve of the coronary sinus** and the **valve of inferior vena cava**, respectively). During development, the valve of the inferior vena cava helps direct incoming oxygenated blood through the foramen ovale and into the left atrium.

Separating the right atrium from the left atrium is the **interatrial septum**, which faces forward and to the right because the left atrium lies posteriorly and to the left of the right atrium. A depression is clearly visible in the septum just above the orifice of the inferior vena cava. This is the **fossa ovalis** (**oval fossa**), with its prominent margin, the **limbus fossa ovalis** (**border of the oval fossa**).

The fossa ovalis marks the location of the embryonic **foramen ovale**, which is an important part of fetal circulation. The foramen ovale allows oxygenated blood entering the right atrium through the inferior vena cava to pass directly to the left atrium and so bypass the lungs, which are nonfunctional before birth.

Finally, numerous small openings—the **openings of the smallest cardiac veins** (the **foramina of the venae cordis minimae**)—are scattered along the walls of the right atrium. These are small veins that drain the myocardium directly into the right atrium.

Right ventricle

In the anatomical position, the right ventricle forms most of the anterior surface of the heart and a portion of the diaphragmatic surface. The right atrium is to the right of the right ventricle and the right ventricle is located in front of and to the left of the right atrioventricular orifice. Blood entering the right ventricle from the right atrium therefore moves in a horizontal and forward direction.

The outflow tract of the right ventricle, which leads to the pulmonary trunk, is the **conus arteriosus** (**infundibulum**). This area has smooth walls and derives from the embryonic bulbus cordis.

The walls of the inflow portion of the right ventricle have numerous muscular, irregular structures called **trabeculae carneae** (Fig. 3.64). Most of these are either attached to the ventricular walls throughout their length, forming ridges, or attached at both ends, forming bridges.

A few trabeculae carneae (**papillary muscles**) have only one end attached to the ventricular surface, while the other end serves as the point of attachment for tendon-like fibrous cords (the **chordae tendineae**), which connect to the free edges of the cusps of the tricuspid valve.

There are three papillary muscles in the right ventricle. Named relative to their point of origin on the ventricular surface, they are the anterior, posterior, and septal papillary muscles:

- The **anterior papillary muscle** is the largest and most constant papillary muscle, and arises from the anterior wall of the ventricle.
- The **posterior papillary muscle** may consist of one, two, or three structures, with some chordae tendineae arising directly from the ventricular wall.
- The **septal papillary muscle** is the most inconsistent papillary muscle, being either small or absent, with chordae tendineae emerging directly from the septal wall.

A single specialized trabeculum, the **septomarginal trabecula** (**moderator band**), forms a bridge between the lower portion of the **interventricular septum** and the base of the anterior papillary muscle. The septomarginal trabecula carries a portion of the cardiac conduction system, the right bundle of the atrioventricular bundle, to the anterior wall of the right ventricle.

Tricuspid valve

The right atrioventricular orifice is closed during ventricular contraction by the **tricuspid valve** (**right atrioventricular valve**), so named because it usually consists of three cusps or leaflets (Fig. 3.64). The base of each cusp is secured to the fibrous ring that surrounds the atrioventricular orifice. This fibrous ring helps to maintain the shape of the opening. The cusps are continuous with each other near their bases at sites termed **commissures**.

The naming of the three cusps, the **anterior**, **septal**, and **posterior cusps**, is based on their relative position in the right ventricle. The free margins of the cusps are attached to the chordae tendineae, which arise from the tips of the papillary muscles.

During filling of the right ventricle, the tricuspid valve is open, and the three cusps project into the right ventricle.

Without the presence of a compensating mechanism, when the ventricular musculature contracts, the valve cusps would be forced upward with the flow of blood and

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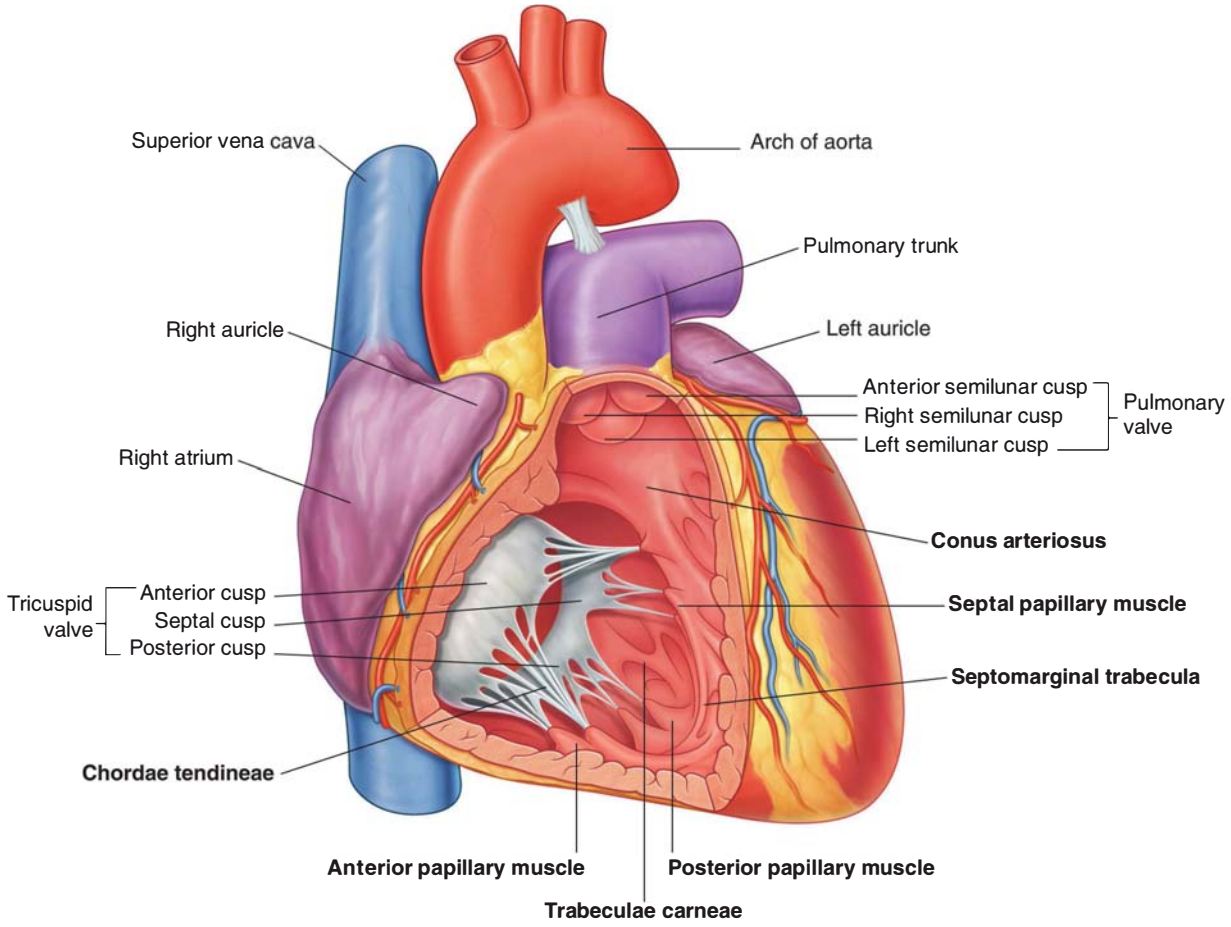


Fig. 3.64 Internal view of the right ventricle.

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blood would move back into the right atrium. However, contraction of the papillary muscles attached to the cusps by chordae tendineae prevents the cusps from being everted into the right atrium.

Simply put, the papillary muscles and associated chordae tendineae keep the valves closed during the dramatic changes in ventricular size that occur during contraction.

In addition, chordae tendineae from two papillary muscles attach to each cusp. This helps prevent separation of the cusps during ventricular contraction. Proper closing of the tricuspid valve causes blood to exit the right ventricle and move into the pulmonary trunk.

Necrosis of a papillary muscle following a myocardial infarction (heart attack) may result in prolapse of the related valve.

Pulmonary valve

At the apex of the infundibulum, the outflow tract of the right ventricle, the opening into the pulmonary trunk is closed by the **pulmonary valve** (Fig. 3.64), which consists of three **semilunar cusps** with free edges projecting upward into the lumen of the pulmonary trunk. The free superior edge of each cusp has a middle, thickened portion, the **nodule of the semilunar cusp**, and a thin lateral portion, the **lunula of the semilunar cusp** (Fig. 3.65).

The cusps are named the **left**, **right,** and **anterior semilunar cusps**, relative to their fetal position before rotation of the outflow tracts from the ventricles is complete. Each cusp forms a pocket-like sinus (Fig. 3.65)—a dilation in the wall of the initial portion of the pulmonary trunk. After ventricular contraction, the recoil of blood fills these **pulmonary sinuses** and forces the cusps closed. This prevents blood in the pulmonary trunk from refilling the right ventricle.

Left atrium

The **left atrium** forms most of the base or posterior surface of the heart.

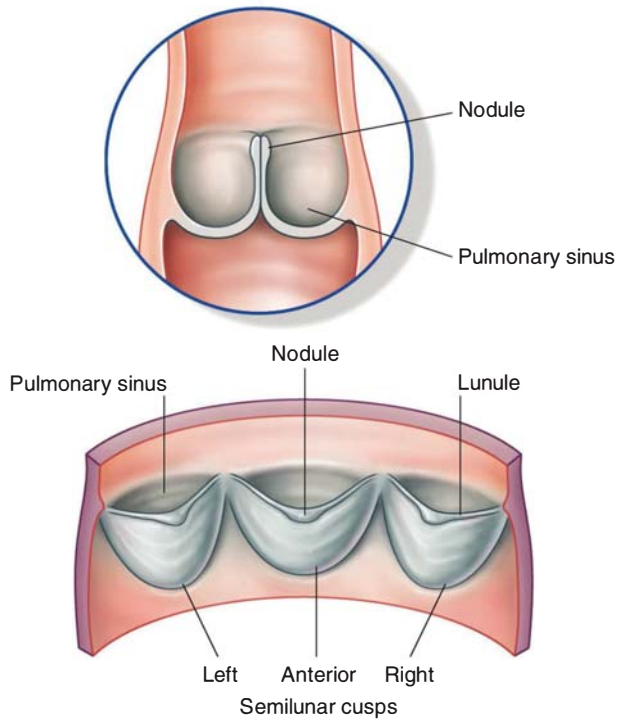


Fig. 3.65 Posterior view of the pulmonary valve.

As with the right atrium, the left atrium is derived embryologically from two structures.

- The posterior half, or inflow portion, receives the four pulmonary veins (Fig. 3.66). It has smooth walls and derives from the proximal parts of the pulmonary veins that are incorporated into the left atrium during development.
- The anterior half is continuous with the left auricle. It contains musculi pectinati and derives from the embryonic primitive atrium. Unlike the crista terminalis in the right atrium, no distinct structure separates the two components of the left atrium.

The interatrial septum is part of the anterior wall of the left atrium. The thin area or depression in the septum is the valve of the foramen ovale and is opposite the floor of the fossa ovalis in the right atrium.

During development, the **valve of the foramen ovale** prevents blood from passing from the left atrium to the right atrium. This valve may not be completely fused in some adults, leaving a "probe patent" passage between the right atrium and the left atrium.

Left ventricle

The left ventricle lies anterior to the left atrium. It contributes to the anterior, diaphragmatic, and left pulmonary surfaces of the heart, and forms the apex.

Blood enters the ventricle through the **left atrioventricular orifice** and flows in a forward direction to the apex. The chamber itself is conical, is longer than the right ventricle, and has the thickest layer of **myocardium**. The outflow tract (the **aortic vestibule**) is posterior to the infundibulum of the right ventricle, has smooth walls, and is derived from the embryonic bulbus cordis.

The **trabeculae carneae** in the left ventricle are fine and delicate in contrast to those in the right ventricle. The general appearance of the trabeculae with muscular ridges and bridges is similar to that of the right ventricle (Fig. 3.67).

Papillary muscles, together with chordae tendineae, are also observed and their structure is as described above for the right ventricle. Two papillary muscles, the **anterior** and **posterior papillary muscles**, are usually found in the left ventricle and are larger than those of the right ventricle.

In the anatomical position, the left ventricle is somewhat posterior to the right ventricle. The interventricular septum therefore forms the anterior wall and some of the wall on the right side of the left ventricle. The septum is described as having two parts:

- a **muscular part**, and
- a **membranous part**.

The muscular part is thick and forms the major part of the septum, whereas the membranous part is the thin, upper part of the septum. A third part of the septum may be considered an atrioventricular part because of its position above the septal cusp of the tricuspid valve. This superior location places this part of the septum between the left ventricle and right atrium.

Mitral valve

The left atrioventricular orifice opens into the posterior right side of the superior part of the left ventricle. It is closed during ventricular contraction by the **mitral valve** (**left atrioventricular valve**), which is also referred to as the bicuspid valve because it has two cusps, the **anterior** and **posterior cusps** (Fig. 3.67). The bases of the cusps are secured to a fibrous ring surrounding the opening, and the cusps are continuous with each other at the commissures. The coordinated action of the papillary muscles and chordae tendineae is as described for the right ventricle.

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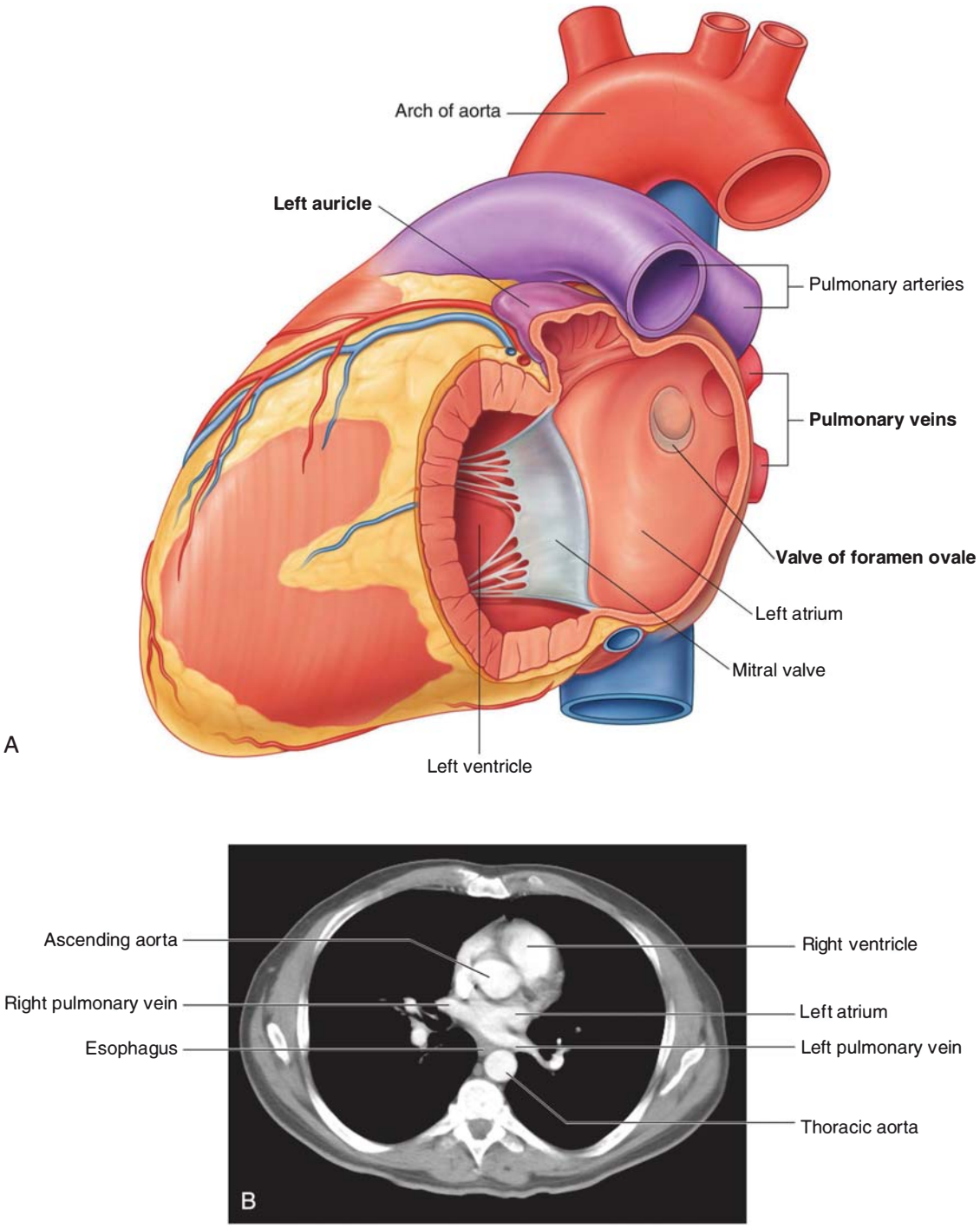


Fig. 3.66 Left atrium. **A.** Internal view. **B.** Axial computed tomography image showing the pulmonary veins entering the left atrium.

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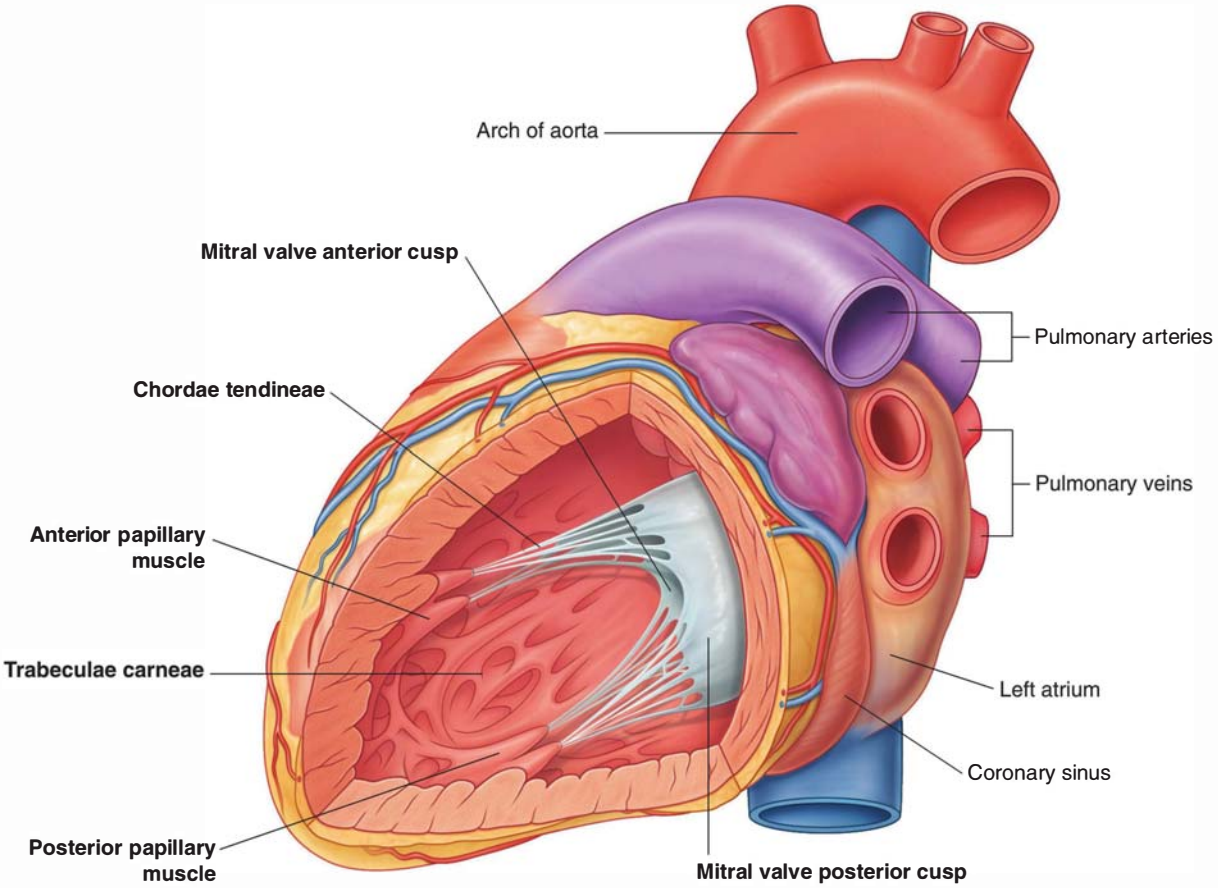


Fig. 3.67 Internal view of the left ventricle.

Aortic valve

The aortic vestibule, or outflow tract of the left ventricle, is continuous superiorly with the ascending aorta. The opening from the left ventricle into the aorta is closed by the **aortic valve**. This valve is similar in structure to the pulmonary valve. It consists of three **semilunar cusps** with the free edge of each projecting upward into the lumen of the ascending aorta (Fig. 3.68).

Between the semilunar cusps and the wall of the ascending aorta are pocket-like sinuses—the **right**, **left**, and **posterior aortic sinuses**. The right and left coronary arteries originate from the right and left aortic sinuses. Because of this, the posterior aortic sinus and cusp are sometimes referred to as the **noncoronary sinus and cusp**.

The functioning of the aortic valve is similar to that of the pulmonary valve with one important additional process: as blood recoils after ventricular contraction and fills the aortic sinuses, it is automatically forced into the coronary arteries because these vessels originate from the right and left aortic sinuses.

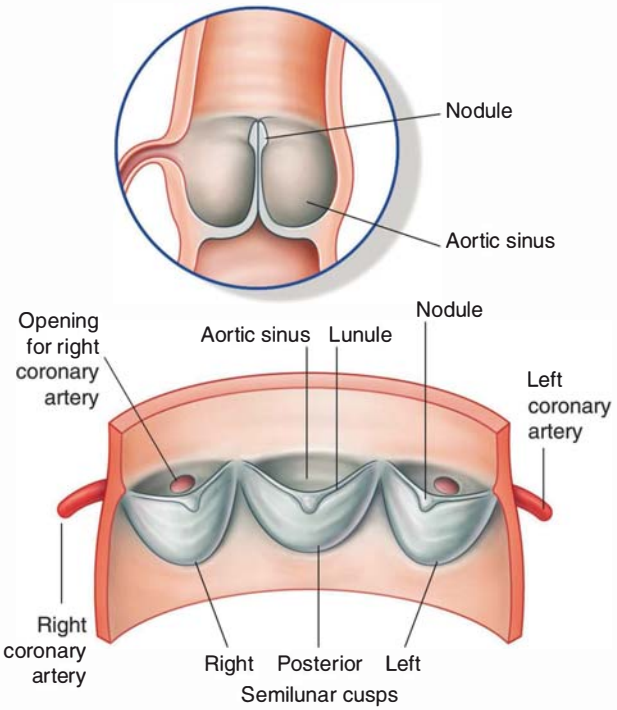


Fig. 3.68 Anterior view of the aortic valve.

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In the clinic

Valve disease

Valve problems consist of two basic types:

- incompetence (insufficiency), which results from poorly functioning valves; and
- stenosis, a narrowing of the orifice, caused by the valve's inability to open fully.

Mitral valve disease is usually a mixed pattern of stenosis and incompetence, one of which usually predominates. Both stenosis and incompetence lead to a poorly functioning valve and subsequent heart changes, which include:

- left ventricular hypertrophy (this is appreciably less marked in patients with mitral stenosis);
- increased pulmonary venous pressure;
- pulmonary edema; and
- enlargement (dilation) and hypertrophy of the left atrium.

Aortic valve disease—both aortic stenosis and aortic regurgitation (backflow) can produce marked heart failure.

Valve disease in the right side of the heart (**affecting the tricuspid or pulmonary valve**) is most likely caused by infection. The resulting valve dysfunction produces abnormal pressure changes in the right atrium and right ventricle, and these can induce cardiac failure.

Cardiac skeleton

The cardiac skeleton is a collection of dense, fibrous connective tissue in the form of four rings with interconnecting areas in a plane between the atria and the ventricles. The four rings of the cardiac skeleton surround the two atrioventricular orifices, the aortic orifice and opening of the pulmonary trunks. They are the **anulus fibrosus**. The interconnecting areas include:

- the **right fibrous trigone**, which is a thickened area of connective tissue between the aortic ring and right atrioventricular ring; and
- the **left fibrous trigone**, which is a thickened area of connective tissue between the aortic ring and the left atrioventricular ring (Fig. 3.69).

The cardiac skeleton helps maintain the integrity of the openings it surrounds and provides points of attachment for the cusps. It also separates the atrial musculature from the ventricular musculature. The atrial myocardium originates from the upper border of the rings, whereas the ventricular myocardium originates from the lower border of the rings.

The cardiac skeleton also serves as a dense connective tissue partition that electrically isolates the atria from

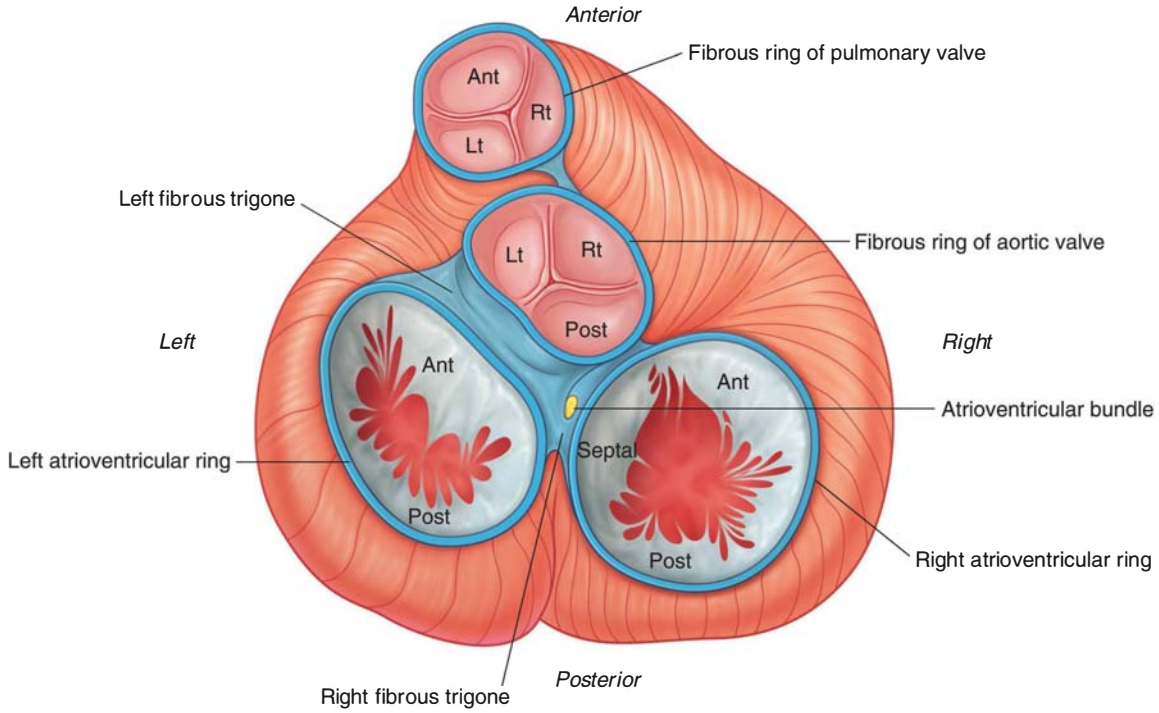


Fig. 3.69 Cardiac skeleton (atria removed).

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the ventricles. The atrioventricular bundle, which passes through the anulus, is the single connection between these two groups of myocardium.

Coronary vasculature

Two coronary arteries arise from the aortic sinuses in the initial portion of the ascending aorta and supply the muscle and other tissues of the heart. They circle the heart in the coronary sulcus, with marginal and interventricular branches, in the interventricular sulci, converging toward the apex of the heart (Fig. 3.70).

The returning venous blood passes through cardiac veins, most of which empty into the coronary sinus. This large venous structure is located in the coronary sulcus on the posterior surface of the heart between the left atrium and left ventricle. The coronary sinus empties into the right atrium between the opening of the inferior vena cava and the right atrioventricular orifice.

Coronary arteries

Right coronary artery. The right coronary artery originates from the right aortic sinus of the ascending aorta. It passes anteriorly and then descends vertically in the coronary sulcus, between the right atrium and right ventricle (Fig. 3.71A). On reaching the inferior margin of the heart, it turns posteriorly and continues in the sulcus onto the diaphragmatic surface and base of the heart. During this course, several branches arise from the main stem of the vessel:

- An early **atrial branch** passes in the groove between the right auricle and ascending aorta, and gives off the **sinu-atrial nodal branch** (Fig. 3.71A), which passes posteriorly around the superior vena cava to supply the sinu-atrial node.
- A **right marginal branch** is given off as the right coronary artery approaches the inferior (acute) margin of the heart (Fig. 3.71A,B) and continues along this border toward the apex of the heart.
- As the right coronary artery continues on the base/diaphragmatic surface of the heart, it supplies a small branch to the atrioventricular node before giving off its final major branch, the **posterior interventricular branch** (Fig. 3.71A), which lies in the posterior interventricular sulcus.

The right coronary artery supplies the right atrium and right ventricle, the sinu-atrial and atrioventricular nodes, the interatrial septum, a portion of the left atrium, the posteroinferior one third of the interventricular septum, and a portion of the posterior part of the left ventricle.

Left coronary artery. The left coronary artery originates from the left aortic sinus of the ascending aorta. It passes between the pulmonary trunk and the left auricle before entering the coronary sulcus. Emerging from behind the pulmonary trunk, the artery divides into its two terminal branches, the anterior interventricular and the circumflex (Fig. 3.71A).

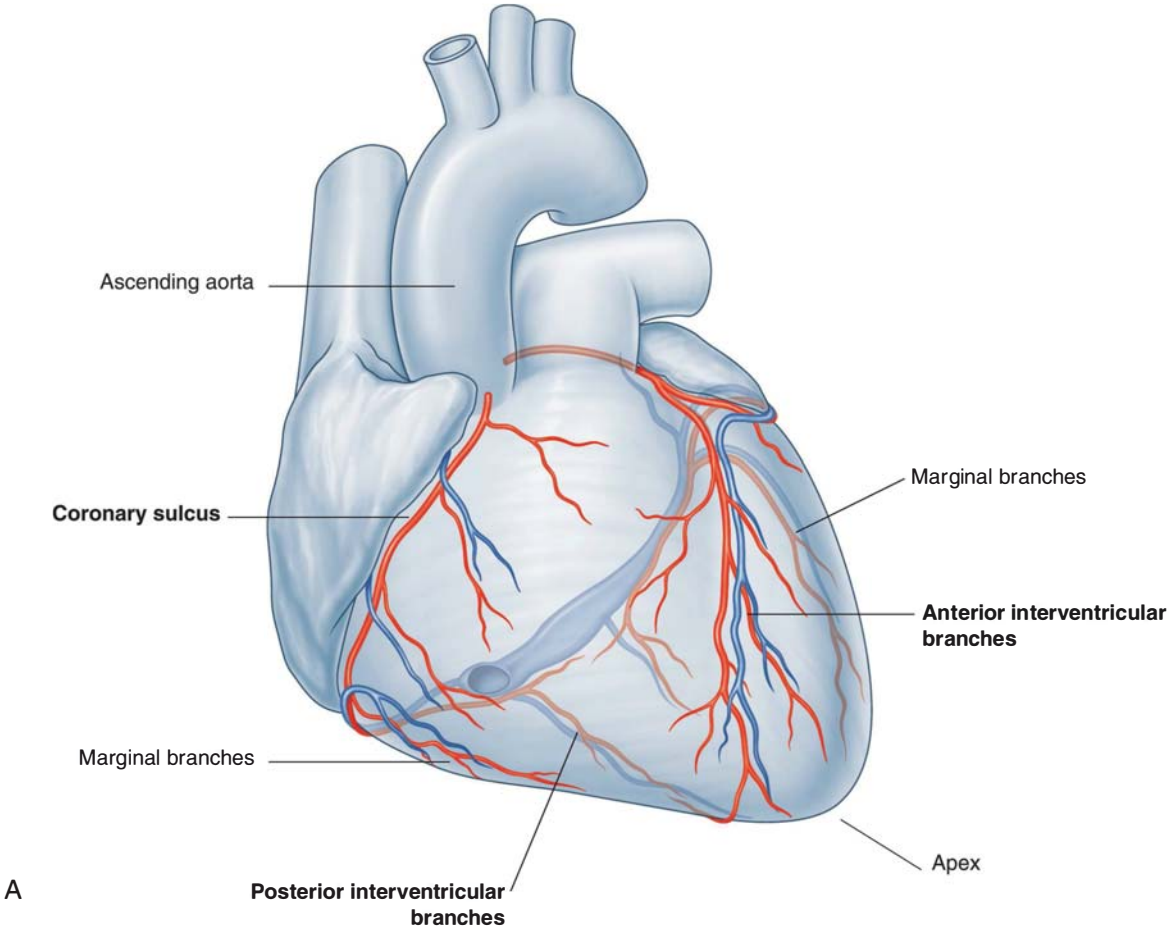
- The **anterior interventricular branch** (**left anterior descending artery—LAD**) (Fig. 3.71A,C) continues around the left side of the pulmonary trunk and descends obliquely toward the apex of the heart in the anterior interventricular sulcus (Fig. 3.71A,C). During its course, one or two large **diagonal branches** may arise and descend diagonally across the anterior surface of the left ventricle.
- The **circumflex branch** (Fig. 3.71A,C) courses toward the left, in the coronary sulcus and onto the base/diaphragmatic surface of the heart, and usually ends before reaching the posterior interventricular sulcus. A large branch, the **left marginal artery** (Fig. 3.71A,C), usually arises from it and continues across the rounded obtuse margin of the heart.

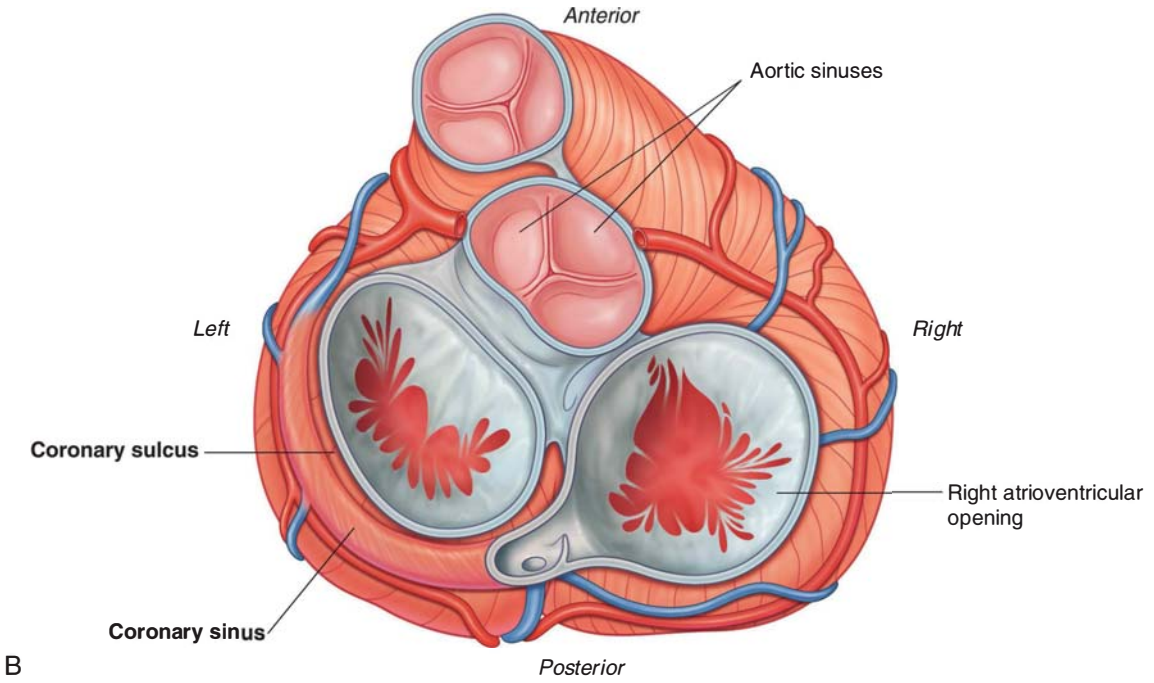
The distribution pattern of the left coronary artery enables it to supply most of the left atrium and left ventricle, and most of the interventricular septum, including the atrioventricular bundle and its branches.

Variations in the distribution patterns of coronary arteries. Several major variations in the basic distribution patterns of the coronary arteries occur.

- The distribution pattern described above for both right and left coronary arteries is the most common and consists of a right dominant coronary artery. This means that the posterior interventricular branch arises from the right coronary artery. The right coronary artery therefore supplies a large portion of the posterior wall of the left ventricle and the circumflex branch of the left coronary artery is relatively small.

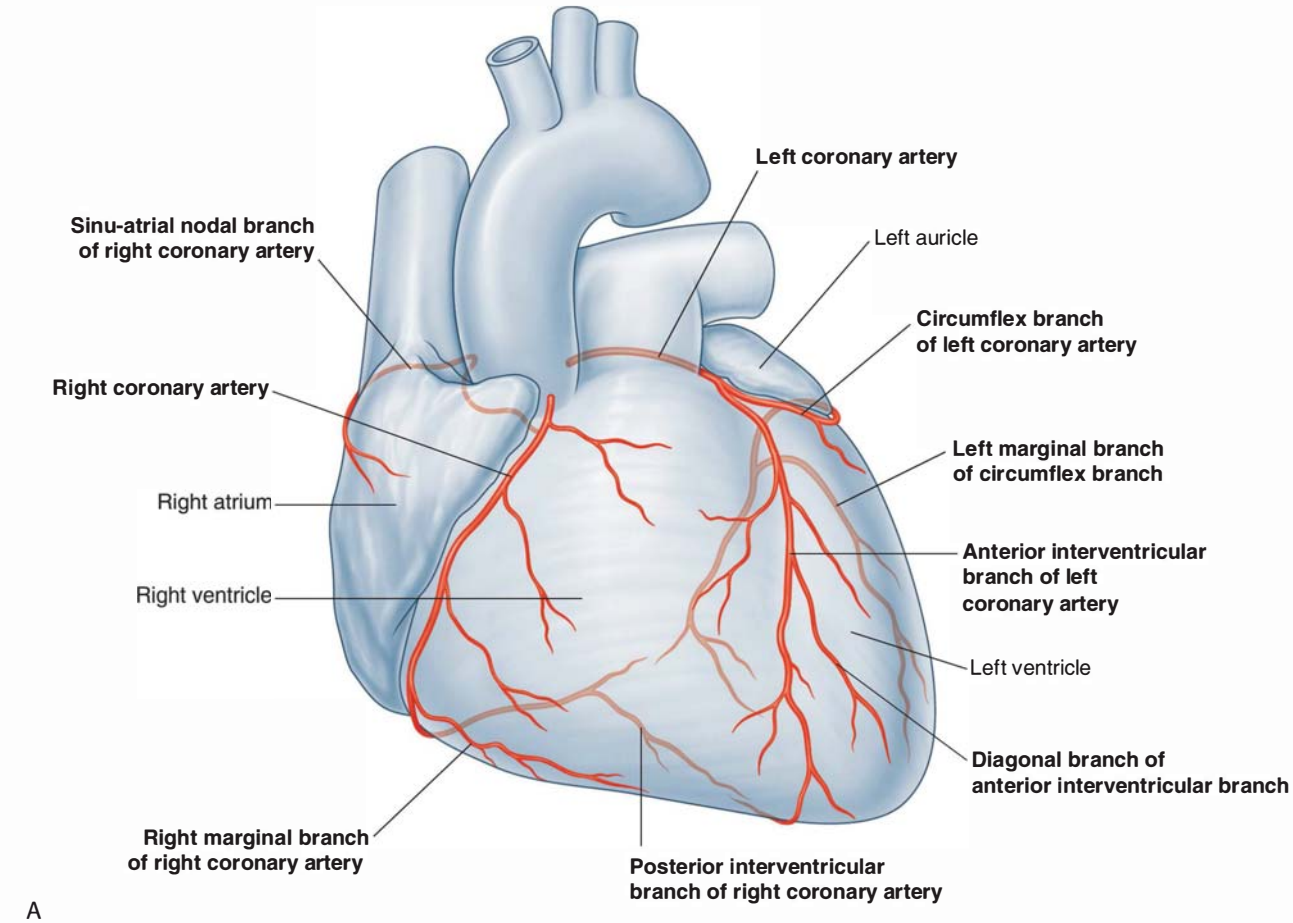
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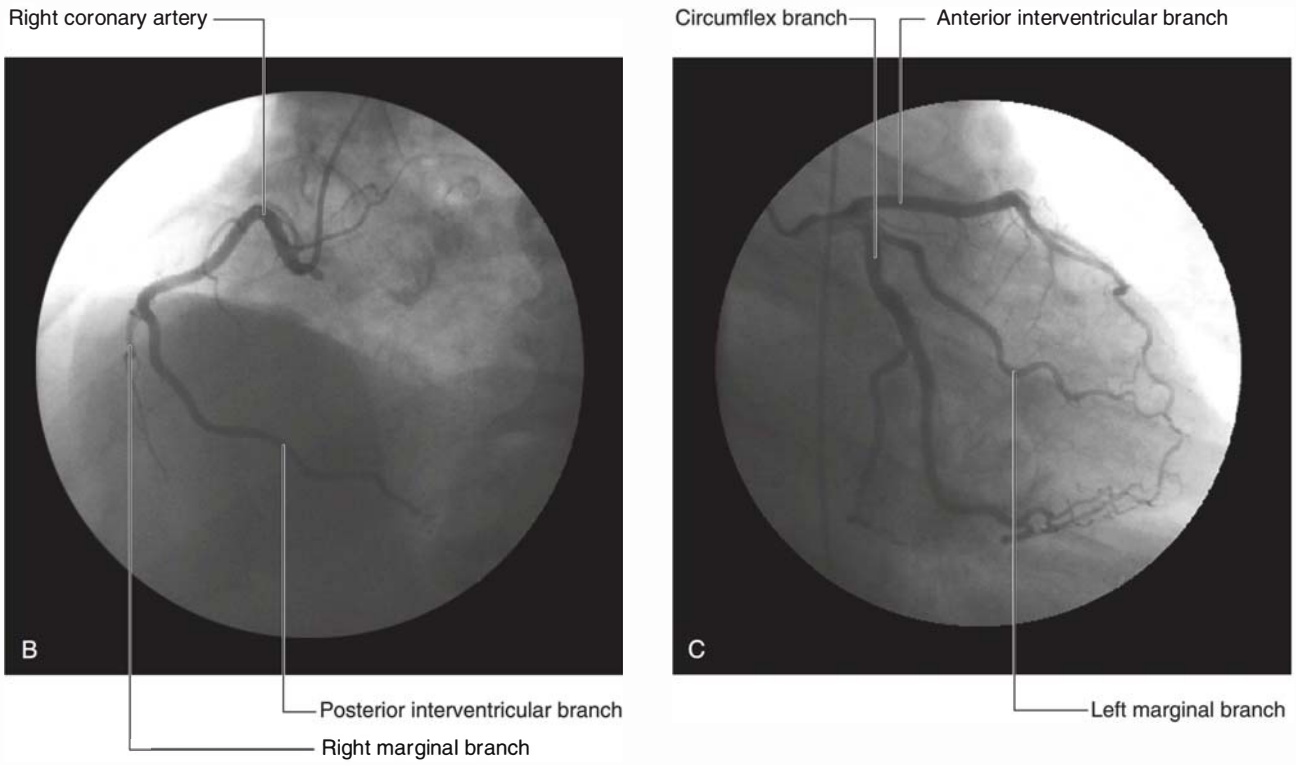




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Fig. 3.70 Cardiac vasculature. **A.** Anterior view. **B.** Superior view (atria removed).





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Fig. 3.71 A. Anterior view of coronary arterial system. Right dominant coronary artery. **B.** Left anterior oblique view of right coronary artery. **C.** Right anterior oblique view of left coronary artery.

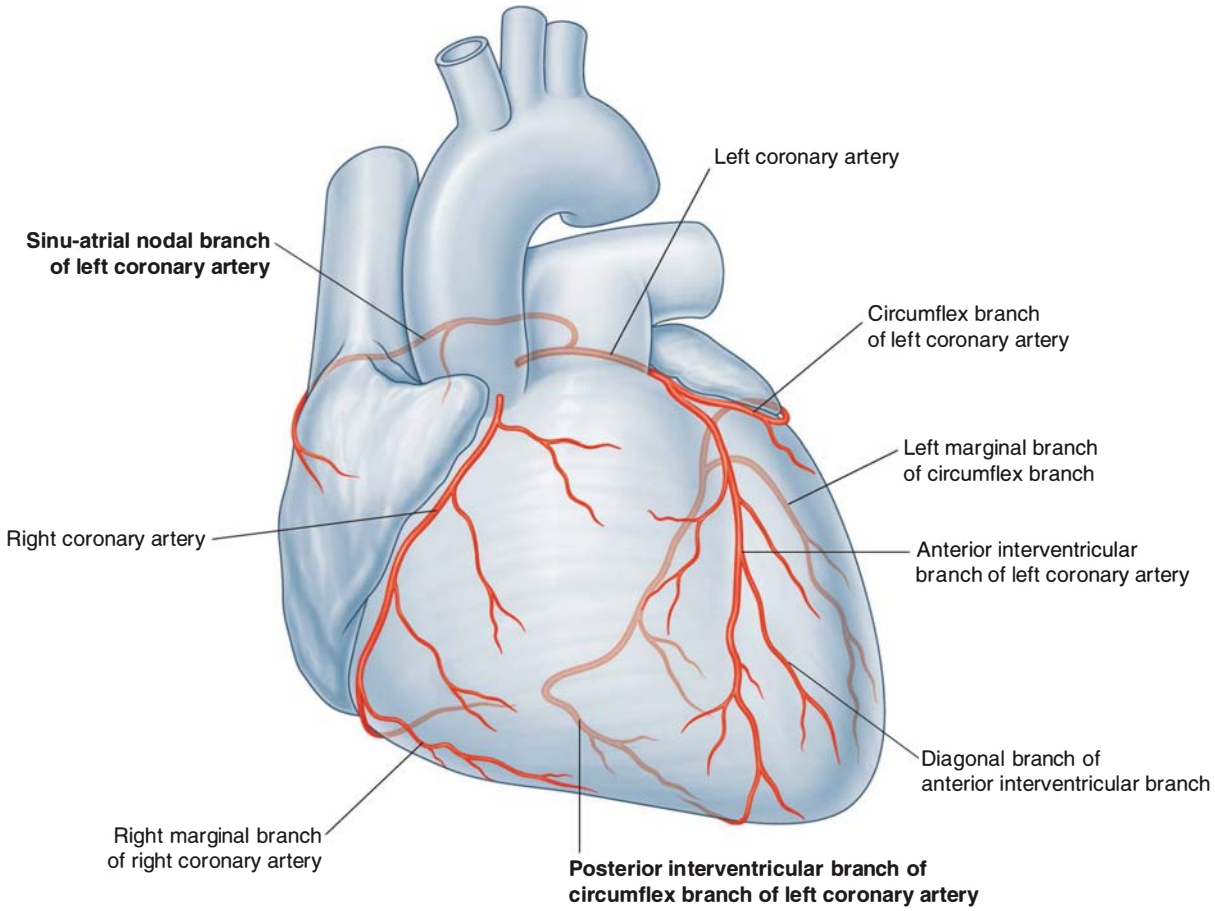


Fig. 3.72 Left dominant coronary artery.

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- In contrast, in hearts with a left dominant coronary artery, the posterior interventricular branch arises from an enlarged circumflex branch and supplies most of the posterior wall of the left ventricle (Fig. 3.72).
- Another point of variation relates to the arterial supply to the sinu-atrial and atrioventricular nodes. In most cases, these two structures are supplied by the right coronary artery. However, vessels from the circumflex branch of the left coronary artery occasionally supply these structures.

In the clinic

Clinical terminology for coronary arteries

In practice, physicians use alternative names for the coronary vessels. The short left coronary artery is referred to as the **left main stem vessel**. One of its primary branches, the anterior interventricular artery, is termed the **left anterior descending artery** (**LAD**). Similarly, the terminal branch of the right coronary artery, the posterior interventricular artery, is termed the **posterior descending artery** (**PDA**).

In the clinic

Heart attack

A heart attack occurs when the perfusion to the myocardium is insufficient to meet the metabolic needs of the tissue, leading to irreversible tissue damage. The most common cause is a total occlusion of a major coronary artery.

Coronary Artery Disease

Occlusion of a major coronary artery, usually due to atherosclerosis, leads to inadequate oxygenation of an area of myocardium and cell death (Fig. 3.73). The severity of the problem will be related to the size and location of the artery involved, whether or not the blockage is complete, and whether there are collateral vessels to provide perfusion to the territory from other vessels. Depending on the severity, patients can develop pain (angina) or a myocardial infarction (MI).

Percutaneous Coronary Intervention

This is a technique in which a long fine tube (a catheter) is inserted into the femoral artery in the thigh and passed through the external and common iliac arteries and into the abdominal aorta. It continues to be moved upward through the thoracic aorta to the origins of the coronary arteries. The coronaries may also be approached via the radial or brachial arteries. A fine wire is then passed into the coronary artery and is used to cross the stenosis. A fine balloon is then passed over the wire and may be inflated at the level of the obstruction, thus widening it; this is termed angioplasty. More commonly, this is augmented by placement of a fine wire mesh (a stent) inside the obstruction to hold it open. Other percutaneous interventions are suction extraction of a coronary thrombus and rotary ablation of a plaque.

Coronary Artery Bypass Grafts

If coronary artery disease is too extensive to be treated by percutaneous intervention, surgical coronary artery bypass grafting may be necessary. The great saphenous vein, in the lower limb, is harvested and used as a graft. It is divided into several pieces, each of which is used to bypass blocked sections of the coronary arteries. The internal thoracic and radial arteries can also be used.

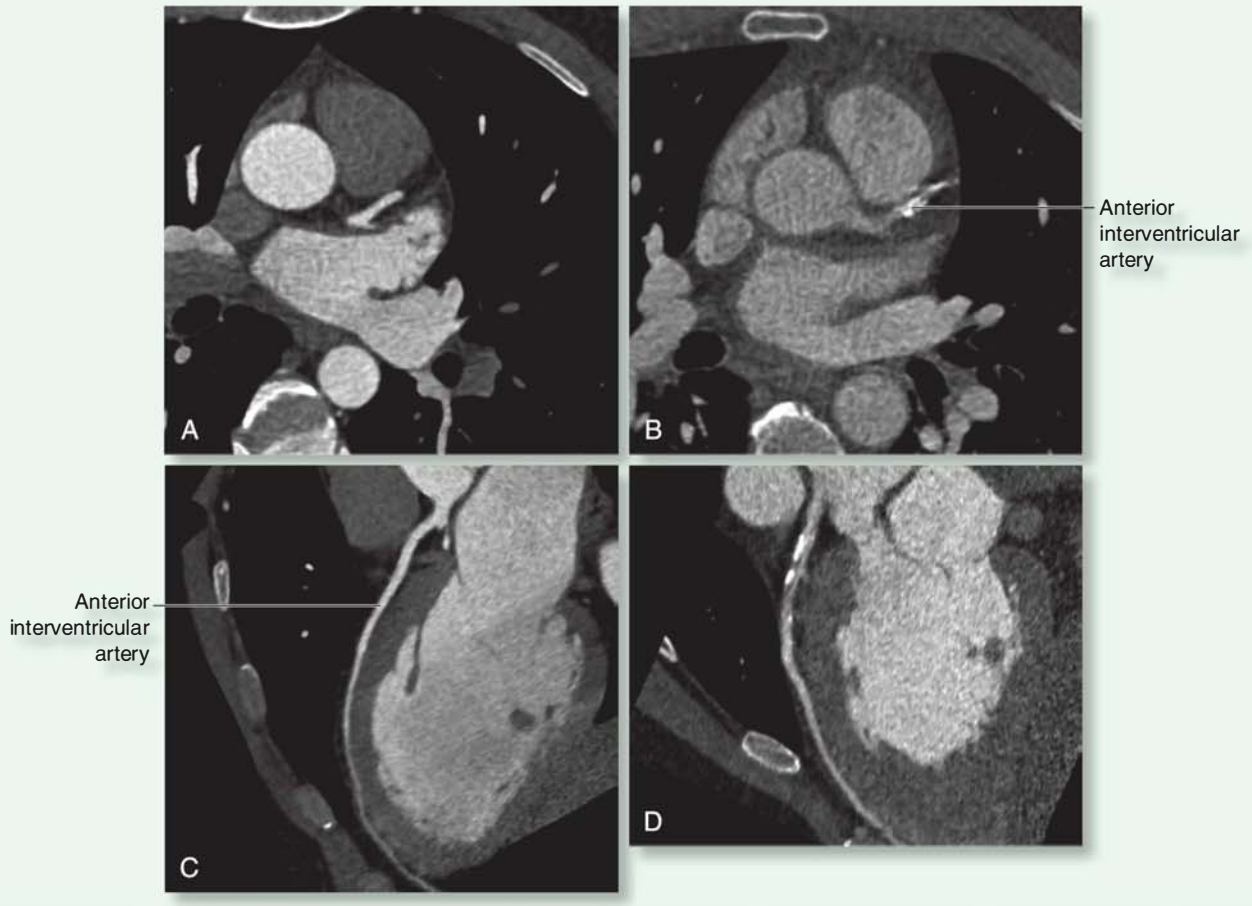


Fig. 3.73 **A** and **B.** Axial maximum intensity projection (MIP) CT image through the heart. **A.** Normal anterior interventricular (left anterior descending) artery. **B.** Stenotic (calcified) anterior interventricular (left anterior descending) artery. **C** and **D.** Vertical long axis multiplanar reformation (MRP) CT image through the heart. **C.** Normal anterior interventricular (left anterior descending) artery. **D.** Stenotic (calcified) anterior interventricular (left anterior descending) artery.

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In the clinic

Classic symptoms of heart attack

The typical symptoms are chest heaviness or pressure, which can be severe, lasting more than 20 minutes, and often associated with sweating. The pain in the chest (which may be described as an "elephant sitting on my chest" or by using a clenched fist to describe the pain [Levine sign]) often radiates to the arms (left more common than the right), and can be associated with nausea. The severity of ischemia and infarction depends on the rate at which the occlusion or stenosis has occurred and whether or not collateral channels have had a chance to develop.

In the clinic

Are heart attack symptoms the same in men and women?

Although men and women can experience the typical symptoms of severe chest pain, cold sweats, and pain in the left arm, women are more likely than men to have subtler, less recognizable symptoms. These may include abdominal pain, achiness in the jaw or back, nausea, shortness of breath, or simply fatigue. The mechanism of this difference is not understood, but it is important to consider cardiac ischemia for a wide range of symptoms.

In the clinic

Common congenital heart defects

The most common abnormalities that occur during development are those produced by a defect in the atrial and ventricular septa.

A **defect in the interatrial septum** allows blood to pass from one side of the heart to the other from the chamber with the higher pressure; this is clinically referred to as a **shunt**. An **atrial septal defect (ASD)** allows oxygenated blood to flow from the left atrium (higher pressure) across the ASD into the right atrium (lower pressure). Many patients with ASD are asymptomatic, but in some cases the ASD may need to be closed surgically or by endovascular devices. Occasionally, increased blood flow into the right atrium over many years leads to right atrial and right ventricular hypertrophy and enlargement of the pulmonary trunk, resulting in pulmonary arterial hypertension.

The most common of all congenital heart defects are those that occur in the ventricular septum—**ventriculoseptal defect** (**VSD**). These lesions are most frequent in the membranous portion of the septum and they allow blood to move from the left ventricle (higher pressure) to the right ventricle (lower pressure); this leads to right ventricular hypertrophy and pulmonary arterial hypertension. If large enough and left untreated, VSDs can produce marked clinical problems that might require surgery.

Occasionally, the **ductus arteriosus**, which connects the left branch of the pulmonary artery to the inferior aspect of the aortic arch, fails to close at birth. When this occurs, the oxygenated blood in the aortic arch (higher pressure) passes into the left branch of the pulmonary artery (lower pressure) and produces pulmonary hypertension. This is termed a **patent** or **persistent ductus arteriosus** (**PDA**).

All of these defects produce a left-to-right shunt, indicating that oxygenated blood from the left heart is being mixed with deoxygenated blood from the right heart before being recirculated into the pulmonary circulation. These shunts are normally compatible with life, but surgery or endovascular treatment may be necessary.

Rarely, a shunt is right-to-left. In isolation this is fatal; however, this type of shunt is often associated with other anomalies, so some deoxygenated blood is returned to the lungs and the systemic circulation.

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In the clinic

Cardiac auscultation

Auscultation of the heart reveals the normal audible cardiac cycle, which allows the clinician to assess heart rate, rhythm, and regularity. Furthermore, cardiac murmurs that have characteristic sounds within the phases of the cardiac cycle can be demonstrated (Fig. 3.74).

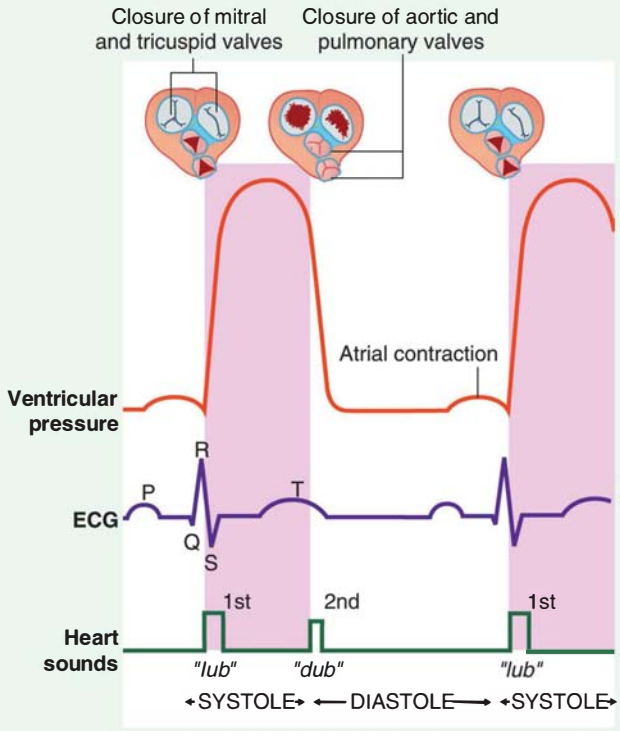


Fig. 3.74 Heart sounds and how they relate to valve closure, the electrocardiogram (ECG), and ventricular pressure.

Cardiac veins

The **coronary sinus** receives four major tributaries: the great, middle, small, and posterior cardiac veins.

Great cardiac vein. The great cardiac vein begins at the apex of the heart (Fig. 3.75A). It ascends in the anterior interventricular sulcus, where it is related to the anterior interventricular artery and is often termed the anterior interventricular vein. Reaching the coronary sulcus, the great cardiac vein turns to the left and continues onto the base/diaphragmatic surface of the heart. At this point, it is associated with the circumflex branch of the left coronary artery. Continuing along its path in the coronary sulcus, the great cardiac vein gradually enlarges to form the coronary sinus, which enters the right atrium (Fig. 3.75B).

Middle cardiac vein. The middle cardiac vein (posterior interventricular vein) begins near the apex of the heart and ascends in the posterior interventricular sulcus toward the coronary sinus (Fig. 3.75B). It is associated with the posterior interventricular branch of the right or left coronary artery throughout its course.

Small cardiac vein. The small cardiac vein begins in the lower anterior section of the coronary sulcus between the right atrium and right ventricle (Fig. 3.75A). It continues in this groove onto the base/diaphragmatic surface of the heart where it enters the coronary sinus at its atrial end. It is a companion of the right coronary artery throughout its course and may receive the right marginal vein (Fig. 3.75A). This small vein accompanies the marginal branch of the right coronary artery along the acute margin of the heart. If the right marginal vein does not join the small cardiac vein, it enters the right atrium directly.

Posterior cardiac vein. The posterior cardiac vein lies on the posterior surface of the left ventricle just to the left of the middle cardiac vein (Fig. 3.75B). It either enters the coronary sinus directly or joins the great cardiac vein.

Other cardiac veins. Two additional groups of cardiac veins are also involved in the venous drainage of the heart.

- The **anterior veins of the right ventricle** (**anterior cardiac veins**) are small veins that arise on the anterior surface of the right ventricle (Fig. 3.75A). They cross the coronary sulcus and enter the anterior wall of the right atrium. They drain the anterior portion of the right ventricle. The right marginal vein may be part of this group if it does not enter the small cardiac vein.
- A group of smallest cardiac veins (**venae cordis minimae** or **veins of Thebesius**) have also been described. Draining directly into the cardiac chambers, they are numerous in the right atrium and right ventricle, are occasionally associated with the left atrium, and are rarely associated with the left ventricle.

Coronary lymphatics

The lymphatic vessels of the heart follow the coronary arteries and drain mainly into:

- brachiocephalic nodes, anterior to the brachiocephalic veins; and
- tracheobronchial nodes, at the inferior end of the trachea.

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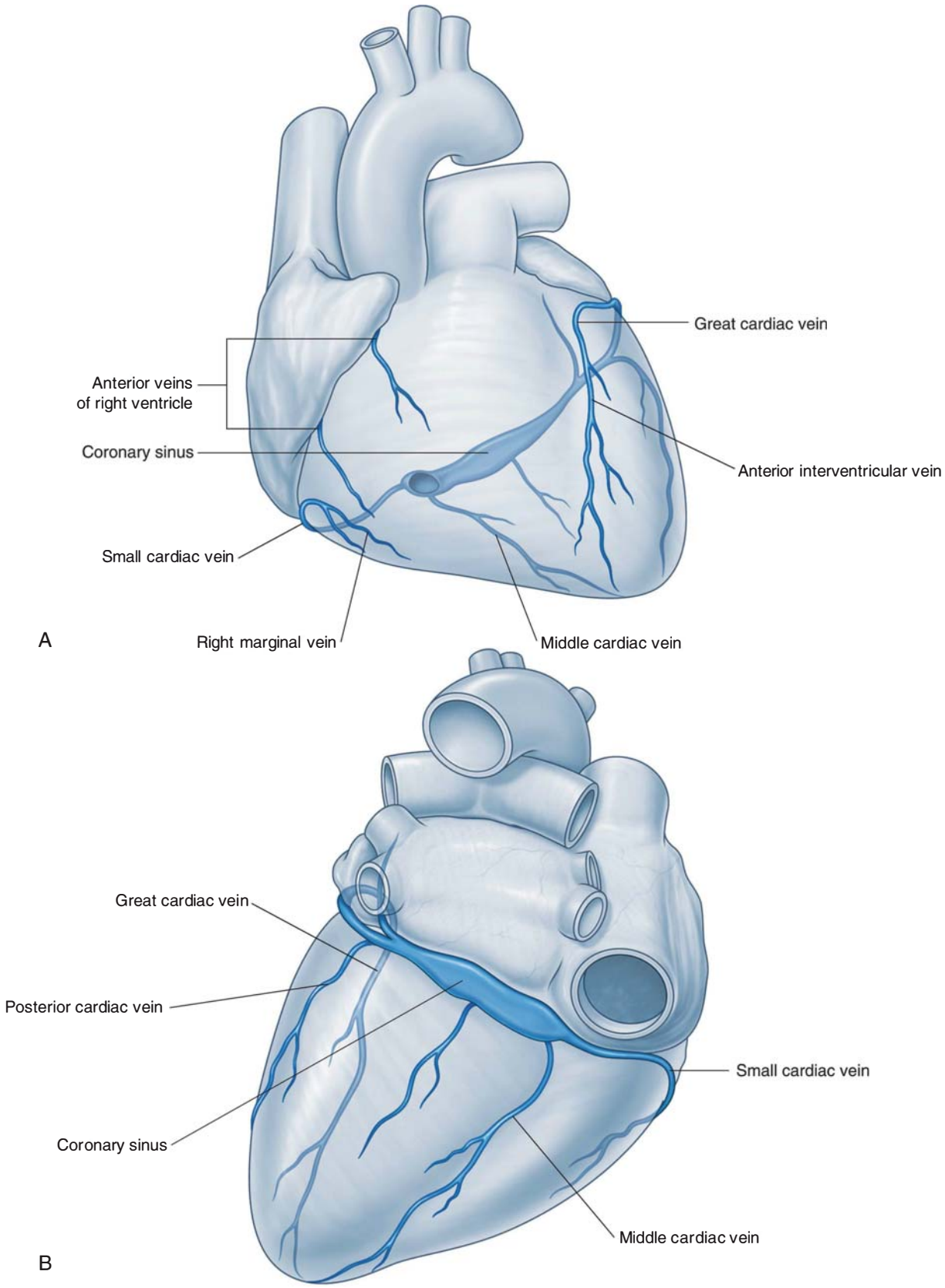


Fig. 3.75 Major cardiac veins. **A.** Anterior view of major cardiac veins. **B.** Posteroinferior view of major cardiac veins.

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Cardiac conduction system

The musculature of the atria and ventricles is capable of contracting spontaneously. The cardiac conduction system initiates and coordinates contraction. The conduction system consists of nodes and networks of specialized cardiac muscle cells organized into four basic components:

- the sinu-atrial node,
- the atrioventricular node,
- the atrioventricular bundle with its right and left bundle branches, and
- the subendocardial plexus of conduction cells (the Purkinje fibers).

The unique distribution pattern of the cardiac conduction system establishes an important unidirectional pathway of excitation/contraction. Throughout its course, large branches of the conduction system are insulated from the surrounding myocardium by connective tissue. This tends to decrease inappropriate stimulation and contraction of cardiac muscle fibers.

The number of functional contacts between the conduction pathway and cardiac musculature greatly increases in the subendocardial network.

Thus, a unidirectional wave of excitation and contraction is established, which moves from the papillary muscles and apex of the ventricles to the arterial outflow tracts.

In the clinic

Cardiac conduction system

The cardiac conduction system can be affected by coronary artery disease. The normal rhythm may be disturbed if the blood supply to the coronary conduction system is disrupted. If a dysrhythmia affects the heart rate or the order in which the chambers contract, heart failure and death may ensue.

Sinu-atrial node

Impulses begin at the **sinu-atrial node**, the cardiac pacemaker. This collection of cells is located at the superior end of the crista terminalis at the junction of the superior vena cava and the right atrium (Fig. 3.76A). This is also the junction between the parts of the right atrium derived from the embryonic sinus venosus and the atrium proper.

The excitation signals generated by the sinu-atrial node spread across the atria, causing the muscle to contract.

Atrioventricular node

Concurrently, the wave of excitation in the atria stimulates the **atrioventricular node**, which is located near the opening of the coronary sinus, close to the attachment of the septal cusp of the tricuspid valve, and within the atrioventricular septum (Fig. 3.76A).

The atrioventricular node is a collection of specialized cells that forms the beginning of an elaborate system of conducting tissue, the atrioventricular bundle, which extends the excitatory impulse to all ventricular musculature.

Atrioventricular bundle

The **atrioventricular bundle** is a direct continuation of the atrioventricular node (Fig. 3.76A). It follows along the lower border of the membranous part of the interventricular septum before splitting into right and left bundles.

The **right bundle branch** continues on the right side of the interventricular septum toward the apex of the right ventricle. From the septum it enters the septomarginal trabecula to reach the base of the anterior papillary muscle. At this point, it divides and is continuous with the final component of the cardiac conduction system, the subendocardial plexus of ventricular conduction cells or Purkinje fibers. This network of specialized cells spreads throughout the ventricle to supply the ventricular musculature, including the papillary muscles.

The **left bundle branch** passes to the left side of the muscular interventricular septum and descends to the apex of the left ventricle (Fig. 3.76B). Along its course it gives off branches that eventually become continuous with the **subendocardial plexus of conduction cells** (**Purkinje fibers**). As with the right side, this network of specialized cells spreads the excitation impulses throughout the left ventricle.

Cardiac innervation

The autonomic division of the peripheral nervous system is directly responsible for regulating:

- heart rate,
- force of each contraction, and
- cardiac output.

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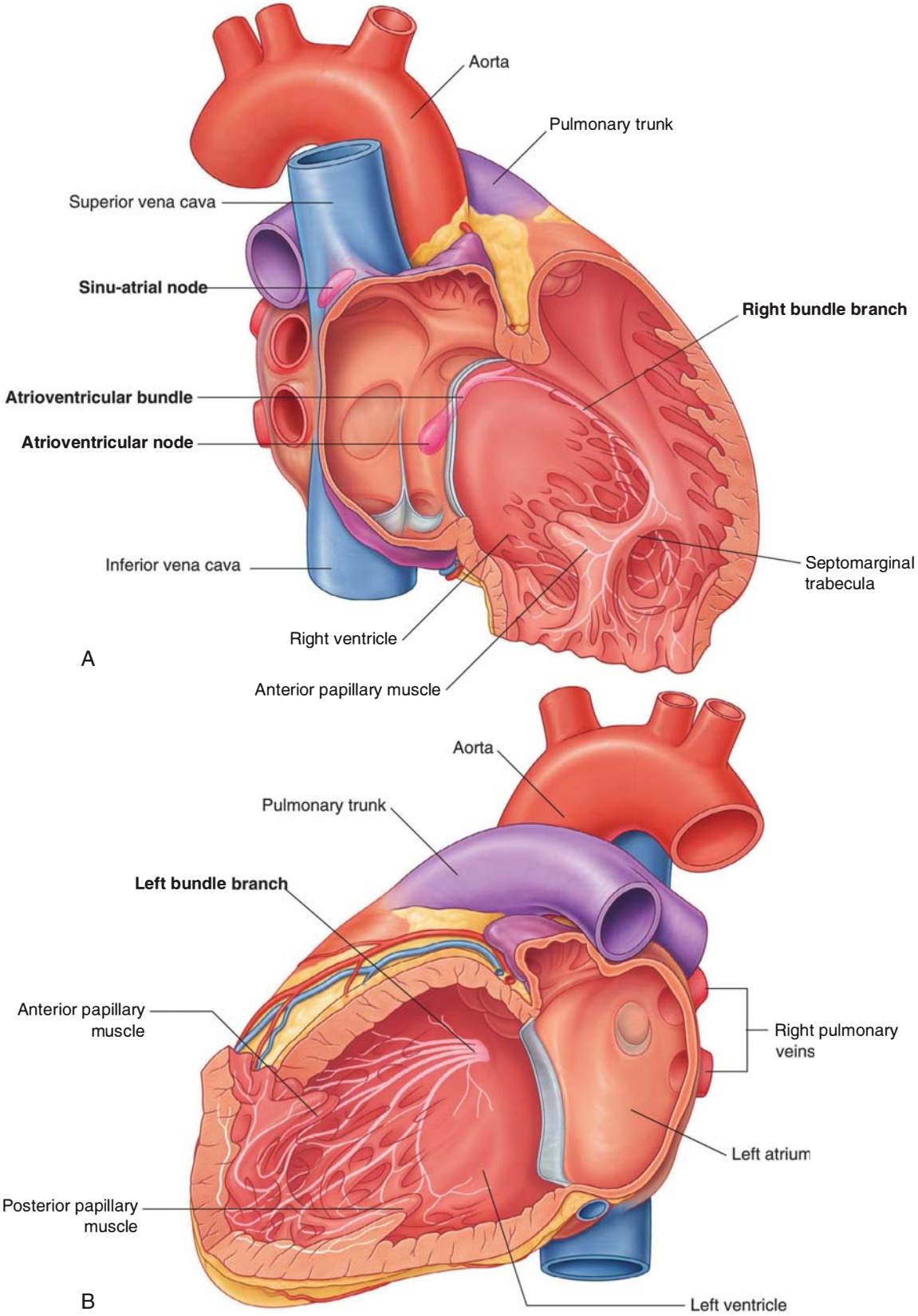


Fig. 3.76 Conduction system of the heart. **A.** Right chambers. **B.** Left chambers.

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Branches from both the parasympathetic and sympathetic systems contribute to the formation of the **cardiac plexus**. This plexus consists of a **superficial part**, inferior to the aortic arch and between it and the pulmonary trunk (Fig. 3.77A), and a **deep part**, between the aortic arch and the tracheal bifurcation (Fig. 3.77B).

From the cardiac plexus, small branches that are mixed nerves containing both sympathetic and parasympathetic fibers supply the heart. These branches affect nodal tissue and other components of the conduction system, coronary blood vessels, and atrial and ventricular musculature.

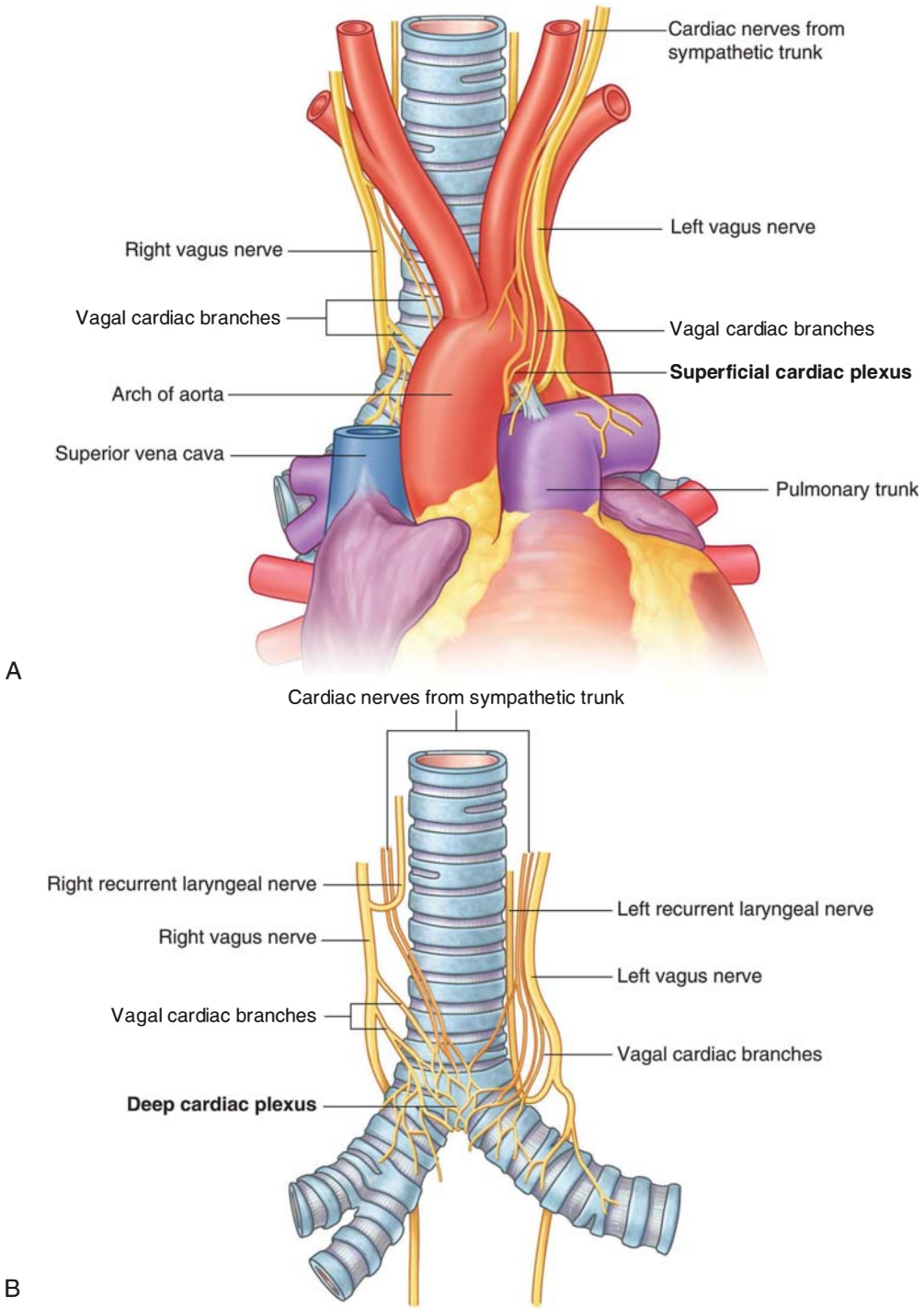


Fig. 3.77 Cardiac plexus. **A.** Superficial. **B.** Deep.

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Parasympathetic innervation

Stimulation of the parasympathetic system:

- decreases heart rate,
- reduces force of contraction, and
- constricts the coronary arteries.

The preganglionic parasympathetic fibers reach the heart as cardiac branches from the right and left vagus nerves. They enter the cardiac plexus and synapse in ganglia located either within the plexus or in the walls of the atria.

Sympathetic innervation

Stimulation of the sympathetic system:

- increases heart rate, and
- increases the force of contraction.

Sympathetic fibers reach the cardiac plexus through the cardiac nerves from the sympathetic trunk. Preganglionic sympathetic fibers from the upper four or five segments of the thoracic spinal cord enter and move through the sympathetic trunk. They synapse in cervical and upper thoracic sympathetic ganglia, and postganglionic fibers proceed as bilateral branches from the sympathetic trunk to the cardiac plexus.

Visceral afferents

Visceral afferents from the heart are also a component of the cardiac plexus. These fibers pass through the cardiac plexus and return to the central nervous system in the cardiac nerves from the sympathetic trunk and in the vagal cardiac branches.

The afferents associated with the vagal cardiac nerves return to the vagus nerve [X]. They sense alterations in blood pressure and blood chemistry and are therefore primarily concerned with cardiac reflexes.

The afferents associated with the cardiac nerves from the sympathetic trunks return to either the cervical or the thoracic portions of the sympathetic trunk. If they are in the cervical portion of the trunk, they normally descend to the thoracic region, where they reenter the upper four or five thoracic spinal cord segments, along with the afferents from the thoracic region of the sympathetic trunk. Visceral afferents associated with the sympathetic system conduct pain sensation from the heart, which is detected at the cellular level as tissue-damaging events (i.e., cardiac ischemia). This pain is often "referred" to cutaneous regions supplied by the same spinal cord levels (see "In the clinic: Referred pain." p. 48, and "Case 4," pp. 243–245).

Pulmonary trunk

The **pulmonary trunk** is contained within the pericardial sac (Fig. 3.78), is covered by the visceral layer of serous

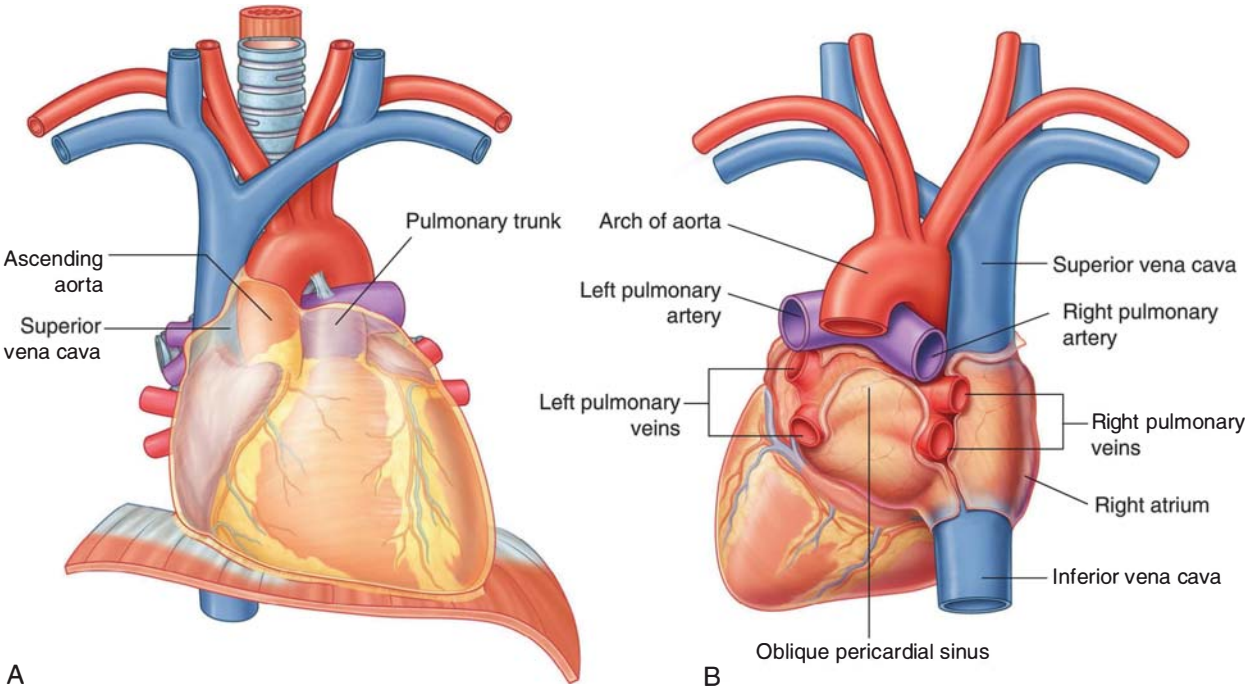


Fig. 3.78 Major vessels within the middle mediastinum. **A.** Anterior view. **B.** Posterior view.

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pericardium, and is associated with the ascending aorta in a common sheath. It arises from the conus arteriosus of the right ventricle at the opening of the pulmonary trunk slightly anterior to the aortic orifice and ascends, moving posteriorly and to the left, lying initially anterior and then to the left of the ascending aorta. At approximately the level of the intervertebral disc between vertebrae TV and TVI, opposite the left border of the sternum and posterior to the third left costal cartilage, the pulmonary trunk divides into:

- the right pulmonary artery, which passes to the right, posterior to the ascending aorta and the superior vena cava, to enter the right lung; and
- the left pulmonary artery, which passes inferiorly to the arch of the aorta and anteriorly to the descending aorta to enter the left lung.

Ascending aorta

The **ascending aorta** is contained within the pericardial sac and is covered by a visceral layer of serous pericardium, which also surrounds the pulmonary trunk in a common sheath (Fig. 3.78A).

The origin of the ascending aorta is the aortic orifice at the base of the left ventricle, which is level with the lower edge of the third left costal cartilage, posterior to the left half of the sternum. Moving superiorly, slightly forward and to the right, the ascending aorta continues to the level of the second right costal cartilage. At this point, it enters the superior mediastinum and is then referred to as the **arch of the aorta**.

Immediately superior to the point where the ascending aorta arises from the left ventricle are three small outward bulges opposite the semilunar cusps of the aortic valve. These are the posterior, right, and left aortic sinuses. The right and left coronary arteries originate from the right and left aortic sinuses, respectively.

Other vasculature

The inferior half of the **superior vena cava** is located within the pericardial sac (Fig. 3.78B). It passes through the fibrous pericardium at approximately the level of the second costal cartilage and enters the right atrium at the lower level of the third costal cartilage. The portion within the pericardial sac is covered with serous pericardium except for a small area on its posterior surface.

After passing through the diaphragm, at approximately the level of vertebra TVIII, the **inferior vena cava** enters the fibrous pericardium. A short portion of this vessel is within the pericardial sac before entering the right atrium. While within the pericardial sac, it is covered by serous pericardium except for a small portion of its posterior surface (Fig. 3.78B).

A very short segment of each of the pulmonary veins is also within the pericardial sac. These veins, usually two from each lung, pass through the fibrous pericardium and enter the superior region of the left atrium on its posterior surface. In the pericardial sac, all but a portion of the posterior surface of these veins is covered by serous pericardium. In addition, the **oblique pericardial sinus** is between the right and left pulmonary veins, within the pericardial sac (Fig. 3.78B).

Superior mediastinum

The **superior mediastinum** is posterior to the manubrium of the sternum and anterior to the bodies of the first four thoracic vertebrae (see Fig. 3.52).

- Its superior boundary is an oblique plane passing from the jugular notch upward and posteriorly to the superior border of vertebra TI.
- Inferiorly, a transverse plane passing from the sternal angle to the intervertebral disc between vertebra TIV/V separates it from the inferior mediastinum.
- Laterally, it is bordered by the mediastinal part of the parietal pleura on either side.

The superior mediastinum is continuous with the neck above and with the inferior mediastinum below.

The major structures found in the superior mediastinum (Figs. 3.79 and 3.80) include the:

- thymus,
- right and left brachiocephalic veins,
- left superior intercostal vein,
- superior vena cava,
- arch of the aorta with its three large branches,
- trachea,
- esophagus,
- phrenic nerves,
- vagus nerves,
- left recurrent laryngeal branch of the left vagus nerve,
- thoracic duct, and
- other small nerves, blood vessels, and lymphatics.

Thymus

The **thymus** is the most anterior component of the superior mediastinum, lying immediately posterior to the

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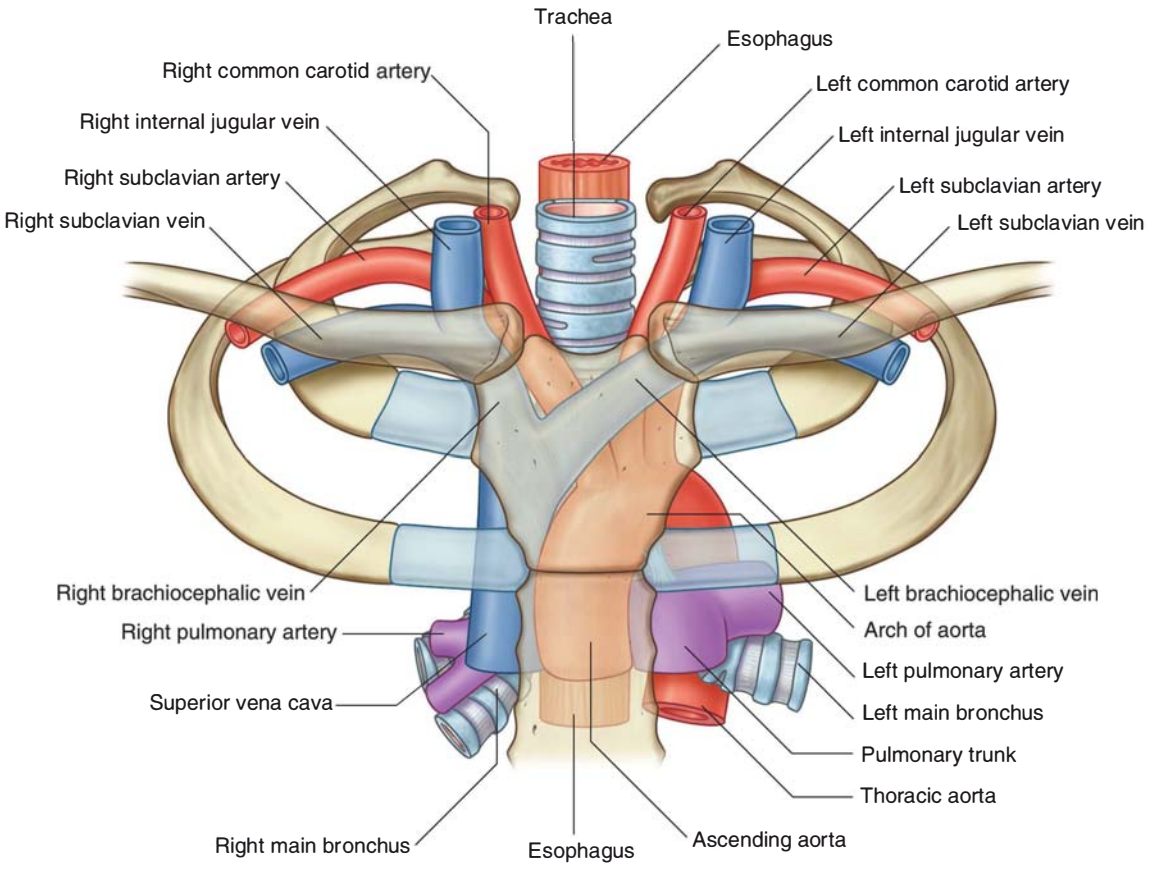


Fig. 3.79 Structures in the superior mediastinum.

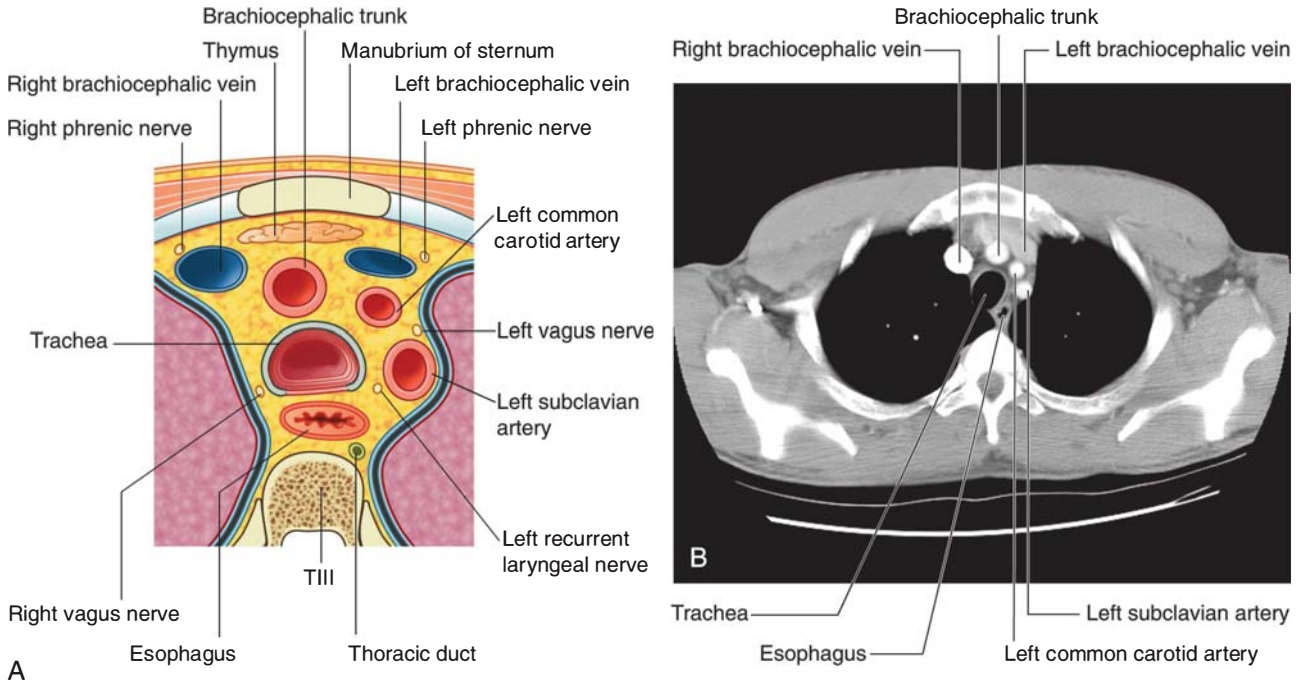


Fig. 3.80 Cross section through the superior mediastinum at the level of vertebra TIII. **A.** Diagram. **B.** Axial computed tomography image.

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manubrium of the sternum. It is an asymmetrical, bilobed structure (Fig. 3.81).

The upper extent of the thymus can reach into the neck as high as the thyroid gland; a lower portion typically extends into the anterior mediastinum over the pericardial sac.

Involved in the early development of the immune system, the thymus is a large structure in the child, begins to atrophy after puberty, and shows considerable size variation in the adult. In the elderly adult, it is barely identifiable as an organ, consisting mostly of fatty tissue that is sometimes arranged as two lobulated fatty structures.

Arteries to the thymus consist of small branches originating from the internal thoracic arteries. Venous drainage is usually into the left brachiocephalic vein and possibly into the internal thoracic veins.

Lymphatic drainage returns to multiple groups of nodes at one or more of the following locations:

- along the internal thoracic arteries (parasternal);
- at the tracheal bifurcation (tracheobronchial); and
- in the root of the neck.

In the clinic

Ectopic parathyroid glands in the thymus

The parathyroid glands develop from the third pharyngeal pouch, which also forms the thymus. The thymus is therefore a common site for ectopic parathyroid glands and, potentially, ectopic parathyroid hormone production.

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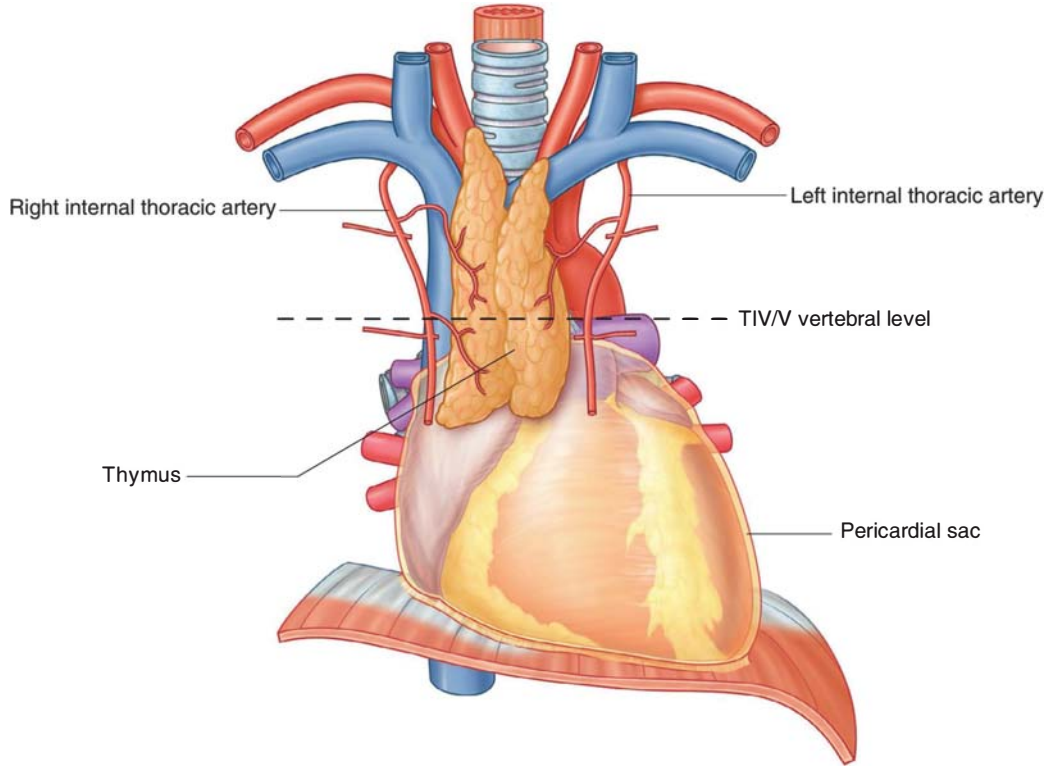


Fig. 3.81 Thymus.

Right and left brachiocephalic veins

The left and right brachiocephalic veins are located immediately posterior to the thymus. They form on each side at the junction between the internal jugular and subclavian veins (see Fig. 3.79). The left brachiocephalic vein crosses the midline and joins with the right brachiocephalic vein to form the superior vena cava (Fig. 3.82).

- The **right brachiocephalic vein** begins posterior to the medial end of the right clavicle and passes vertically downward, forming the superior vena cava when it is joined by the left brachiocephalic vein. Venous tributaries include the vertebral, first posterior intercostal, and internal thoracic veins. The inferior thyroid and thymic veins may also drain into it.
- The **left brachiocephalic vein** begins posterior to the medial end of the left clavicle. It crosses to the right, moving in a slightly inferior direction, and joins with the right brachiocephalic vein to form the superior vena cava posterior to the lower edge of the right first costal cartilage close to the right sternal border. Venous tributaries include the vertebral, first posterior intercostal, left superior intercostal, inferior thyroid, and internal thoracic veins. It may also receive thymic and pericardial veins. The left brachiocephalic vein crosses the midline posterior to the manubrium in the adult. In infants and children the left brachiocephalic vein rises above the superior border of the manubrium and therefore is less protected.

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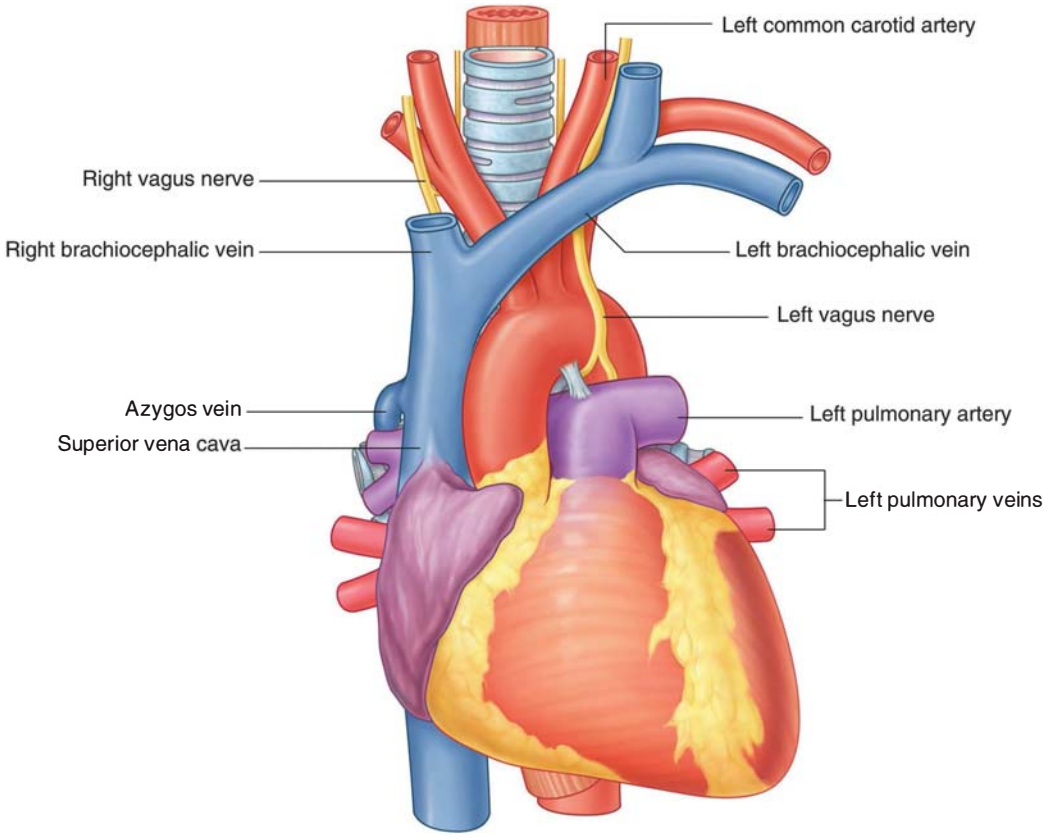


Fig. 3.82 Superior mediastinum with thymus removed.

Left superior intercostal vein

The **left superior intercostal vein** receives the second, third, and sometimes the fourth posterior intercostal veins, usually the left bronchial veins, and sometimes the left pericardiacophrenic vein. It passes over the left side of the aortic arch, lateral to the left vagus nerve and medial to the left phrenic nerve, before entering the left brachiocephalic vein (Fig. 3.83). Inferiorly, it may connect with the **accessory hemiazygos vein** (**superior hemiazygos vein**).

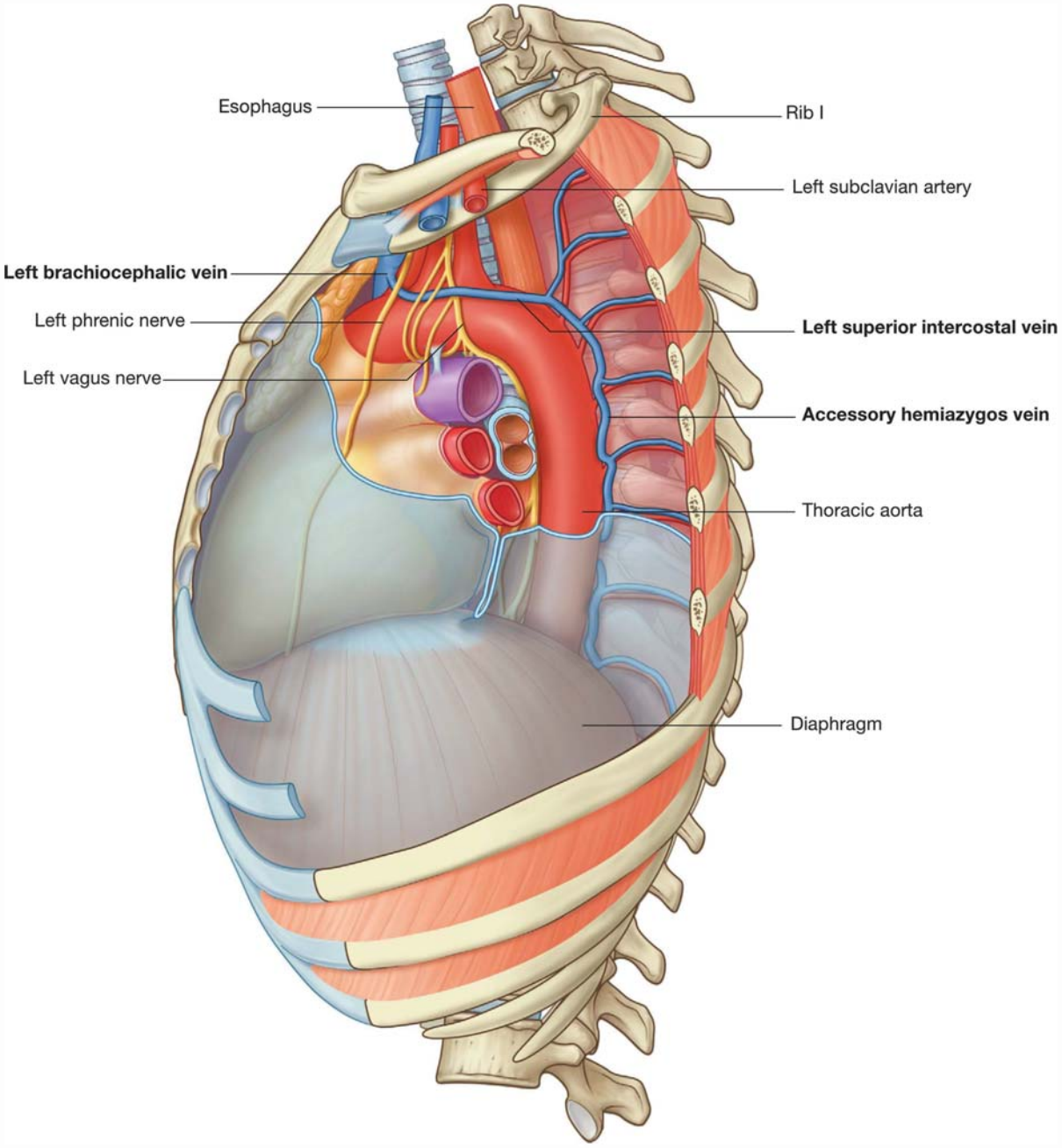


Fig. 3.83 Left superior intercostal vein.

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Superior vena cava

The vertically oriented superior vena cava begins posterior to the lower edge of the right first costal cartilage, where the right and left brachiocephalic veins join, and terminates at the lower edge of the right third costal cartilage, where it joins the right atrium (see Fig. 3.79).

The lower half of the superior vena cava is within the pericardial sac and is therefore contained in the middle mediastinum.

The superior vena cava receives the azygos vein immediately before entering the pericardial sac and may also receive pericardial and mediastinal veins.

The superior vena cava can be easily visualized forming part of the right superolateral border of the mediastinum on a chest radiograph (see Fig. 3.60A).

In the clinic

Venous access for central and dialysis lines

Large systemic veins are used to establish central venous access for administering large amounts of fluid, drugs, and blood. Most of these lines (small-bore tubes) are introduced through venous puncture into the axillary, subclavian, or internal jugular veins. The lines are then passed through the main veins of the superior mediastinum, with the tips of the lines usually residing in the distal portion of the superior vena cava or in the right atrium.

Similar devices, such as dialysis lines, are inserted into patients who have renal failure, so that a large volume of blood can be aspirated through one channel and reinfused through a second channel.

In the clinic

Using the superior vena cava to access the inferior vena cava

Because the superior and inferior venae cavae are oriented along the same vertical axis, a guidewire, catheter, or line can be passed from the superior vena cava through the right atrium and into the inferior vena cava. This is a common route of access for such procedures as:

- transjugular liver biopsy,
- transjugular intrahepatic portosystemic shunts (TIPS), and
- insertion of an inferior vena cava filter to catch emboli dislodged from veins in the lower limb and pelvis (i.e., patients with deep vein thrombosis [DVT]).

Arch of aorta and its branches

The thoracic portion of the aorta can be divided into **ascending aorta**, **arch of the aorta**, and **thoracic** (**descending**) **aorta**. Only the arch of the aorta is in the superior mediastinum. It begins when the ascending aorta emerges from the pericardial sac and courses upward, backward, and to the left as it passes through the superior mediastinum, ending on the left side at vertebral level TIV/V (see Fig. 3.79). Extending as high as the midlevel of the manubrium of the sternum, the arch is initially anterior and finally lateral to the trachea.

Three branches arise from the superior border of the arch of the aorta; at their origins, all three are crossed anteriorly by the left brachiocephalic vein.

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The first branch

Beginning on the right, the first branch of the arch of the aorta is the **brachiocephalic trunk** (Fig. 3.84). It is the largest of the three branches and, at its point of origin behind the manubrium of the sternum, is slightly anterior to the other two branches. It ascends slightly posteriorly and to the right. At the level of the upper edge of the right sternoclavicular joint, the brachiocephalic trunk divides into:

- the **right common carotid artery,** and
- the **right subclavian artery** (see Fig. 3.79).

The arteries mainly supply the right side of the head and neck and the right upper limb, respectively.

Occasionally, the brachiocephalic trunk has a small branch, the **thyroid ima artery**, which contributes to the vascular supply of the thyroid gland.

The second branch

The second branch of the arch of the aorta is the **left common carotid artery** (Fig. 3.84). It arises from the arch immediately to the left and slightly posterior to the brachiocephalic trunk and ascends through the superior mediastinum along the left side of the trachea.

The left common carotid artery supplies the left side of the head and neck.

The third branch

The third branch of the arch of the aorta is the **left subclavian artery** (Fig. 3.84). It arises from the arch of the aorta immediately to the left of, and slightly posterior to, the left common carotid artery and ascends through the superior mediastinum along the left side of the trachea.

The left subclavian artery is the major blood supply to the left upper limb.

Ligamentum arteriosum

The **ligamentum arteriosum** is also in the superior mediastinum and is important in embryonic circulation, when it is a patent vessel (the **ductus arteriosus**). It connects the pulmonary trunk with the arch of the aorta and allows blood to bypass the lungs during development (Fig. 3.84). The vessel closes soon after birth and forms the ligamentous connection observed in the adult.

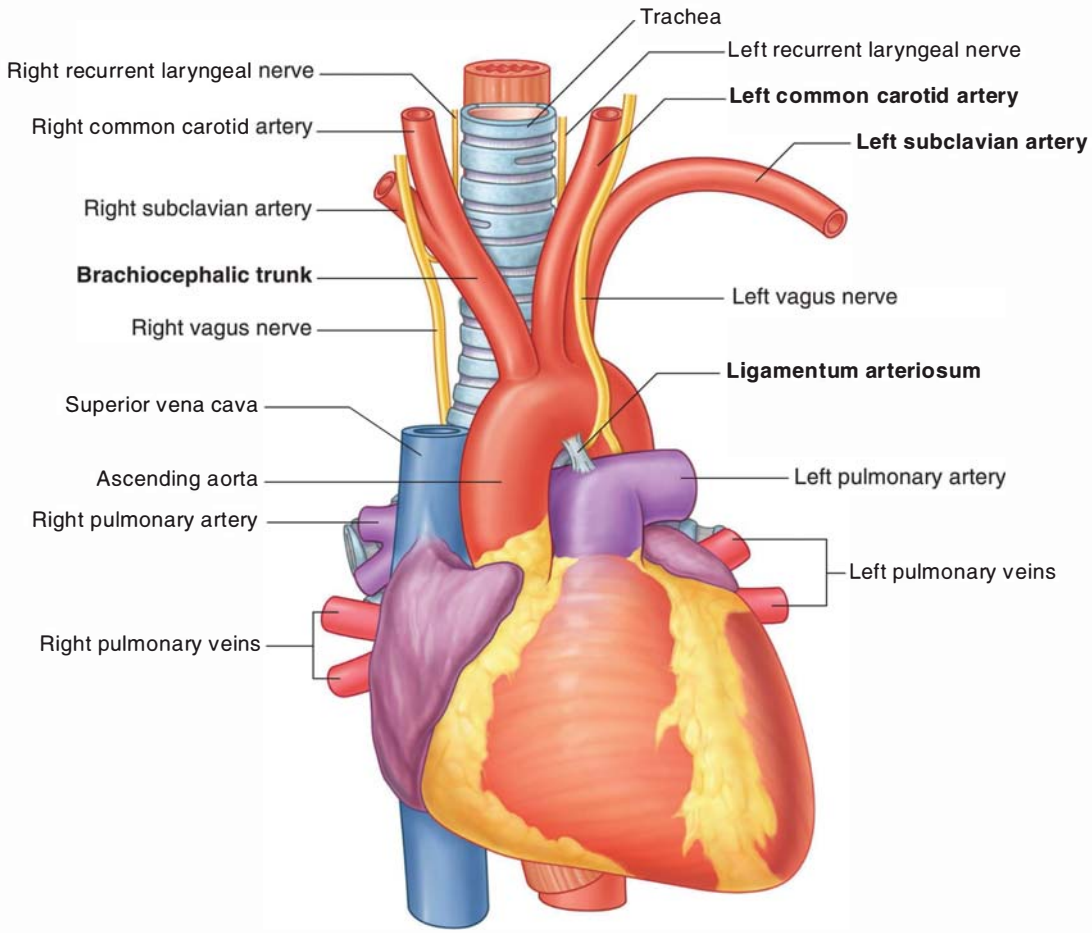


Fig. 3.84 Superior mediastinum with thymus and venous channels removed.

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In the clinic

Coarctation of the aorta

Coarctation of the aorta is a congenital abnormality in which the aortic lumen is constricted just distal to the origin of the left subclavian artery. At this point, the aorta becomes significantly narrowed and the blood supply to the lower limbs and abdomen is diminished. Over time, collateral vessels develop around the chest wall and abdomen to supply the lower body. The coarctation also affects the heart, which has to pump the blood at higher pressure to maintain peripheral perfusion. This in turn may produce cardiac failure.

In the clinic

Aortic arch and its anomalies

A right-sided arch of aorta occasionally occurs and may be asymptomatic. It can be associated with **dextrocardia** (right-sided heart) and, in some instances, with complete **situs inversus** (left-to-right inversion of the body's organs). It can also be associated with abnormal branching of the great vessels.

In the clinic

Thoracic aorta

Diffuse atherosclerosis of the thoracic aorta may occur in patients with vascular disease, but this rarely produces symptoms. There are, however, two clinical situations in which aortic pathology can produce life-threatening situations.

Trauma

The aorta has three fixed points of attachment:

- the aortic valve,
- the ligamentum arteriosum, and
- the point of passing behind the median arcuate ligament of the diaphragm to enter the abdomen.

The rest of the aorta is relatively free from attachment to other structures of the mediastinum. A serious deceleration injury (e.g., in a road traffic accident) is most likely to cause aortic trauma at these fixed points.

Aortic Dissection

In certain conditions, such as in severe arteriovascular disease, the wall of the aorta can split longitudinally, creating a false channel, which may or may not rejoin into the true lumen distally. This aortic dissection occurs between the intima and media anywhere along its length. If it occurs in the ascending aorta or arch of the aorta, blood flow in the coronary and cerebral arteries may be disrupted, resulting in myocardial infarction or stroke. In the abdomen the visceral vessels may be disrupted, producing ischemia to the gut or kidneys.

In the clinic

Abnormal origin of great vessels

Great vessels occasionally have an abnormal origin, including:

- a common origin of the brachiocephalic trunk and the left common carotid artery,
- the left vertebral artery originating from the aortic arch, and
- the right subclavian artery originating from the distal portion of the aortic arch and passing behind the esophagus to supply the right arm—as a result, the great vessels form a vascular ring around the trachea and the esophagus, which can potentially produce difficulty swallowing.

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Trachea and esophagus

The trachea is a midline structure that is palpable in the jugular notch as it enters the superior mediastinum. Posterior to it is the esophagus, which is immediately anterior to the vertebral column (Fig. 3.85, and see Figs. 3.79 and 3.80). Significant mobility exists in the vertical positioning of these structures as they pass through the superior mediastinum. Swallowing and breathing cause positional shifts, as may disease and the use of specialized instrumentation.

As the trachea and esophagus pass through the superior mediastinum, they are crossed laterally by the azygos vein on the right side and the arch of the aorta on the left side.

The trachea divides into the right and left main bronchi at, or just inferior to, the transverse plane between the sternal angle and vertebral level TIV/V (Fig. 3.86), whereas the esophagus continues into the posterior mediastinum.

Nerves of the superior mediastinum

Vagus nerves

The **vagus nerves** [X] pass through the superior and posterior divisions of the mediastinum on their way to the abdominal cavity. As they pass through the thorax, they provide parasympathetic innervation to the thoracic viscera and carry visceral afferents from the thoracic viscera.

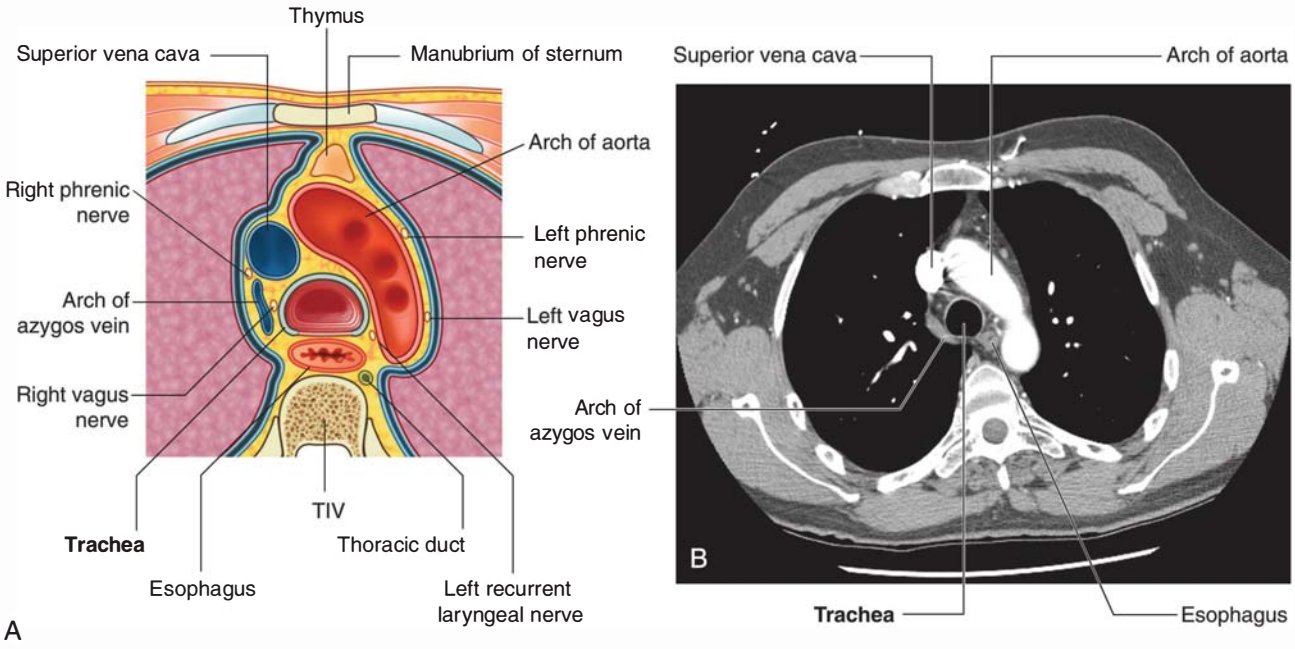


Fig. 3.85 Cross section through the superior mediastinum at the level of vertebra TIV. **A.** Diagram. **B.** Axial computed tomography image.

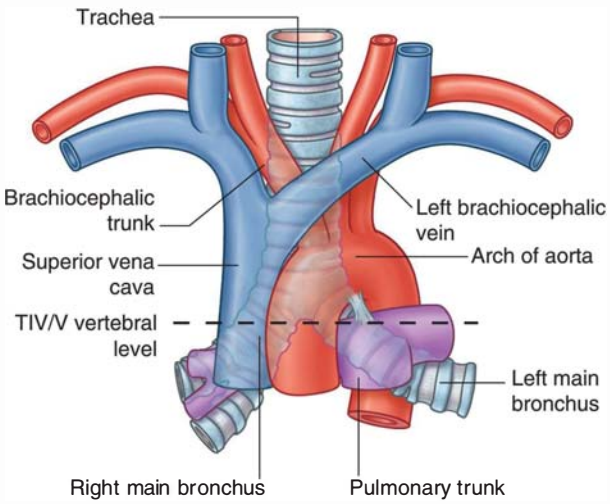


Fig. 3.86 Trachea in the superior mediastinum.

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Visceral afferents in the vagus nerves relay information to the central nervous system about normal physiological processes and reflex activities. They do not transmit pain sensation.

Right vagus nerve

The **right vagus nerve** enters the superior mediastinum and lies between the right brachiocephalic vein and the brachiocephalic trunk. It descends in a posterior direction toward the trachea (Fig. 3.87), crosses the lateral surface of the trachea, and passes posteriorly to the root of the right lung to reach the esophagus. Just before the esophagus, it is crossed by the arch of the azygos vein.

As the right vagus nerve passes through the superior mediastinum, it gives branches to the esophagus, cardiac plexus, and pulmonary plexus.

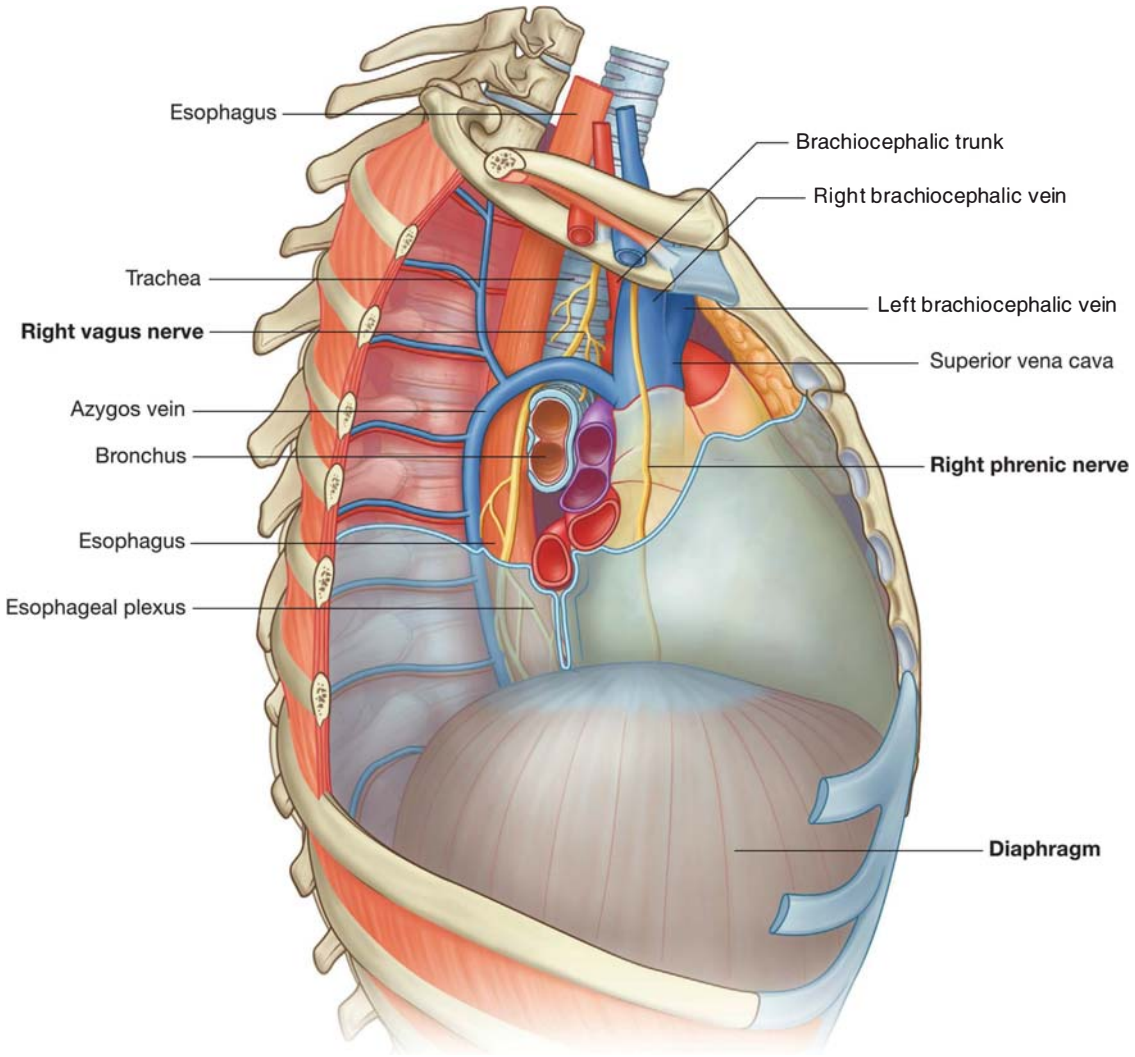


Fig. 3.87 Right vagus nerve passing through the superior mediastinum.

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Left vagus nerve

The **left vagus nerve** enters the superior mediastinum posterior to the left brachiocephalic vein and between the left common carotid and left subclavian arteries (Fig. 3.88). As it passes into the superior mediastinum, it lies just deep to the mediastinal part of the parietal pleura and crosses the left side of the arch of the aorta. It continues to descend in a posterior direction and passes posterior to the root of the left lung to reach the esophagus in the posterior mediastinum.

As the left vagus nerve passes through the superior mediastinum, it gives branches to the esophagus, the cardiac plexus, and the pulmonary plexus.

The left vagus nerve also gives rise to the **left recurrent laryngeal nerve**, which arises from it at the inferior margin of the arch of the aorta just lateral to the ligamentum arteriosum. The left recurrent laryngeal nerve passes inferior to the arch of the aorta before ascending on its medial surface. Entering a groove between the trachea and esophagus, the left recurrent laryngeal nerve continues

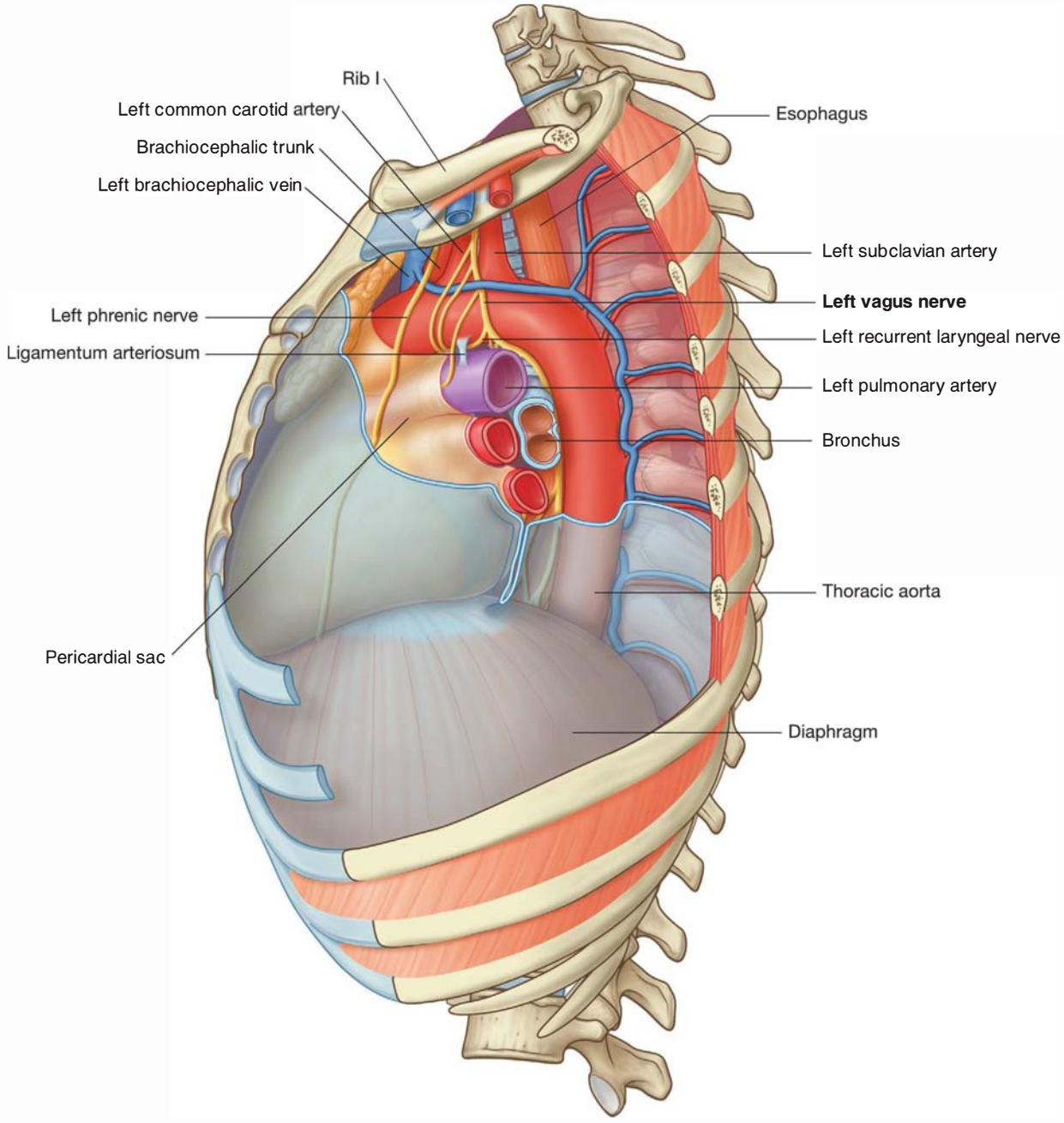


Fig. 3.88 Left vagus nerve passing through the superior mediastinum.

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superiorly to enter the neck and terminate in the larynx (Fig. 3.89).

Phrenic nerves

The phrenic nerves arise in the cervical region mainly from the fourth, but also from the third and fifth, cervical spinal cord segments.

The phrenic nerves descend through the thorax to supply motor and sensory innervation to the diaphragm and its associated membranes. As they pass through the thorax, they provide innervation through somatic afferent fibers to the mediastinal pleura, fibrous pericardium, and parietal layer of serous pericardium.

Right phrenic nerve

The **right phrenic nerve** enters the superior mediastinum lateral to the right vagus nerve and lateral and slightly posterior to the beginning of the right brachiocephalic vein (see Fig. 3.87). It continues inferiorly along the right side of this vein and the right side of the superior vena cava.

On entering the middle mediastinum, the right phrenic nerve descends along the right side of the pericardial sac, within the fibrous pericardium, anterior to the root of the right lung. The pericardiacophrenic vessels accompany it through most of its course in the thorax (see Fig. 3.54). It leaves the thorax by passing through the diaphragm with the inferior vena cava.

Left phrenic nerve

The **left phrenic nerve** enters the superior mediastinum in a position similar to the path taken by the right phrenic nerve. It lies lateral to the left vagus nerve and lateral and slightly posterior to the beginning of the left brachiocephalic vein (see Fig. 3.83), and continues to descend across the left lateral surface of the arch of the aorta, passing superficially to the left vagus nerve and the left superior intercostal vein.

On entering the middle mediastinum, the left phrenic nerve follows the left side of the pericardial sac, within the fibrous pericardium, anterior to the root of the left lung, and is accompanied by the pericardiacophrenic vessels (see Fig. 3.54). It leaves the thorax by piercing the diaphragm near the apex of the heart.

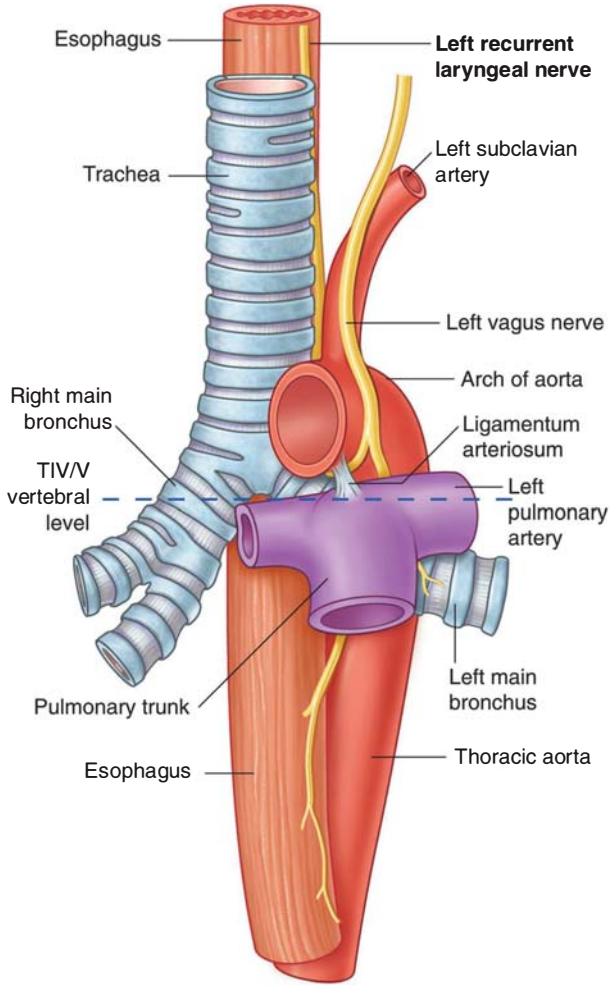


Fig. 3.89 Left recurrent laryngeal nerve passing through the superior mediastinum.

In the clinic

The vagus nerves, recurrent laryngeal nerves, and hoarseness

The left recurrent laryngeal nerve is a branch of the left vagus nerve. It passes between the pulmonary artery and the aorta, a region known clinically as the **aortopulmonary window**, and may be compressed in any patient with a pathological mass in this region. This compression results in vocal cord paralysis and hoarseness of the voice. Lymph node enlargement, often associated with the spread of lung cancer, is a common condition that may produce compression. Chest radiography is therefore usually carried out for all patients whose symptoms include a hoarse voice.

More superiorly, in the root of the neck, the right vagus nerve gives off the right recurrent laryngeal nerve, which "hooks" around the right subclavian artery as it passes over the cervical pleura. If a patient has a hoarse voice and a right vocal cord palsy is demonstrated at laryngoscopy, chest radiography with an apical lordotic view should be obtained to assess for cancer in the right lung apex (**Pancoast's tumor**).

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Thoracic duct in the superior mediastinum

The **thoracic duct**, which is the major lymphatic vessel in the body, passes through the posterior portion of the superior mediastinum (see Figs. 3.80 and 3.85). It:

- enters the superior mediastinum inferiorly, slightly to the left of the midline, having moved to this position just before leaving the posterior mediastinum opposite vertebral level TIV/V; and
- continues through the superior mediastinum, posterior to the arch of the aorta, and the initial portion of the left subclavian artery, between the esophagus and the left mediastinal part of the parietal pleura.

Posterior mediastinum

The **posterior mediastinum** is posterior to the pericardial sac and diaphragm and anterior to the bodies of the mid and lower thoracic vertebrae (see Fig. 3.52).

- Its superior boundary is a transverse plane passing from the sternal angle to the intervertebral disc between vertebrae TIV and TV.
- Its inferior boundary is the diaphragm.
- Laterally, it is bordered by the mediastinal part of parietal pleura on either side.
- Superiorly, it is continuous with the superior mediastinum.

Major structures in the posterior mediastinum include the:

- esophagus and its associated nerve plexus,
- thoracic aorta and its branches,
- azygos system of veins,
- thoracic duct and associated lymph nodes,
- sympathetic trunks, and
- thoracic splanchnic nerves.

Esophagus

The **esophagus** is a muscular tube passing between the pharynx in the neck and the stomach in the abdomen. It begins at the inferior border of the cricoid cartilage, opposite vertebra CVI, and ends at the cardiac opening of the stomach, opposite vertebra TXI.

The esophagus descends on the anterior aspect of the bodies of the vertebrae, generally in a midline position as it moves through the thorax (Fig. 3.90). As it approaches the diaphragm, it moves anteriorly and to the left, crossing from the right side of the thoracic aorta to eventually assume a position anterior to it. It then passes through the esophageal hiatus, an opening in the muscular part of the diaphragm, at vertebral level TX.

The esophagus has a slight anterior-to-posterior curvature that parallels the thoracic portion of the vertebral column, and is secured superiorly by its attachment to the pharynx and inferiorly by its attachment to the diaphragm.

Relationships to important structures in the posterior mediastinum

In the posterior mediastinum, the esophagus is related to a number of important structures. The right side is covered by the mediastinal part of the parietal pleura.

Posterior to the esophagus, the thoracic duct is on the right side inferiorly, but crosses to the left more superiorly. Also on the left side of the esophagus is the thoracic aorta.

Anterior to the esophagus, below the level of the tracheal bifurcation, are the right pulmonary artery and the left main bronchus. The esophagus then passes immediately posteriorly to the left atrium, separated from it only by pericardium. Inferior to the left atrium, the esophagus is related to the diaphragm.

Structures other than the thoracic duct posterior to the esophagus include portions of the hemiazygos veins, the right posterior intercostal vessels, and, near the diaphragm, the thoracic aorta.

The esophagus is a flexible, muscular tube that can be compressed or narrowed by surrounding structures at four locations (Fig. 3.91):

- the junction of the esophagus with the pharynx in the neck;
- in the superior mediastinum where the esophagus is crossed by the arch of the aorta;
- in the posterior mediastinum where the esophagus is compressed by the left main bronchus;
- in the posterior mediastinum at the esophageal hiatus in the diaphragm.

These constrictions have important clinical consequences. For example, a swallowed object is most likely to lodge at a constricted area. An ingested corrosive substance would move more slowly through a narrowed region, causing more damage at this site than elsewhere along the esophagus. Also, constrictions present problems during the passage of instruments.

Arterial supply and venous and lymphatic drainage

The arterial supply and venous drainage of the esophagus in the posterior mediastinum involve many vessels. Esophageal arteries arise from the thoracic aorta, bronchial arteries, and ascending branches of the left gastric artery in the abdomen.

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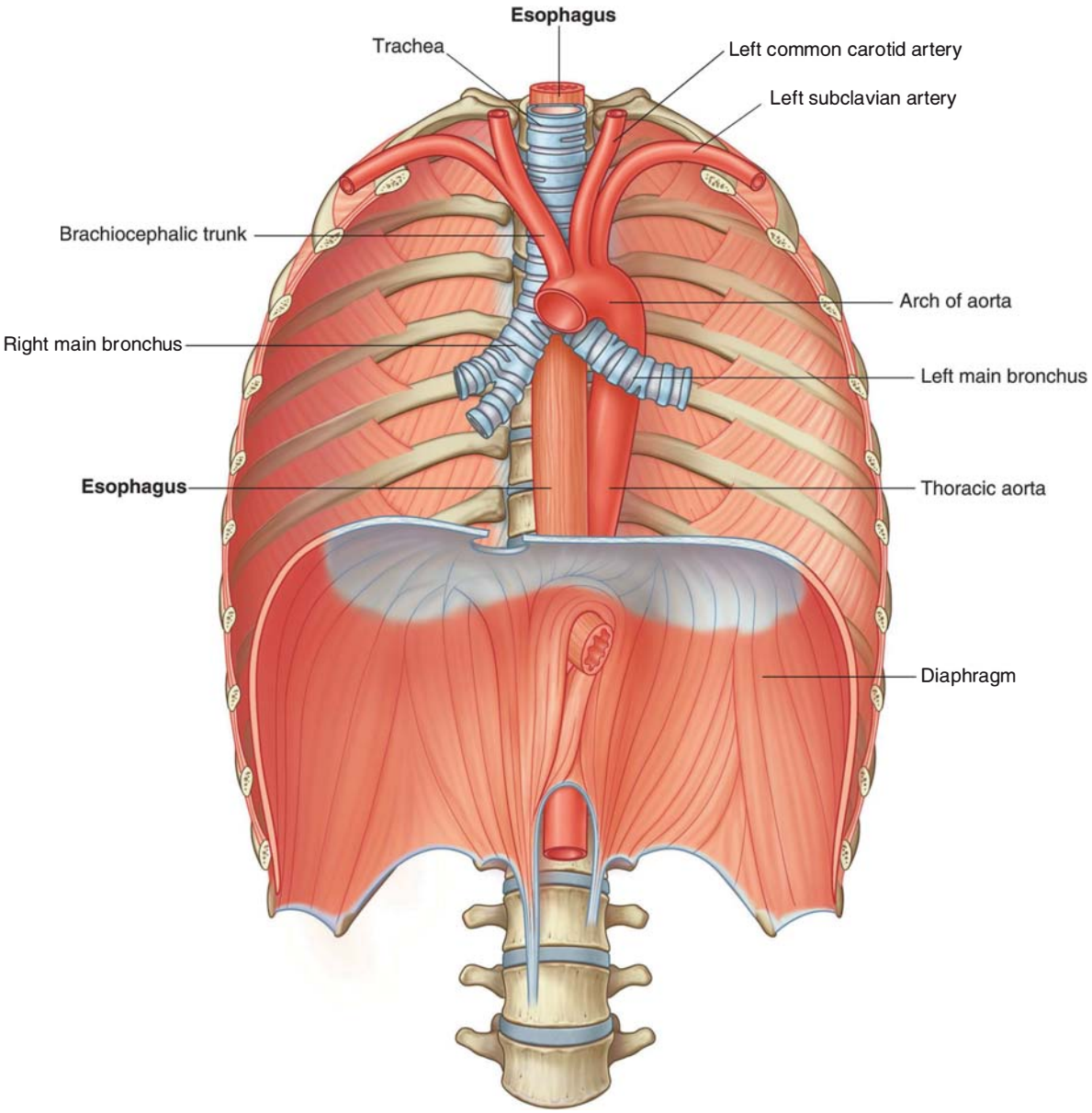


Fig. 3.90 Esophagus.

Venous drainage involves small vessels returning to the azygos vein, hemiazygos vein, and esophageal branches to the left gastric vein in the abdomen.

Lymphatic drainage of the esophagus in the posterior mediastinum returns to posterior mediastinal and left gastric nodes.

Innervation

Innervation of the esophagus, in general, is complex. Esophageal branches arise from the vagus nerves and sympathetic trunks.

Striated muscle fibers in the superior portion of the esophagus originate from the branchial arches and are innervated by branchial efferents from the vagus nerves.

Smooth muscle fibers are innervated by components of the parasympathetic part of the autonomic division of the peripheral nervous system, visceral efferents from the vagus nerves. These are preganglionic fibers that synapse in the myenteric and submucosal plexuses of the enteric nervous system in the esophageal wall.

Sensory innervation of the esophagus involves visceral afferent fibers originating in the vagus nerves, sympathetic trunks, and splanchnic nerves.

The visceral afferents from the vagus nerves are involved in relaying information back to the central nervous system about normal physiological processes and reflex activities. They are not involved in the relay of pain recognition.

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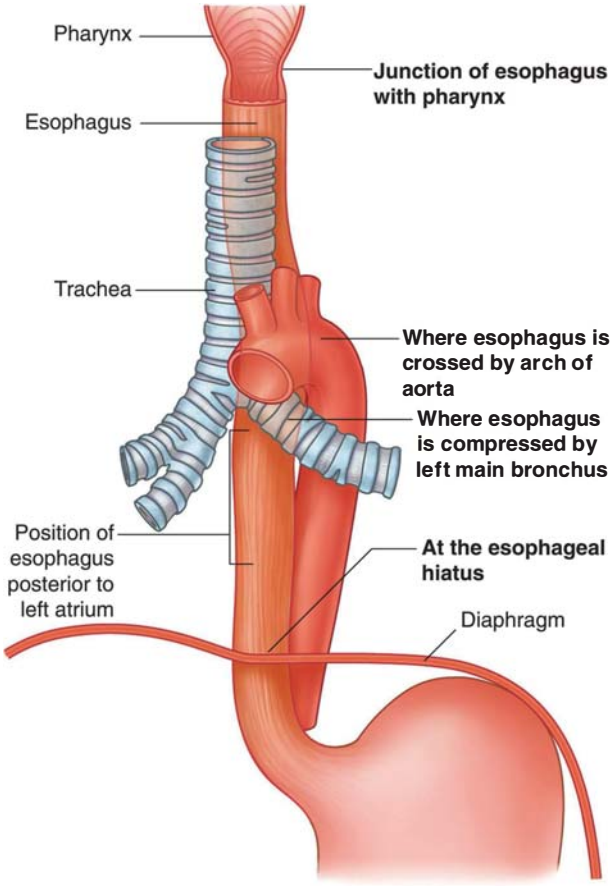


Fig. 3.91 Sites of normal esophageal constrictions.

The visceral afferents that pass through the sympathetic trunks and the splanchnic nerves are the primary participants in detection of esophageal pain and transmission of this information to various levels of the central nervous system.

Esophageal plexus

After passing posteriorly to the root of the lungs, the right and left vagus nerves approach the esophagus. As they reach the esophagus, each nerve divides into several branches that spread over this structure, forming the **esophageal plexus** (Fig. 3.92). There is some mixing of fibers from the two vagus nerves as the plexus continues inferiorly on the esophagus toward the diaphragm. Just above the diaphragm, fibers of the plexus converge to form two trunks:

- the **anterior vagal trunk** on the anterior surface of the esophagus, mainly from fibers originally in the left vagus nerve;
- the **posterior vagal trunk** on the posterior surface of the esophagus, mainly from fibers originally in the right vagus nerve.

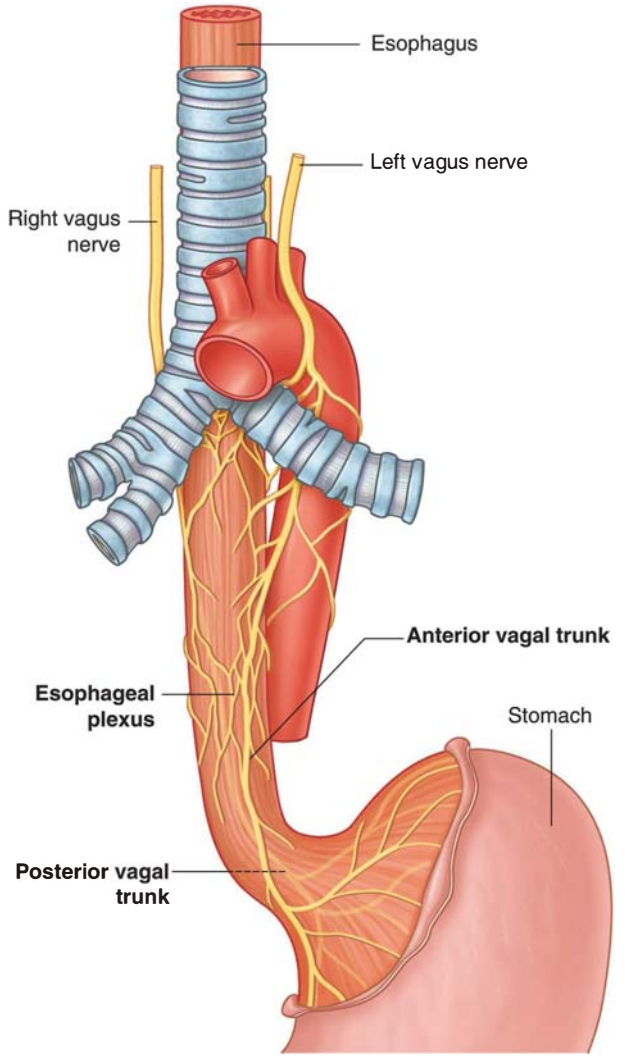


Fig. 3.92 Esophageal plexus.

The vagal trunks continue on the surface of the esophagus as it passes through the diaphragm into the abdomen.

In the clinic

Esophageal cancer

When patients present with esophageal cancer, it is important to note which portion of the esophagus contains the tumor because tumor location determines the sites to which the disease will spread.

Esophageal cancer spreads quickly to lymphatics, draining to lymph nodes in the neck and around the celiac artery. Endoscopy or barium swallow is used to assess the site. CT and MRI may be necessary to stage the disease.

Once the extent of the disease has been assessed, treatment can be planned.

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In the clinic

Esophageal rupture

The first case of esophageal rupture was described by Herman Boerhaave in 1724. This case was fatal, but early diagnosis has increased the survival rate up to 65%. If the disease is left untreated, mortality is 100%.

Typically, the rupture occurs in the lower third of the esophagus with a sudden rise in intraluminal esophageal pressure produced by vomiting secondary to an uncoordination and failure of the cricopharyngeus muscle to relax. Because the tears typically occur on the left, they are often associated with a large left pleural effusion that contains the gastric contents. In some patients, subcutaneous emphysema may be demonstrated.

Treatment is optimal with urgent surgical repair.

Thoracic aorta

The thoracic portion of the descending aorta (**thoracic aorta**) begins at the lower edge of vertebra TIV, where it is continuous with the arch of the aorta. It ends anterior to the lower edge of vertebra TXII, where it passes through the aortic hiatus posterior to the diaphragm. Situated to the left of the vertebral column superiorly, it approaches the midline inferiorly, lying directly anterior to the lower thoracic vertebral bodies (Fig. 3.93). Throughout its course, it gives off a number of branches, which are summarized in Table 3.3.

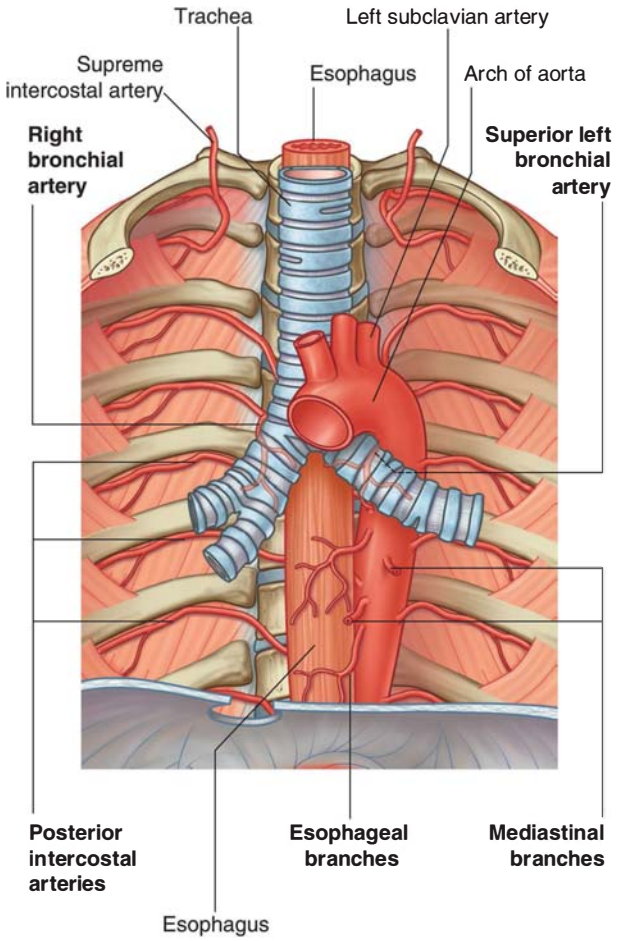


Fig. 3.93 Thoracic aorta and branches.

Table 3.3 Branches of the thoracic aorta

Branches	Origin and course
Pericardial branches	A few small vessels to the posterior surface of the pericardial sac
Bronchial branches	Vary in number, size, and origin—usually, two left bronchial arteries from the thoracic aorta and one right bronchial artery from the third posterior intercostal artery or the superior left bronchial artery
Esophageal branches	Four or five vessels from the anterior aspect of the thoracic aorta, which form a continuous anastomotic chain—anastomotic connections include esophageal branches of the inferior thyroid artery superiorly, and esophageal branches of the left inferior phrenic and the left gastric arteries inferiorly
Mediastinal branches	Several small branches supplying lymph nodes, vessels, nerves, and areolar tissue in the posterior mediastinum
Posterior intercostal arteries	Usually, nine pairs of vessels branching from the posterior surface of the thoracic aorta—usually supply lower nine intercostal spaces (first two spaces are supplied by the supreme intercostal artery—a branch of the costocervical trunk)
Superior phrenic arteries	Small vessels from the lower part of the thoracic aorta supplying the posterior part of the superior surface of the diaphragm—they anastomose with the musculophrenic and pericardiacophrenic arteries
Subcostal artery	The lowest pair of branches from the thoracic aorta located inferior to rib XII

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Azygos system of veins

The azygos system of veins consists of a series of longitudinal vessels on each side of the body that drain blood from the body wall and move it superiorly to empty into the superior vena cava. Blood from some of the thoracic viscera may also enter the system, and there are anastomotic connections with abdominal veins.

The longitudinal vessels may or may not be continuous and are connected to each other from side to side at various points throughout their course (Fig. 3.94).

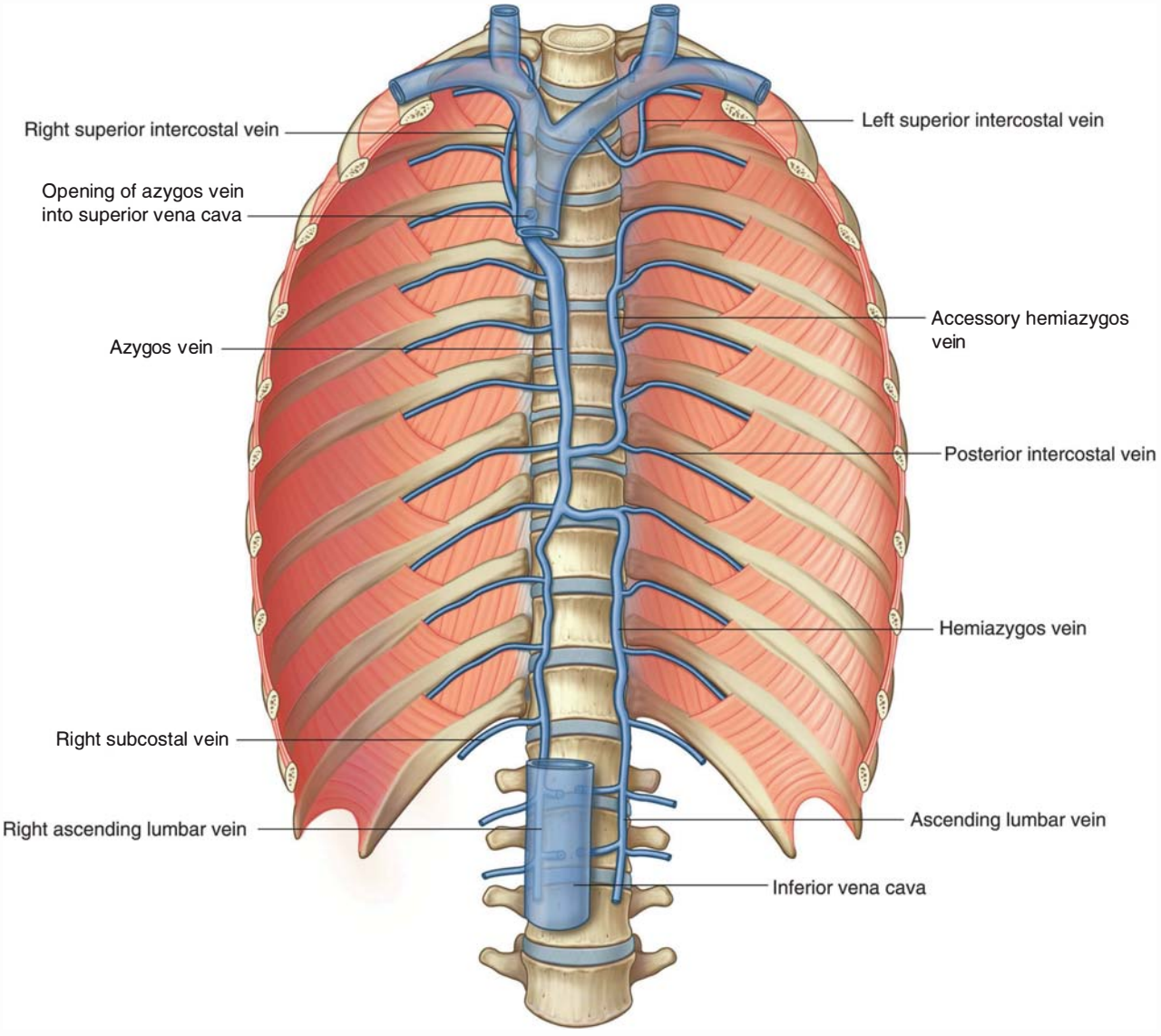


Fig. 3.94 Azygos system of veins.

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The azygos system of veins serves as an important anastomotic pathway capable of returning venous blood from the lower part of the body to the heart if the inferior vena cava is blocked.

The major veins in the system are:

- the azygos vein, on the right; and
- the hemiazygos vein and the accessory hemiazygos vein, on the left.

There is significant variation in the origin, course, tributaries, anastomoses, and termination of these vessels.

Azygos vein

The **azygos vein** arises opposite vertebra LI or LII at the junction between the **right ascending lumbar vein** and the **right subcostal vein** (Fig. 3.94). It may also arise as a direct branch of the inferior vena cava, which is joined by a common trunk from the junction of the right ascending lumbar vein and the right subcostal vein.

The azygos vein enters the thorax through the aortic hiatus of the diaphragm, or it enters through or posterior to the right crus of the diaphragm. It ascends through the posterior mediastinum, usually to the right of the thoracic duct. At approximately vertebral level TIV, it arches anteriorly, over the root of the right lung, to join the superior vena cava before the superior vena cava enters the pericardial sac.

Tributaries of the azygos vein include:

- the **right superior intercostal vein** (a single vessel formed by the junction of the second, third, and fourth intercostal veins),
- fifth to eleventh right posterior intercostal veins,
- the hemiazygos vein,
- the accessory hemiazygos vein,
- esophageal veins,
- mediastinal veins,
- pericardial veins, and
- right bronchial veins.

Hemiazygos vein

The **hemiazygos vein** (**inferior hemiazygos vein**) usually arises at the junction between the **left ascending lumbar vein** and the **left subcostal vein** (Fig. 3.94). It may also arise from either of these veins alone and often has a connection to the left renal vein.

The hemiazygos vein usually enters the thorax through the left crus of the diaphragm, but may enter through the aortic hiatus. It ascends through the posterior mediastinum, on the left side, to approximately vertebral level TIX. At this point, it crosses the vertebral column, posterior to the thoracic aorta, esophagus, and thoracic duct, to enter the azygos vein.

Tributaries joining the hemiazygos vein include:

- the lowest four or five left posterior intercostal veins,
- esophageal veins, and
- mediastinal veins.

Accessory hemiazygos vein

The **accessory hemiazygos vein** (**superior hemiazygos vein**) descends on the left side from the superior portion of the posterior mediastinum to approximately vertebral level TVIII (Fig. 3.94). At this point, it crosses the vertebral column to join the azygos vein, or ends in the hemiazygos vein, or has a connection to both veins. Usually, it also has a connection superiorly to the **left superior intercostal vein**.

Vessels that drain into the accessory hemiazygos vein include:

- the fourth to eighth left posterior intercostal veins, and
- sometimes, the left bronchial veins.

Thoracic duct in the posterior mediastinum

The thoracic duct is the principal channel through which lymph from most of the body is returned to the venous system. It begins as a confluence of lymph trunks in the abdomen, sometimes forming a saccular dilation referred to as the **cisterna chyli** (**chyle cistern**), which drains the abdominal viscera and walls, pelvis, perineum, and lower limbs.

The thoracic duct extends from vertebra LII to the root of the neck.

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Entering the thorax, posterior to the aorta, through the aortic hiatus of the diaphragm, the thoracic duct ascends through the posterior mediastinum to the right of midline between the thoracic aorta on the left and the azygos vein on the right (Fig. 3.95). It lies posterior to the diaphragm and the esophagus and anterior to the bodies of the vertebrae.

At vertebral level TV, the thoracic duct moves to the left of midline and enters the superior mediastinum. It continues through the superior mediastinum and into the neck.

After being joined, in most cases, by the **left jugular trunk**, which drains the left side of the head and neck, and the **left subclavian trunk**, which drains the left upper limb, the thoracic duct empties into the junction of the left subclavian and left internal jugular veins.

The thoracic duct usually receives the contents from:

- the confluence of lymph trunks in the abdomen,
- descending thoracic lymph trunks draining the lower six or seven intercostal spaces on both sides,
- upper intercostal lymph trunks draining the upper left five or six intercostal spaces,
- ducts from posterior mediastinal nodes, and
- ducts from posterior diaphragmatic nodes.

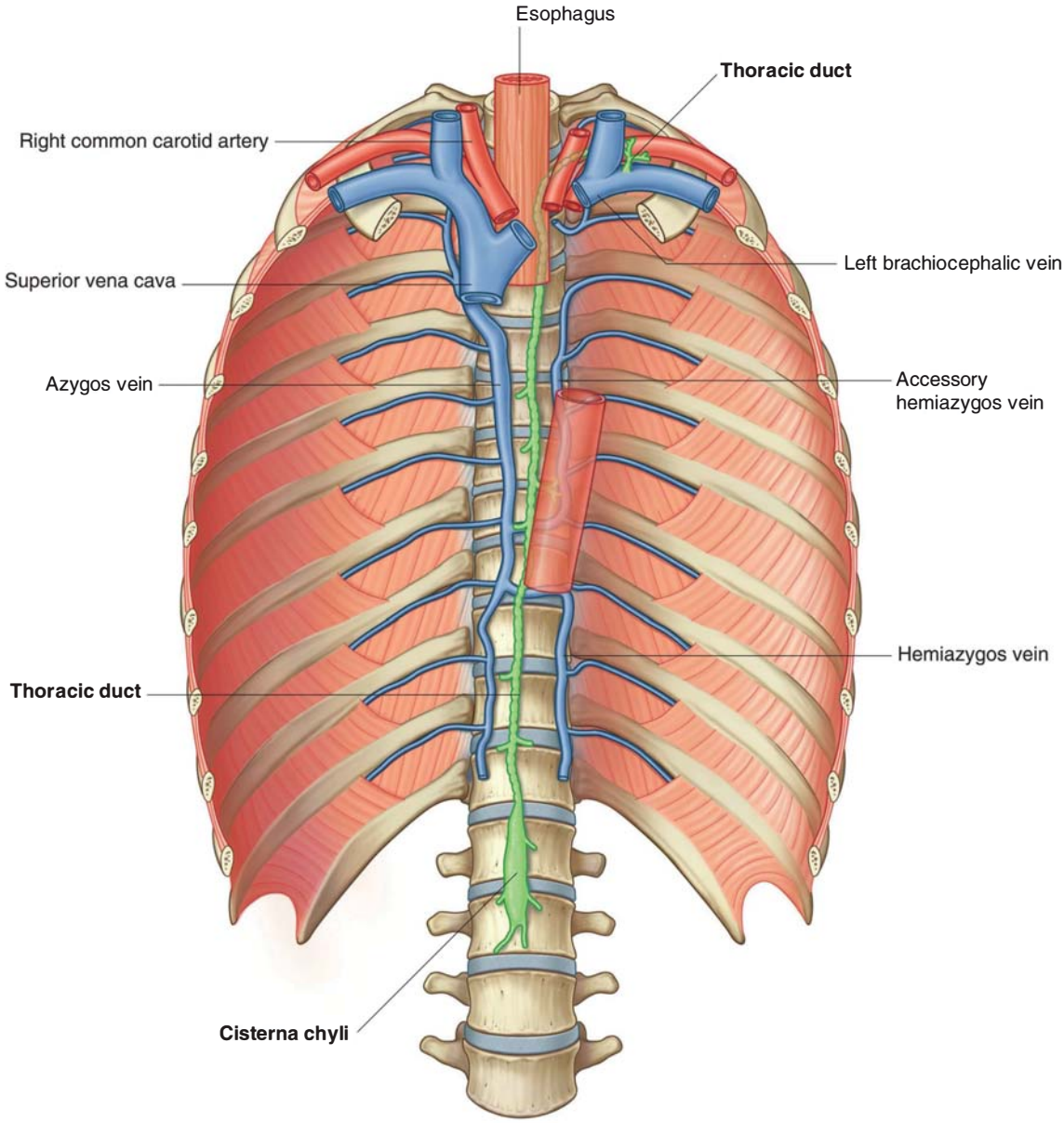


Fig. 3.95 Thoracic duct.

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Sympathetic trunks

The **sympathetic trunks** are an important component of the sympathetic part of the autonomic division of the peripheral nervous system and are usually considered a component of the posterior mediastinum as they pass through the thorax.

This portion of the sympathetic trunks consists of two parallel cords punctuated by 11 or 12 **ganglia** (Fig. 3.96). The ganglia are connected to adjacent thoracic spinal nerves by **white** and **gray rami communicantes** and are numbered according to the thoracic spinal nerve with which they are associated.

In the superior portion of the posterior mediastinum, the trunks are anterior to the neck of the ribs. Inferiorly, they become more medial in position until they lie on the lateral aspect of the vertebral bodies. The sympathetic trunks leave the thorax by passing posterior to the diaphragm under the medial arcuate ligament or through the crura of the diaphragm. Throughout their course the trunks are covered by parietal pleura.

Branches from the ganglia

Two types of medial branches are given off by the ganglia:

- The first type includes branches from the upper five ganglia.
- The second type includes branches from the lower seven ganglia.

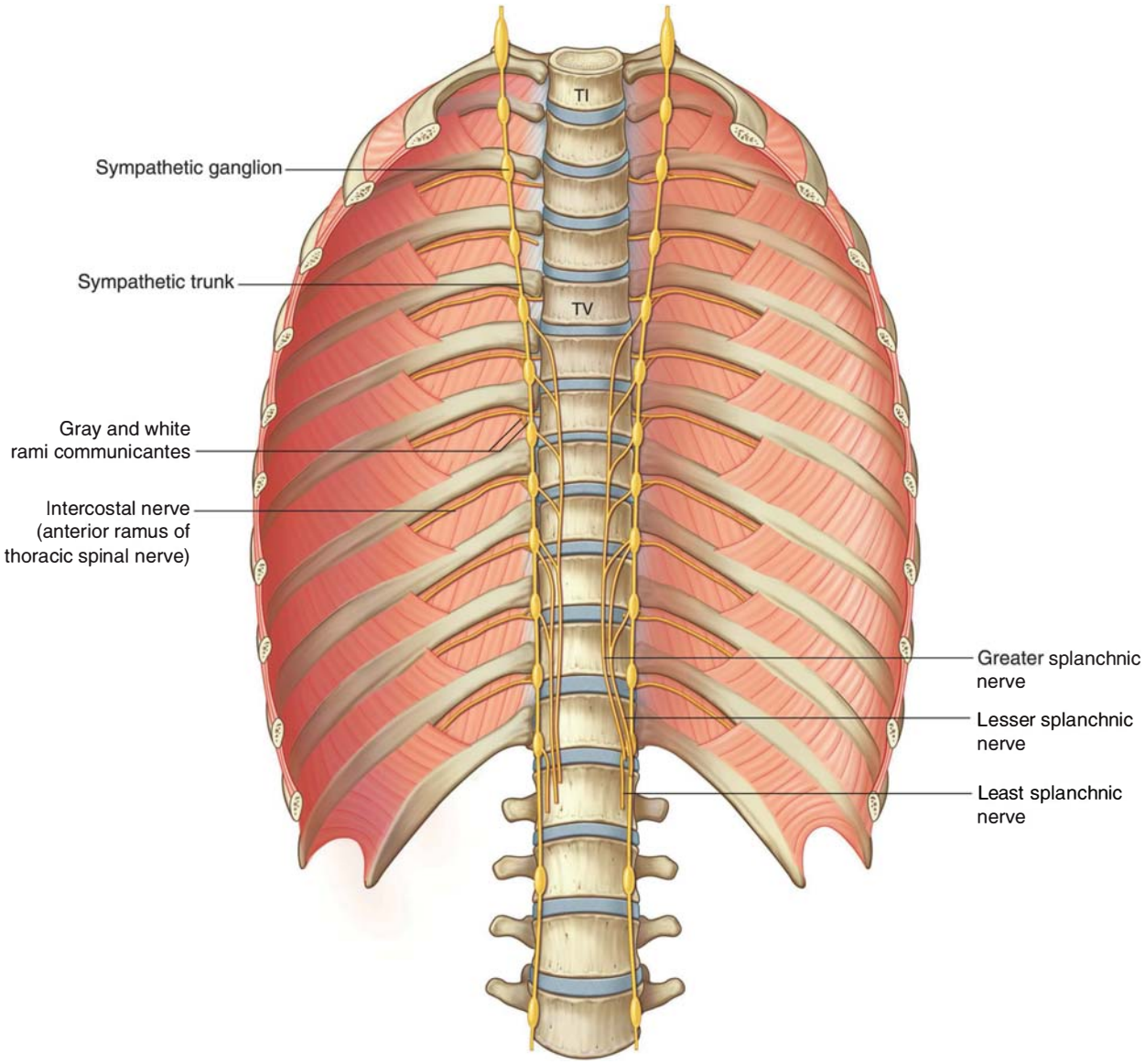


Fig. 3.96 Thoracic portion of sympathetic trunks.

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The first type, which includes branches from the upper five ganglia, consists mainly of postganglionic sympathetic fibers, which supply the various thoracic viscera. These branches are relatively small, and also contain visceral afferent fibers.

The second type, which includes branches from the lower seven ganglia, consists mainly of preganglionic sympathetic fibers, which supply the various abdominal and pelvic viscera. These branches are large, also carry visceral afferent fibers, and form the three thoracic splanchnic nerves referred to as the greater, lesser, and least splanchnic nerves (Fig. 3.96).

- The **greater splanchnic nerve** on each side usually arises from the fifth to ninth or tenth thoracic ganglia. It descends across the vertebral bodies moving in a medial direction, passes into the abdomen through the crus of the diaphragm, and ends in the celiac ganglion.
- The **lesser splanchnic nerve** usually arises from the ninth and tenth, or tenth and eleventh thoracic ganglia. It descends across the vertebral bodies moving in a medial direction, and passes into the abdomen through the crus of the diaphragm to end in the aorticorenal ganglion.
- The **least splanchnic nerve** (**lowest splanchnic nerve**) usually arises from the twelfth thoracic ganglion. It descends and passes into the abdomen through the crus of the diaphragm to end in the renal plexus.

Anterior mediastinum

The **anterior mediastinum** is posterior to the body of the sternum and anterior to the pericardial sac (see Fig. 3.52).

- Its superior boundary is a transverse plane passing from the sternal angle to the intervertebral disc between vertebra TIV and TV, separating it from the superior mediastinum.
- Its inferior boundary is the diaphragm.
- Laterally, it is bordered by the mediastinal part of parietal pleura on either side.

The major structure in the anterior mediastinum is a portion of thymus, described previously (see Fig. 3.81). Also present are fat, connective tissue, lymph nodes, mediastinal branches of the internal thoracic vessels, and sternopericardial ligaments, which pass from the posterior surface of the body of the sternum to the fibrous pericardium.

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Surface anatomy

Thorax surface anatomy

The ability to visualize how anatomical structures in the thorax are related to surface features is fundamental to a physical examination. Landmarks on the body's surface can be used to locate deep structures and to assess function by auscultation and percussion.

How to count ribs

Knowing how to count ribs is important because different ribs provide palpable landmarks for the positions of deeper structures. To determine the location of specific ribs, palpate the **jugular notch** at the superior extent of the manubrium of the sternum. Move down the sternum until a ridge is felt. This ridge is the **sternal angle**, which identifies the articulation between the manubrium of the sternum and the body of the sternum. The costal cartilage of rib II articulates with the sternum at this location. Identify rib II. Then continue counting the ribs, moving in a downward and lateral direction (Fig. 3.97).

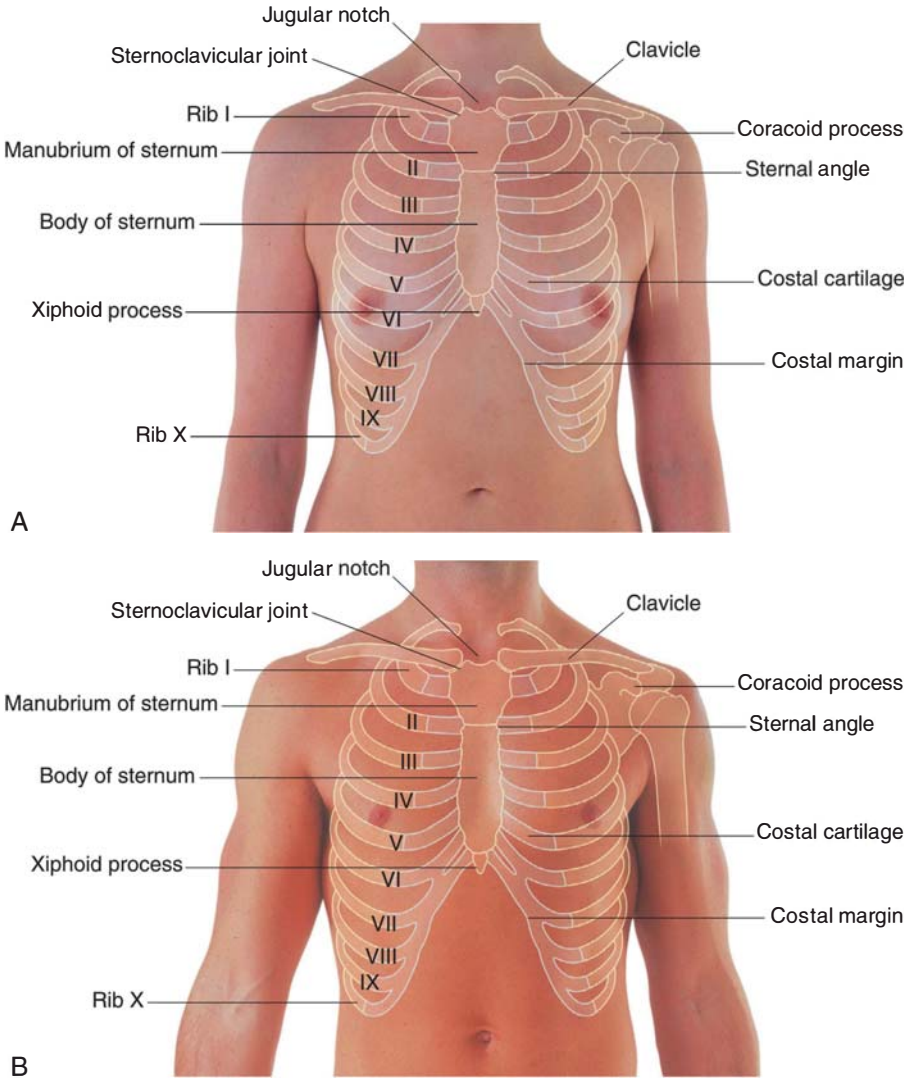


Fig. 3.97 Anterior view of chest wall with the locations of skeletal structures shown. **A.** In women. The location of the nipple relative to a specific intercostal space varies depending on the size of the breasts, which may not be symmetrical. **B.** In men. Note the location of the nipple in the fourth intercostal space.

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Surface anatomy of the breast in women

Although breasts vary in size, they are normally positioned on the thoracic wall between ribs II and VI and overlie the pectoralis major muscles. Each mammary gland extends superolaterally around the lower margin of the pectoralis major muscle and enters the axilla (Fig. 3.98). This portion of the gland is the axillary tail or axillary process. The positions of the nipple and areola vary relative to the chest wall depending on breast size.

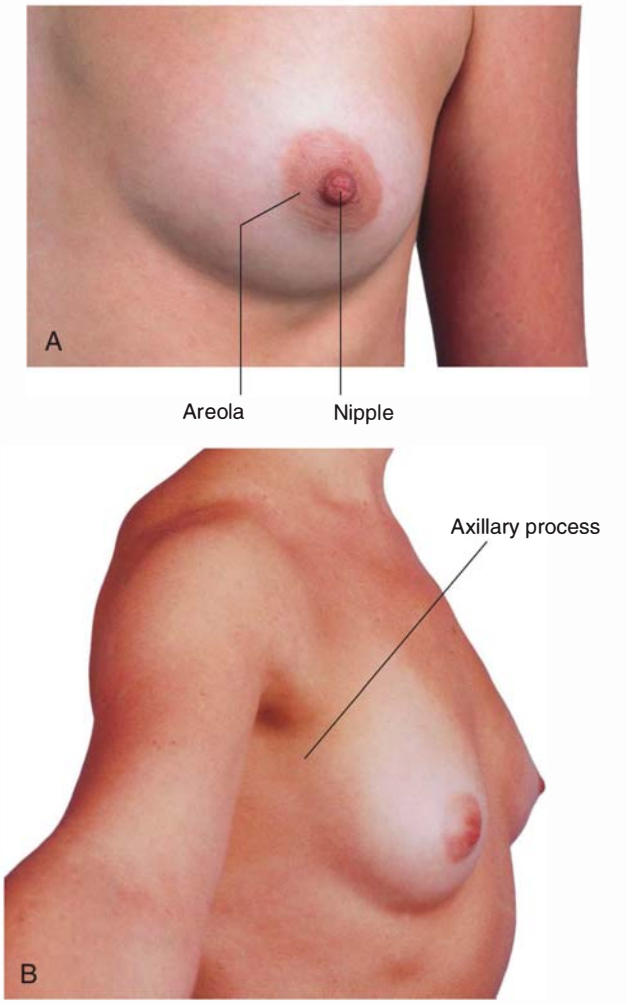


Fig. 3.98 A. Close-up view of nipple and surrounding areola of the breast. **B.** Lateral view of the chest wall of a woman showing the axillary process of the breast.

Visualizing structures at the TIV/V vertebral level

The TIV/V vertebral level is a transverse plane that passes through the sternal angle on the anterior chest wall and the intervertebral disc between TIV and TV vertebrae posteriorly. This plane can easily be located, because the joint between the manubrium of the sternum and the body of the sternum forms a distinct bony protuberance that can be palpated. At the TIV/V level (Fig. 3.99):

- The costal cartilage of rib II articulates with the sternum.
- The superior mediastinum is separated from the inferior mediastinum.
- The ascending aorta ends and the arch of the aorta begins.
- The arch of the aorta ends and the thoracic aorta begins.
- The trachea bifurcates.

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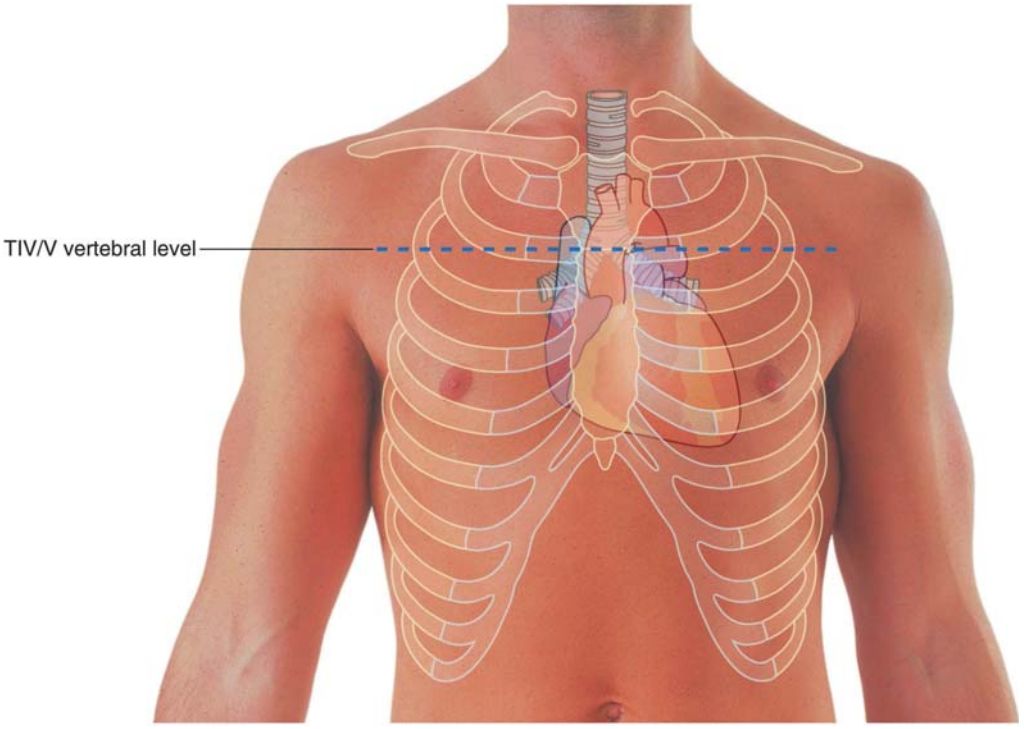


Fig. 3.99 Anterior view of the chest wall of a man showing the locations of various structures related to the TIV/V level.

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Visualizing structures in the superior mediastinum

A number of structures in the superior mediastinum in adults can be visualized based on their positions relative to skeletal landmarks that can be palpated through the skin (Fig. 3.100).

- On each side, the internal jugular and subclavian veins join to form the brachiocephalic veins behind the sternal ends of the clavicles near the sternoclavicular joints.
- The left brachiocephalic vein crosses from left to right behind the manubrium of the sternum.
- The brachiocephalic veins unite to form the superior vena cava behind the lower border of the costal cartilage of the right first rib.
- The arch of the aorta begins and ends at the transverse plane between the sternal angle anteriorly and vertebral level TIV/V posteriorly. The arch may reach as high as the midlevel of the manubrium of the sternum.

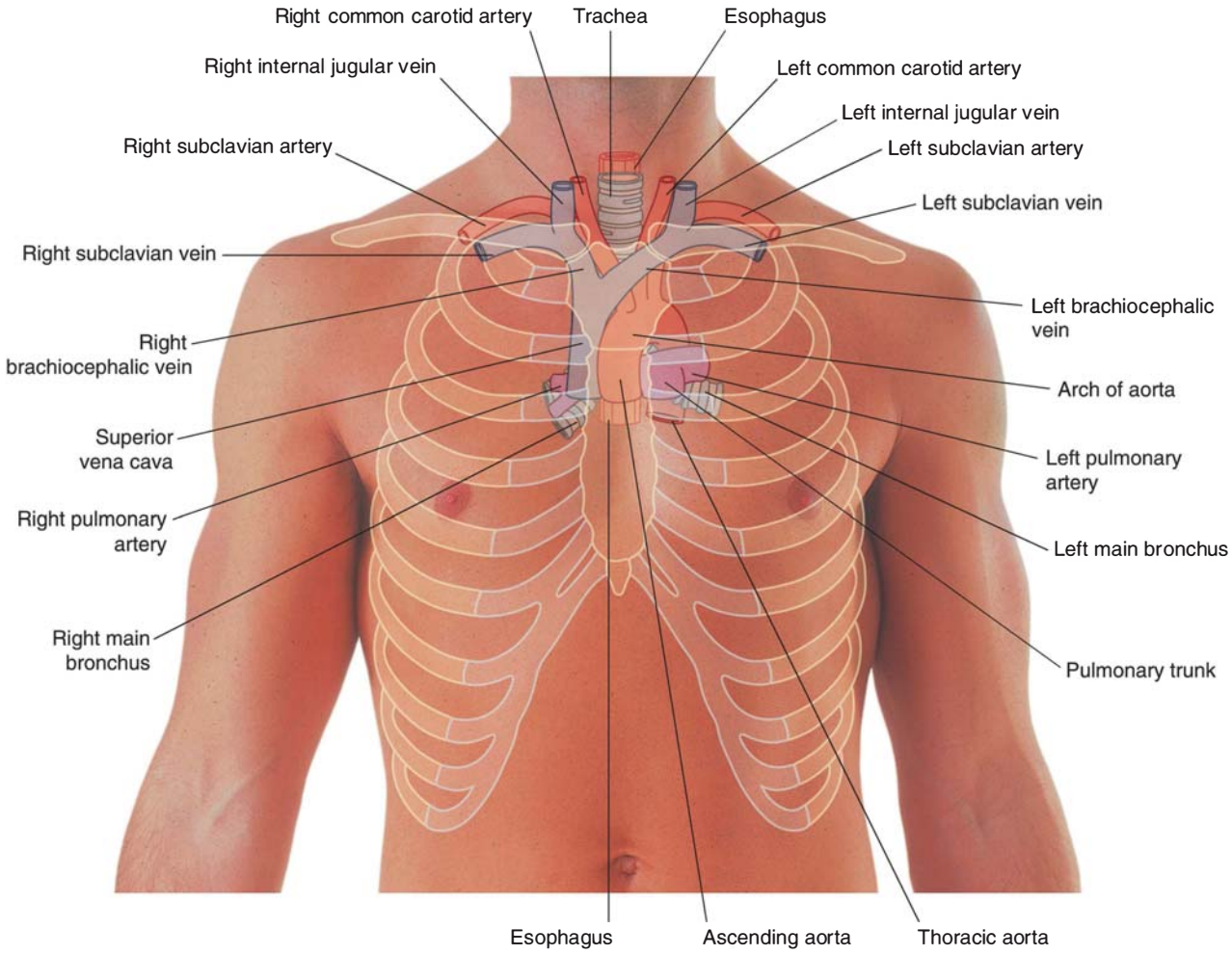


Fig. 3.100 Anterior view of the chest wall of a man showing the locations of different structures in the superior mediastinum as they relate to the skeleton.

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Visualizing the margins of the heart

Surface landmarks can be palpated to visualize the outline of the heart (Fig. 3.101).

- The upper limit of the heart reaches as high as the third costal cartilage on the right side of the sternum and the second intercostal space on the left side of the sternum.
- The right margin of the heart extends from the right third costal cartilage to near the right sixth costal cartilage.
- The left margin of the heart descends laterally from the second intercostal space to the apex located near the midclavicular line in the fifth intercostal space.
- The lower margin of the heart extends from the sternal end of the right sixth costal cartilage to the apex in the fifth intercostal space near the midclavicular line.

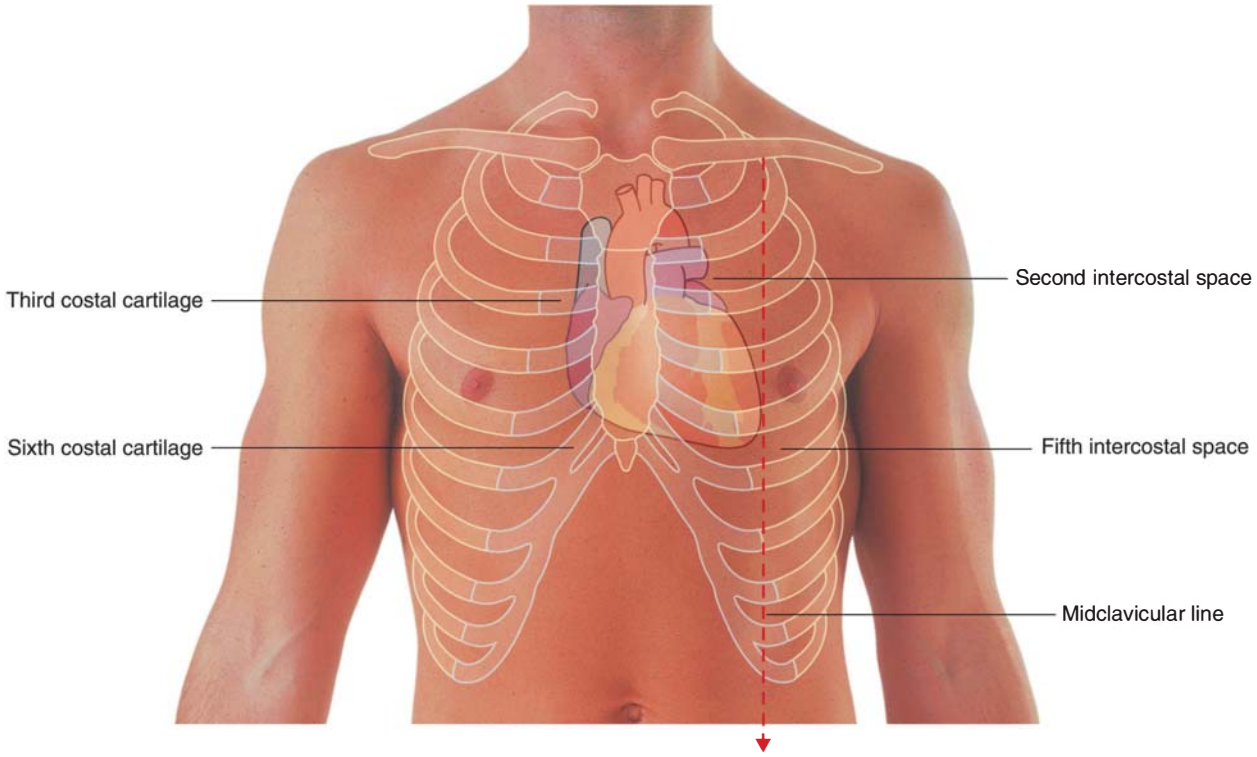


Fig. 3.101 Anterior view of the chest wall of a man showing skeletal structures and the surface projection of the heart.

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Where to listen for heart sounds

To listen for valve sounds, position the stethoscope downstream from the flow of blood through the valves (Fig. 3.102).

- The tricuspid valve is heard just to the left of the lower part of the sternum near the fifth intercostal space.
- The mitral valve is heard over the apex of the heart in the left fifth intercostal space at the midclavicular line.
- The pulmonary valve is heard over the medial end of the left second intercostal space.
- The aortic valve is heard over the medial end of the right second intercostal space.

Visualizing the pleural cavities and lungs, pleural recesses, and lung lobes and fissures

Palpable surface landmarks can be used to visualize the normal outlines of the pleural cavities and the lungs and to determine the positions of the pulmonary lobes and fissures.

Superiorly, the parietal pleura projects above the first costal cartilage. Anteriorly, the costal pleura approaches the midline posterior to the upper portion of the sternum. Posterior to the lower portion of the sternum, the left parietal pleura does not come as close to the midline as it does on the right side. This is because the heart bulges onto the left side (Fig. 3.103A).

Inferiorly, the pleura reflects onto the diaphragm above the costal margin and courses around the thoracic wall following an VIII, X, XII contour (i.e., rib VIII in the midclavicular line, rib X in the midaxillary line, and vertebra TXII posteriorly).

The lungs do not completely fill the area surrounded by the pleural cavities, particularly anteriorly and inferiorly.

- Costomediastinal recesses occur anteriorly, particularly on the left side in relationship to the heart bulge.
- Costodiaphragmatic recesses occur inferiorly between the lower lung margin and the lower margin of the pleural cavity.

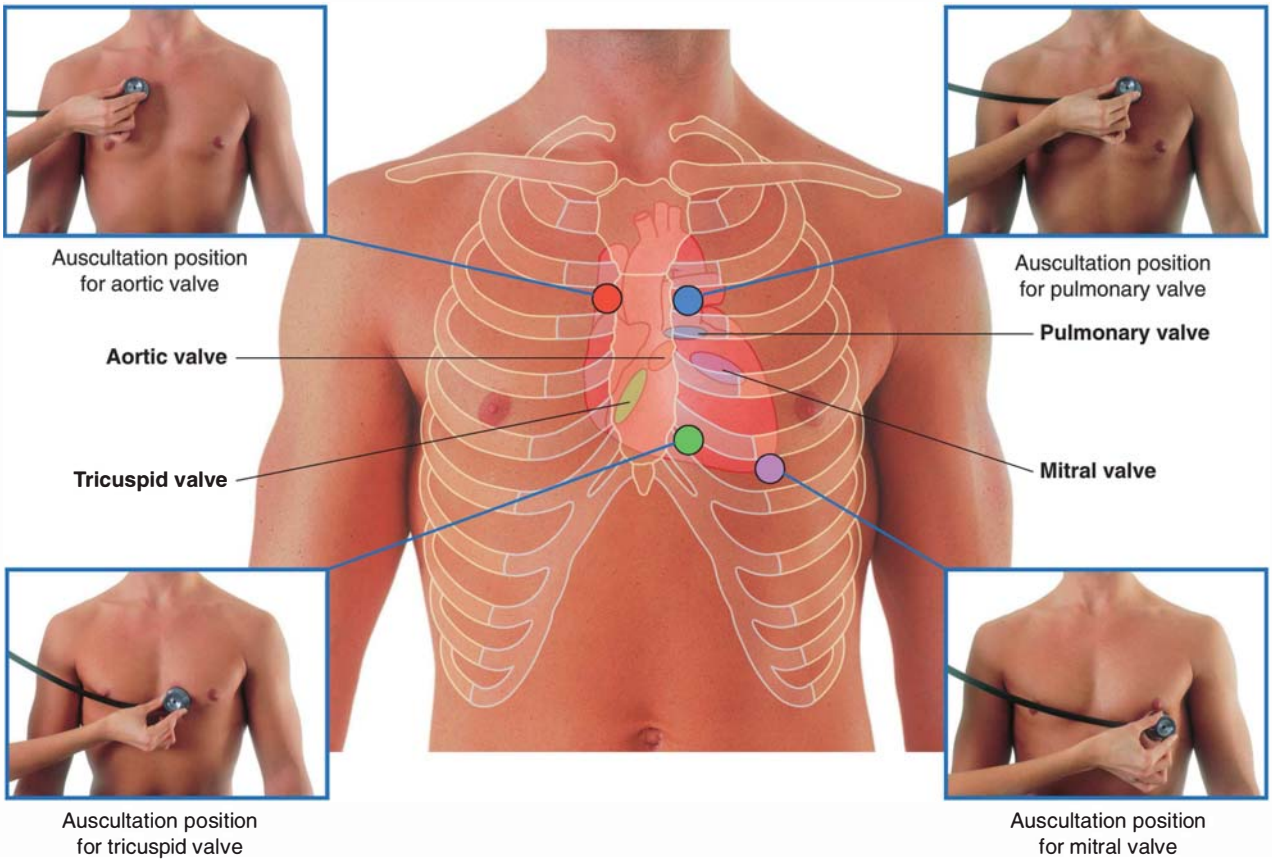


Fig. 3.102 Anterior view of the chest wall of a man showing skeletal structures, heart, location of the heart valves, and auscultation points.

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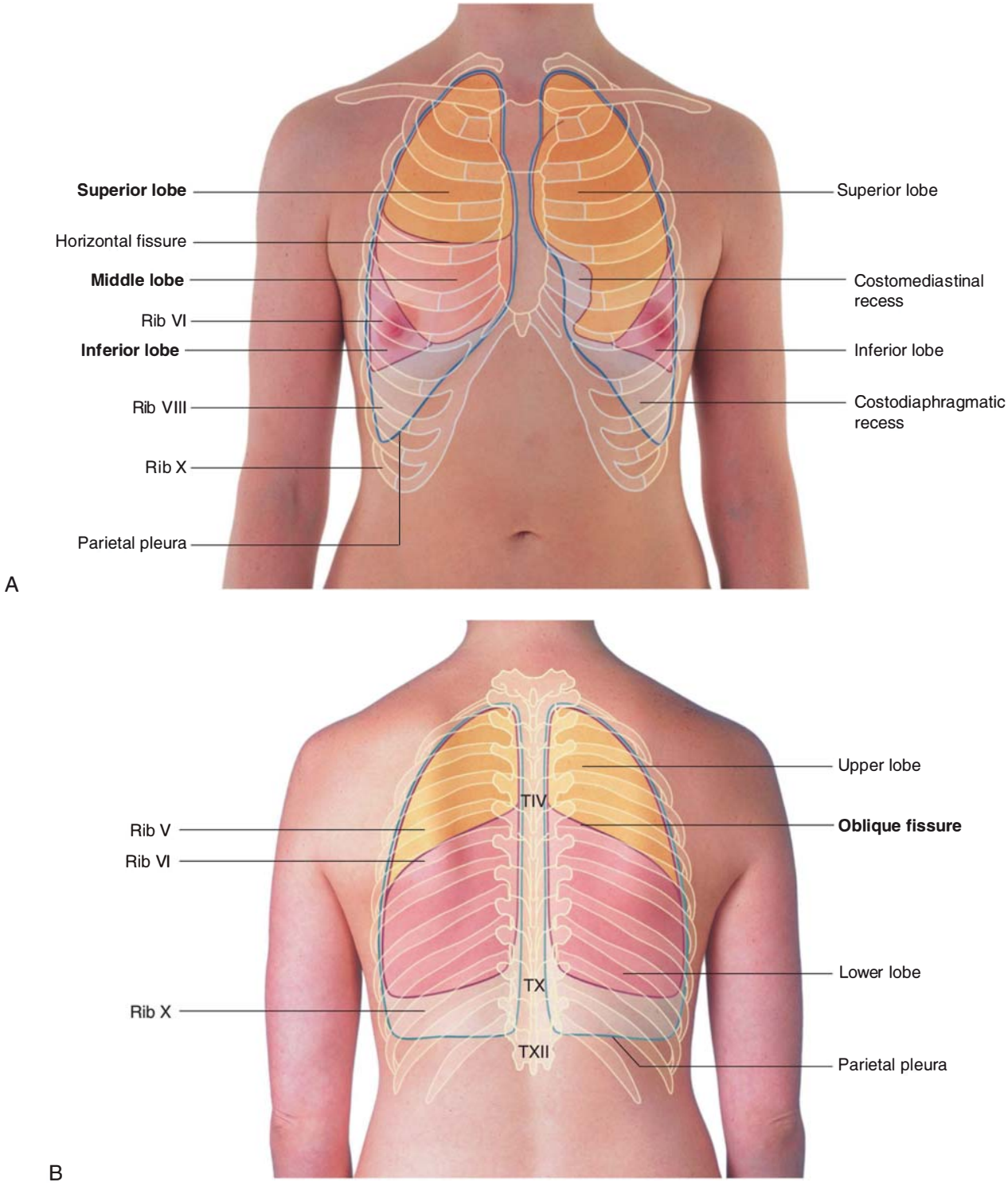


Fig. 3.103 Views of the chest wall showing the surface projections of the lobes and the fissures of the lungs. **A.** Anterior view in a woman. On the right side, the superior, middle, and inferior lobes are illustrated. On the left side, the superior and inferior lobes are illustrated. **B.** Posterior view in a woman. On both sides, the superior and inferior lobes are illustrated. The middle lobe on the right side is not visible in this view.

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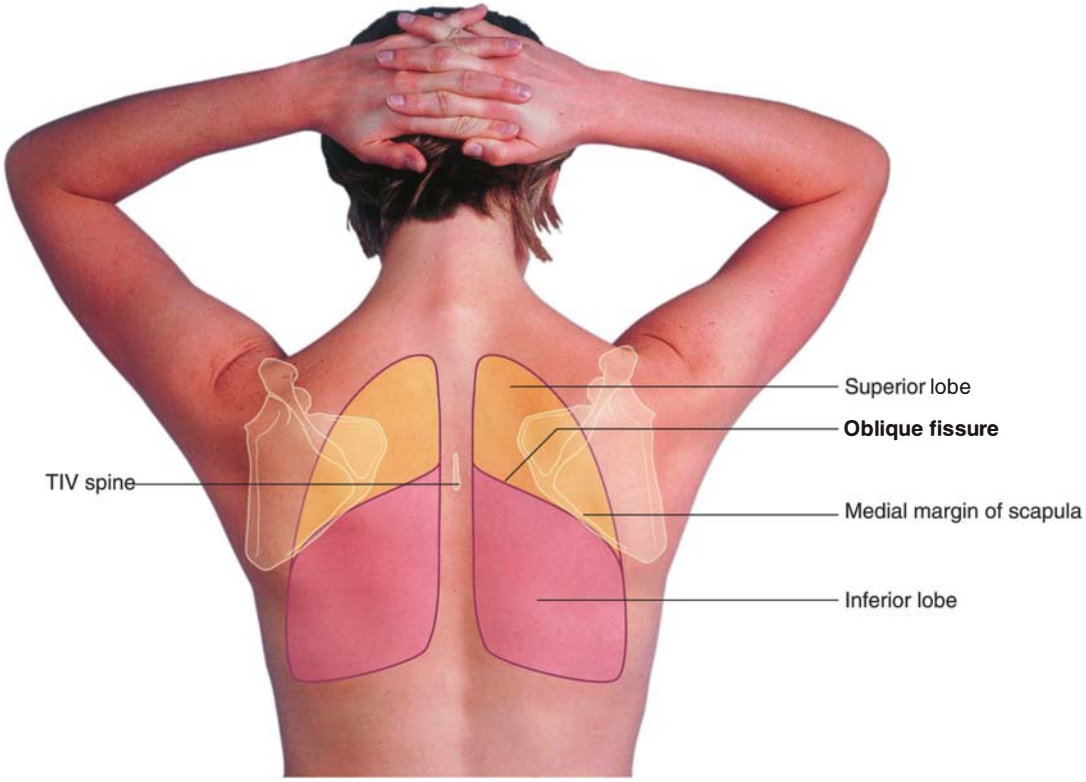
In quiet respiration, the inferior margin of the lungs travels around the thoracic wall following a VI, VIII, X contour (i.e., rib VI in the midclavicular line, rib VIII in the midaxillary line, and vertebra TX posteriorly).

In the posterior view, the oblique fissure on both sides is located in the midline near the spine of vertebra TIV (Figs. 3.103B and 3.104A). It moves laterally in a downward direction, crossing the fourth and fifth intercostal spaces and reaches rib VI laterally.

In the anterior view, the horizontal fissure on the right side follows the contour of rib IV and its costal cartilage and the oblique fissures on both sides follow the contour of rib VI and its costal cartilage (Fig. 3.104B).

Where to listen for lung sounds

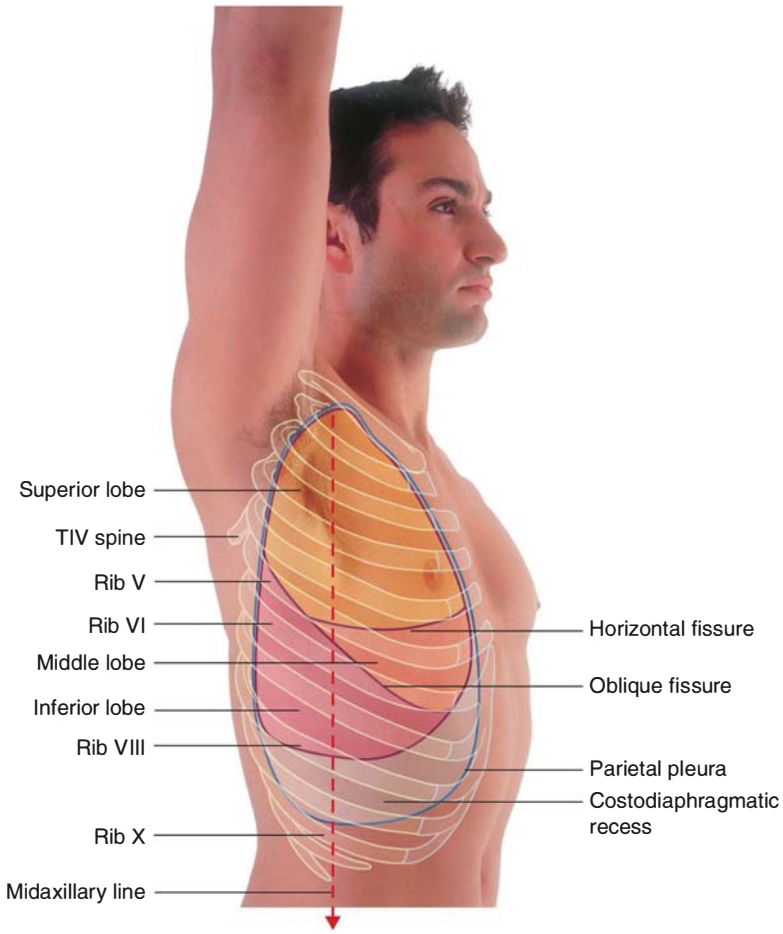
The stethoscope placements for listening for lung sounds are shown in Fig. 3.105.



A

Fig. 3.104 Views of the chest wall. **A.** Posterior view in a woman with arms abducted and hands positioned behind her head. On both sides, the superior and inferior lobes of the lungs are illustrated. When the scapula is rotated into this position, the medial border of the scapula parallels the position of the oblique fissure and can be used as a guide for determining the surface projection of the superior and inferior lobes of the lungs.

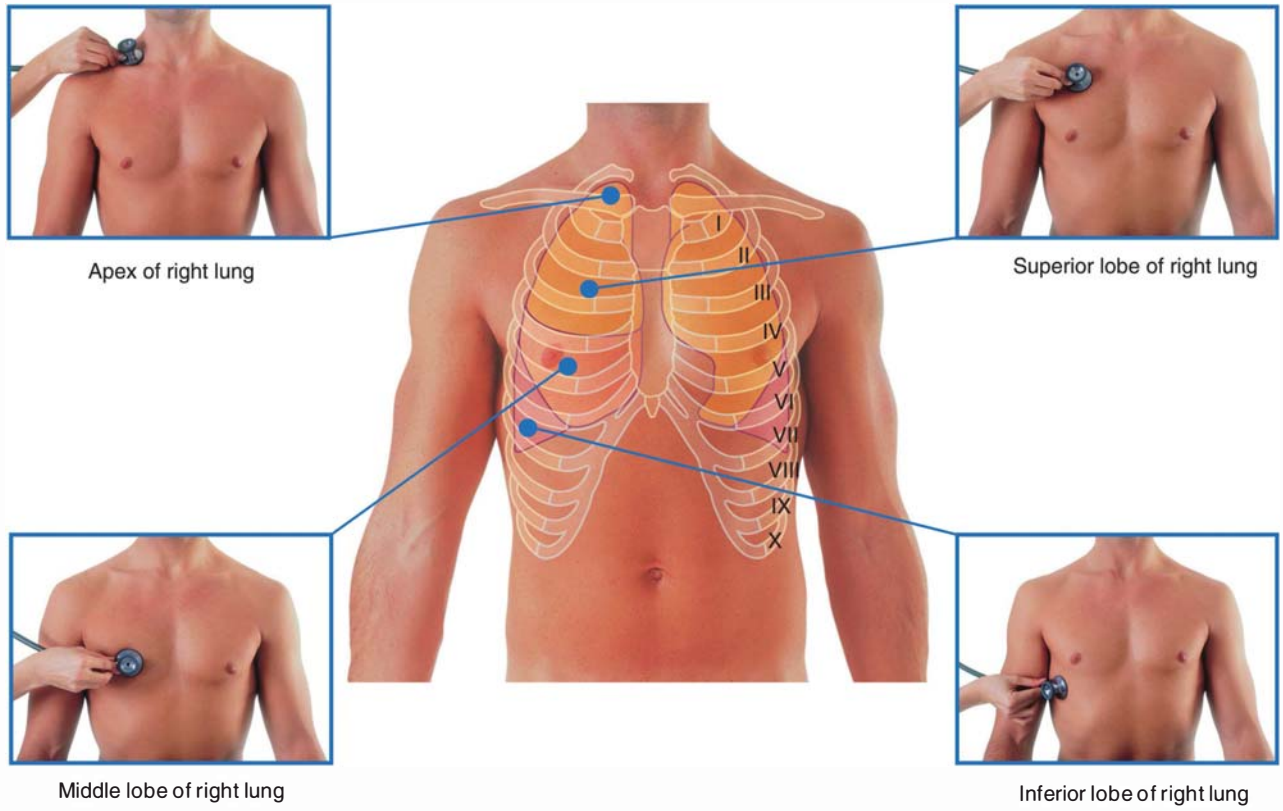
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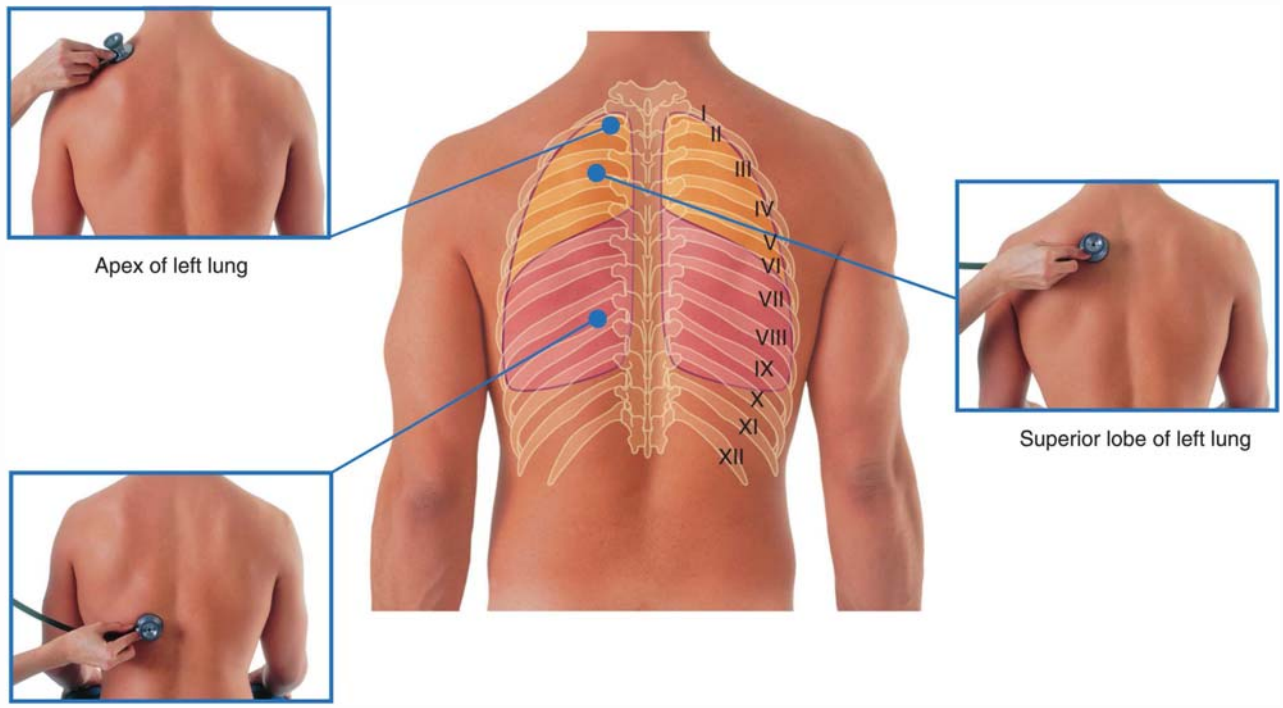
B

Fig. 3.104, cont'd B. Lateral view in a man with his right arm abducted. The superior, middle, and inferior lobes of the right lung are illustrated. The oblique fissure begins posteriorly at the level of the spine of vertebra TIV, passes inferiorly crossing rib IV, the fourth intercostal space, and rib V. It crosses the fifth intercostal space at the midaxillary line and continues anteriorly along the contour of rib VI. The horizontal fissure crosses rib V in the midaxillary space and continues anteriorly, crossing the fourth intercostal space and following the contour of rib IV and its costal cartilage to the sternum.

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A



B

Fig. 3.105 Views of the chest wall of a man with stethoscope placements for listening to the lobes of the lungs. **A.** Anterior views. **B.** Posterior views.

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Clinical cases

Case 1

CERVICAL RIB

A young man has black areas of skin on the tips of his fingers of his left hand. A clinical diagnosis of platelet emboli was made and a source of the emboli sought.

Emboli can arise from many sources. They are clots and plugs of tissue, usually platelets, that are carried from a source to eventually reside in small vessels which they may occlude. Arterial emboli may arise in the heart or in the arteries that supply the region affected. In cases of infected emboli, bacteria grow on the valve and are showered off into the peripheral circulation.

A neck radiograph and coronal CT image of the neck demonstrates a cervical rib (Fig. 3.106).

Cervical ribs may produce three distinct disease entities:

- *Arterial compression and embolization—the cervical rib (or band) on the undersurface of the distal portion of the subclavian artery reduces the diameter of the vessel and allows eddy currents to form. Platelets aggregate and atheroma may develop in this region. This debris can be dislodged and flow distally within the upper limb vessels to block off blood flow to the fingers and the hand, a condition called distal embolization.*
- *Tension on the T1 nerve—the T1 nerve, which normally passes over rib I, is also elevated by the presence of a cervical rib; thus the patient may experience a sensory disturbance over the medial aspect of the forearm, and develop wasting of the intrinsic muscles of the hand.*
- *Compression of the subclavian vein—this may induce axillary vein thrombosis.*

A Doppler ultrasound scan revealed marked stenosis of the subclavian artery at the outer border of the rib with abnormal flow distal to the narrowing. Within this region of abnormal flow there was evidence of thrombus adherent to the vessel wall.

This patient underwent surgical excision of the cervical rib and had no further symptoms.

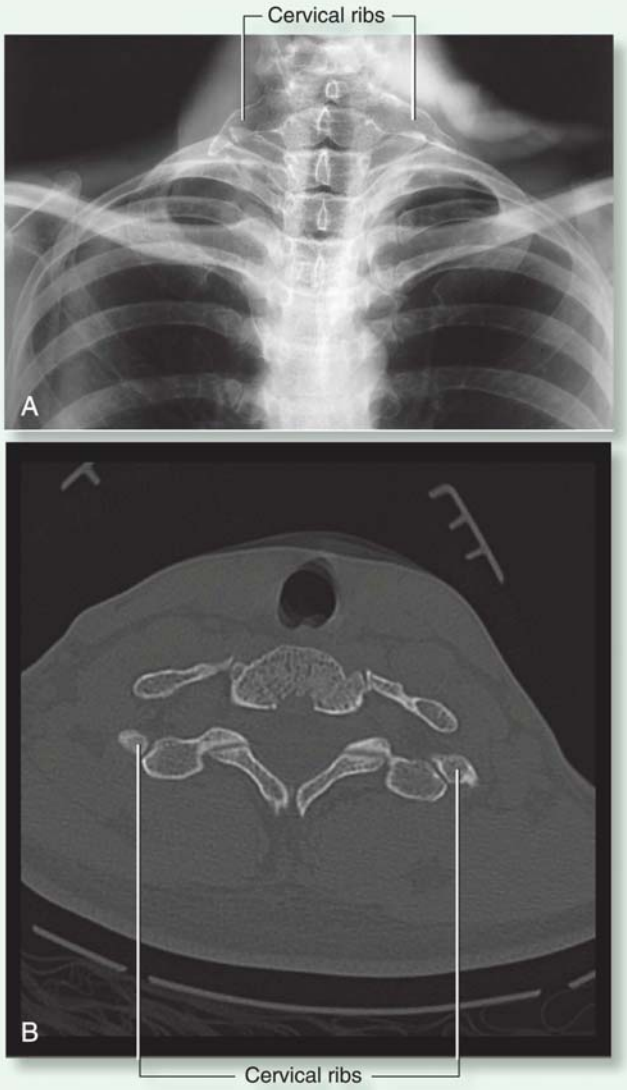


Fig. 3.106 Cervical ribs. **A.** Neck radiograph demonstrating bilateral cervical ribs. **B.** Coronal computed tomography image showing cervical ribs.

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Case 2

LUNG CANCER

A 52-year-old man presented with headaches and shortness of breath. He also complained of coughing up small volumes of blood. Clinical examination revealed multiple dilated veins around the neck. A chest radiograph demonstrated an elevated diaphragm on the right and a tumor mass, which was believed to be a primary bronchogenic carcinoma.

By observing the clinical findings and applying anatomical knowledge, the site of the tumor can be inferred.

The multiple dilated veins around the neck are indicative of venous obstruction. The veins are dilated on both sides of the neck, implying that the obstruction must be within a common vessel, the superior vena cava. Anterior to the superior vena cava in the right side of the chest is the phrenic nerve, which supplies the diaphragm. Because the diaphragm is elevated, suggesting paralysis, it is clear that the phrenic nerve has been involved with the tumor.

Case 3

CHEST WOUND

A 35-year-old man was shot during an armed robbery. The bullet entry wound was in the right fourth intercostal space, above the nipple. A chest radiograph obtained on admission to the emergency room demonstrated complete collapse of the lung.

A further chest radiograph performed 20 minutes later demonstrated an air/fluid level in the pleural cavity (Fig. 3.107**).**



Fig. 3.107 Chest radiograph demonstrating an air/fluid level in the pleural cavity.

Three common pathological processes may occur in the pleural cavity.

- *If air is introduced into the pleural cavity, a* **pneumothorax** *develops and the lung collapses because of its own elastic recoil. The pleural space fills with air, which may further compress the lung. Most patients with a collapsed lung are unlikely to have respiratory impairment. Under certain conditions, air may enter the pleural cavity at such a rate that it shifts and pushes the mediastinum to the opposite side of the chest. This is called* **tension pneumothorax** *and is potentially lethal, requiring urgent treatment by insertion of an intercostal tube to remove the air. The commonest causes of pneumothorax are rib fractures and positive pressure ventilation lung damage.*
- *The pleural cavity may fill with fluid (a pleural effusion) and this can be associated with many diseases (e.g., lung infection, cancer, abdominal sepsis). It is important to aspirate fluid from these patients to relieve any respiratory impairment and to carry out laboratory tests on the fluid to determine its nature.*
- *Severe chest trauma can lead to development of hemopneumothorax. A tube must be inserted to remove the blood and air that has entered the pleural space and prevent respiratory impairment.*

This man needs treatment to drain either the air or fluid or both.

The pleural space can be accessed by passing a needle between the ribs into the pleural cavity. In a normal healthy adult, the pleural space is virtually nonexistent; therefore, any attempt to introduce a needle into this

(continues)

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Case 3 *(continued)*

space is unlikely to succeed and the procedure may damage the underlying lung.

Before any form of chest tube is inserted, the rib must be well anesthetized by infiltration because its periosteum is extremely sensitive. The intercostal drain should pass directly on top of the rib. Insertion adjacent to the lower part of the rib may damage the artery, vein, and nerve, which lie within the neurovascular bundle.

Appropriate sites for insertion of a chest drain are either in the fourth or fifth intercostal space between the anterior axillary and midaxillary anatomical lines.

This position is determined by palpating the sternal angle, which is the point of articulation of rib II. Counting inferiorly will determine the rib number and simple observation will determine the positions of the anterior axillary and midaxillary lines. Insertion of any tube or needle below the fifth interspace runs an appreciable risk of crossing the pleural recesses and placing the needle or the drain into either the liver or the spleen, depending upon which side the needle is inserted.

Case 4

MYOCARDIAL INFARCTION

A 65-year-old man was admitted to the emergency room with severe central chest pain that radiated to the neck and predominantly to the left arm. He was overweight and a known heavy smoker.

On examination he appeared gray and sweaty. His blood pressure was 74/40 mm Hg (normal range 120/80 mm Hg). An electrocardiogram (ECG) was performed and demonstrated anterior myocardial infarction. An urgent echocardiograph demonstrated poor left ventricular function. The cardiac angiogram revealed an occluded vessel (Fig. 3.108A,B). Another approach to evaluating coronary arteries in patients is to perform maximum intensity projection (MIP) CT studies (Fig. 3.109A,B).

This patient underwent an emergency coronary artery bypass graft and made an excellent recovery. He has now lost weight, stopped smoking, and exercises regularly.

When cardiac cells die during a myocardial infarction, pain fibers (visceral afferents) are stimulated. These visceral sensory fibers follow the course of sympathetic fibers that innervate the heart and enter the spinal cord

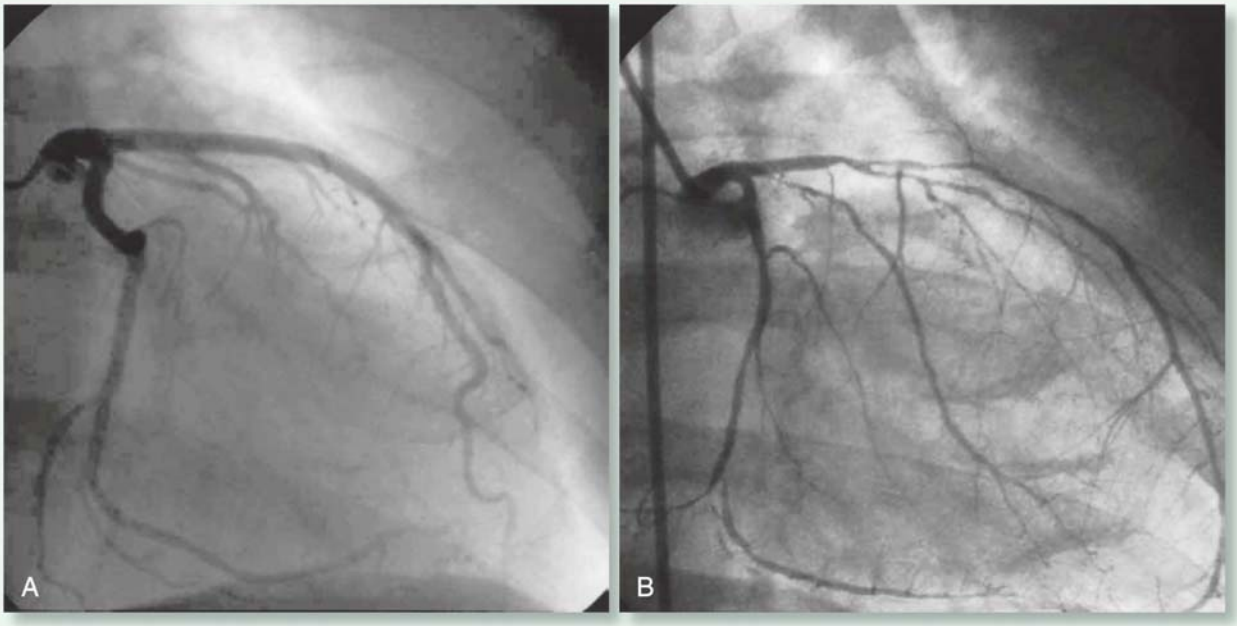


Fig. 3.108 **A.** Normal left coronary artery angiogram. **B.** Left coronary artery angiogram showing decreased flow due to blockages.

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Case 4 *(continued)*

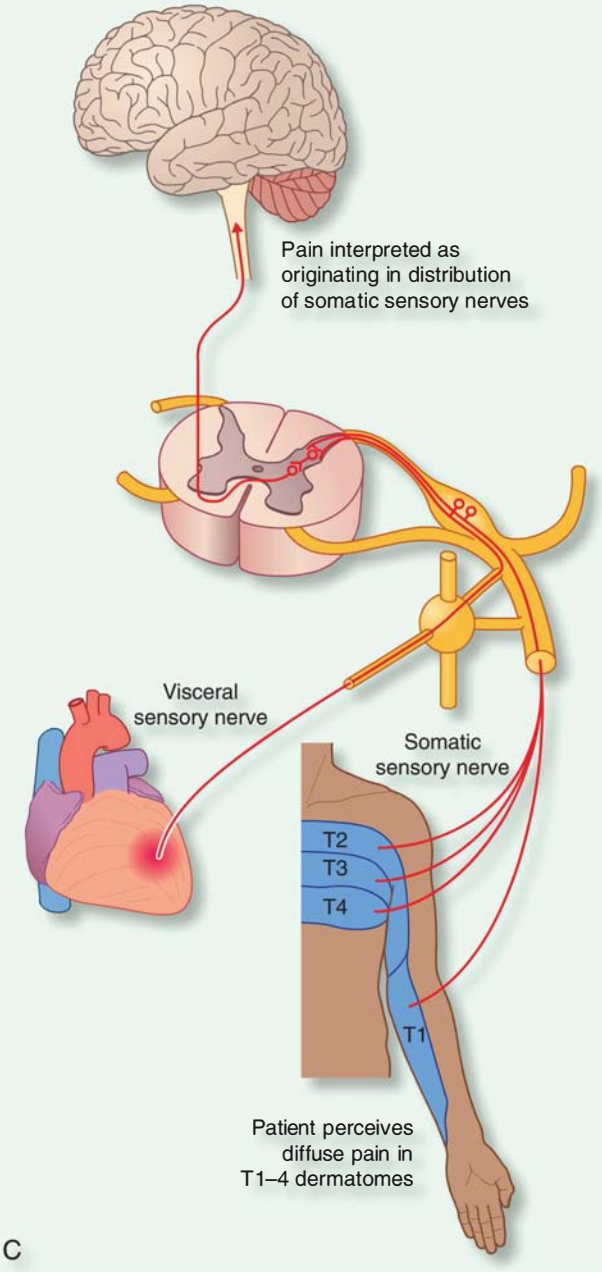


Fig. 3.108, cont'd C. Mechanism for perceiving heart pain in T1-4 dermatomes.

between the TI and TIV levels. At this level, somatic afferent nerves from spinal nerves T1 to T4 also enter the spinal cord via the posterior roots. Both types of afferents (visceral and somatic) synapse with interneurons, which then synapse with second neurons whose fibers pass across the cord and then ascend to the somatosensory areas of the brain that represent the T1 to T4 levels. The brain is unable to distinguish clearly between the visceral sensory distribution and the somatic sensory distribution and therefore the pain is interpreted as arising from the somatic regions rather than the visceral organ (i.e., the heart; Fig. 3.108C).

The patient was breathless because his left ventricular function was poor.

When the left ventricle fails, it produces two effects.

- *First, the contractile force is reduced. This reduces the pressure of the ejected blood and lowers the blood pressure.*
- *The left atrium has to work harder to fill the failing left ventricle. This extra work increases left atrial pressure, which is reflected in an increased pressure in the pulmonary veins, and this subsequently creates a higher pulmonary venular pressure. This rise in pressure will cause fluid to leak from the capillaries into the pulmonary interstitium and then into the alveoli. Such fluid is called pulmonary edema and it markedly restricts gas exchange. This results in shortness of breath.*

This man had a blocked left coronary artery, as shown in Fig. 3.108B.

It is important to know which coronary artery is blocked.

- *The left coronary artery supplies the majority of the left side of the heart. The left main stem vessel is approximately 2 cm long and divides into the circumflex artery, which lies between the atrium and the ventricle in the coronary sulcus, and the anterior interventricular artery, which is often referred to as the left anterior descending artery (LAD).*
- *When the right coronary artery is involved with arterial disease and occludes, associated disorders of cardiac rhythm often result because the sinu-atrial and the atrioventricular nodes derive their blood supplies predominantly from the right coronary artery.*

(continues)

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Case 4 *(continued)*

When this patient sought medical care, his myocardial function was assessed using ECG, echocardiography, and angiography.

During a patient's initial examination, the physician will usually assess myocardial function.

After obtaining a clinical history and carrying out a physical examination, a differential diagnosis for the cause of the malfunctioning heart is made. Objective assessment of myocardial and valve function is obtained in the following ways:

- **ECG/EKG (electrocardiography)**—*a series of electrical traces taken around the long and short axes of the heart that reveal heart rate and rhythm and conduction defects. In addition, it demonstrates the overall function of the right and left sides of the heart and points of dysfunction. Specific changes in the ECG relate to the areas of the heart that have been involved in a myocardial infarction. For example, a right coronary artery occlusion produces infarction in the area of myocardium it supplies, which is predominantly the inferior aspect; the infarct is therefore called an inferior myocardial infarction. The ECG changes are demonstrated in the leads that visualize the inferior aspect of the myocardium (namely, leads II, III, and aVF).*
- **Chest radiography**—*reveals the size of the heart and chamber enlargement. Careful observation of the lungs will demonstrate excess fluid (pulmonary edema), which builds up when the left ventricle fails and can produce marked respiratory compromise and death unless promptly treated.*
- **Blood tests**—*the heart releases enzymes during myocardial infarction, namely lactate dehydrogenase (LDH), creatine kinase (CK), and aspartate transaminase (AST). These plasma enzymes are easily measured in the hospital laboratory and used to determine the diagnosis at an early stage. Further specific enzymes termed isoenzymes can also be determined (creatine kinase MB isoenzyme [CKMB]). Newer tests include an assessment for troponin (a specific component of the myocardium), which is released when cardiac cells die during myocardial infarction.*
- **Exercise testing**—*patients are connected to an ECG monitor and exercised on a treadmill. Areas of ischemia, or poor blood flow, can be demonstrated, so localizing the vascular abnormality.*
- **Nuclear medicine**—*thallium (a radioactive X-ray emitter) and its derivatives are potassium analogs. They are used to determine areas of coronary ischemia. If no areas of myocardial uptake are demonstrated when these substances are administered to a patient the myocardium is dead.*
- **Coronary angiography**—*small arterial catheters are maneuvered from a femoral artery puncture site through the femoral artery and aorta and up to the origins of the coronary vessels. X-ray contrast medium is then injected to demonstrate the coronary vessels and their important branches. If there is any narrowing (stenosis), angioplasty may be carried out. In angioplasty tiny balloons are passed across the narrowed areas and inflated to refashion the vessel and so prevent further coronary ischemia and myocardial infarction.*

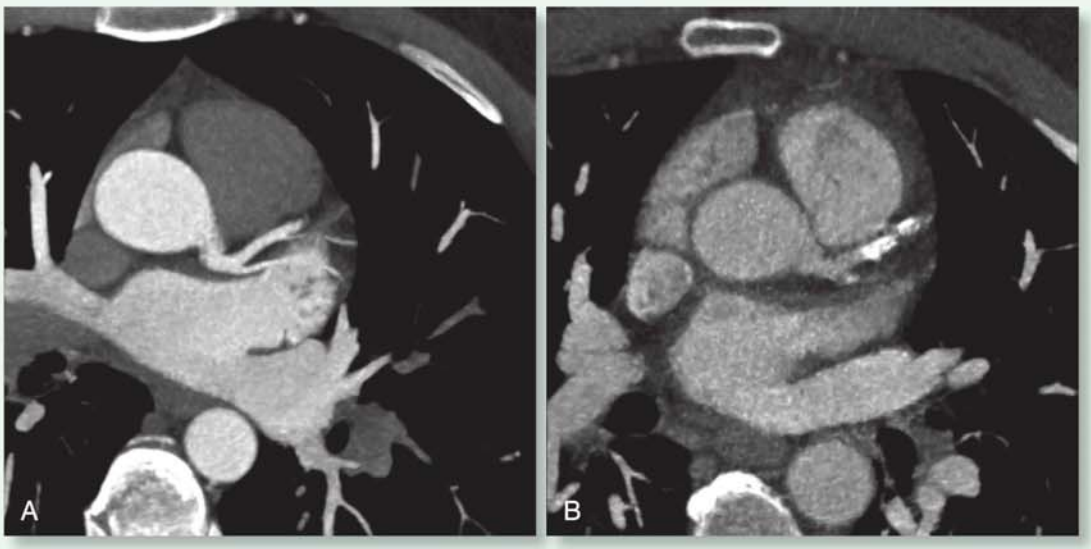


Fig. 3.109 Axial maximum intensity projection (MIP) CT image through the heart. **A.** Normal anterior interventricular (left anterior descending) artery. **B.** Stenotic (calcified) anterior interventricular (left anterior descending) artery.

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Case 5

BROKEN PACEMAKER

An elderly woman was admitted to the emergency room with severe cardiac failure. She had a left-sided pacemaker box, which had been inserted for a cardiac rhythm disorder (fast atrial fibrillation) many years previously. An ECG demonstrated fast atrial fibrillation. A chest radiograph showed that the wire from the pacemaker had broken under the clavicle.

Anatomical knowledge of this region of the chest explains why the wire broke.

Many patients have cardiac pacemakers. A wire arises from the pacemaker, which lies within the subcutaneous tissue over the pectoralis major muscle and travels from the pacemaker under the skin to pierce the axillary vein just beneath the clavicle, lateral to the subclavius muscle. The wire then passes through the subclavian vein, the brachiocephalic vein, the superior vena cava, and the right atrium, and lies on the wall of the right ventricle (where it can stimulate the heart to contract) (Fig. 3-110). If the wire pierces the axillary vein directly adjacent to the subclavius muscle, it is possible that after many years of shoulder movement the subclavius muscle stresses and breaks the wire, causing the pacemaker to fail. Every effort is made to place the insertion point of the wire as far laterally as feasible within the first part of the axillary vein.

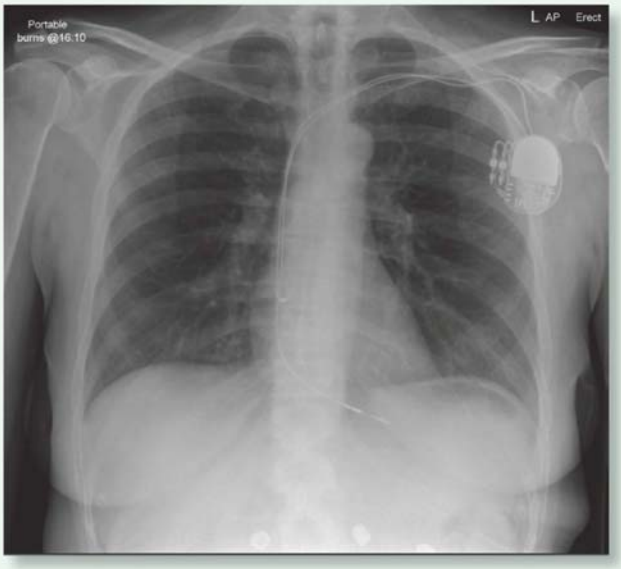


Fig. 3.110 Chest radiograph of an individual with a pacemaker. The pacemaker wires *(2)* can be seen traveling through the venous system to the heart where one ends in the right atrium and the other ends in the right ventricle.

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Case 6

COARCTATION OF THE AORTA

A 20-year-old man visited his family doctor because he had a cough. A chest radiograph demonstrated translucent notches along the inferior border of ribs III to VI (Fig. 3.111). He was referred to a cardiologist and a diagnosis of coarctation of the aorta was made. The rib notching was caused by dilated collateral intercostal arteries.

Coarctation of the aorta is a narrowing of the aorta distal to the left subclavian artery. This narrowing can markedly reduce blood flow to the lower body. Many of the vessels above the narrowing therefore enlarge due to the increased pressure so that blood can reach the aorta below the level of the narrowing. Commonly, the internal thoracic, superior epigastric, and musculophrenic arteries enlarge anteriorly. These arteries supply the anterior intercostal arteries, which anastomose with the posterior intercostal arteries that allow blood to flow retrogradely into the aorta. Enlargement of the intertcostal vessels results in notching of the ribs.

The first and second posterior intercostal vessels are supplied from the costocervical trunk, which arises from the subclavian artery proximal to the coarctation, so do not enlarge and do not induce rib notching.



Fig. 3.111 Chest radiograph demonstrating translucent notches along the inferior border of ribs III to VI.

Case 7

AORTIC DISSECTION

A 62-year-old man was admitted to the emergency room with severe interscapular pain. His past medical history indicated that he was otherwise fit and well; however, it was noted he was 6′ 9″ and had undergone previous eye surgery for dislocating lenses.

On examination the man was pale, clammy, and hypotensive. The pulse in his right groin was weak. An ECG demonstrated an inferior myocardial infarction. Serum blood tests revealed poor kidney function and marked acidosis.

The patient was transferred to the CT scanner and a diagnosis of aortic dissection was made.

Aortic dissection is an uncommon disorder in which a small tear occurs within the aortic wall (Fig. 3.112). The aortic wall contains three layers, an intima, a media, and an adventitia. A tear in the intima extends into the media and peels it away, forming a channel within the wall of the vessel. Usually the blood reenters the main vessel wall distal to its point of entry.

The myocardial infarction

Aortic dissection may extend retrogradely to involve the coronary sinus of the right coronary artery. Unfortunately, in this patient's case the right coronary artery became occluded as the dissection passed into the origin. In normal individuals the right coronary artery supplies the anterior inferior aspect of the myocardium, and this is evident as an anterior myocardial infarct on an ECG.

The ischemic left leg

The two channels within the aorta have extended throughout the length of the aorta into the right iliac system and to the level of the right femoral artery. Although blood flows through these structures it often

(continues)

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Case 7 *(continued)*

causes reduced blood flow. Hence the reduced blood flow into the left lower limb renders it ischemic.

The patient became acidotic.

All cells in the body produce acid, which is excreted in the urine or converted into water with the production of carbon dioxide, which is removed with ventilation. Unfortunately, when organs become extremely ischemic they release significant amounts of hydrogen ions. Typically, this occurs when the gut becomes ischemic. With the pattern of dissection, (1) the celiac trunk, superior mesenteric artery, and inferior mesenteric artery can be effectively removed from the circulation or (2) the blood flow within these vessels can be significantly impeded, rendering the gut ischemic and hence accounting for the relatively high hydrogen ion levels.

Kidney ischemia

Similarly the dissection can impair blood flow to the kidneys, which decreases their ability to function.

Treatment

The patient underwent emergency surgery and survived. Interestingly, the height of the patient and the previous lens surgery would suggest a diagnosis of Marfan syndrome, and a series of blood tests and review of the family history revealed this was so.

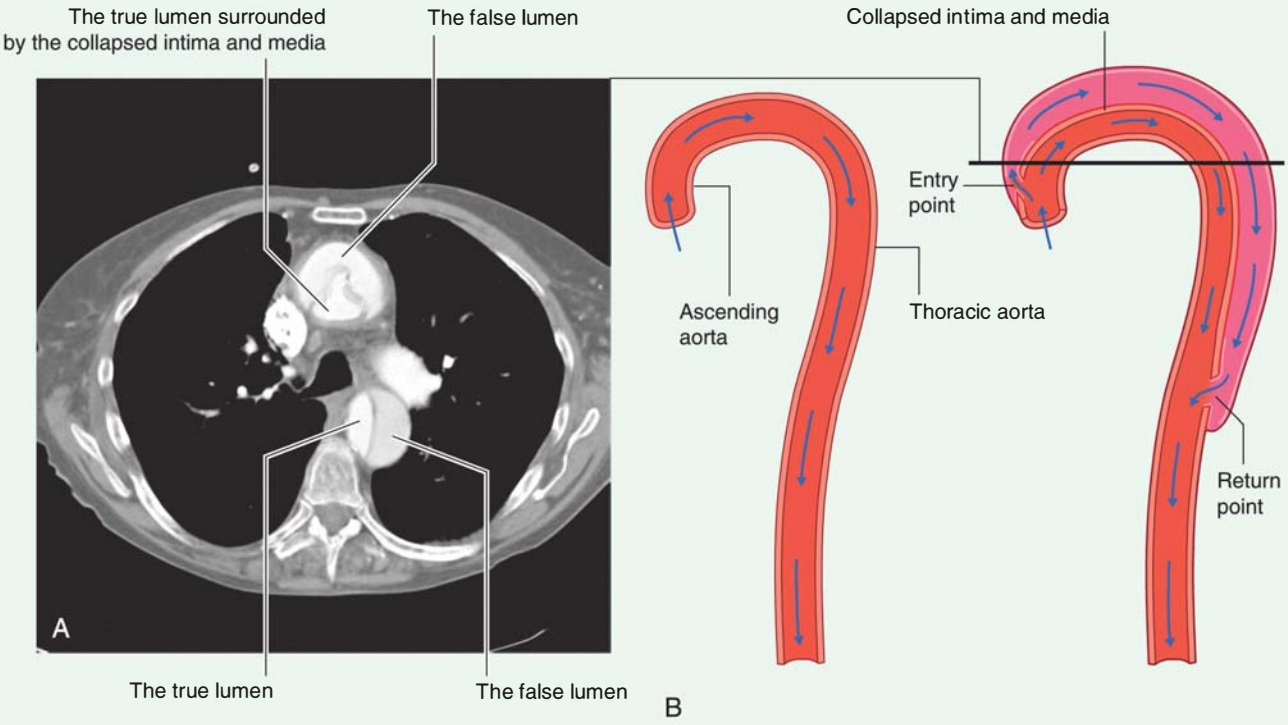


Fig. 3.112 A. CT image of aortic dissection. **B.** Normal aorta (*left*) and an aortic dissection (*right*). The line in the right figure indicates the plane of the CT scan shown in **A.**

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Case 8

PNEUMONIA

A 35-year-old male patient presented to his family practitioner because of recent weight loss (14 lb over the previous 2 months). He also complained of a cough with streaks of blood in the sputum (hemoptysis) and left-sided chest pain. Recently, he noticed significant sweating, especially at night, which necessitated changing his sheets.

On examination the patient had a low-grade temperature and was tachypneic (breathing fast). There was reduced expansion of the left side of the chest. When the chest was percussed it was noted that the anterior aspect of the left chest was dull, compared to the resonant percussion note of the remainder of the chest. Auscultation (listening with a stethoscope) revealed decreased breath sounds, which were hoarse in nature (bronchial breathing).

A diagnosis of chest infection was made.

Chest infection is a common disease. In most patients the infection affects the large airways and bronchi. If the infection continues, exudates and transudates are produced, filling the alveoli and the secondary pulmonary lobules. The diffuse patchy nature of this type of infection is termed bronchial pneumonia.

Given the patient's specific clinical findings, bronchial pneumonia was unlikely.

From the clinical findings it was clear that the patient was likely to have a pneumonia confined to a lobe. Because there are only two lobes in the left lung, the likely diagnosis was a left upper lobe pneumonia.

A chest radiograph was obtained (Fig. 3.113). The posteroanterior view of the chest demonstrated an area of veil-like opacification throughout the whole of the left lung.

Knowing the position of the oblique fissure, any consolidation within the left upper lobe will produce this veil-like shadowing. Lateral radiographs are usually not necessary but would demonstrate opacification anteriorly and superiorly that ends abruptly at the oblique fissure.

Upper lobe pneumonias are unusual because most patients develop gravity-dependent infection. Certain infections, however, are typical within the middle and upper lobes, commonly, tuberculosis (TB) and histoplasmosis.

A review of the patient's history suggested a serious and chronic illness and the patient was admitted to hospital.

After admission a bronchoscopy was carried out and sputum was aspirated from the left upper lobe bronchus. This was cultured in the laboratory and also viewed under the microscope and tuberculous bacilli (TB) were identified.

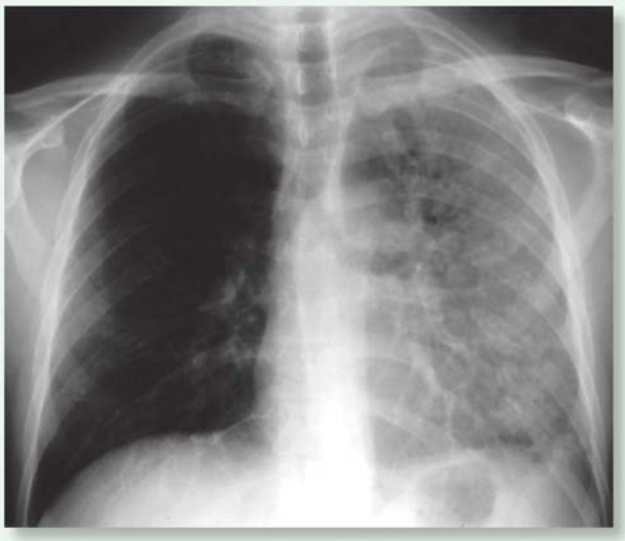


Fig. 3.113 Chest radiograph showing left upper lobe infection.

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Case 9

ESOPHAGEAL CANCER

A 68-year-old man came to his family physician complaining of discomfort when swallowing (dysphagia). The physician examined the patient and noted since his last visit he had lost approximately 18 lb over 6 months. Routine blood tests revealed the patient was anemic and he was referred to the gastroenterology unit. A diagnosis of esophageal cancer was made and the patient underwent a resection, which involved a chest and abdominal incision. After 4 years the patient remains well though still subject to follow-up.

The patient underwent a flexible endoscopic examination of the esophagus in which a tube is placed through the mouth and into the esophagus and a camera is placed on the end of the tube. It is also possible to use biopsy forceps to obtain small portions of tissue for adequate diagnosis.

The diagnosis of esophageal carcinoma was made (squamous cell type) and the patient underwent a staging procedure.

Staging of any malignancy is important because it determines the extent of treatment and allows the physician to determine the patient's prognosis. In this case our patient underwent a CT scan of the chest and abdomen, which revealed no significant lymph nodes around the lower third esophageal tumor.

The abdominal scan revealed no evidence of spread to the nodes around the celiac trunk and no evidence of spread to the liver.

Bleeding was the cause of the anemia.

Many tumors of the gastrointestinal system are remarkably friable, and with the passage of digested material across the tumor, low-grade chronic bleeding occurs. Over a period of time the patient is rendered anemic, which in the first instance is asymptomatic; however, it can be diagnosed on routine blood tests.

Complex surgery is planned.

The length of the esophagus is approximately 22 cm. Tumor spread can occur through the submucosal route and also through locoregional lymph nodes. The lymph nodes drain along the arterial supply to the esophagus, which is predominantly supplied by the inferior thyroid artery, esophageal branches from the thoracic aorta, and branches from the left gastric artery. The transthoracic esophagectomy procedure involves placing the patient supine. A laparotomy is performed to assess for any evidence of disease in the abdominal cavity. The stomach is mobilized, with preservation of the right gastric and right gastro-omental arteries. The short gastric vessels and left gastric vessels are divided, and a pyloromyotomy is also performed.

The abdominal wound is then closed and the patient is placed in the left lateral position. A right posterolateral thoracotomy is performed through the fifth intercostal space, and the azygos vein is divided to provide full access to the whole length of the esophagus. The stomach is delivered through the diaphragmatic hiatus. The esophagus is resected and the stomach is anastomosed to the cervical esophagus.

The patient made an uneventful recovery.

Most esophageal cancers are diagnosed relatively late and often have lymph node metastatic spread. A number of patients will also have a spread of tumor to the liver. The overall prognosis for esophageal cancer is poor, with approximately a 25%, 5-year survival rate.

Diagnosing esophageal cancer in its early stages before lymph node spread is ideal and can produce a curative procedure.

Our patient went on to have chemotherapy and enjoys a good quality of life 4 years after his operation.

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Case 10

VENOUS ACCESS

A 45-year-old woman, with a history of breast cancer in the left breast, returned to her physician. Unfortunately the disease had spread to the axillary lymph nodes and bones (bony metastatic disease). A surgeon duly resected the primary breast tumor with a wide local excision and then performed an axillary nodal clearance. The patient was then referred to an oncologist for chemotherapy. Chemotherapy was delivered through a portacath, which is a subcutaneous reservoir from which a small catheter passes under the skin into the internal jugular vein. The patient duly underwent a portacath insertion without complication, completed her course of chemotherapy, and is currently doing well 5 years later.

The portacath was placed on the patient's right anterior chest wall and the line was placed into the right internal jugular vein. The left internal jugular vein and subcutaneous tissues were not used. The reason for not using this site was that the patient had previously undergone an axillary dissection on the left, and the lymph nodes and lymphatics were removed. Placement of a portacath in this region may produce an inflammatory response and may even get infected. Unfortunately, because there are no lymphatics to drain away infected material and to remove bacteria, severe sepsis and life-threatening infection may ensue.

How was it placed?

The ultrasound shows an axial image across the root of the neck on the right demonstrating the right common carotid artery and the right internal jugular vein. The internal jugular vein is the larger of the two structures and generally demonstrates normal respiratory variation, compressibility, and a size dependence upon the patient's position (when the patient is placed in the head-down position, the vein fills and makes puncture easy).

The risks of the procedure

As with all procedures and operations there is always a small risk of complication. These risks are always balanced against the potential benefits of the procedure. Placing the needle into the internal jugular vein can be performed under ultrasound guidance, which reduces the risk of puncturing the common carotid artery. Furthermore, by puncturing under direct vision it is less likely that the operator will hit the lung apex and pierce the superior pleural fascia, which may produce a pneumothorax.

The position of the indwelling catheter

The catheter is placed through the right internal jugular vein and into the right brachiocephalic vein. The tip of the catheter is then placed more inferiorly at the junction of the right atrium and the superior vena cava. The reason for placing the catheter in such a position relates to the agents that are infused. Most chemotherapeutic agents are severely cytotoxic (kill cells), and enabling good mixing with the blood prevents thrombosis and vein wall irritation.

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4

Abdomen

ADDITIONAL LEARNING RESOURCES for Chapter 4, *Abdomen*, on STUDENT CONSULT (www.studentconsult.com):

- Image Library—illustrations of abdominal anatomy, Chapter 4
- Self-Assessment—National Board style multiple-choice questions, Chapter 4
- Short Questions—these are questions requiring short responses, Chapter 4
- Interactive Surface Anatomy—interactive surface animations, Chapter 4
- Medical Clinical Case Studies, Chapter 4
  - Aorto-iliac occlusive disease
  - Colon cancer
  - Intussusception
  - Zollinger-Ellison syndrome

Free Online Anatomy and Embryology Self-Study Course

- Anatomy modules 10 through 17
- Embryology modules 65 and 66

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Conceptual overview

GENERAL DESCRIPTION

The abdomen is a roughly cylindrical chamber extending from the inferior margin of the thorax to the superior margin of the pelvis and the lower limb (Fig. 4.1A).

The **inferior thoracic aperture** forms the superior opening to the abdomen, and is closed by the diaphragm. Inferiorly, the deep abdominal wall is continuous with the pelvic wall at the **pelvic inlet**. Superficially, the inferior limit of the abdominal wall is the superior margin of the lower limb.

The chamber enclosed by the abdominal wall contains a single large **peritoneal cavity**, which freely communicates with the pelvic cavity.

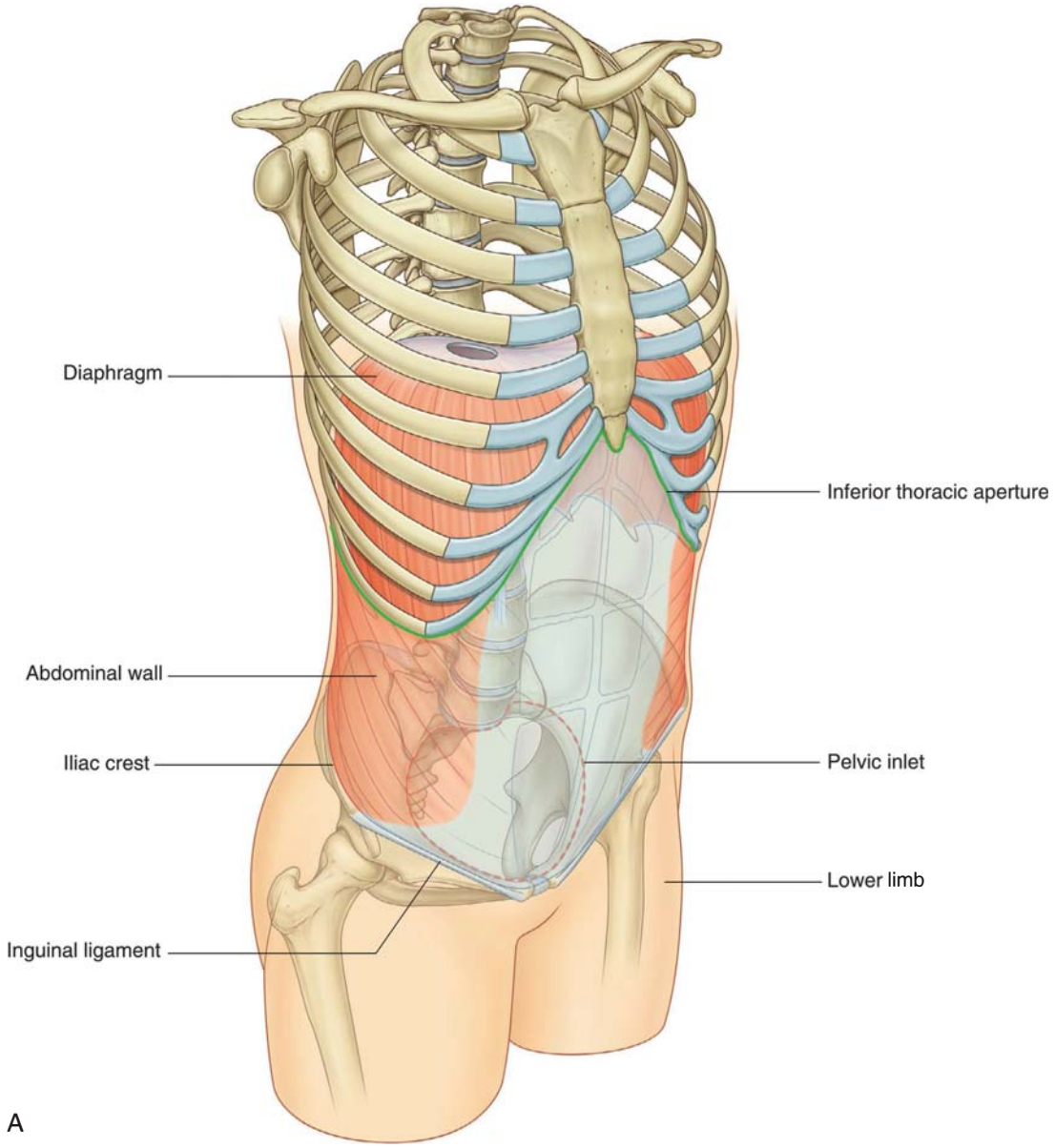


Fig. 4.1 Abdomen. **A.** Boundaries.

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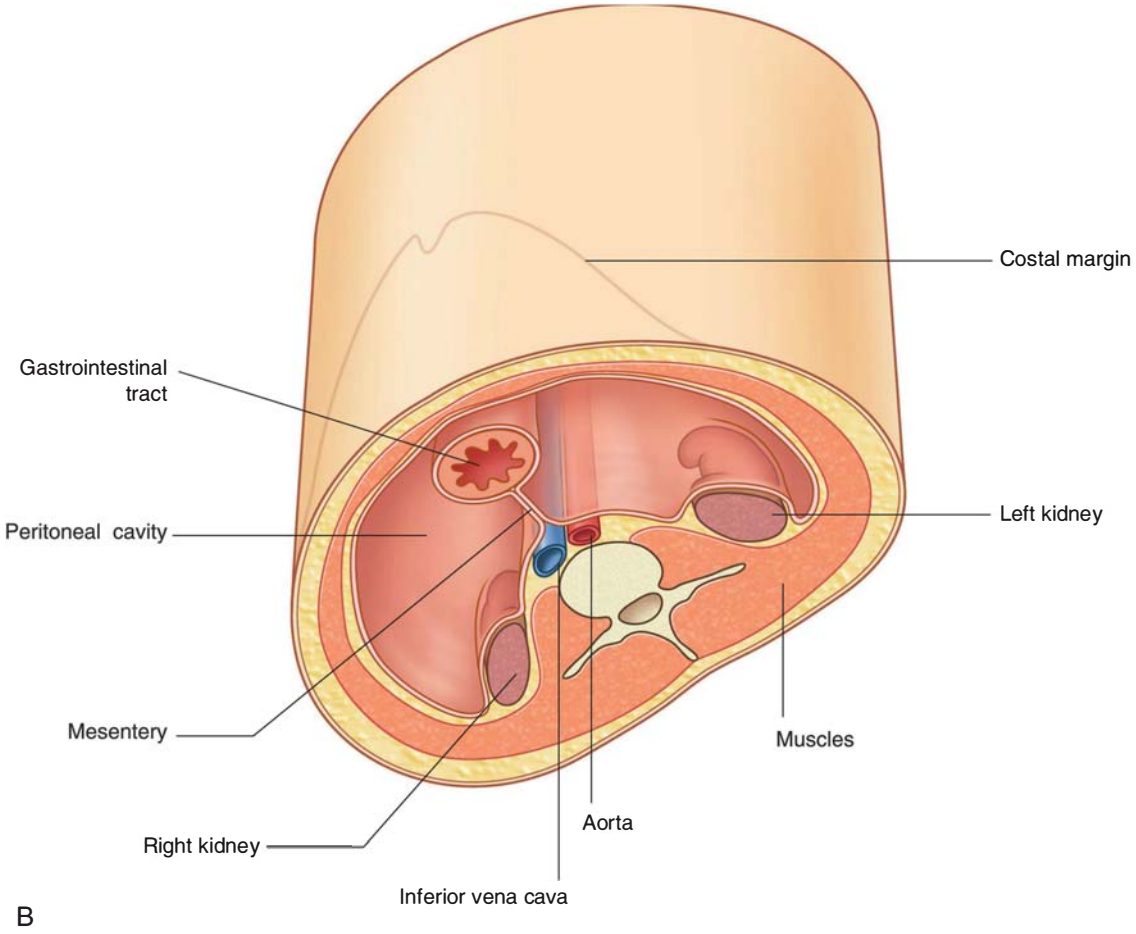


Fig. 4.1, cont'd B. Arrangement of abdominal contents. Inferior view.

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Abdominal viscera are either suspended in the peritoneal cavity by mesenteries or positioned between the cavity and the musculoskeletal wall (Fig. 4.1B).

Abdominal viscera include:

- major elements of the gastrointestinal system—the caudal end of the esophagus, stomach, small and large intestines, liver, pancreas, and gallbladder;
- the spleen;
- components of the urinary system—kidneys and ureters;
- the suprarenal glands; and
- major neurovascular structures.

FUNCTIONS

Houses and protects major viscera

The abdomen houses major elements of the gastrointestinal system (Fig. 4.2), the spleen, and parts of the urinary system.

Much of the liver, gallbladder, stomach, and spleen and parts of the colon are under the domes of the diaphragm, which project superiorly above the costal margin of the thoracic wall, and as a result these abdominal viscera are protected by the thoracic wall. The superior poles of the kidneys are deep to the lower ribs.

Viscera not under the domes of the diaphragm are supported and protected predominantly by the muscular walls of the abdomen.

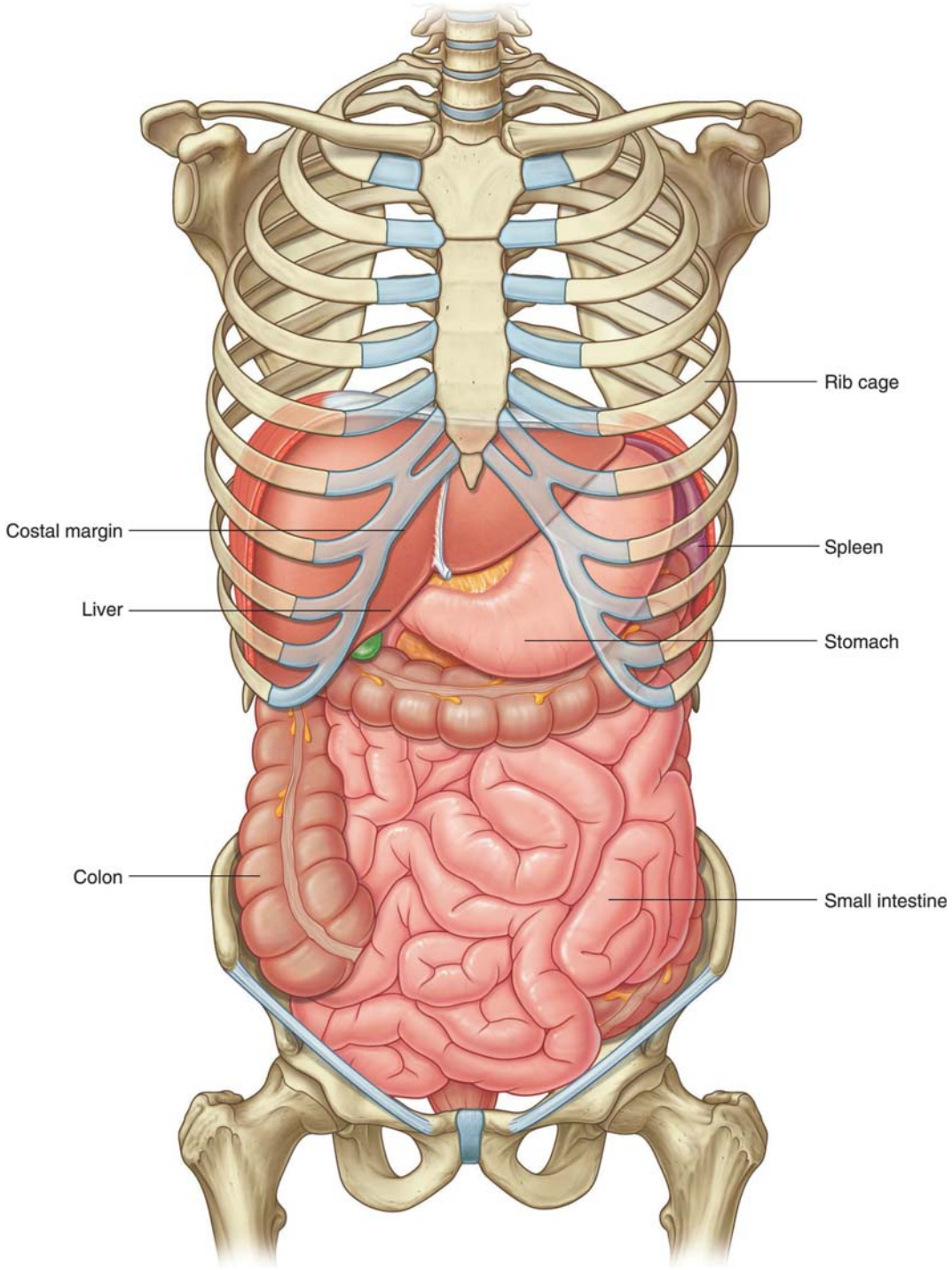


Fig. 4.2 The abdomen contains and protects the abdominal viscera.

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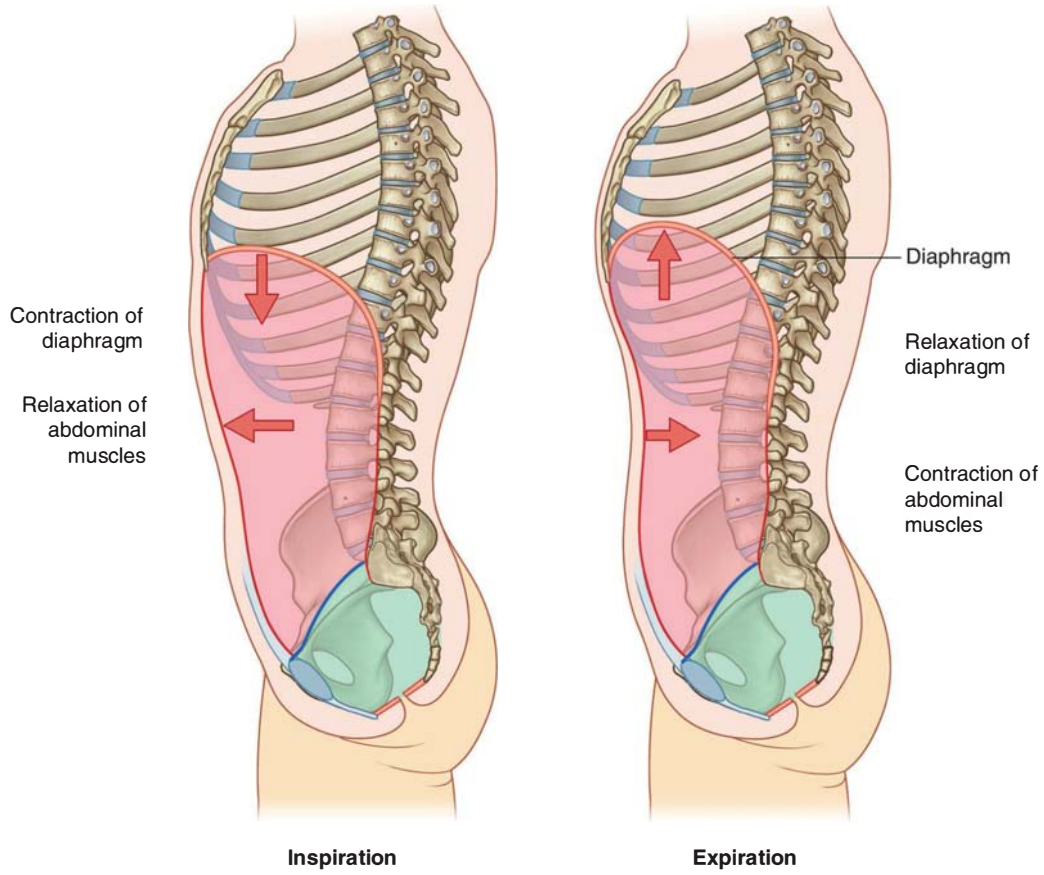


Fig. 4.3 The abdomen assists in breathing.

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Breathing

One of the most important roles of the abdominal wall is to assist in breathing:

- It relaxes during inspiration to accommodate expansion of the thoracic cavity and the inferior displacement of abdominal viscera during contraction of the diaphragm (Fig. 4.3).
- During expiration, it contracts to assist in elevating the domes of the diaphragm, thus reducing thoracic volume.

Material can be expelled from the airway by forced expiration using the abdominal muscles, as in coughing or sneezing.

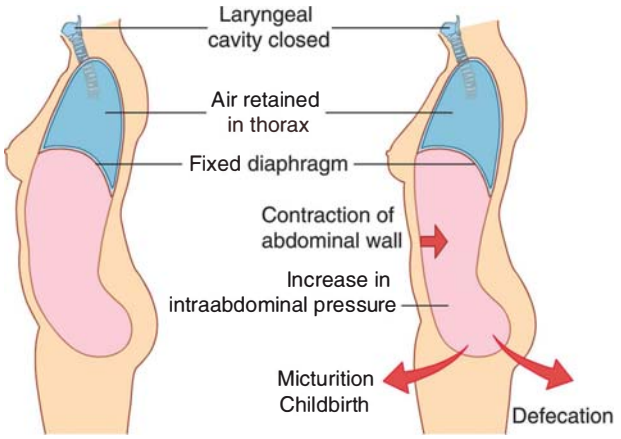


Fig. 4.4 Increasing intraabdominal pressure to assist in micturition, defecation, and childbirth.

Changes in intraabdominal pressure

Contraction of abdominal wall muscles can dramatically increase intraabdominal pressure when the diaphragm is in a fixed position (Fig. 4.4). Air is retained in the lungs by closing valves in the larynx in the neck. Increased intraabdominal pressure assists in voiding the contents of the bladder and rectum and in giving birth.

COMPONENT PARTS

Wall

The abdominal wall consists partly of bone but mainly of muscle (Fig. 4.5). The skeletal elements of the wall (Fig. 4.5A) are:

- the five lumbar vertebrae and their intervening intervertebral discs,
- the superior expanded parts of the pelvic bones, and
- bony components of the inferior thoracic wall, including the costal margin, rib XII, the end of rib XI, and the xiphoid process.

Muscles make up the rest of the abdominal wall (Fig. 4.5B):

- Lateral to the vertebral column, the quadratus lumborum, psoas major, and iliacus muscles reinforce the posterior aspect of the wall. The distal ends of the psoas major and iliacus muscles pass into the thigh and are major flexors of the hip joint.
- Lateral parts of the abdominal wall are predominantly formed by three layers of muscles, which are similar in orientation to the intercostal muscles of the thorax—transversus abdominis, internal oblique, and external oblique.

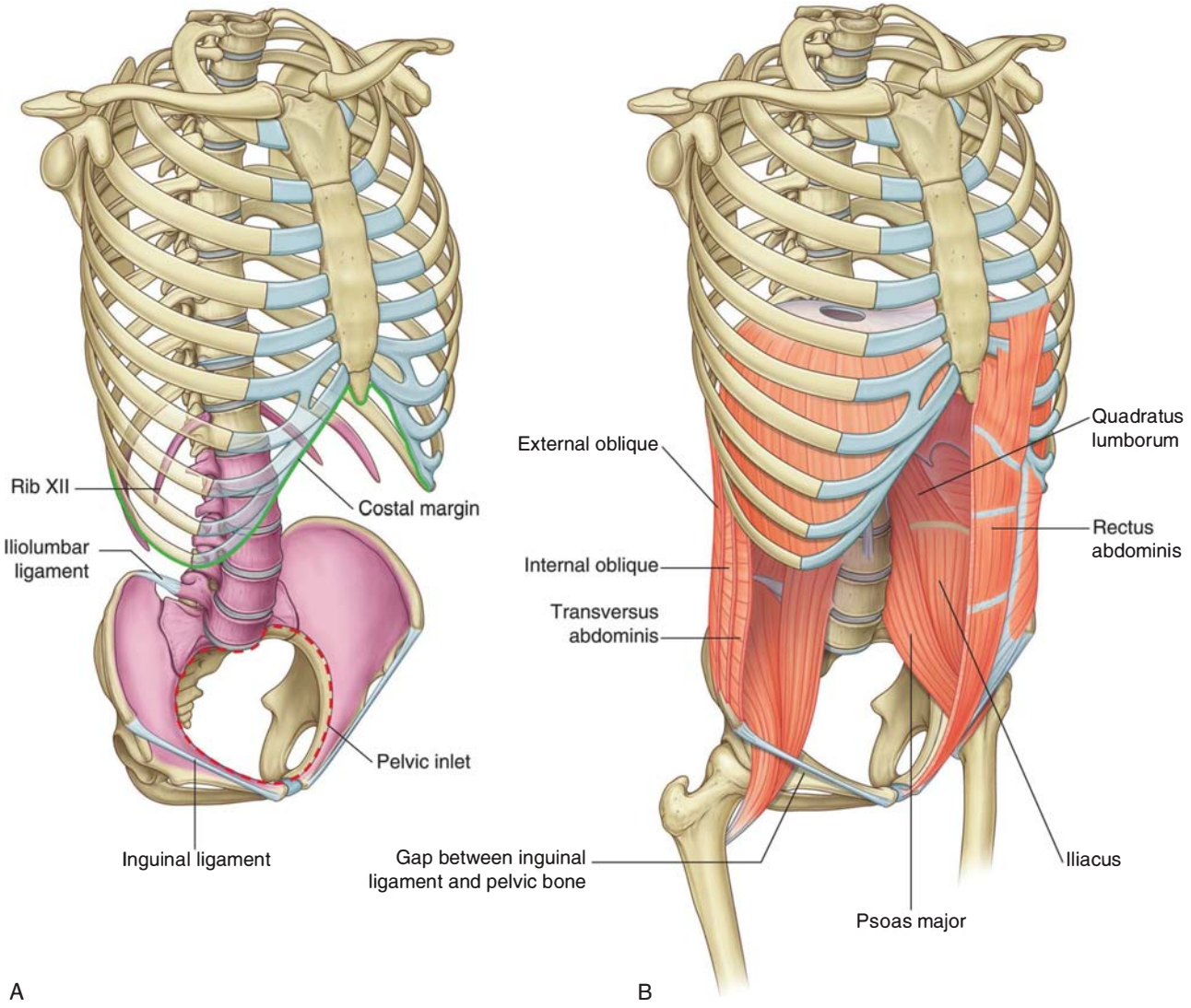


Fig. 4.5 Abdominal wall. **A.** Skeletal elements. **B.** Muscles.

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- Anteriorly, a segmented muscle (the rectus abdominis) on each side spans the distance between the inferior thoracic wall and the pelvis.

Structural continuity between posterior, lateral, and anterior parts of the abdominal wall is provided by thick fascia posteriorly and by flat tendinous sheets (aponeuroses) derived from muscles of the lateral wall. A fascial layer of varying thickness separates the abdominal wall from the peritoneum, which lines the abdominal cavity.

Abdominal cavity

The general organization of the abdominal cavity is one in which a central gut tube (gastrointestinal system) is suspended from the posterior abdominal wall and partly from the anterior abdominal wall by thin sheets of tissue (**mesenteries**; Fig. 4.6):

- a ventral (anterior) mesentery for proximal regions of the gut tube;
- a dorsal (posterior) mesentery along the entire length of the system.

Different parts of these two mesenteries are named according to the organs they suspend or with which they are associated.

Major viscera, such as the kidneys, that are not suspended in the abdominal cavity by mesenteries are associated with the abdominal wall.

The abdominal cavity is lined by **peritoneum**, which consists of an epithelial-like single layer of cells (the **mesothelium**) together with a supportive layer of connective tissue. Peritoneum is similar to the pleura and serous pericardium in the thorax.

The peritoneum reflects off the abdominal wall to become a component of the mesenteries that suspend the viscera.

- **Parietal peritoneum** lines the abdominal wall.
- **Visceral peritoneum** covers suspended organs.

Normally, elements of the gastrointestinal tract and its derivatives completely fill the abdominal cavity, making the peritoneal cavity a potential space, and visceral peritoneum on organs and parietal peritoneum on the adjacent abdominal wall slide freely against one another.

Abdominal viscera are either intraperitoneal or retroperitoneal:

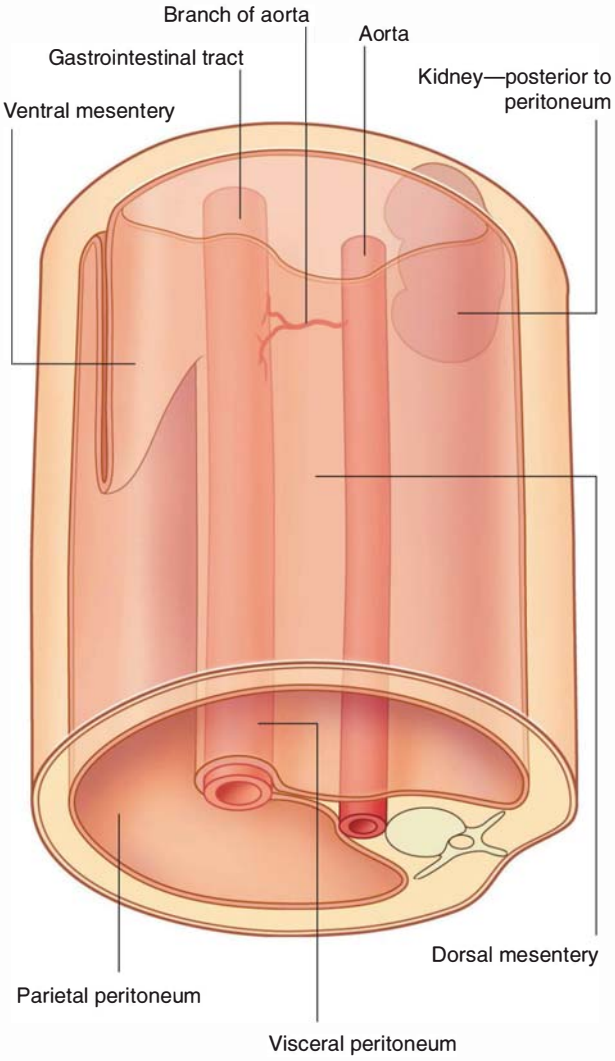


Fig. 4.6 The gut tube is suspended by mesenteries.

- **Intraperitoneal** structures, such as elements of the gastrointestinal system, are suspended from the abdominal wall by mesenteries;
- Structures that are not suspended in the abdominal cavity by a mesentery and that lie between the parietal peritoneum and abdominal wall are **retroperitoneal** in position.

Retroperitoneal structures include the kidneys and ureters, which develop in the region between the peritoneum and the abdominal wall and remain in this position in the adult.

During development, some organs, such as parts of the small and large intestines, are suspended initially in the abdominal cavity by a mesentery, and later become retroperitoneal secondarily by fusing with the abdominal wall (Fig. 4.7).

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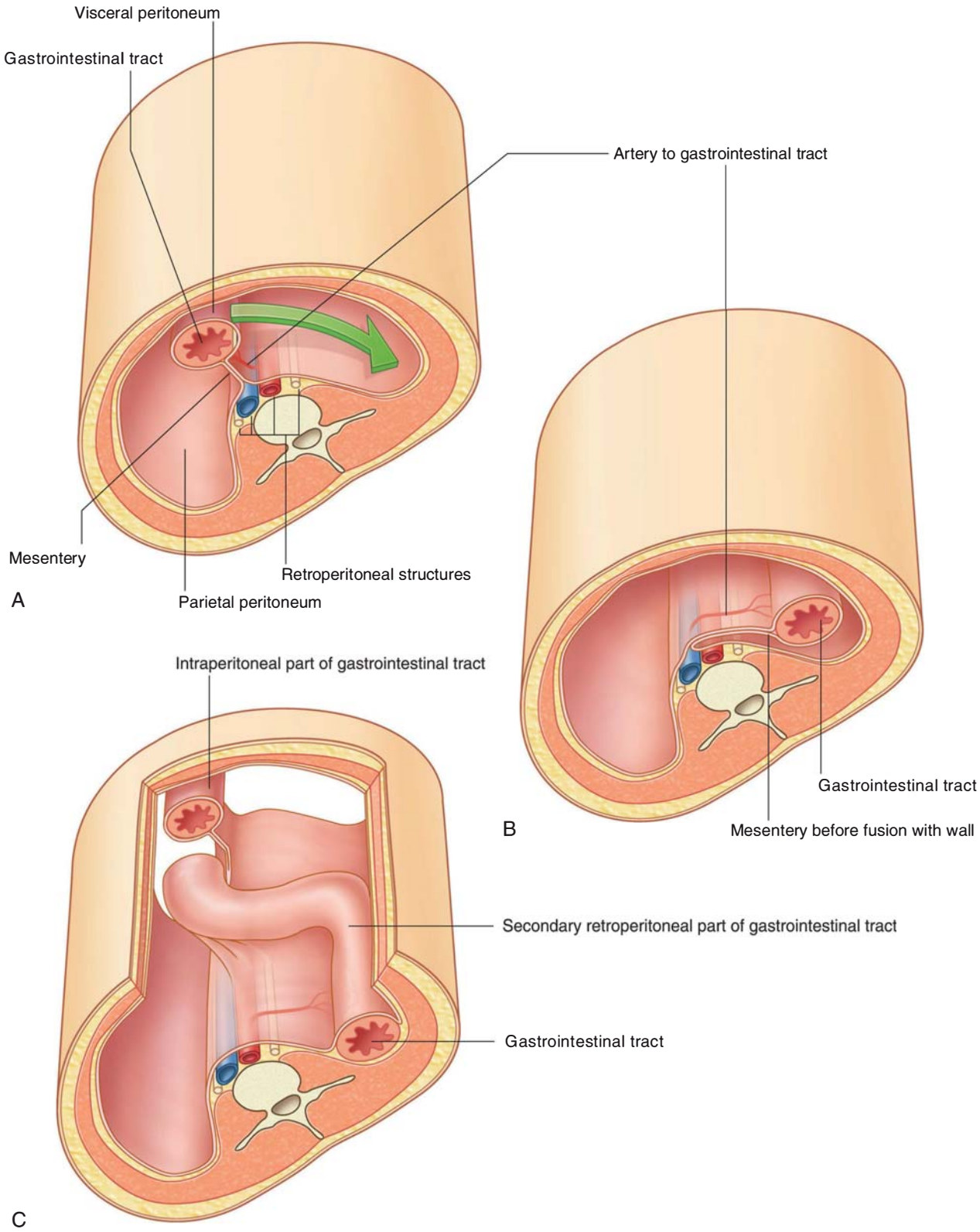


Fig. 4.7 A series showing the progression (**A** to **C**) from an intraperitoneal organ to a secondarily retroperitoneal organ.

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Large vessels, nerves, and lymphatics are associated with the posterior abdominal wall along the median axis of the body in the region where, during development, the peritoneum reflects off the wall as the dorsal mesentery, which supports the developing gut tube. As a consequence, branches of the neurovascular structures that pass to parts of the gastrointestinal system are unpaired, originate from the anterior aspects of their parent structures, and travel in mesenteries or pass retroperitoneally in areas where the mesenteries secondarily fuse to the wall.

Generally, vessels, nerves, and lymphatics to the abdominal wall and to organs that originate as retroperitoneal structures branch laterally from the central neurovascular structures and are usually paired, one on each side.

Inferior thoracic aperture

The superior aperture of the abdomen is the inferior thoracic aperture, which is closed by the diaphragm (see pp. 126-127). The margin of the inferior thoracic aperture consists of vertebra TXII, rib XII, the distal end of rib XI, the costal margin, and the xiphoid process of the sternum.

Diaphragm

The musculotendinous diaphragm separates the abdomen from the thorax.

The diaphragm attaches to the margin of the inferior thoracic aperture, but the attachment is complex posteriorly and extends into the lumbar area of the vertebral column (Fig. 4.8). On each side, a muscular extension (crus) firmly anchors the diaphragm to the anterolateral surface of the vertebral column as far down as vertebra LIII on the right and vertebra LII on the left.

Because the costal margin is not complete posteriorly, the diaphragm is anchored to arch-shaped (arcuate) ligaments, which span the distance between available bony points and the intervening soft tissues:

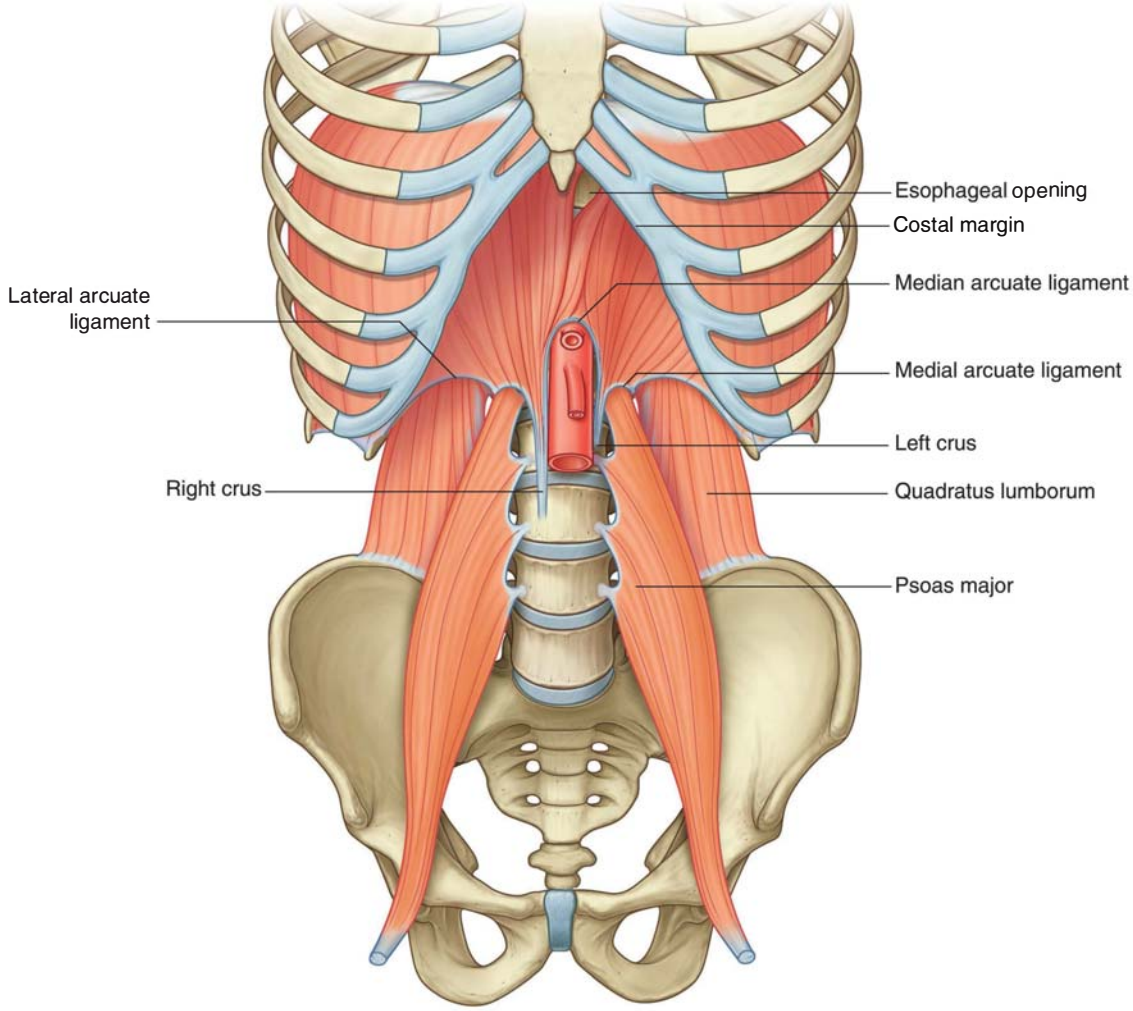


Fig. 4.8 Inferior thoracic aperture and the diaphragm.

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- **Medial** and **lateral arcuate ligaments** cross muscles of the posterior abdominal wall and attach to vertebrae, the transverse processes of vertebra LI and rib XII, respectively.
- A **median arcuate ligament** crosses the aorta and is continuous with the crus on each side.

The posterior attachment of the diaphragm extends much farther inferiorly than the anterior attachment. Consequently, the diaphragm is an important component of the posterior abdominal wall, to which a number of viscera are related.

Pelvic inlet

The abdominal wall is continuous with the pelvic wall at the pelvic inlet, and the abdominal cavity is continuous with the pelvic cavity.

The circular margin of the pelvic inlet is formed entirely by bone:

- posteriorly by the sacrum,
- anteriorly by the pubic symphysis, and
- laterally, on each side, by a distinct bony rim on the pelvic bone (Fig. 4.9).

Because of the way in which the sacrum and attached pelvic bones are angled posteriorly on the vertebral column, the pelvic cavity is not oriented in the same vertical plane as the abdominal cavity. Instead, the pelvic cavity projects posteriorly, and the inlet opens anteriorly and somewhat superiorly (Fig. 4.10).

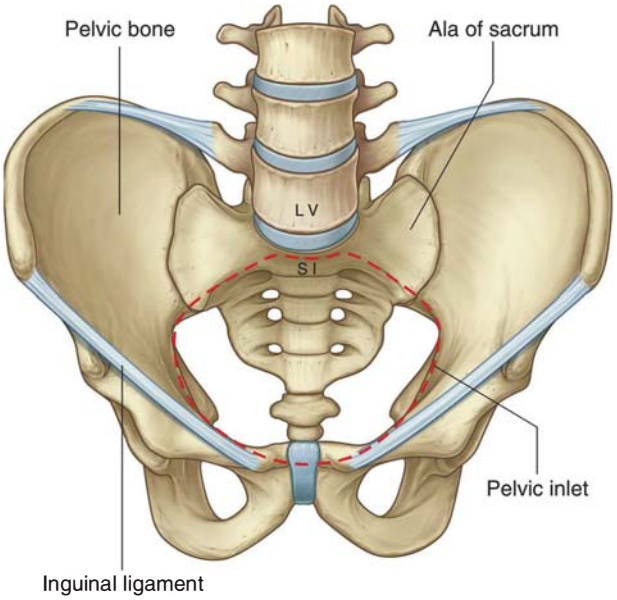


Fig. 4.9 Pelvic inlet.

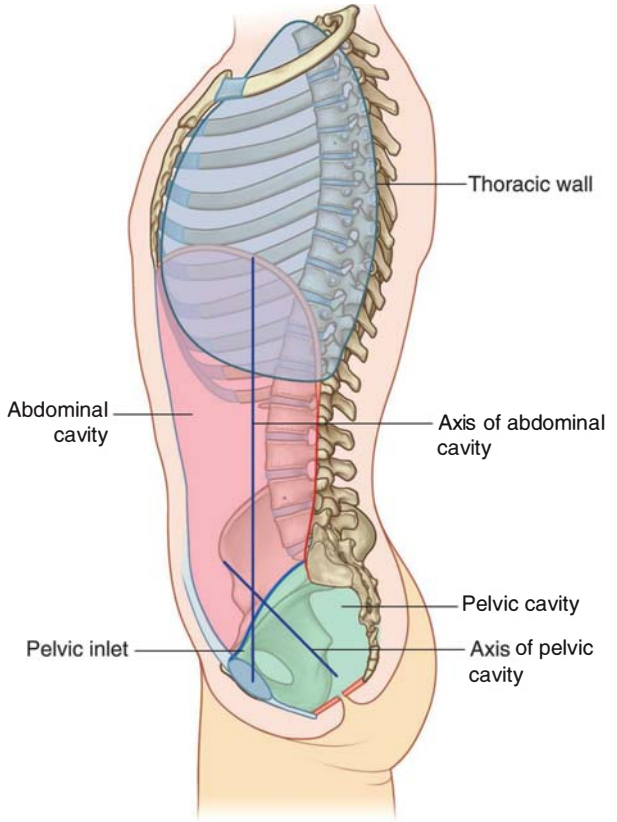


Fig. 4.10 Orientation of abdominal and pelvic cavities.

RELATIONSHIP TO OTHER REGIONS

Thorax

The abdomen is separated from the thorax by the diaphragm. Structures pass between the two regions through or posterior to the diaphragm (see Fig. 4.8).

Pelvis

The pelvic inlet opens directly into the abdomen and structures pass between the abdomen and pelvis through it.

The peritoneum lining the abdominal cavity is continuous with the peritoneum in the pelvis. Consequently, the

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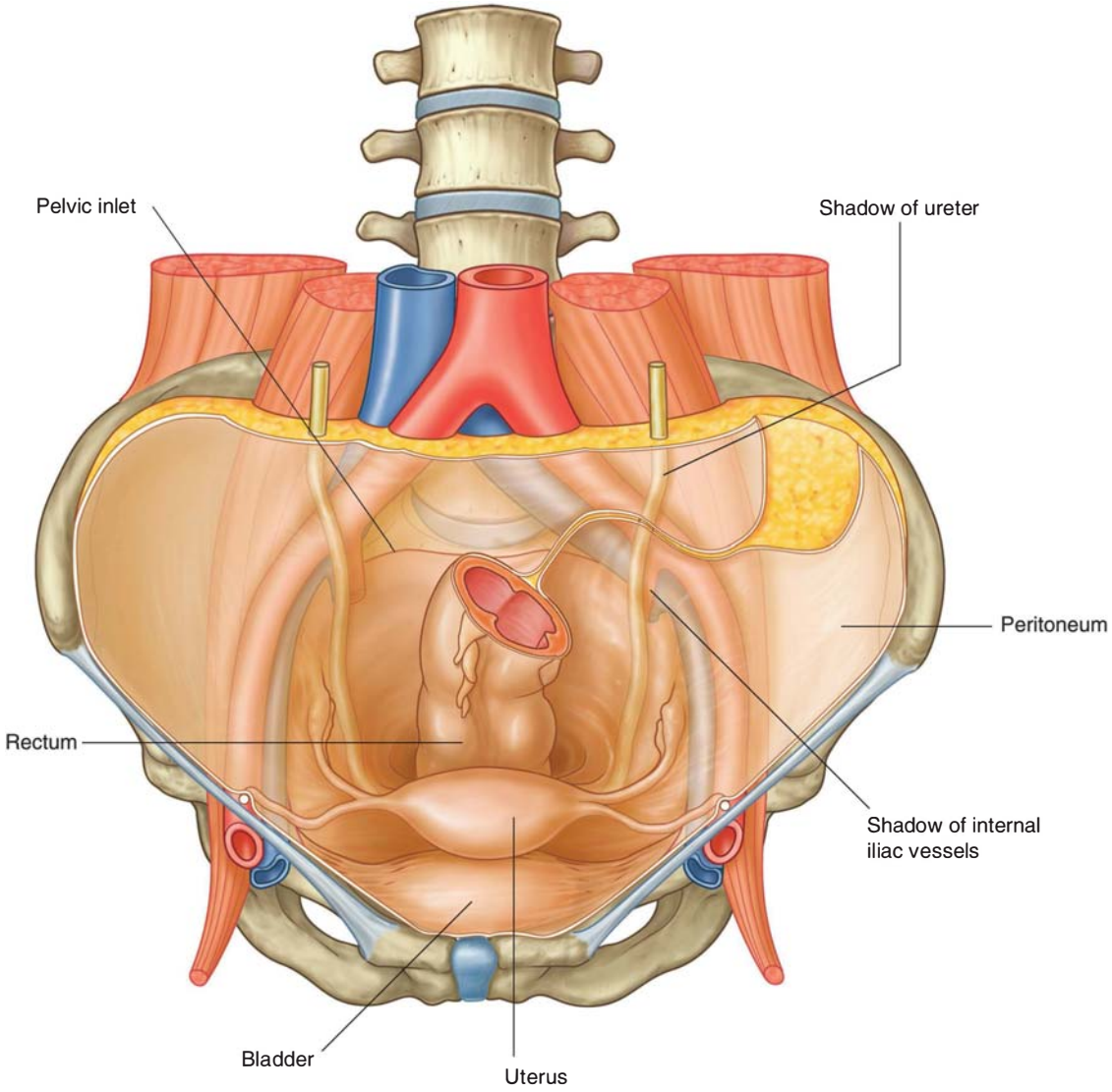


Fig. 4.11 The abdominal cavity is continuous with the pelvic cavity.

abdominal cavity is entirely continuous with the pelvic cavity (Fig. 4.11). Infections in one region can therefore freely spread into the other.

The bladder expands superiorly from the pelvic cavity into the abdominal cavity and, during pregnancy, the uterus expands freely superiorly out of the pelvic cavity into the abdominal cavity.

Lower limb

The abdomen communicates directly with the thigh through an aperture formed anteriorly between the inferior margin of the abdominal wall (marked by the inguinal ligament) and the pelvic bone (Fig. 4.12). Structures that pass through this aperture are:

- the major artery and vein of the lower limb;
- the femoral nerve, which innervates the quadriceps femoris muscle, which extends the knee;
- lymphatics; and
- the distal ends of psoas major and iliacus muscles, which flex the thigh at the hip joint.

As vessels pass inferior to the inguinal ligament, their names change—the external iliac artery and vein of the abdomen become the femoral artery and vein of the thigh.

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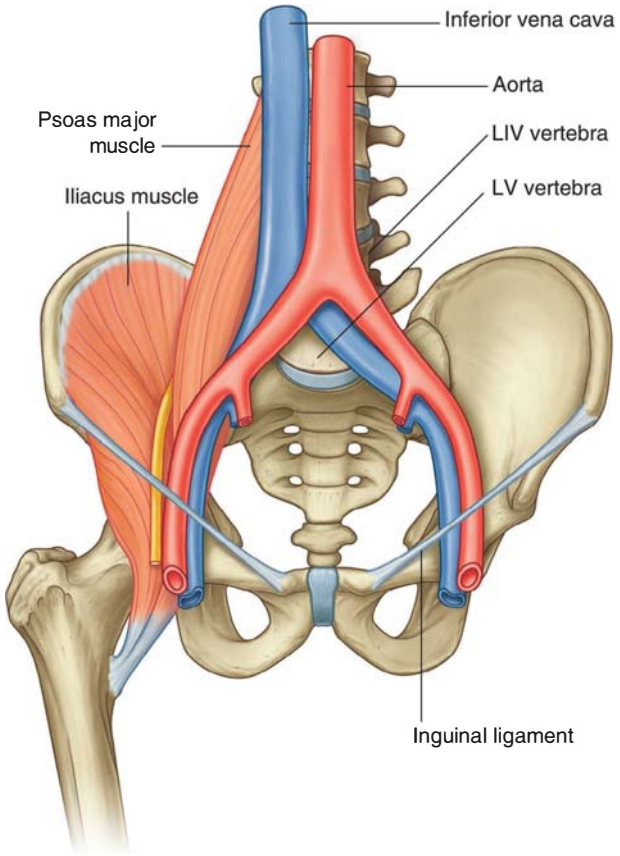


Fig. 4.12 Structures passing between the abdomen and thigh.

KEY FEATURES

Arrangement of abdominal viscera in the adult

A basic knowledge of the development of the gastrointestinal tract is needed to understand the arrangement of viscera and mesenteries in the abdomen (Fig. 4.13).

The early gastrointestinal tract is oriented longitudinally in the body cavity and is suspended from surrounding walls by a large dorsal mesentery and a much smaller ventral mesentery.

Superiorly, the dorsal and ventral mesenteries are anchored to the diaphragm.

The primitive gut tube consists of the foregut, the midgut, and the hindgut. Massive longitudinal growth of the gut tube, rotation of selected parts of the tube, and secondary fusion of some viscera and their associated mesenteries to the body wall participate in generating the adult arrangement of abdominal organs.

Development of the foregut

In abdominal regions, the **foregut** gives rise to the distal end of the esophagus, the stomach, and the proximal part of the duodenum. The foregut is the only part of the gut tube suspended from the wall by both the ventral and dorsal mesenteries.

A diverticulum from the anterior aspect of the foregut grows into the ventral mesentery, giving rise to the liver and gallbladder, and, ultimately, to the ventral part of the pancreas.

The dorsal part of the pancreas develops from an outgrowth of the foregut into the dorsal mesentery. The spleen develops in the dorsal mesentery in the region between the body wall and presumptive stomach.

In the foregut, the developing stomach rotates clockwise and the associated dorsal mesentery, containing the spleen, moves to the left and greatly expands. During this process, part of the mesentery becomes associated with, and secondarily fuses with, the left side of the body wall.

At the same time, the duodenum, together with its dorsal mesentery and an appreciable part of the pancreas, swings to the right and fuses to the body wall.

Secondary fusion of the duodenum to the body wall, massive growth of the liver in the ventral mesentery, and fusion of the superior surface of the liver to the diaphragm restrict the opening to the space enclosed by the ballooned dorsal mesentery associated with the stomach. This restricted opening is the **omental foramen** (**epiploic foramen**).

The part of the abdominal cavity enclosed by the expanded dorsal mesentery, and posterior to the stomach, is the **omental bursa** (**lesser sac**). Access, through the omental foramen, to this space from the rest of the peritoneal cavity (**greater sac**) is inferior to the free edge of the ventral mesentery.

Part of the dorsal mesentery that initially forms part of the lesser sac greatly enlarges in an inferior direction, and the two opposing surfaces of the mesentery fuse to form an apron-like structure (the **greater omentum**). The greater omentum is suspended from the greater curvature of the stomach, lies over other viscera in the abdominal cavity, and is the first structure observed when the abdominal cavity is opened anteriorly.

Development of the midgut

The midgut develops into the distal part of the duodenum and the jejunum, ileum, ascending colon, and proximal two-thirds of the transverse colon. A small yolk sac projects anteriorly from the developing midgut into the umbilicus.

Rapid growth of the gastrointestinal system results in a loop of the midgut herniating out of the abdominal cavity and into the umbilical cord. As the body grows in size and the connection with the yolk sac is lost, the midgut returns to the abdominal cavity. While this process is occurring, the two limbs of the midgut loop rotate counterclockwise

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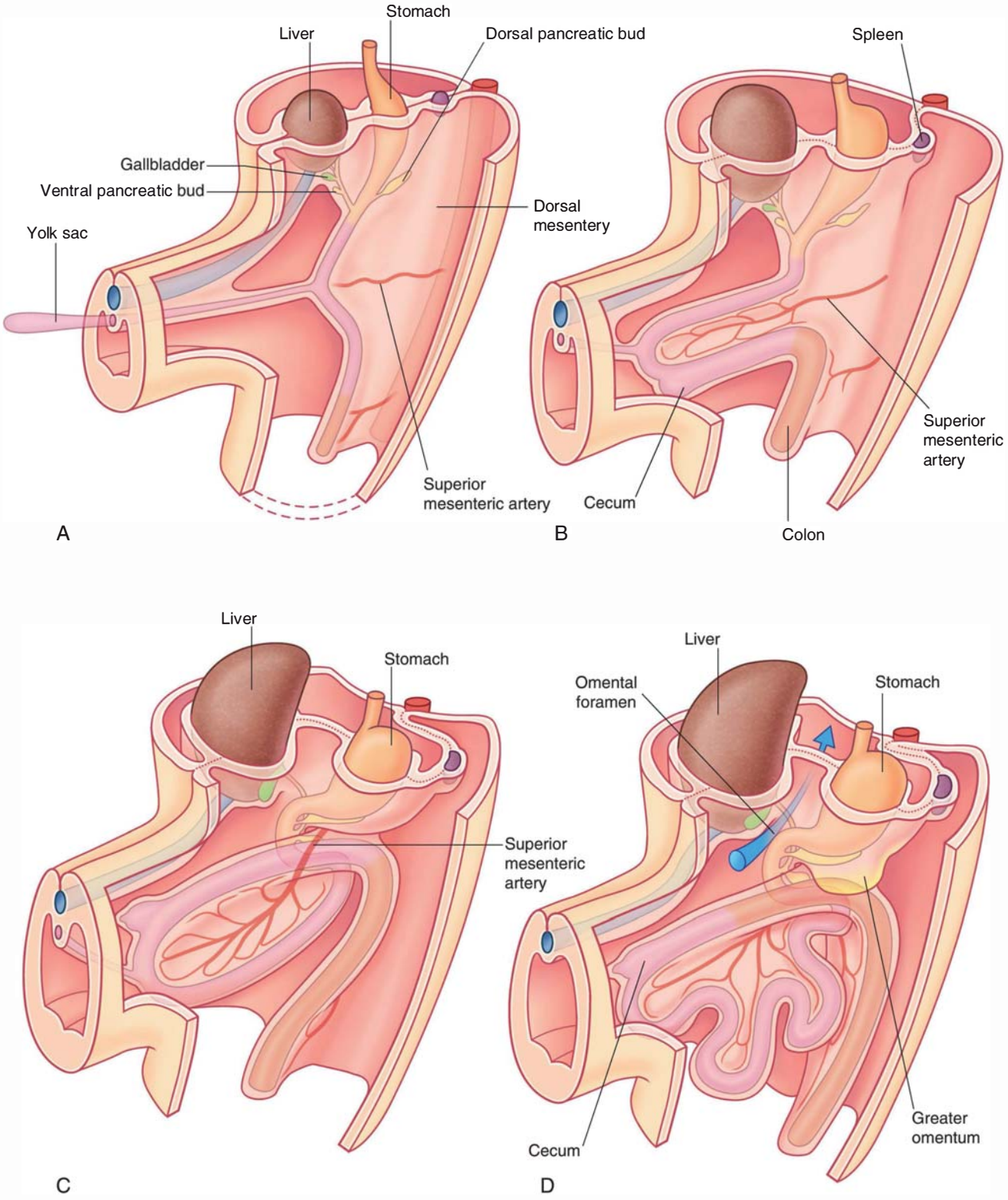


Fig. 4.13 A series (**A** to **H**) showing the development of the gut and mesenteries.

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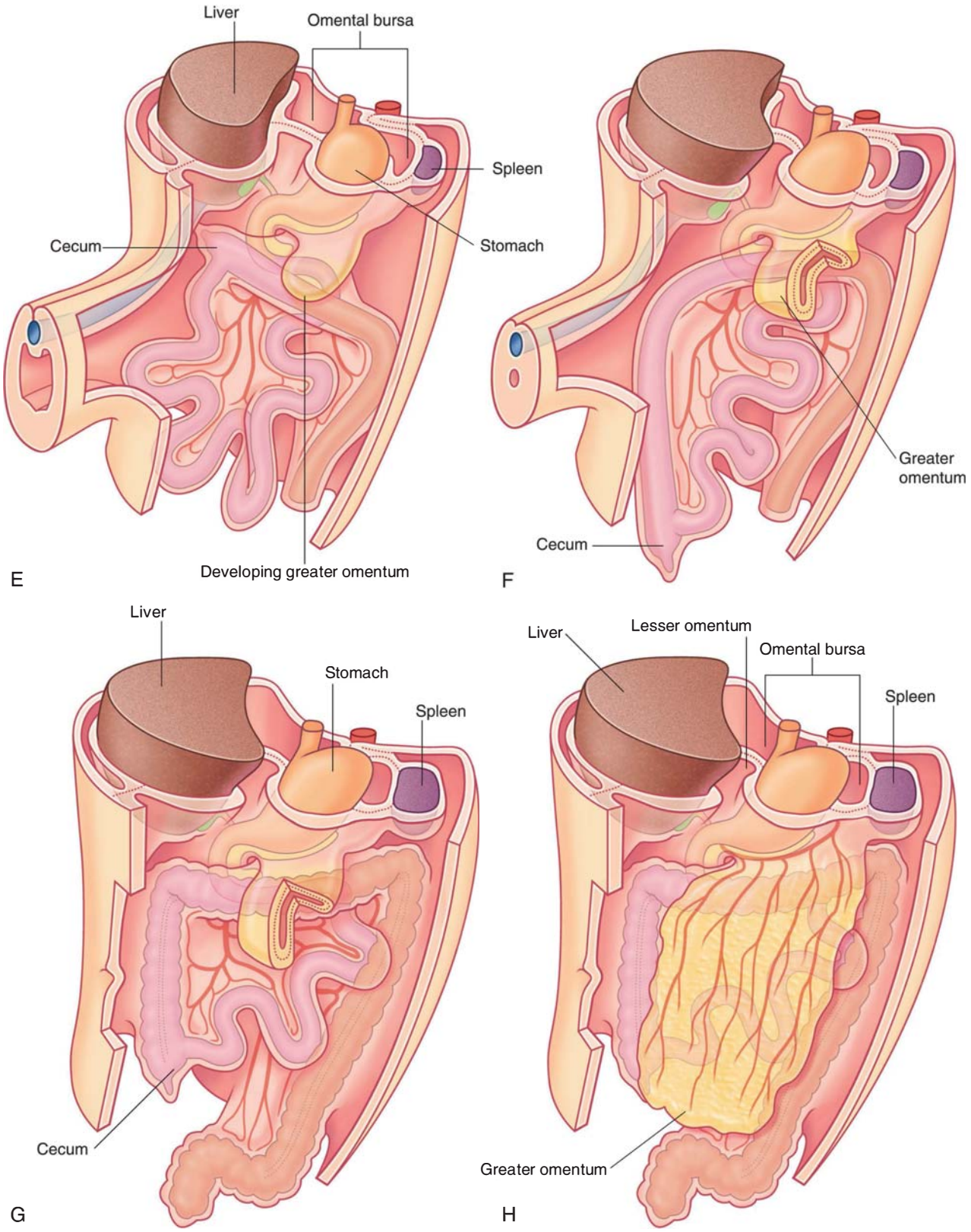


Fig. 4.13, cont'd

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around their combined central axis, and the part of the loop that becomes the cecum descends into the inferior right aspect of the cavity. The superior mesenteric artery, which supplies the midgut, is at the center of the axis of rotation.

The cecum remains intraperitoneal, the ascending colon fuses with the body wall becoming secondarily retroperitoneal, and the transverse colon remains suspended by its dorsal mesentery (transverse mesocolon). The greater omentum hangs over the transverse colon and the mesocolon and usually fuses with these structures.

Development of the hindgut

The distal one-third of the transverse colon, descending colon, sigmoid colon, and superior part of the rectum develop from the hindgut.

Proximal parts of the hindgut swing to the left and become the descending colon and sigmoid colon. The descending colon and its dorsal mesentery fuse to the body wall, while the sigmoid colon remains intraperitoneal. The sigmoid colon passes through the pelvic inlet and is continuous with the rectum at the level of vertebra SIII.

Skin and muscles of the anterior and lateral abdominal wall and thoracic intercostal nerves

The anterior rami of thoracic spinal nerves T7 to T12 follow the inferior slope of the lateral parts of the ribs and cross the costal margin to enter the abdominal wall (Fig. 4.14). Intercostal nerves T7 to T11 supply skin and muscle of the abdominal wall, as does the subcostal nerve T12. In addition, T5 and T6 supply upper parts of the external oblique muscle of the abdominal wall; T6 also supplies cutaneous innervation to skin over the xiphoid.

Skin and muscle in the inguinal and suprapubic regions of the abdominal wall are innervated by L1 and not by thoracic nerves.

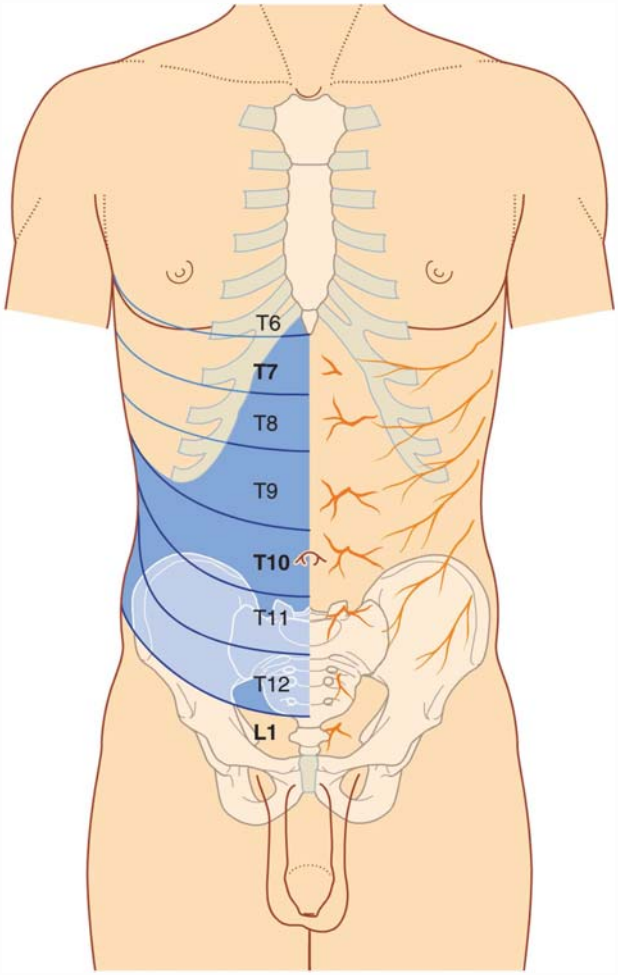


Fig. 4.14 Innervation of the anterior abdominal wall.

Dermatomes of the anterior abdominal wall are indicated in Figure 4.14. In the midline, skin over the infrasternal angle is T6 and that around the umbilicus is T10. L1 innervates skin in the inguinal and suprapubic regions.

Muscles of the abdominal wall are innervated segmentally in patterns that generally reflect the patterns of the overlying dermatomes.

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The groin is a weak area in the anterior abdominal wall

During development, the gonads in both sexes descend from their sites of origin on the posterior abdominal wall into the pelvic cavity in women and the developing scrotum in men (Fig. 4.15).

Before descent, a cord of tissue (the **gubernaculum**) passes through the anterior abdominal wall and connects the inferior pole of each gonad with primordia of the scrotum in men and the labia majora in women (labioscrotal swellings).

A tubular extension (the **processus vaginalis**) of the peritoneal cavity and the accompanying muscular layers of the anterior abdominal wall project along the gubernaculum on each side into the labioscrotal swellings.

In men, the testis, together with its neurovascular structures and its efferent duct (the ductus deferens) descends into the scrotum along a path, initially defined by the gubernaculum, between the processus vaginalis and the accompanying coverings derived from the abdominal wall. All that remains of the gubernaculum is a connective tissue remnant that attaches the caudal pole of the testis to the scrotum.

The **inguinal canal** is the passage through the anterior abdominal wall created by the processus vaginalis. The **spermatic cord** is the tubular extension of the layers of the abdominal wall into the scrotum that contains all structures passing between the testis and the abdomen.

The distal sac-like terminal end of the spermatic cord on each side contains the testis, associated structures, and the now isolated part of the peritoneal cavity (the cavity of the tunica vaginalis).

In women, the gonads descend to a position just inside the pelvic cavity and never pass through the anterior abdominal wall. As a result, the only major structure passing through the inguinal canal is a derivative of the gubernaculum (the round ligament of the uterus).

In both men and women, the groin (inguinal region) is a weak area in the abdominal wall (Fig. 4.15) and is the site of inguinal hernias.

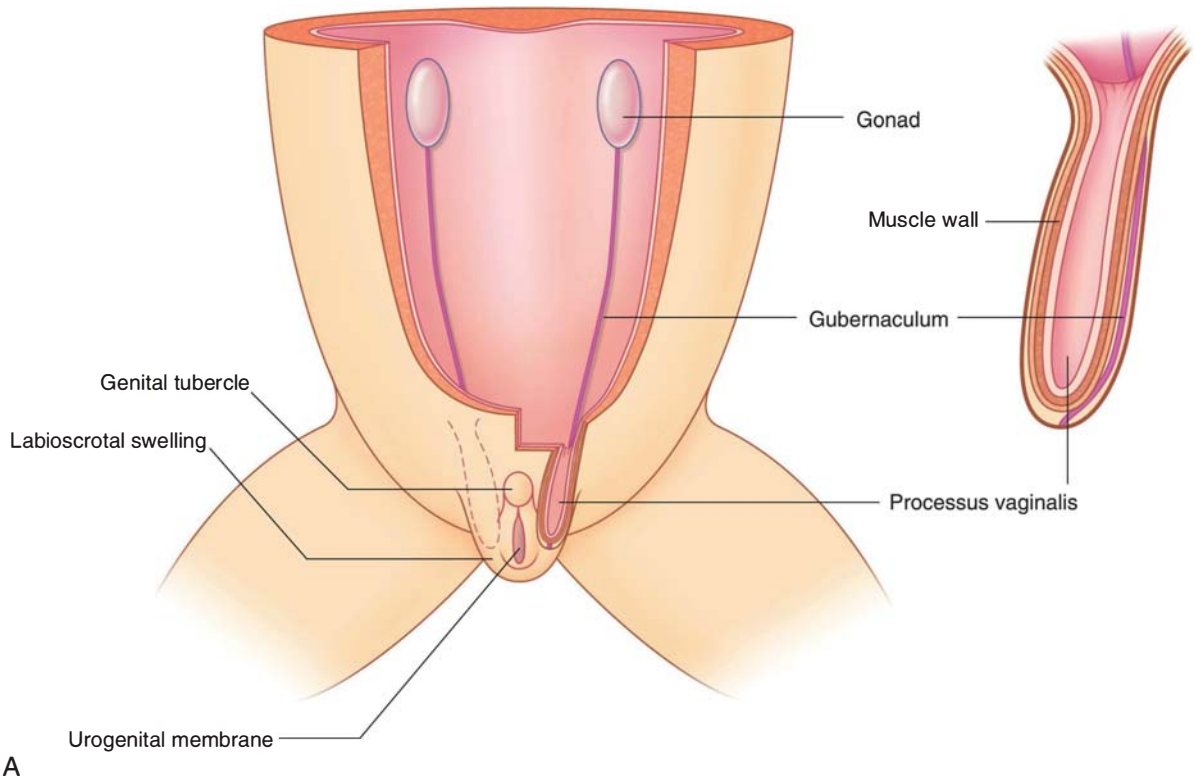


Fig. 4.15 Inguinal region. **A.** Development.

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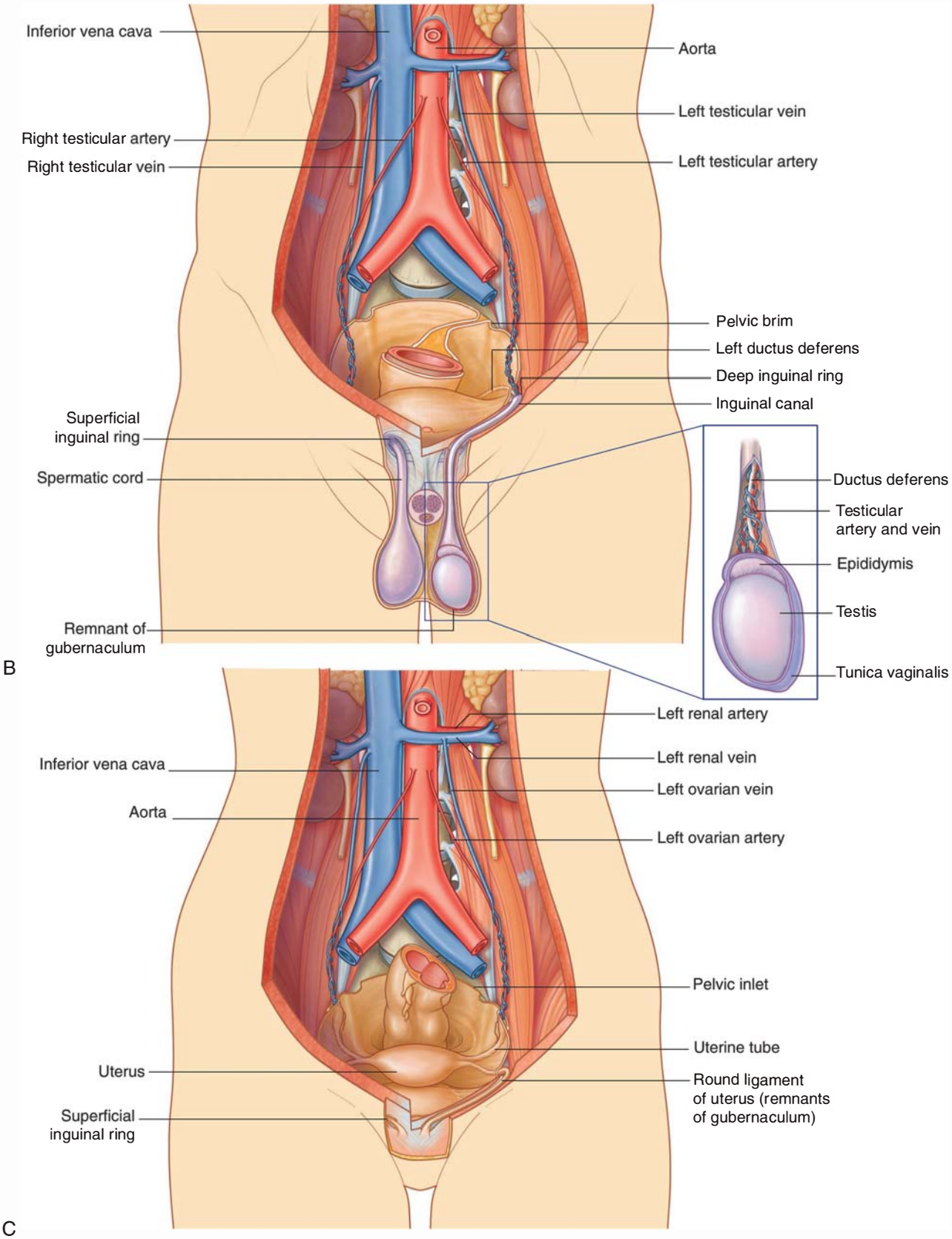


Fig. 4.15, cont'd B. In men. **C.** In women.

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Vertebral level LI

The transpyloric plane is a horizontal plane that transects the body through the lower aspect of vertebra LI (Fig. 4.16). It:

- is about midway between the jugular notch and the pubic symphysis, and crosses the costal margin on each side at roughly the ninth costal cartilage;
- crosses through the opening of the stomach into the duodenum (the pyloric orifice), which is just to the right of the body of LI; the duodenum then makes a characteristic C-shaped loop on the posterior abdominal wall and crosses the midline to open into the jejunum just to the left of the body of vertebra LII, whereas the head of the pancreas is enclosed by the loop of the duodenum, and the body of the pancreas extends across the midline to the left;
- crosses through the body of the pancreas; and
- approximates the position of the hila of the kidneys; though because the left kidney is slightly higher than the right, the transpyloric plane crosses through the inferior aspect of the left hilum and the superior part of the right hilum.

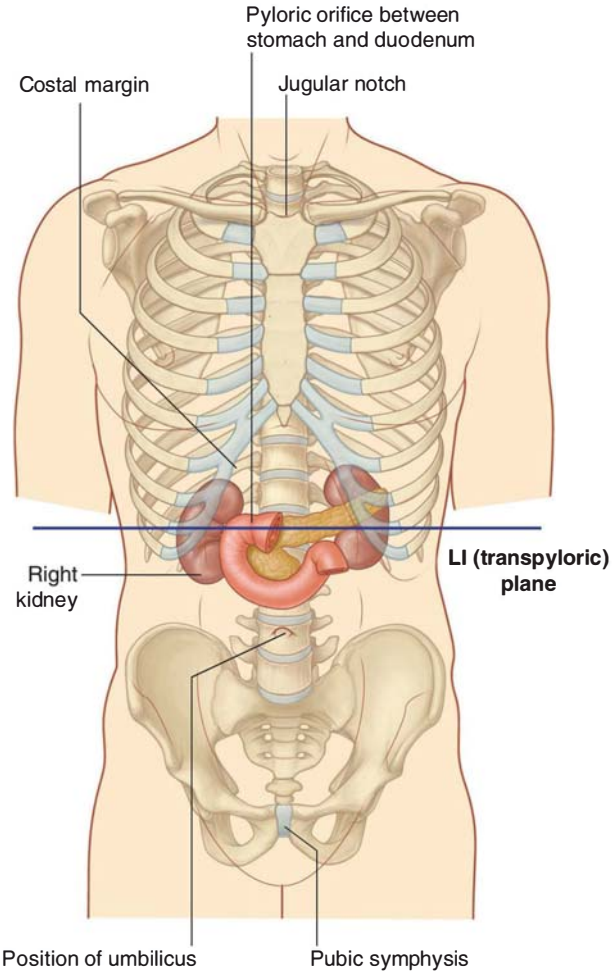


Fig. 4.16 Vertebral level LI.

The gastrointestinal system and its derivatives are supplied by three major arteries

Three large unpaired arteries branch from the anterior surface of the abdominal aorta to supply the abdominal part of the gastrointestinal tract and all of the structures (liver, pancreas, and gallbladder) to which this part of the gut gives rise during development (Fig. 4.17). These arteries pass through derivatives of the dorsal and ventral mesenteries to reach the target viscera. These vessels therefore also supply structures such as the spleen and lymph nodes that develop in the mesenteries. These three arteries are:

- the **celiac artery**, which branches from the abdominal aorta at the upper border of vertebra LI and supplies the foregut;
- the **superior mesenteric artery**, which arises from the abdominal aorta at the lower border of vertebra LI and supplies the midgut; and
- the **inferior mesenteric artery**, which branches from the abdominal aorta at approximately vertebral level LIII and supplies the hindgut.

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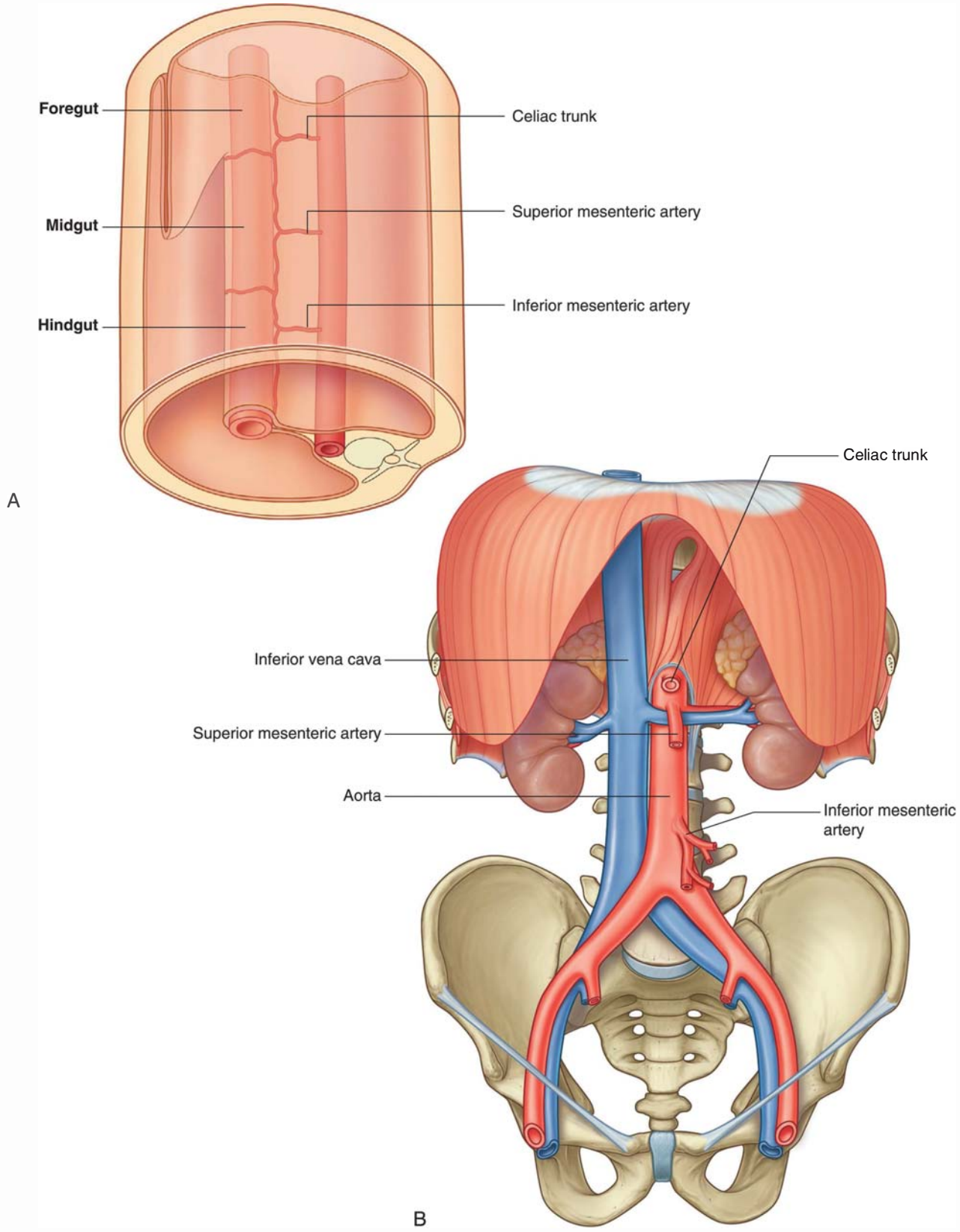


Fig. 4.17 Blood supply of the gut. **A.** Relationship of vessels to the gut and mesenteries. **B.** Anterior view.

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Venous shunts from left to right

All blood returning to the heart from regions of the body other than the lungs flows into the right atrium of the heart. The inferior vena cava is the major systemic vein in the abdomen and drains this region together with the pelvis, perineum, and both lower limbs (Fig. 4.18).

The inferior vena cava lies to the right of the vertebral column and penetrates the central tendon of the diaphragm at approximately vertebral level TVIII. A number of large vessels cross the midline to deliver blood from the left side of the body to the inferior vena cava.

- One of the most significant is the left renal vein, which drains the kidney, suprarenal gland, and gonad on the same side.

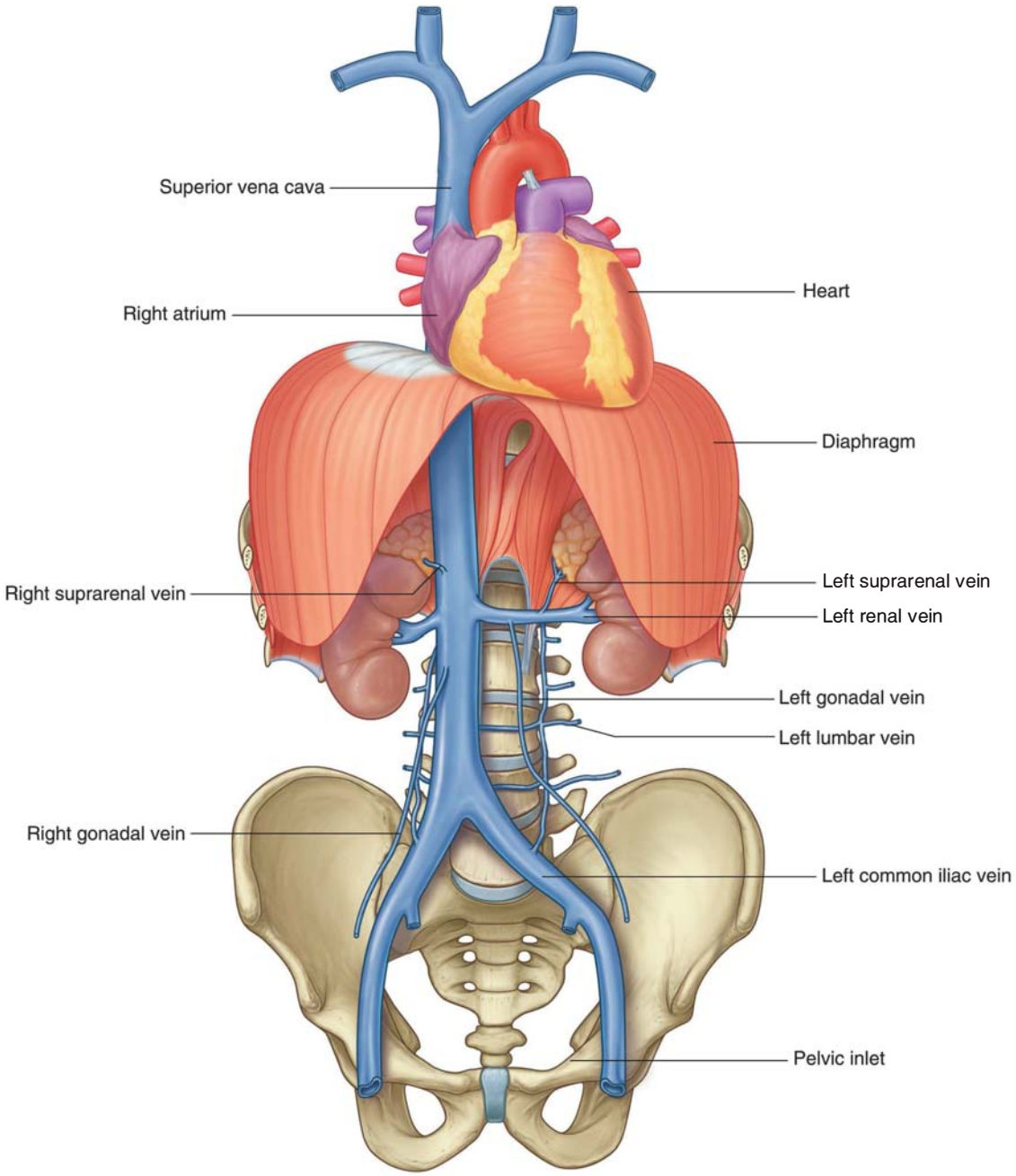


Fig. 4.18 Left-to-right venous shunts.

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- Another is the left common iliac vein, which crosses the midline at approximately vertebral level LV to join with its partner on the right to form the inferior vena cava. These veins drain the lower limbs, the pelvis, the perineum, and parts of the abdominal wall.
- Other vessels crossing the midline include the left lumbar veins, which drain the back and posterior abdominal wall on the left side.

All venous drainage from the gastrointestinal system passes through the liver

Blood from abdominal parts of the gastrointestinal system and the spleen passes through a second vascular bed, in the liver, before ultimately returning to the heart (Fig. 4.19).

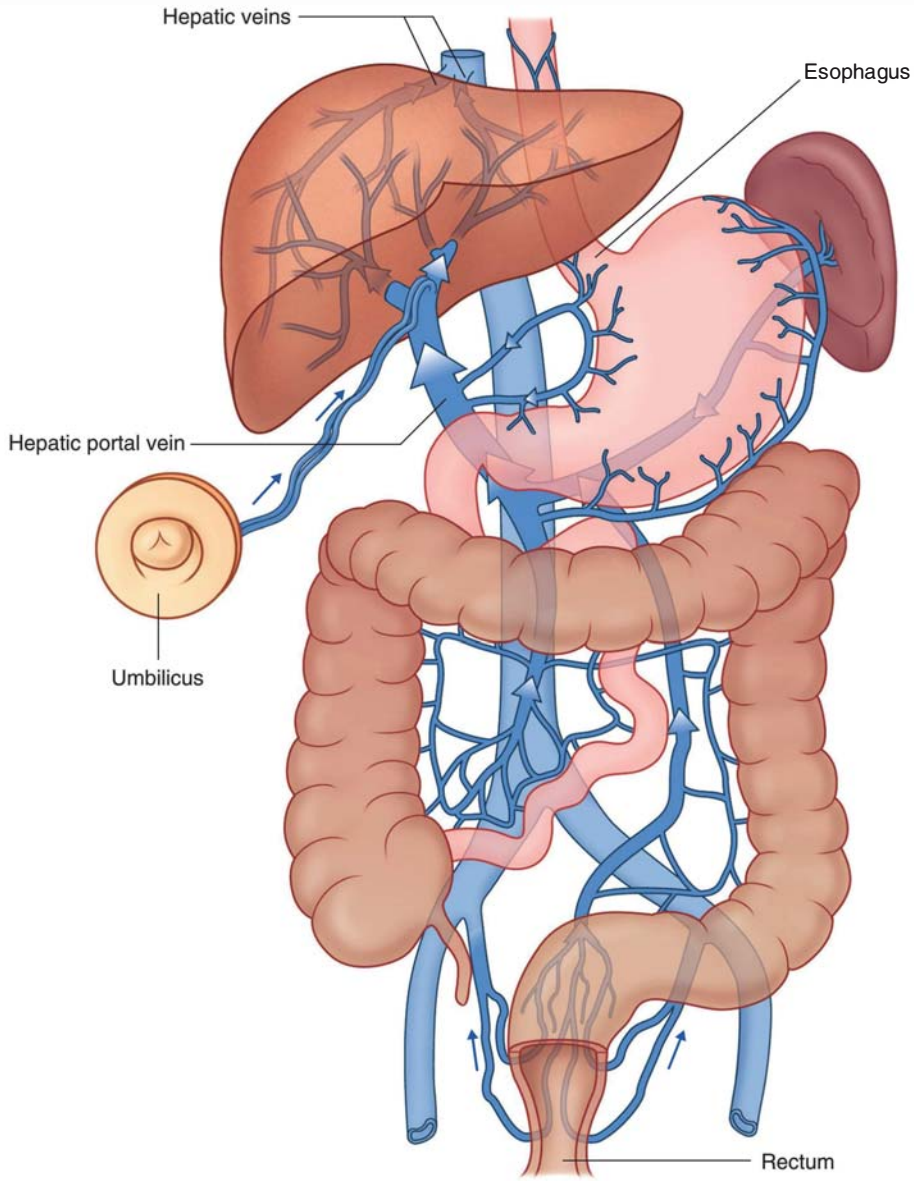


Fig. 4.19 Hepatic portal system.

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Venous blood from the digestive tract, pancreas, gallbladder, and spleen enters the inferior surface of the liver through the large **hepatic portal vein**. This vein then ramifies like an artery to distribute blood to small endothelial-lined hepatic sinusoids, which form the vascular exchange network of the liver.

After passing through the sinusoids, the blood collects in a number of short **hepatic veins**, which drain into the inferior vena cava just before the inferior vena cava penetrates the diaphragm and enters the right atrium of the heart.

Normally, vascular beds drained by the hepatic portal system interconnect, through small veins, with beds drained by systemic vessels, which ultimately connect directly with either the superior or inferior vena cava.

Portacaval anastomoses

Among the clinically most important regions of overlap between the portal and caval systems are those at each end of the abdominal part of the gastrointestinal system:

- around the inferior end of the esophagus;
- around the inferior part of the rectum.

Small veins that accompany the degenerate umbilical vein (**round ligament of the liver**) establish another important portacaval anastomosis.

The round ligament of the liver connects the umbilicus of the anterior abdominal wall with the left branch of the portal vein as it enters the liver. The small veins that accompany this ligament form a connection between the portal system and para-umbilical regions of the abdominal wall, which drain into systemic veins.

Other regions where portal and caval systems interconnect include:

- where the liver is in direct contact with the diaphragm (the bare area of the liver);
- where the wall of the gastrointestinal tract is in direct contact with the posterior abdominal wall (retroperitoneal areas of the large and small intestine); and
- the posterior surface of the pancreas (much of the pancreas is secondarily retroperitoneal).

Blockage of the hepatic portal vein or of vascular channels in the liver

Blockage of the hepatic portal vein or of vascular channels in the liver can affect the pattern of venous return from abdominal parts of the gastrointestinal system. Vessels that interconnect the portal and caval systems can become greatly enlarged and tortuous, allowing blood in tributaries of the portal system to bypass the liver, enter the caval system, and thereby return to the heart. Portal hypertension can result in esophageal and rectal varices and in caput medusae in which systemic vessels that radiate from para-umbilical veins enlarge and become visible on the abdominal wall.

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Abdominal viscera are supplied by a large prevertebral plexus

Innervation of the abdominal viscera is derived from a large prevertebral plexus associated mainly with the anterior and lateral surfaces of the aorta (Fig. 4.20). Branches are distributed to target tissues along vessels that originate from the abdominal aorta.

The prevertebral plexus contains sympathetic, parasympathetic, and visceral sensory components:

- Sympathetic components originate from spinal cord levels T5 to L2.
- Parasympathetic components are from the vagus nerve [X] and spinal cord levels S2 to S4.
- Visceral sensory fibers generally parallel the motor pathways.

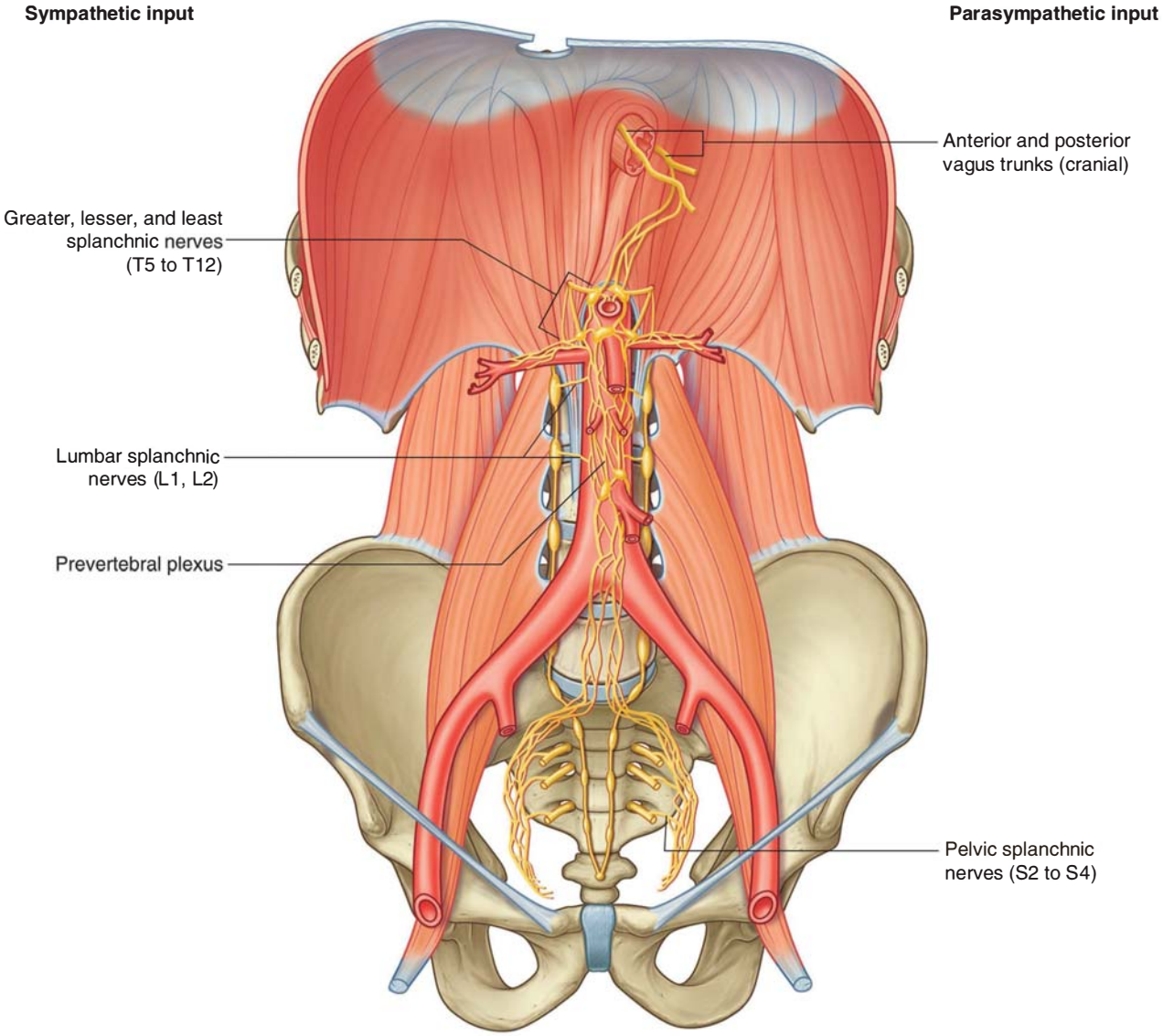


Fig. 4.20 Prevertebral plexus.

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Regional anatomy

The abdomen is the part of the trunk inferior to the thorax (Fig. 4.21). Its musculomembranous walls surround a large cavity (the **abdominal cavity**), which is bounded superiorly by the diaphragm and inferiorly by the pelvic inlet.

The abdominal cavity may extend superiorly as high as the fourth intercostal space, and is continuous inferiorly with the pelvic cavity. It contains the **peritoneal cavity** and the abdominal viscera.

SURFACE TOPOGRAPHY

Topographical divisions of the abdomen are used to describe the location of abdominal organs and the pain associated with abdominal problems. The two schemes most often used are:

- a four-quadrant pattern and
- a nine-region pattern.

Four-quadrant pattern

A horizontal transumbilical plane passing through the umbilicus and the intervertebral disc between vertebrae LIII and LIV and intersecting with the vertical median plane divides the abdomen into four quadrants—the right upper, left upper, right lower, and left lower quadrants (Fig. 4.22).

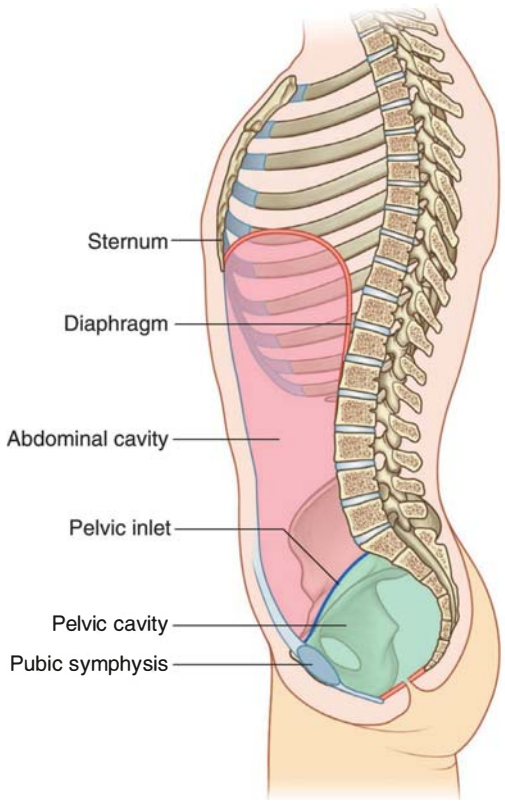


Fig. 4.21 Boundaries of the abdominal cavity.

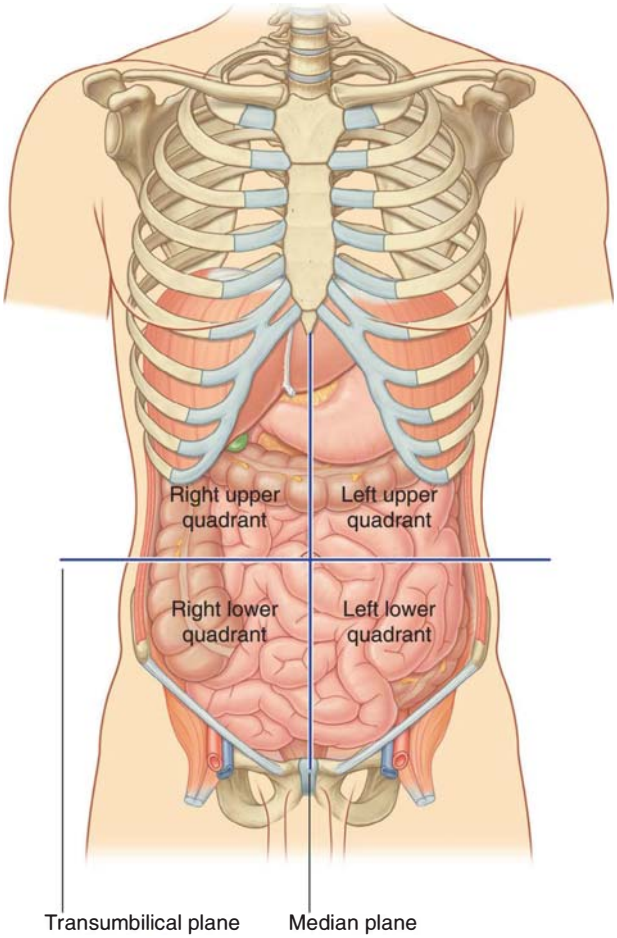


Fig. 4.22 Four-quadrant topographical pattern.

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Nine-region pattern

The nine-region pattern is based on two horizontal and two vertical planes (Fig. 4.23).

- The superior horizontal plane (the **subcostal plane**) is immediately inferior to the costal margins, which places it at the lower border of the costal cartilage of rib X and passing posteriorly through the body of vertebra LIII. (Note, however, that sometimes the **transpyloric plane,** halfway between the jugular notch and the symphysis pubis or halfway between the umbilicus and the inferior end of the body of the sternum, passing posteriorly through the lower border of vertebra LI and intersecting with the costal margin at the ends of the ninth costal cartilages, is used instead.)
- The inferior horizontal plane (the **intertubercular plane**) connects the tubercles of the iliac crests, which are palpable structures 5 cm posterior to the anterior superior iliac spines, and passes through the upper part of the body of vertebra LV.
- The vertical planes pass from the midpoint of the clavicles inferiorly to a point midway between the anterior superior iliac spine and pubic symphysis.

These four planes establish the topographical divisions in the nine-region organization. The following designations are used for each region: superiorly the right hypochondrium, the epigastric region, and the left hypochondrium; inferiorly the right groin (inguinal region), pubic region, and left groin (inguinal region); and in the middle the right flank (lateral region), the umbilical region, and the left flank (lateral region) (Fig. 4.23).

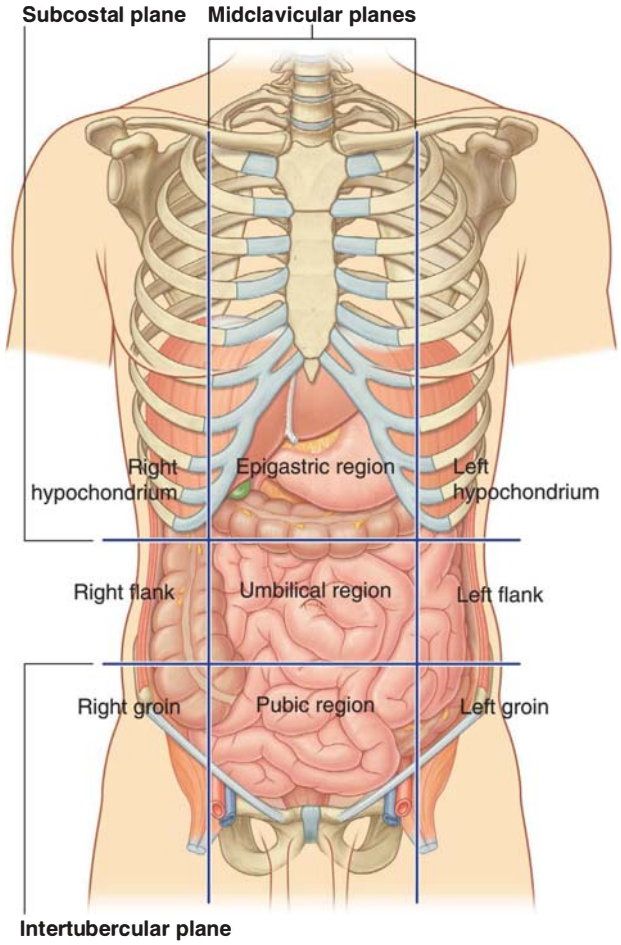


Fig. 4.23 Nine-region organizational pattern.

In the clinic

Surgical incisions

Access to the abdomen and its contents is usually obtained through incisions in the anterior abdominal wall. Traditionally, incisions have been placed at and around the region of surgical interest. The size of these incisions was usually large to allow good access and optimal visualization of the abdominal cavity. As anesthesia has developed and muscle-relaxing drugs have become widely used, the abdominal incisions have become smaller.

Currently, the most commonly used large abdominal incision is a central craniocaudad incision from the xiphoid process to the symphysis pubis, which provides wide access to the whole of the abdominal contents and allows an exploratory procedure to be performed (laparotomy).

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In the clinic

Laparoscopic surgery

Laparoscopic surgery, also known as minimally invasive or keyhole surgery, is performed by operating through a series of small incisions no more than 1 to 2 cm in length. As the incisions are much smaller than those used in traditional abdominal surgery, patients experience less postoperative pain and have shorter recovery times. There is also a favorable cosmetic outcome with smaller scars. Several surgical procedures such as appendicectomy, cholecystectomy, and hernia repair, as well as numerous orthopaedic, urological, and gynecological procedures, are now commonly performed laparoscopically.

During the operation, a camera known as a laparoscope is used to transmit live, magnified images of the surgical field to a monitor viewed by the surgeon. The camera is inserted into the abdominal cavity through a small incision, called a port-site, usually at the umbilicus. In order to create enough space to operate, the abdominal wall is elevated by inflating the cavity with gas, typically carbon dioxide. Other long, thin surgical instruments are then introduced through additional port-sites, which can be used by the surgeon to operate. The placement of these port-sites is carefully planned to allow optimal access to the surgical field.

Laparoscopic surgery has been further enhanced with the use of surgical robots. Using these systems the surgeon moves the surgical instruments indirectly by controlling robotic arms, which are inserted into the operating field through small incisions. Robot-assisted surgery is now routinely used worldwide and has helped overcome some of the limitations of laparoscopy by enhancing the surgeon's dexterity. The robotic system is precise, provides the surgeon with a 3D view of the surgical field, and allows improved degree of rotation and manipulation of the surgical instruments. Several procedures such as prostatectomy and cholecystectomy can now be performed with this method.

Laparoendoscopic single-site surgery, also known as single-port laparoscopy, is the most recent advance in laparoscopic surgery. This method uses a single incision, usually umbilical, to introduce a port with several operating channels and can be performed with or without robotic assistance. Benefits include less postoperative pain, a faster recovery time, and an even better cosmetic result than traditional laparoscopic surgery.

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ABDOMINAL WALL

The abdominal wall covers a large area. It is bounded superiorly by the xiphoid process and costal margins, posteriorly by the vertebral column, and inferiorly by the upper parts of the pelvic bones. Its layers consist of skin, superficial fascia (subcutaneous tissue), muscles and their associated deep fascias, extraperitoneal fascia, and parietal peritoneum (Fig. 4.24).

Superficial fascia

The superficial fascia of the abdominal wall (subcutaneous tissue of abdomen) is a layer of fatty connective tissue. It is usually a single layer similar to, and continuous with, the superficial fascia throughout other regions of the body. However, in the lower region of the anterior part of the abdominal wall, below the umbilicus, it forms two layers: a superficial fatty layer and a deeper membranous layer.

Superficial layer

The superficial fatty layer of superficial fascia (**Camper's fascia**) contains fat and varies in thickness (Figs. 4.25 and 4.26). It is continuous over the inguinal ligament with the superficial fascia of the thigh and with a similar layer in the perineum.

In men, this superficial layer continues over the penis and, after losing its fat and fusing with the deeper layer of superficial fascia, continues into the scrotum where it forms a specialized fascial layer containing smooth muscle fibers (the **dartos fascia**). In women, this superficial layer retains some fat and is a component of the labia majora.

Deeper layer

The deeper membranous layer of superficial fascia (**Scarpa's fascia**) is thin and membranous, and contains little or no fat (Fig. 4.25). Inferiorly, it continues into the thigh, but just below the inguinal ligament, it fuses with the deep fascia of the thigh (the **fascia lata**; Fig. 4.26). In the midline, it is firmly attached to the linea alba and the symphysis pubis. It continues into the anterior part of the perineum where it is firmly attached to the ischiopubic rami and to the posterior margin of the perineal membrane. Here, it is referred to as the **superficial perineal fascia** (**Colles' fascia**).

In men, the deeper membranous layer of superficial fascia blends with the superficial layer as they both pass over the penis, forming the superficial fascia of the penis, before they continue into the scrotum where they form the dartos fascia (Fig. 4.25). Also in men, extensions of the deeper membranous layer of superficial fascia attached to the pubic symphysis pass inferiorly onto the dorsum and

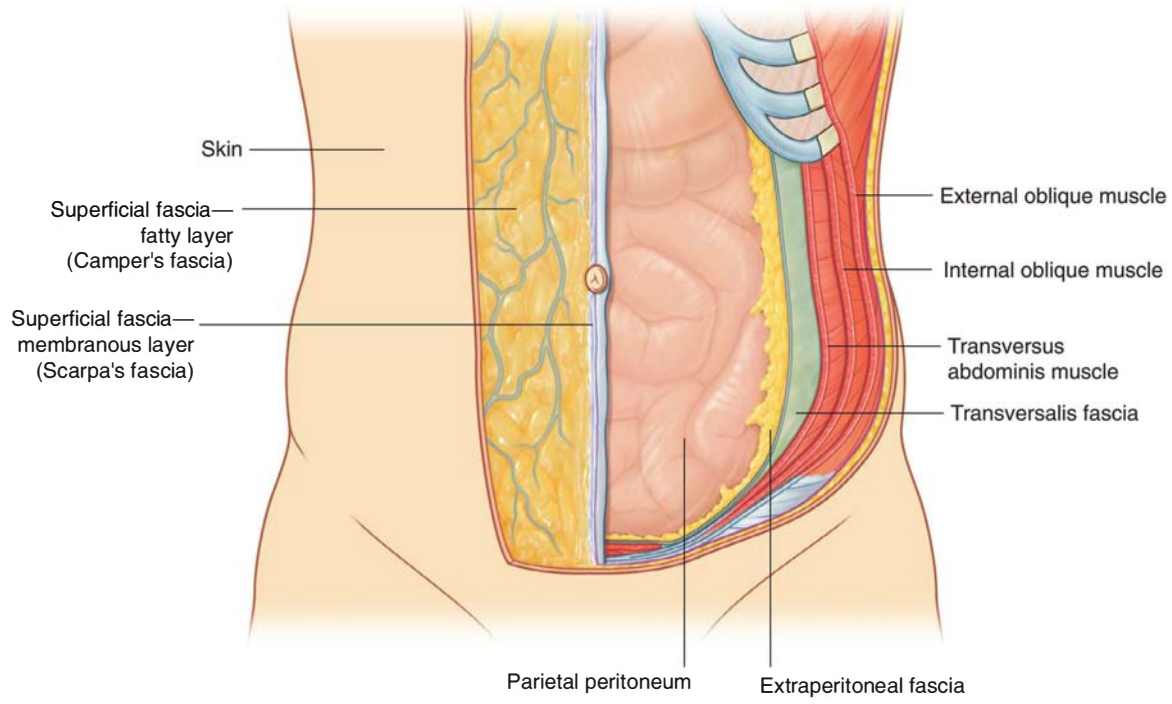


Fig. 4.24 Layers of the abdominal wall.

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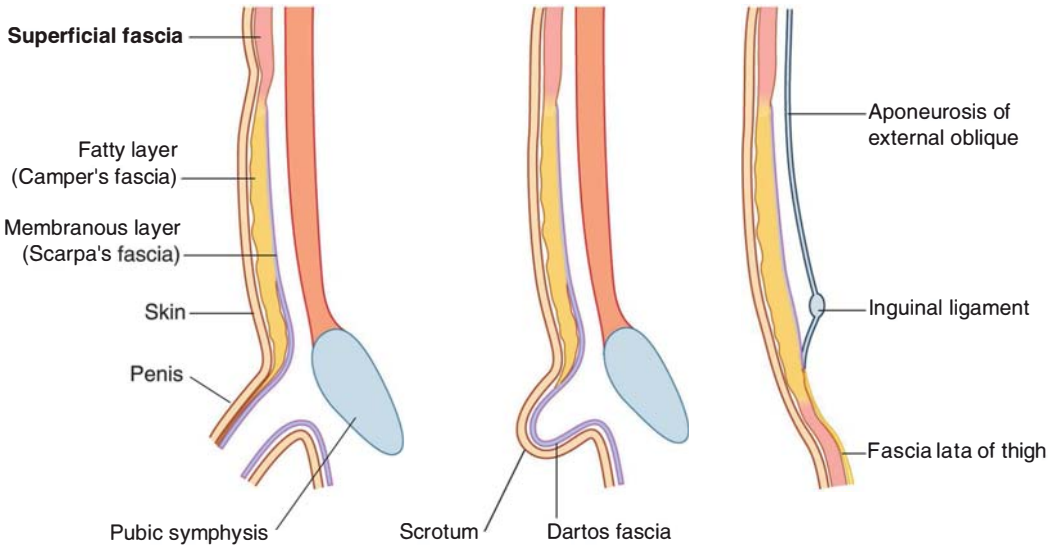


Fig. 4.25 Superficial fascia.

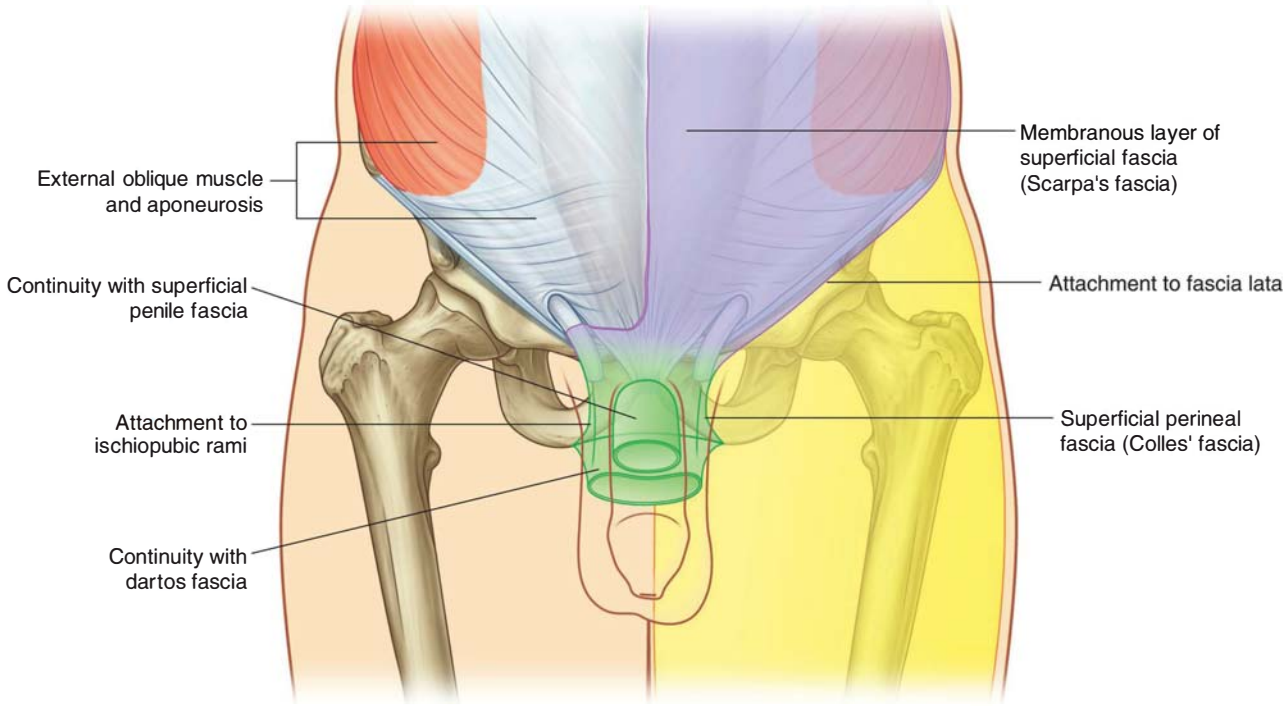


Fig. 4.26 Continuity of membranous layer of superficial fascia into other areas.

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sides of the penis to form the **fundiform ligament of penis**. In women, the membranous layer of the superficial fascia continues into the labia majora and the anterior part of the perineum.

Anterolateral muscles

There are five muscles in the anterolateral group of abdominal wall muscles:

- three flat muscles whose fibers begin posterolaterally, pass anteriorly, and are replaced by an aponeurosis as the muscle continues toward the midline—the external oblique, internal oblique, and transversus abdominis muscles;
- two vertical muscles, near the midline, which are enclosed within a tendinous sheath formed by the aponeuroses of the flat muscles—the rectus abdominis and pyramidalis muscles.

Each of these five muscles has specific actions, but together the muscles are critical for the maintenance of many normal physiological functions. By their positioning, they form a firm, but flexible, wall that keeps the abdominal viscera within the abdominal cavity, protects the viscera from injury, and helps maintain the position of the viscera in the erect posture against the action of gravity.

In addition, contraction of these muscles assists in both quiet and forced expiration by pushing the viscera upward (which helps push the relaxed diaphragm further into the thoracic cavity) and in coughing and vomiting.

All these muscles are also involved in any action that increases intraabdominal pressure, including parturition (childbirth), micturition (urination), and defecation (expulsion of feces from the rectum).

Flat muscles

External oblique

The most superficial of the three flat muscles in the anterolateral group of abdominal wall muscles is the **external oblique**, which is immediately deep to the superficial fascia (Fig. 4.27, Table 4.1). Its laterally placed muscle fibers pass in an inferomedial direction, while its large

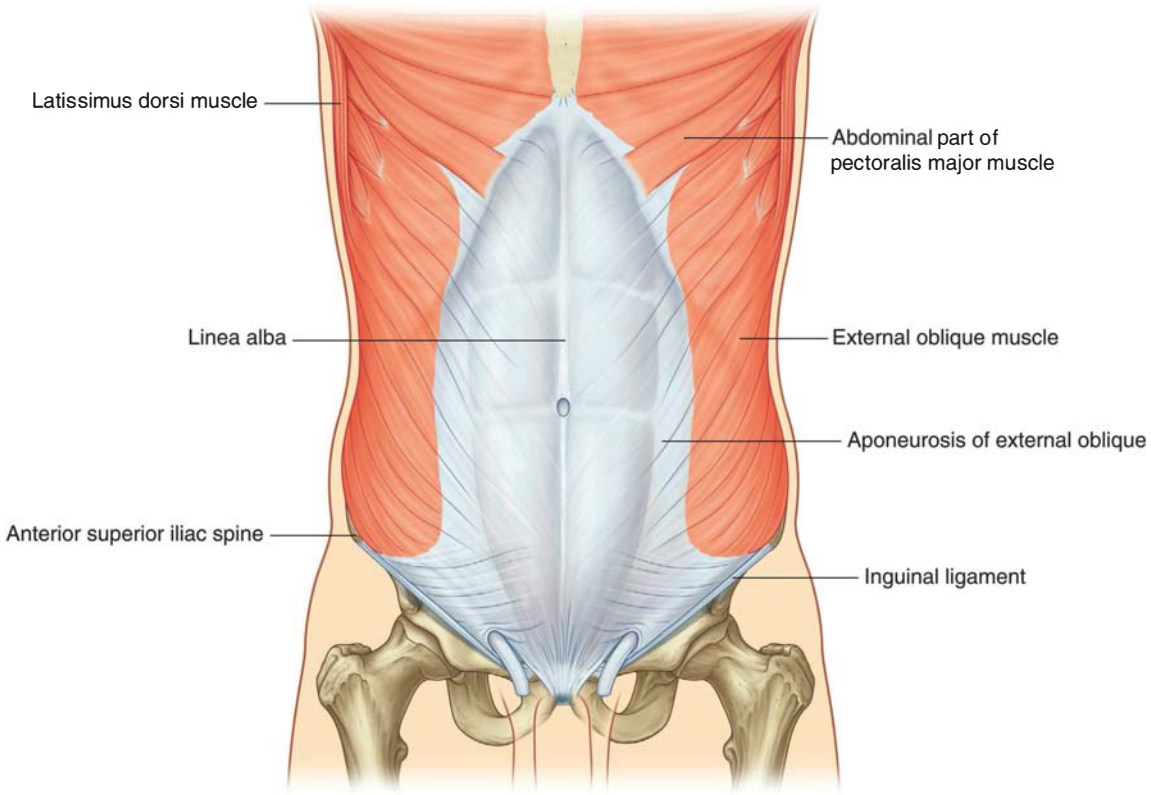


Fig. 4.27 External oblique muscle and its aponeurosis.

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aponeurotic component covers the anterior part of the abdominal wall to the midline. Approaching the midline, the aponeuroses are entwined, forming the linea alba, which extends from the xiphoid process to the pubic symphysis.

Associated ligaments

The lower border of the external oblique aponeurosis forms the **inguinal ligament** on each side (Fig. 4.27). This thickened reinforced free edge of the external oblique aponeurosis passes between the anterior superior iliac spine laterally and the pubic tubercle medially (Fig. 4.28). It folds under itself forming a trough, which plays an important role in the formation of the inguinal canal.

Several other ligaments are also formed from extensions of the fibers at the medial end of the inguinal ligament:

- The **lacunar ligament** is a crescent-shaped extension of fibers at the medial end of the inguinal ligament that pass backward to attach to the **pecten pubis** on the superior ramus of the pubic bone (Figs. 4.28 and 4.29).
- Additional fibers extend from the lacunar ligament along the pecten pubis of the pelvic brim to form the **pectineal** (**Cooper's**) **ligament**.

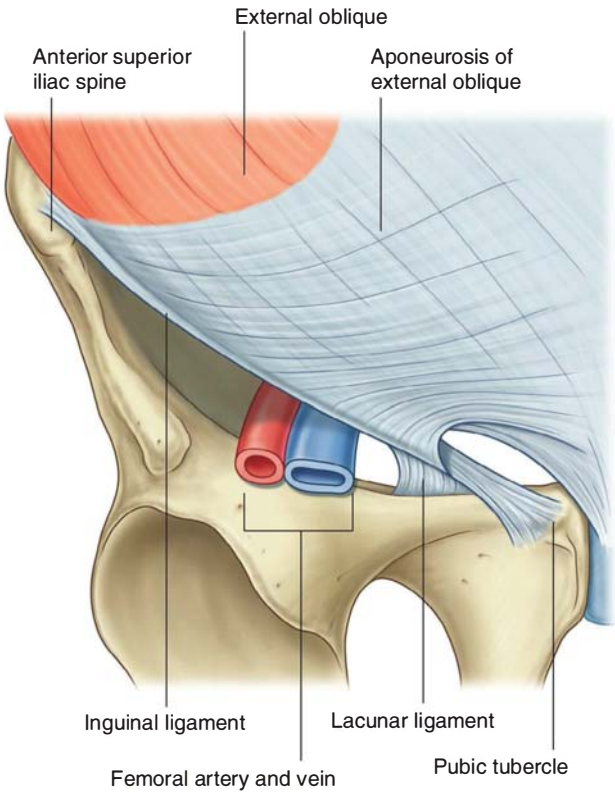


Fig. 4.28 Ligaments formed from the external oblique aponeurosis.

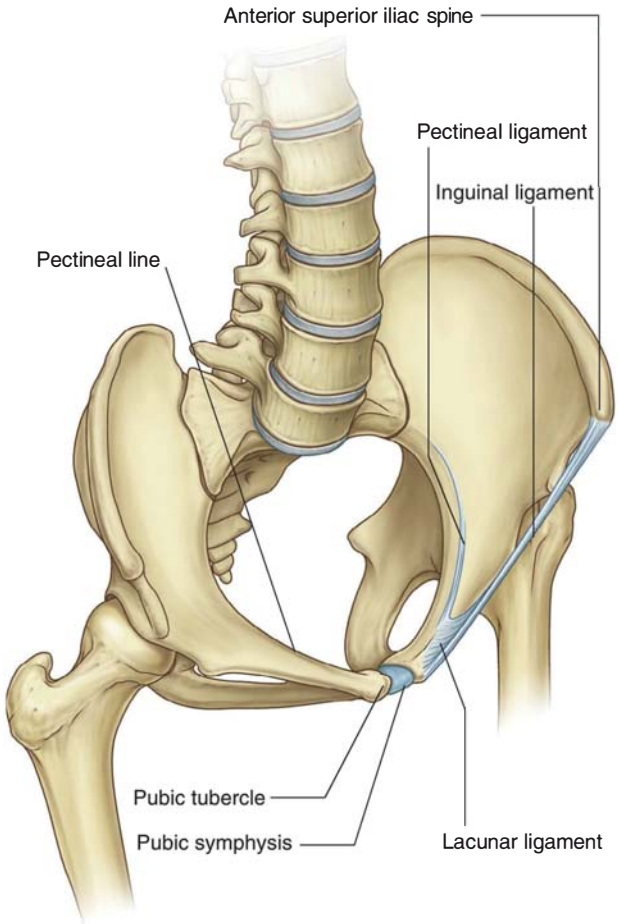


Fig. 4.29 Ligaments of the inguinal region.

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Internal oblique

Deep to the external oblique muscle is the **internal oblique** muscle, which is the second of the three flat muscles (Fig. 4.30, Table 4.1). This muscle is smaller and thinner than the external oblique, with most of its muscle fibers passing in a superomedial direction. Its lateral muscular components end anteriorly as an aponeurosis that blends into the linea alba at the midline.

Transversus abdominis

Deep to the internal oblique muscle is the **transversus abdominis** muscle (Fig. 4.31, Table 4.1), so named because of the direction of most of its muscle fibers. It ends in an anterior aponeurosis, which blends with the linea alba at the midline.

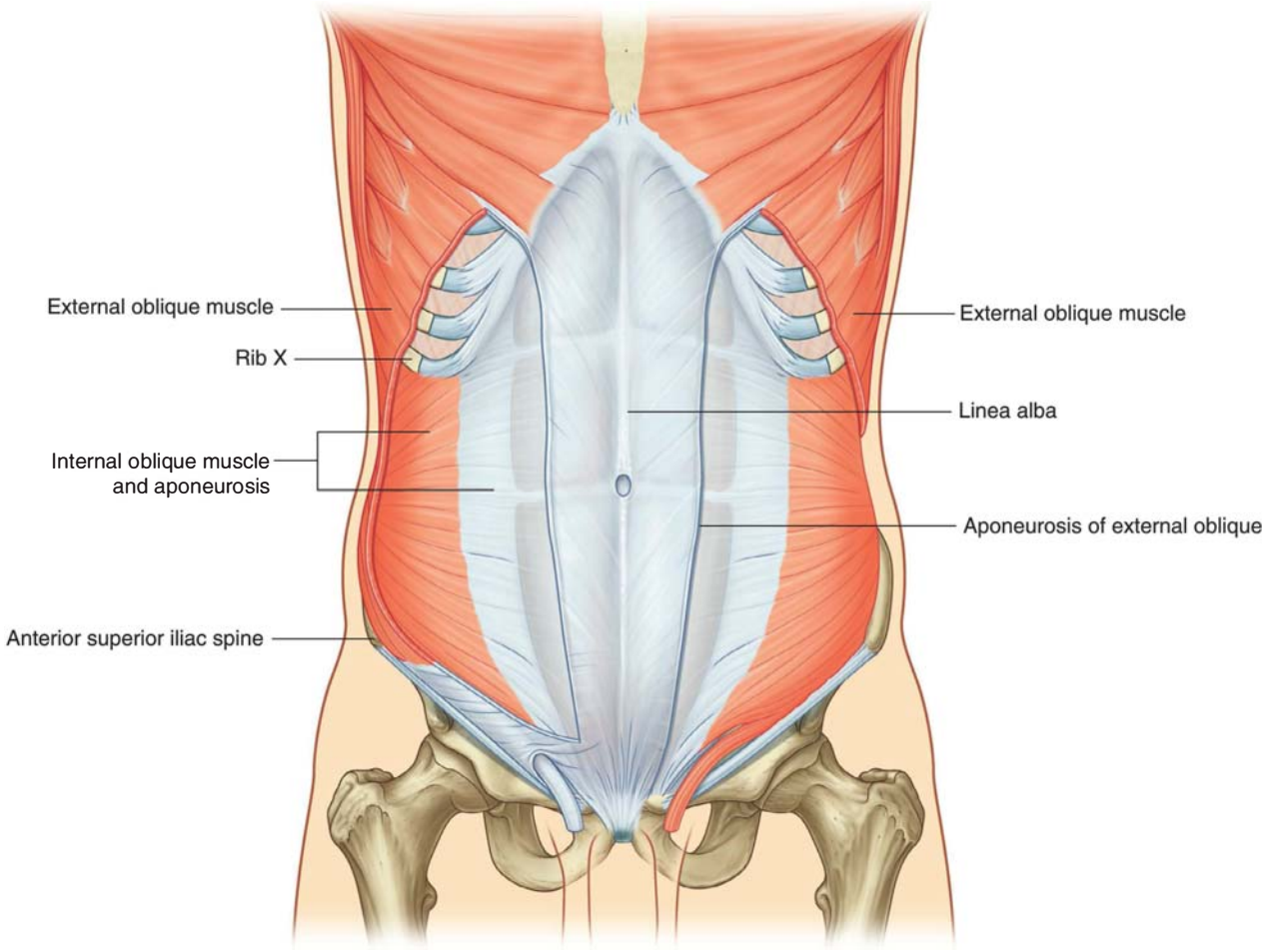


Fig. 4.30 Internal oblique muscle and its aponeurosis.

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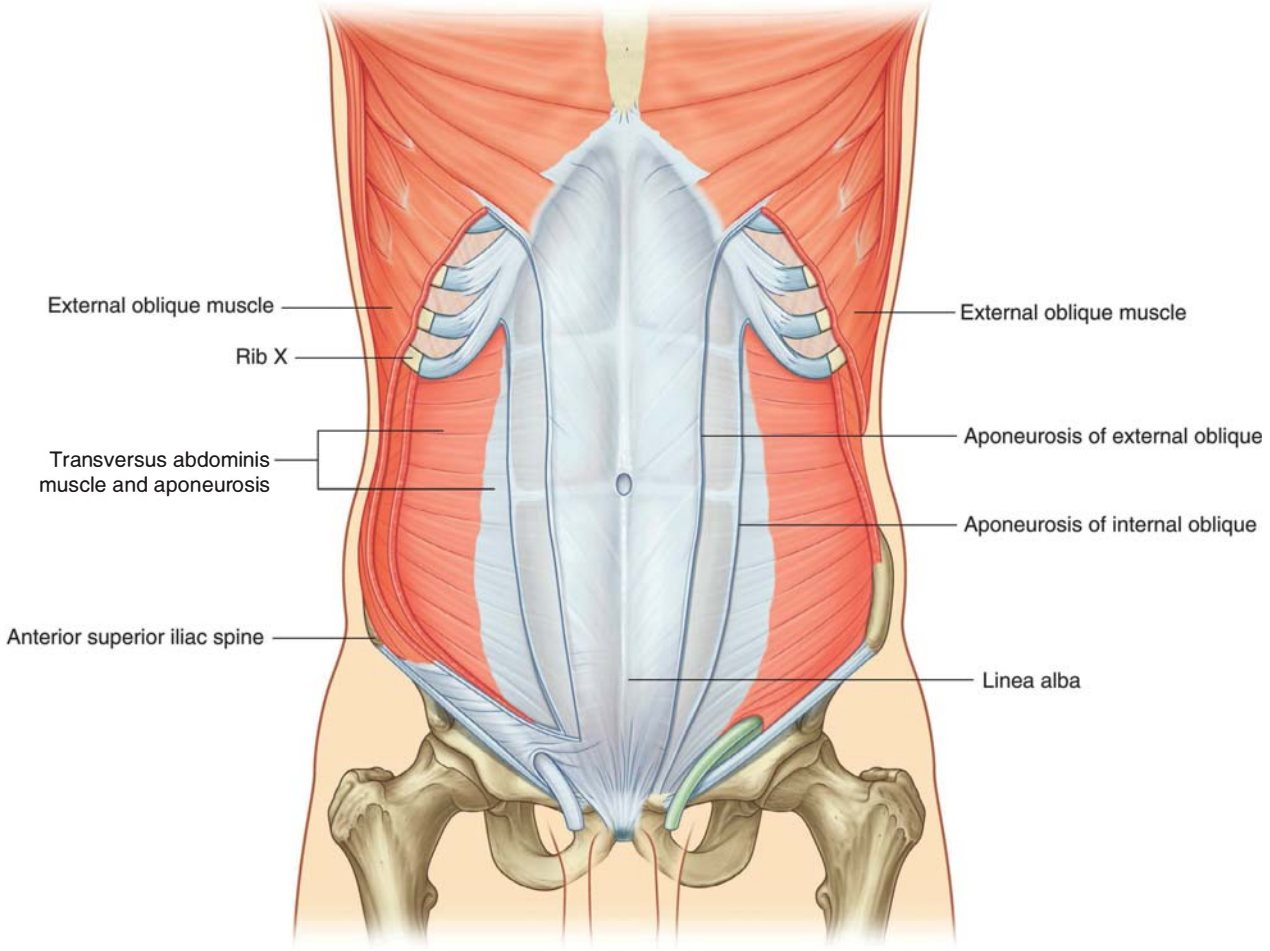


Fig. 4.31 Transversus abdominis muscle and its aponeurosis.

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Transversalis fascia

Each of the three flat muscles is covered on its anterior and posterior surfaces by a layer of deep (or investing) fascia. In general, these layers are unremarkable except for the layer deep to the transversus abdominis muscle (the **transversalis fascia**), which is better developed.

The transversalis fascia is a continuous layer of deep fascia that lines the abdominal cavity and continues into the pelvic cavity. It crosses the midline anteriorly, associating with the transversalis fascia of the opposite side, and is continuous with the fascia on the inferior surface of the diaphragm. It is continuous posteriorly with the deep fascia covering the muscles of the posterior abdominal wall and attaches to the thoracolumbar fascia.

After attaching to the crest of the ilium, the transversalis fascia blends with the fascia covering the muscles associated with the upper regions of the pelvic bones and with similar fascia covering the muscles of the pelvic cavity. At this point, it is referred to as the **parietal pelvic** (or **endopelvic**) **fascia**.

There is therefore a continuous layer of deep fascia surrounding the abdominal cavity that is thick in some areas, thin in others, attached or free, and participates in the formation of specialized structures.

Vertical muscles

The two vertical muscles in the anterolateral group of abdominal wall muscles are the large rectus abdominis and the small pyramidalis (Fig. 4.32, Table 4.1).

Rectus abdominis

The **rectus abdominis** is a long, flat muscle and extends the length of the anterior abdominal wall. It is a paired muscle, separated in the midline by the linea alba, and it

Table 4.1 Abdominal wall muscles

Muscle	Origin	Insertion	Innervation	Function
External oblique	Muscular slips from the outer surfaces of the lower eight ribs (ribs V to XII)	Lateral lip of iliac crest; aponeurosis ending in midline raphe (linea alba)	Anterior rami of lower six thoracic spinal nerves (T7 to T12)	Compress abdominal contents; both muscles flex trunk; each muscle bends trunk to same side, turning anterior part of abdomen to opposite side
Internal oblique	Thoracolumbar fascia; iliac crest between origins of external and transversus; lateral two-thirds of inguinal ligament	Inferior border of the lower three or four ribs; aponeurosis ending in linea alba; pubic crest and pectineal line	Anterior rami of lower six thoracic spinal nerves (T7 to T12) and L1	Compress abdominal contents; both muscles flex trunk; each muscle bends trunk and turns anterior part of abdomen to same side
Transversus abdominis	Thoracolumbar fascia; medial lip of iliac crest; lateral one-third of inguinal ligament; costal cartilages lower six ribs (ribs VII to XII)	Aponeurosis ending in linea alba; pubic crest and pectineal line	Anterior rami of lower six thoracic spinal nerves (T7 to T12) and L1	Compress abdominal contents
Rectus abdominis	Pubic crest, pubic tubercle, and pubic symphysis	Costal cartilages of ribs V to VII; xiphoid process	Anterior rami of lower seven thoracic spinal nerves (T7 to T12)	Compress abdominal contents; flex vertebral column; tense abdominal wall
Pyramidalis	Front of pubis and pubic symphysis	Into linea alba	Anterior ramus of T12	Tenses the linea alba

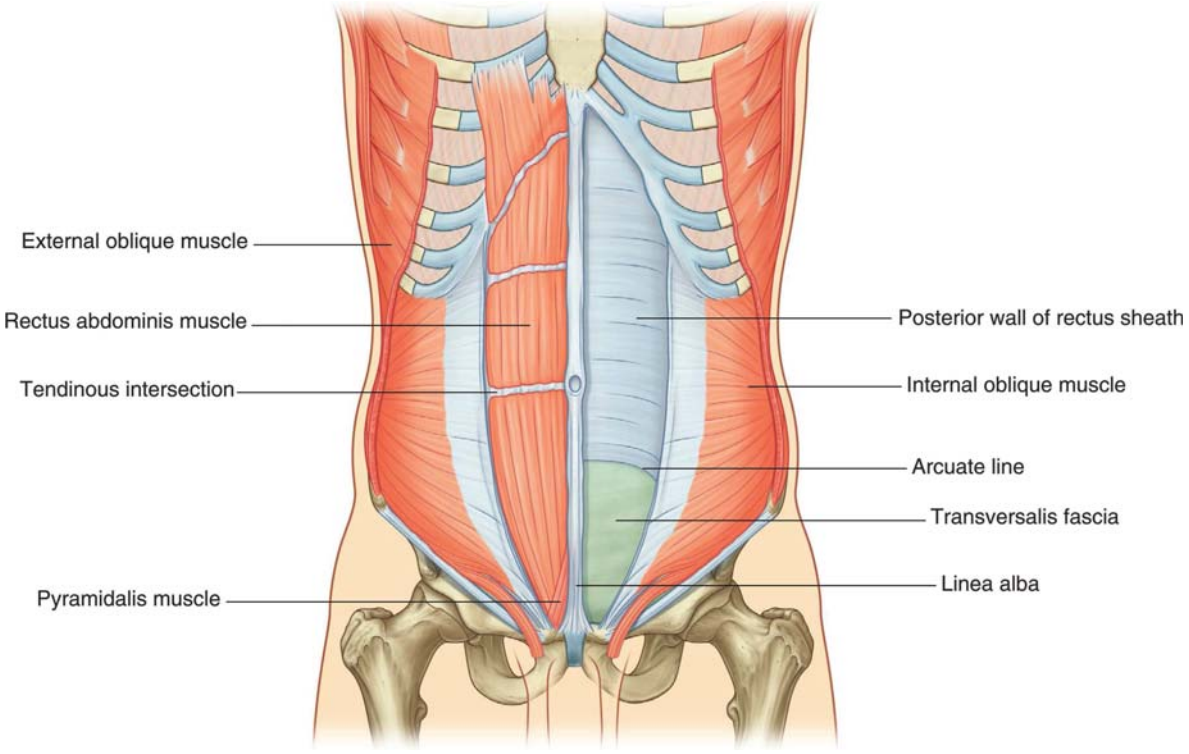


Fig. 4.32 Rectus abdominis and pyramidalis muscles.

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widens and thins as it ascends from the pubic symphysis to the costal margin. Along its course, it is intersected by three or four transverse fibrous bands or **tendinous intersections** (Fig. 4.32). These are easily visible on individuals with a well-developed rectus abdominis.

Pyramidalis

The second vertical muscle is the **pyramidalis**. This small, triangular muscle, which may be absent, is anterior to the rectus abdominis and has its base on the pubis, and its apex is attached superiorly and medially to the linea alba (Fig. 4.32).

Rectus sheath

The rectus abdominis and pyramidalis muscles are enclosed in an aponeurotic tendinous sheath (the **rectus sheath**) formed by a unique layering of the aponeuroses of the external and internal oblique, and transversus abdominis muscles (Fig. 4.33).

The rectus sheath completely encloses the upper three-quarters of the rectus abdominis and covers the anterior surface of the lower one-quarter of the muscle. As no sheath covers the posterior surface of the lower quarter of the rectus abdominis muscle, the muscle at this point is in direct contact with the transversalis fascia.

The formation of the rectus sheath surrounding the upper three-quarters of the rectus abdominis muscle has the following pattern:

- The anterior wall consists of the aponeurosis of the external oblique and half of the aponeurosis of the internal oblique, which splits at the lateral margin of the rectus abdominis.
- The posterior wall of the rectus sheath consists of the other half of the aponeurosis of the internal oblique and the aponeurosis of the transversus abdominis.

At a point midway between the umbilicus and the pubic symphysis, corresponding to the beginning of the lower one-quarter of the rectus abdominis muscle, all of the aponeuroses move anterior to the rectus muscle. There is no posterior wall of the rectus sheath and the anterior wall of the sheath consists of the aponeuroses of the external oblique, the internal oblique, and the transversus abdominis muscles. From this point inferiorly, the rectus abdominis muscle is in direct contact with the transversalis fascia. Marking this point of transition is an arch of fibers (the **arcuate line**; see Fig. 4.32).

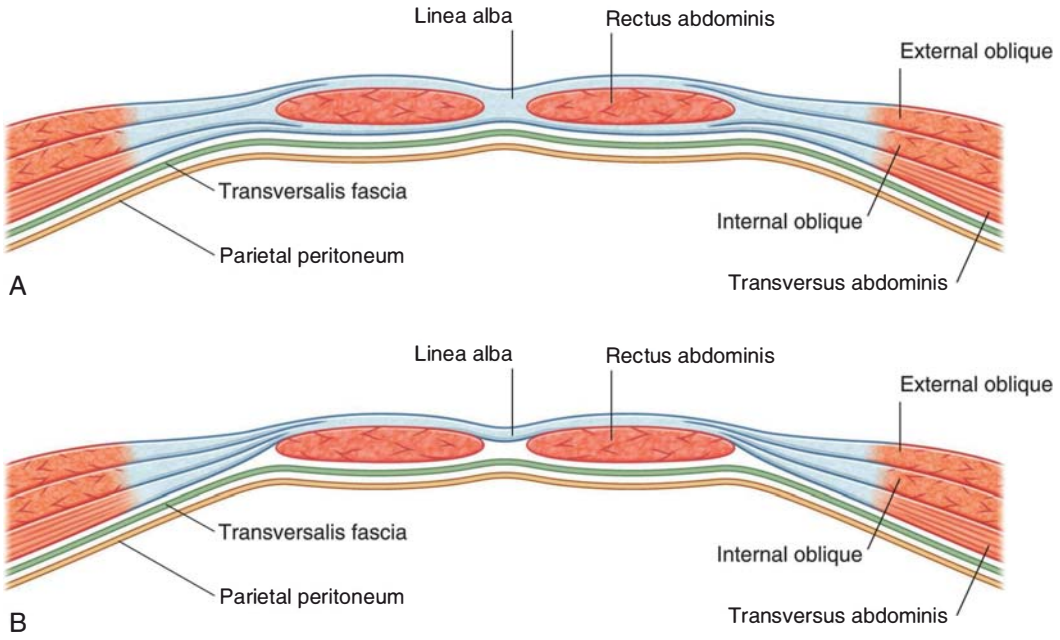


Fig. 4.33 Organization of the rectus sheath. **A.** Transverse section through the upper three-quarters of the rectus sheath. **B.** Transverse section through the lower one-quarter of the rectus sheath.

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Extraperitoneal fascia

Deep to the transversalis fascia is a layer of connective tissue, the **extraperitoneal fascia**, which separates the transversalis fascia from the peritoneum (Fig. 4.34). Containing varying amounts of fat, this layer not only lines the abdominal cavity but is also continuous with a similar layer lining the pelvic cavity. It is abundant on the posterior abdominal wall, especially around the kidneys, continues over organs covered by peritoneal reflections, and, as the vasculature is located in this layer, extends into mesenteries with the blood vessels. Viscera in the extraperitoneal fascia are referred to as **retroperitoneal**.

In the description of specific surgical procedures, the terminology used to describe the extraperitoneal fascia is further modified. The fascia toward the anterior side of the body is described as preperitoneal (or, less commonly, properitoneal) and the fascia toward the posterior side of the body has been described as retroperitoneal (Fig. 4.35). Examples of the use of these terms would be the continuity of fat in the inguinal canal with the preperitoneal fat and a transabdominal preperitoneal laparoscopic repair of an inguinal hernia.

Peritoneum

Deep to the extraperitoneal fascia is the peritoneum (see Figs. 4.6 and 4.7 on pp. 260-261). This thin serous membrane lines the walls of the abdominal cavity and, at various points, reflects onto the abdominal viscera, providing either a complete or a partial covering. The peritoneum lining the walls is the parietal peritoneum; the peritoneum covering the viscera is the visceral peritoneum.

The continuous lining of the abdominal walls by the parietal peritoneum forms a sac. This sac is closed in men but has two openings in women where the uterine tubes provide a passage to the outside. The closed sac in men and the semiclosed sac in women is called the peritoneal cavity.

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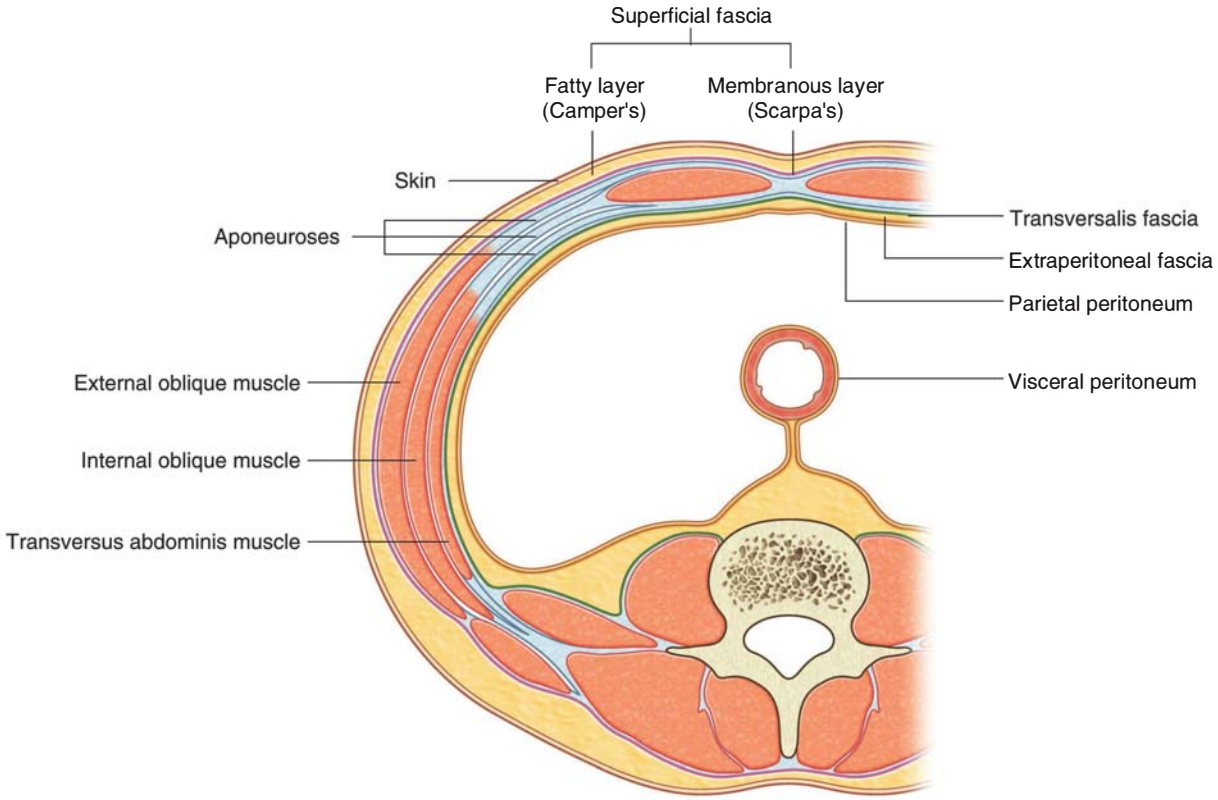


Fig. 4.34 Transverse section showing the layers of the abdominal wall.

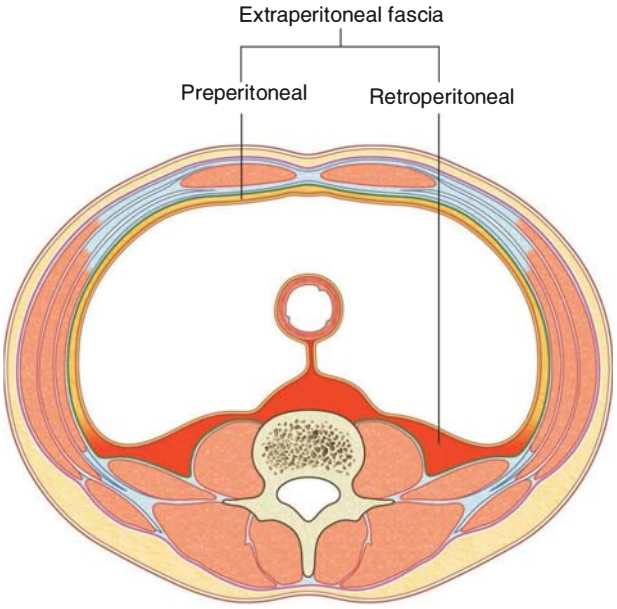


Fig. 4.35 Subdivisions of the extraperitoneal fascia.

Innervation

The skin, muscles, and parietal peritoneum of the anterolateral abdominal wall are supplied by T7 to T12 and L1 spinal nerves. The anterior rami of these spinal nerves pass around the body, from posterior to anterior, in an inferomedial direction (Fig. 4.36). As they proceed, they give off a lateral cutaneous branch and end as an anterior cutaneous branch.

The intercostal nerves (T7 to T11) leave their intercostal spaces, passing deep to the costal cartilages, and continue onto the anterolateral abdominal wall between the internal oblique and transversus abdominis muscles (Fig. 4.37). Reaching the lateral edge of the rectus sheath, they enter the rectus sheath and pass posterior to the lateral aspect of the rectus abdominis muscle. Approaching the midline, an anterior cutaneous branch passes through the rectus abdominis muscle and the anterior wall of the rectus sheath to supply the skin.

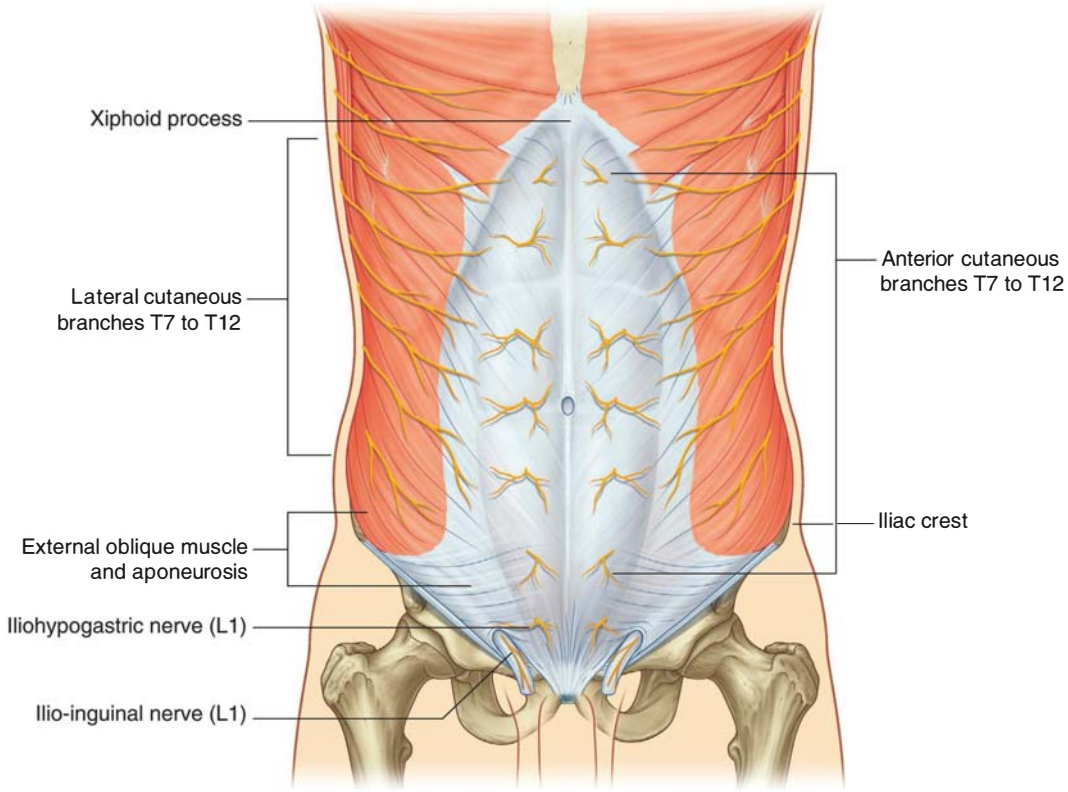


Fig. 4.36 Innervation of the anterolateral abdominal wall.

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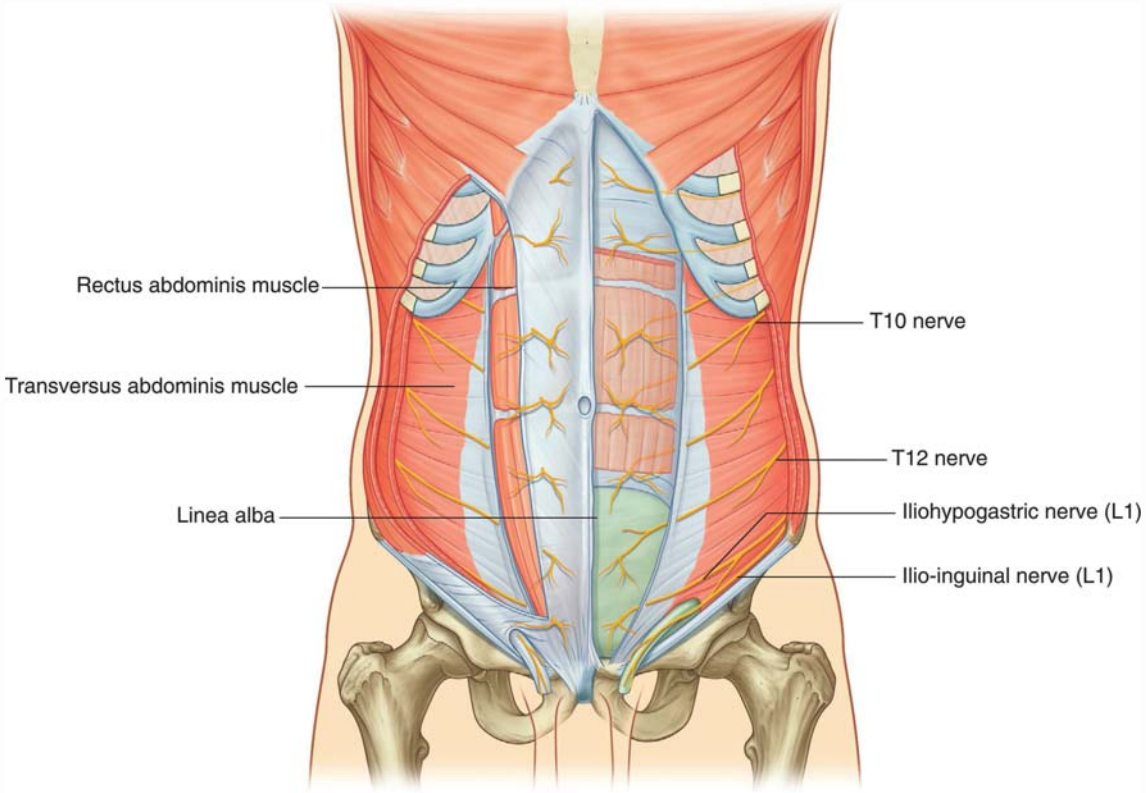


Fig. 4.37 Path taken by the nerves innervating the anterolateral abdominal wall.

Spinal nerve T12 (the **subcostal nerve**) follows a similar course as the intercostals. Branches of L1 (the **ilio-hypogastric nerve** and **ilio-inguinal nerve**), which originate from the lumbar plexus, follow similar courses initially, but deviate from this pattern near their final destination.

Along their course, nerves T7 to T12 and L1 supply branches to the anterolateral abdominal wall muscles and the underlying parietal peritoneum. All terminate by supplying skin:

- Nerves T7 to T9 supply the skin from the xiphoid process to just above the umbilicus.
- T10 supplies the skin around the umbilicus.
- T11, T12, and L1 supply the skin from just below the umbilicus to, and including, the pubic region (Fig. 4.38).
- Additionally, the ilio-inguinal nerve (a branch of L1) supplies the anterior surface of the scrotum or labia majora, and sends a small cutaneous branch to the thigh.

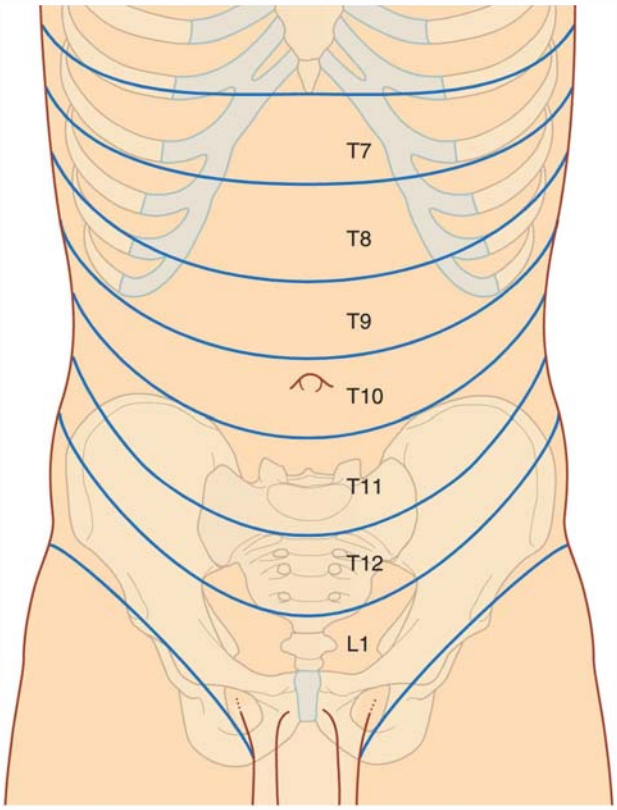


Fig. 4.38 Dermatomes of the anterolateral abdominal wall.

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Arterial supply and venous drainage

Numerous blood vessels supply the anterolateral abdominal wall. Superficially:

- the superior part of the wall is supplied by branches from the **musculophrenic artery**, a terminal branch of the **internal thoracic artery**, and
- the inferior part of the wall is supplied by the medially placed **superficial epigastric artery** and the laterally placed **superficial circumflex iliac artery**, both branches of the **femoral artery** (Fig. 4.39).

At a deeper level:

- the superior part of the wall is supplied by the **superior epigastric artery**, a terminal branch of the internal thoracic artery;
- the lateral part of the wall is supplied by branches of the **tenth** and **eleventh intercostal arteries** and the **subcostal artery**; and
- the inferior part of the wall is supplied by the medially placed **inferior epigastric artery** and the laterally placed **deep circumflex iliac artery**, both branches of the **external iliac artery**.

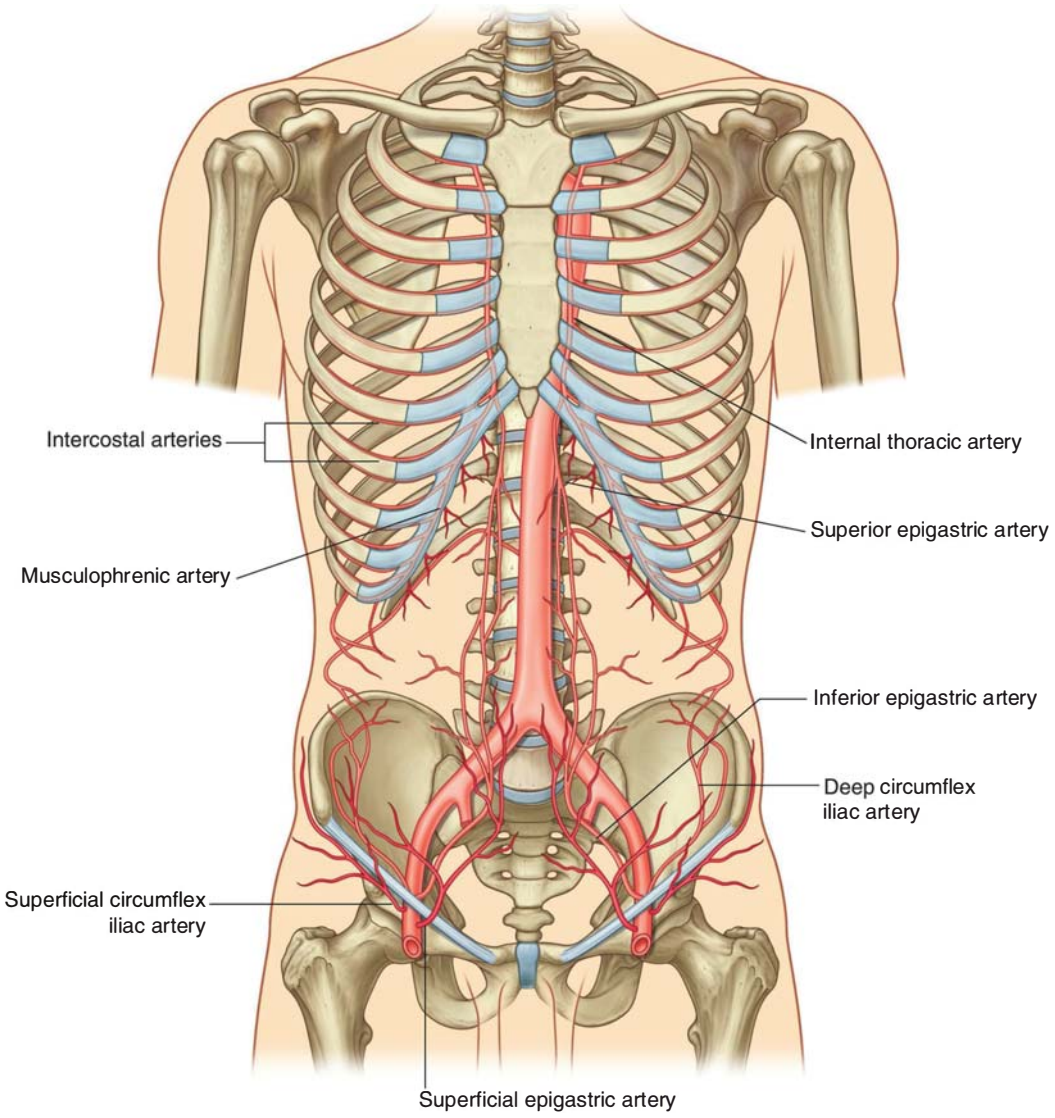


Fig. 4.39 Arterial supply to the anterolateral abdominal wall.

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The superior and inferior epigastric arteries both enter the rectus sheath. They are posterior to the rectus abdominis muscle throughout their course, and anastomose with each other (Fig. 4.40).

Veins of similar names follow the arteries and are responsible for venous drainage.

Lymphatic drainage

Lymphatic drainage of the anterolateral abdominal wall follows the basic principles of lymphatic drainage:

- Superficial lymphatics above the umbilicus pass in a superior direction to the **axillary nodes**, while drainage below the umbilicus passes in an inferior direction to the **superficial inguinal nodes.**
- Deep lymphatic drainage follows the deep arteries back to **parasternal nodes** along the internal thoracic artery, lumbar nodes along the abdominal aorta, and external iliac nodes along the external iliac artery.

GROIN

The groin (inguinal region) is the area of junction between the anterior abdominal wall and the thigh. In this area, the abdominal wall is weakened from changes that occur during development and a peritoneal sac or diverticulum, with or without abdominal contents, can therefore protrude through it, creating an inguinal hernia. This type of hernia can occur in both sexes, but it is most common in males.

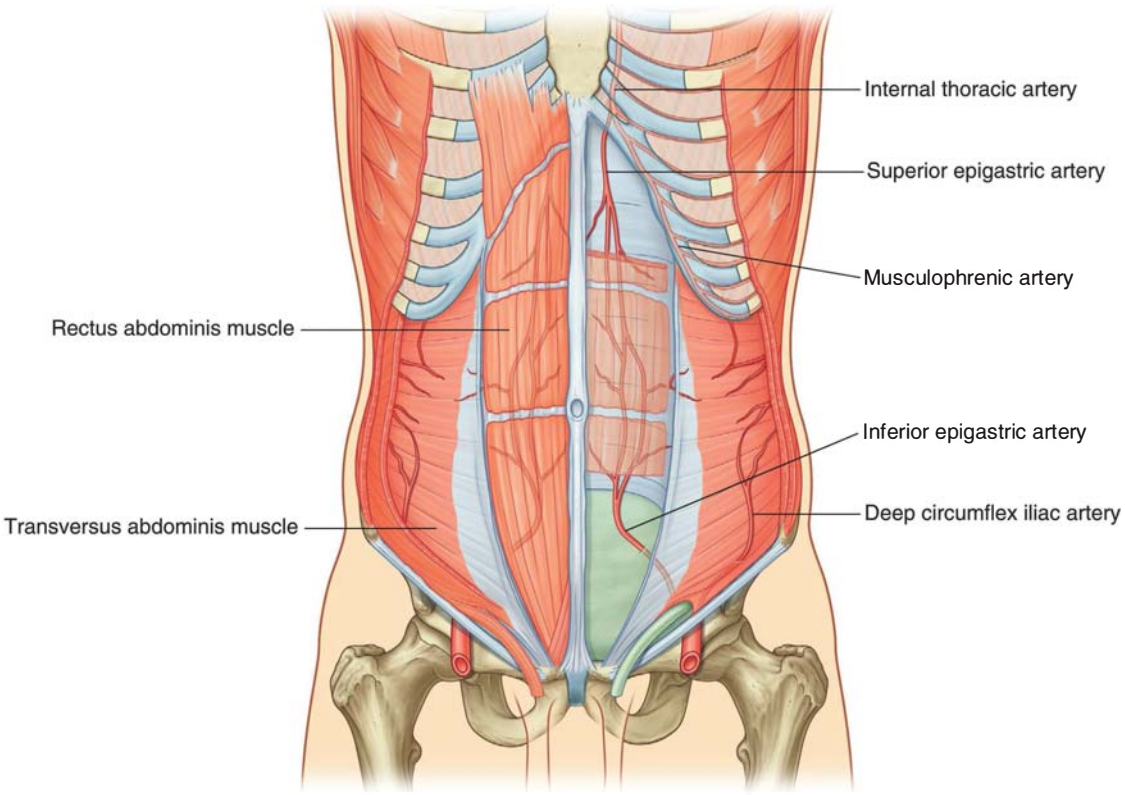


Fig. 4.40 Superior and inferior epigastric arteries.

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The inherent weakness in the anterior abdominal wall in the groin is caused by changes that occur during the development of the gonads. Before the descent of the testes and ovaries from their initial position high in the posterior abdominal wall, a peritoneal outpouching (the processus vaginalis) forms (Fig. 4.41), protruding through the various layers of the anterior abdominal wall and acquiring coverings from each:

- The transversalis fascia forms its deepest covering.
- The second covering is formed by the musculature of the internal oblique (a covering from the transversus abdominis muscle is not acquired because the processus vaginalis passes under the arching fibers of this abdominal wall muscle).
- Its most superficial covering is the aponeurosis of the external oblique.

As a result the processus vaginalis is transformed into a tubular structure with multiple coverings from the layers of the anterior abdominal wall. This forms the basic structure of the **inguinal canal**.

The final event in this development is the descent of the testes into the scrotum or of the ovaries into the pelvic cavity. This process depends on the development of the gubernaculum, which extends from the inferior border of the developing gonad to the labioscrotal swellings (Fig. 4.41).

The processus vaginalis is immediately anterior to the gubernaculum within the inguinal canal.

In men, as the testes descend, the testes and their accompanying vessels, ducts, and nerves pass through the inguinal canal and are therefore surrounded by the same fascial layers of the abdominal wall. Testicular descent completes the formation of the spermatic cord in men.

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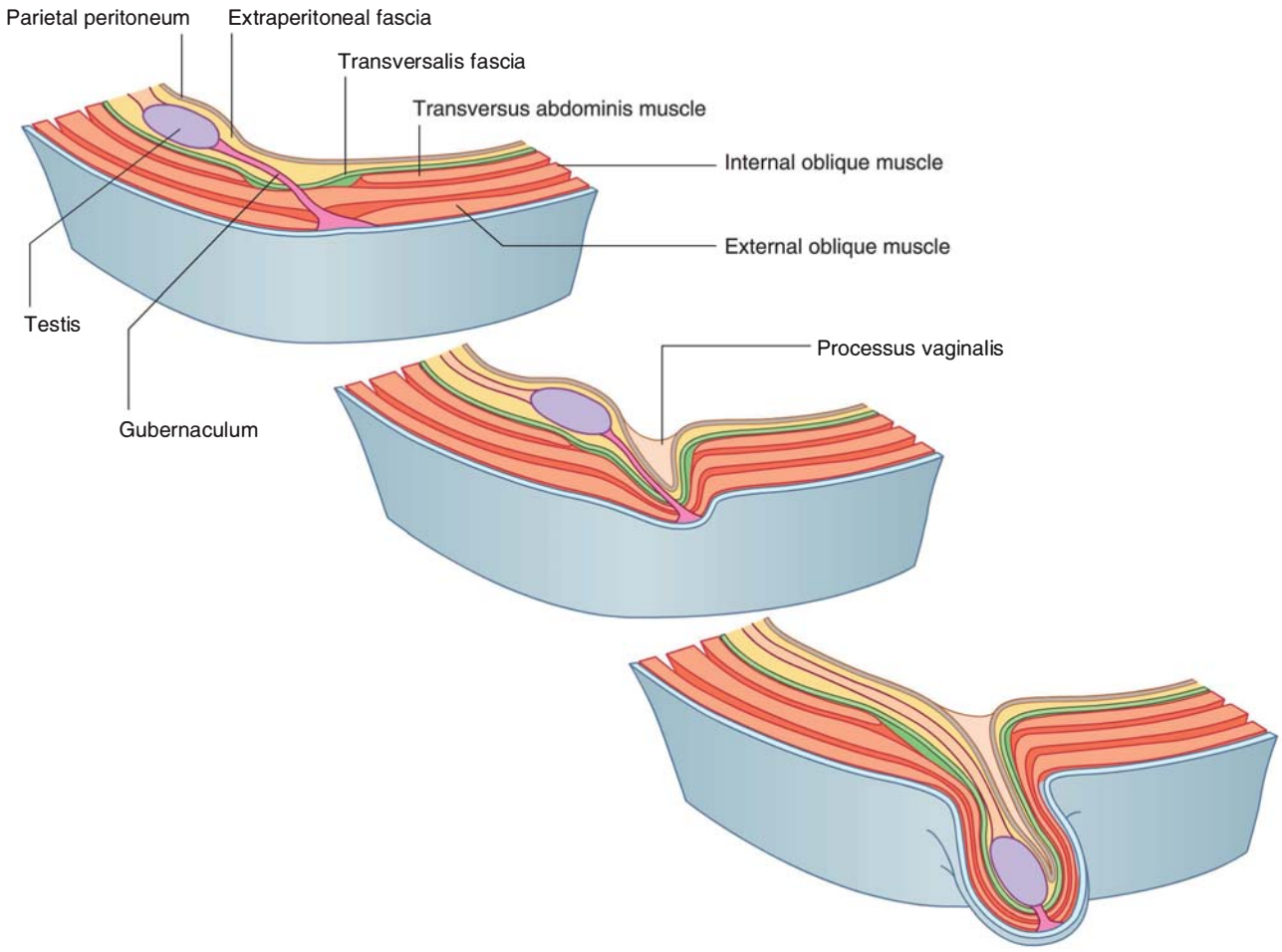


Fig. 4.41 Descent of the testis from week 7 (postfertilization) to birth.

In women, the ovaries descend into the pelvic cavity and become associated with the developing uterus. Therefore, the only remaining structure passing through the inguinal canal is the round ligament of the uterus, which is a remnant of the gubernaculum.

The development sequence is concluded in both sexes when the processus vaginalis obliterates. If this does not occur or is incomplete, a potential weakness exists in the anterior abdominal wall and an inguinal hernia may develop. In males, only proximal regions of the tunica vaginalis obliterate. The distal end expands to enclose most of the testis in the scrotum. In other words, the cavity of the tunica vaginalis in men forms as an extension of the developing peritoneal cavity that becomes separated off during development.

Inguinal canal

The inguinal canal is a slit-like passage that extends in a downward and medial direction, just above and parallel to the lower half of the inguinal ligament. It begins at the deep inguinal ring and continues for approximately 4 cm, ending at the superficial inguinal ring (Fig. 4.42). The contents of the canal are the genital branch of the genitofemoral nerve, the **spermatic cord** in men, and the round ligament of the uterus in women. Additionally, in both sexes, the ilio-inguinal nerve passes through part of the canal, exiting through the superficial inguinal ring with the other contents.

Deep inguinal ring

The deep (internal) inguinal ring is the beginning of the inguinal canal and is at a point midway between the anterior superior iliac spine and the pubic symphysis (Fig. 4.43). It is just above the inguinal ligament and immediately lateral to the inferior epigastric vessels. Although sometimes referred to as a defect or opening in the transversalis fascia, it is actually the beginning of the tubular evagination of transversalis fascia that forms one of the coverings (the **internal spermatic fascia**) of the spermatic cord in men or the round ligament of the uterus in women.

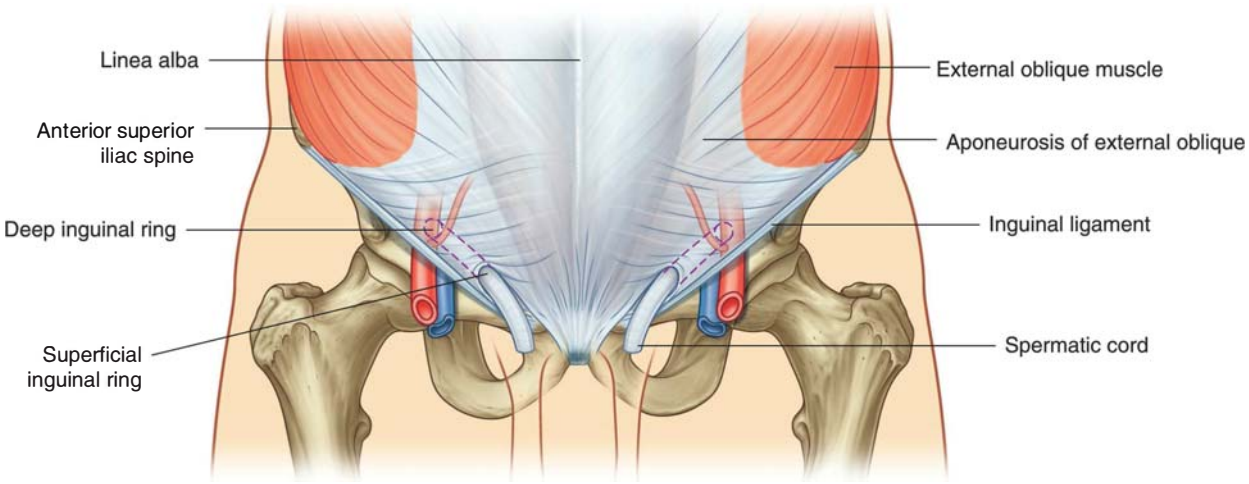


Fig. 4.42 Inguinal canal.

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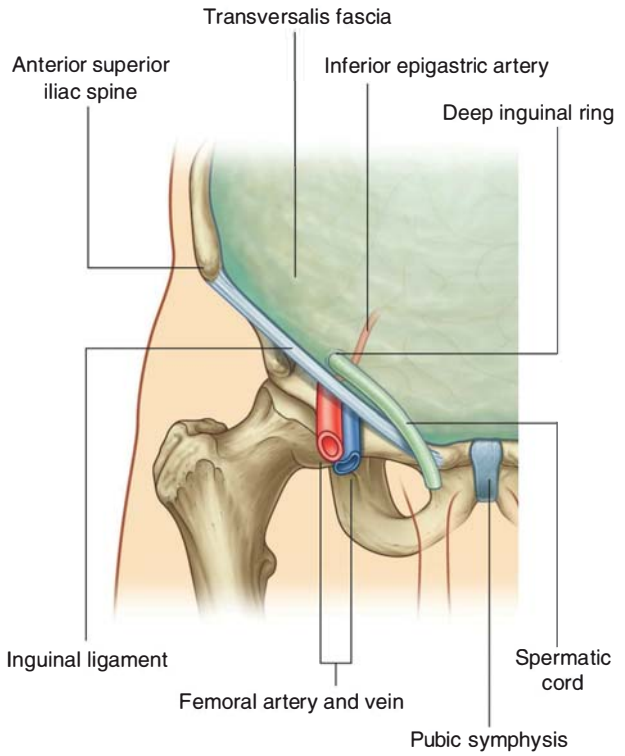


Fig. 4.43 Deep inguinal ring and the transversalis fascia.

Superficial inguinal ring

The superficial (external) inguinal ring is the end of the inguinal canal and is superior to the pubic tubercle (Fig. 4.44). It is a triangular opening in the aponeurosis of the external oblique, with its apex pointing superolaterally and its base formed by the pubic crest. The two remaining sides of the triangle (the **medial crus** and the **lateral crus**) are attached to the pubic symphysis and the pubic tubercle, respectively. At the apex of the triangle the two crura are held together by crossing (intercrural) fibers, which prevent further widening of the superficial ring.

As with the deep inguinal ring, the superficial inguinal ring is actually the beginning of the tubular evagination of the aponeurosis of the external oblique onto the structures traversing the inguinal canal and emerging from the superficial inguinal ring. This continuation of tissue over the spermatic cord is the **external spermatic fascia**.

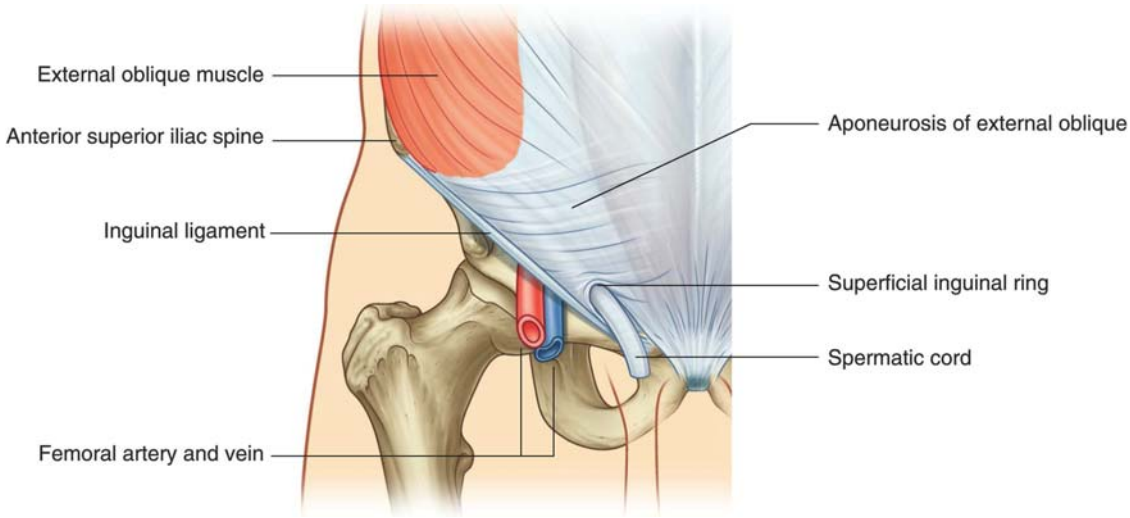


Fig. 4.44 Superficial inguinal ring and the aponeurosis of the external oblique.

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Anterior wall

The anterior wall of the inguinal canal is formed along its entire length by the aponeurosis of the external oblique muscle (Fig. 4.44). It is also reinforced laterally by the lower fibers of the internal oblique that originate from the lateral two-thirds of the inguinal ligament (Fig. 4.45). This adds an additional covering over the deep inguinal ring, which is a potential point of weakness in the anterior abdominal wall. Furthermore, as the internal oblique muscle covers the deep inguinal ring, it also contributes a layer (the **cremasteric fascia** containing the **cremasteric muscle**) to the coverings of the structures traversing the inguinal canal.

Posterior wall

The posterior wall of the inguinal canal is formed along its entire length by the transversalis fascia (see Fig. 4.43). It is reinforced along its medial one-third by the **conjoint tendon** (**inguinal falx**; Fig. 4.45). This tendon is the combined insertion of the transversus abdominis and internal oblique muscles into the pubic crest and pectineal line.

As with the internal oblique muscle's reinforcement of the area of the deep inguinal ring, the position of the conjoint tendon posterior to the superficial inguinal ring provides additional support to a potential point of weakness in the anterior abdominal wall.

Roof

The roof (superior wall) of the inguinal canal is formed by the arching fibers of the transversus abdominis and internal oblique muscles (Figs. 4.45 and 4.46). They pass from their lateral points of origin from the inguinal ligament to their common medial attachment as the conjoint tendon.

Floor

The floor (inferior wall) of the inguinal canal is formed by the medial one-half of the inguinal ligament. This rolled-under, free margin of the lowest part of the aponeurosis of the external oblique forms a gutter or trough on which the contents of the inguinal canal are positioned. The lacunar ligament reinforces most of the medial part of the gutter.

Contents

The contents of the inguinal canal are:

- the spermatic cord in men, and
- the round ligament of the uterus and genital branch of the genitofemoral nerve in women.

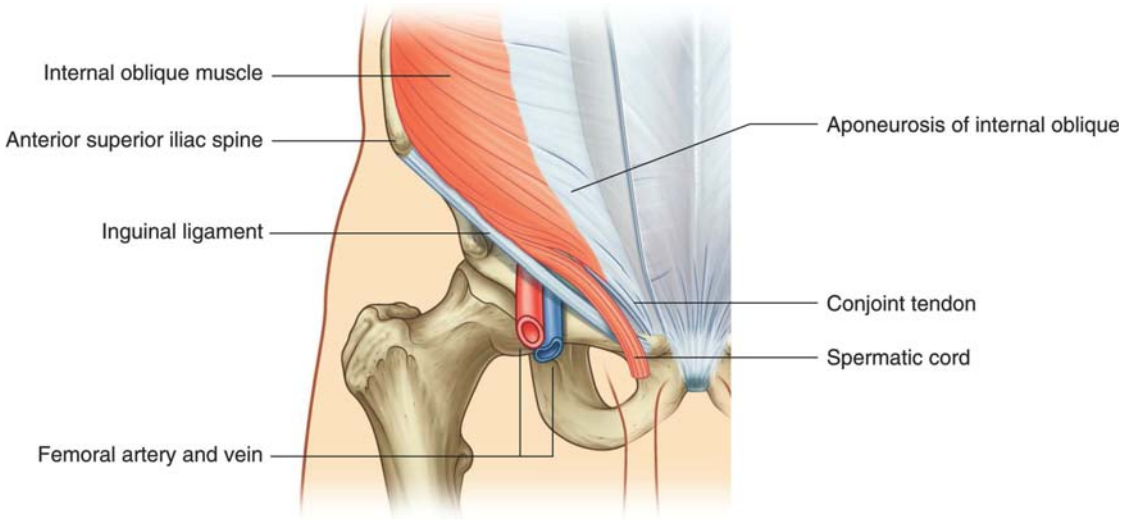


Fig. 4.45 Internal oblique muscle and the inguinal canal.

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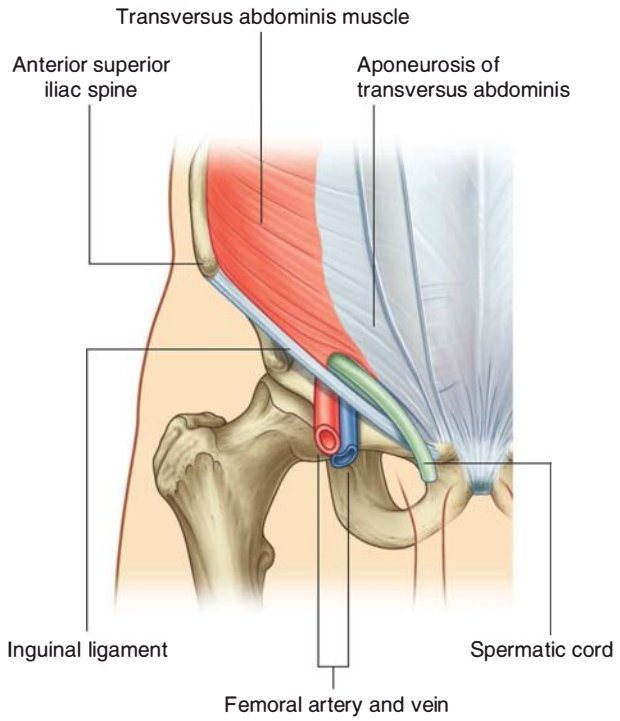


Fig. 4.46 Transversus abdominis muscle and the inguinal canal.

These structures enter the inguinal canal through the deep inguinal ring and exit it through the superficial inguinal ring.

Additionally, the ilio-inguinal nerve (L1) passes through part of the inguinal canal. This nerve is a branch of the lumbar plexus, enters the abdominal wall posteriorly by piercing the internal surface of the transversus abdominis muscle, and continues through the layers of the anterior abdominal wall by piercing the internal oblique muscle. As it continues to pass inferomedially, it enters the inguinal canal. It continues down the canal to exit through the superficial inguinal ring.

Spermatic cord

The spermatic cord begins to form proximally at the deep inguinal ring and consists of structures passing between the abdominopelvic cavities and the testis, and the three fascial coverings that enclose these structures (Fig. 4.47).

The structures in the spermatic cord include:

- the ductus deferens,
- the artery to the ductus deferens (from the inferior vesical artery),
- the testicular artery (from the abdominal aorta),
- the pampiniform plexus of veins (testicular veins),
- the cremasteric artery and vein (small vessels associated with the cremasteric fascia),
- the genital branch of the genitofemoral nerve (innervation to the cremasteric muscle),
- sympathetic and visceral afferent nerve fibers,
- lymphatics, and
- remnants of the processus vaginalis.

These structures enter the deep inguinal ring, proceed down the inguinal canal, and exit from the superficial inguinal ring, having acquired the three fascial coverings during their journey. This collection of structures and fascias continues into the scrotum where the structures connect with the testes and the fascias surround the testes.

Three fascias enclose the contents of the spermatic cord:

- The internal spermatic fascia, which is the deepest layer, arises from the transversalis fascia and is attached to the margins of the deep inguinal ring.
- The cremasteric fascia with the associated cremasteric muscle, which is the middle fascial layer, arises from the internal oblique muscle.
- The external spermatic fascia, which is the most superficial covering of the spermatic cord, arises from the aponeurosis of the external oblique muscle and is attached to the margins of the superficial inguinal ring (Fig. 4.47).

Round ligament of the uterus

The round ligament of the uterus is a cord-like structure that passes from the uterus to the deep inguinal ring where it enters the inguinal canal. It passes down the inguinal canal and exits through the superficial inguinal ring. At this point, it has changed from a cord-like structure to a few strands of tissue, which attach to the connective tissue associated with the labia majora. As it traverses the inguinal canal, it acquires the same coverings as the spermatic cord in men.

The round ligament of the uterus is the long distal part of the original gubernaculum in the fetus that extends from the ovary to the labioscrotal swellings. From its attachment to the uterus, the round ligament of the uterus continues to the ovary as the ligament of the ovary that develops from the short proximal end of the gubernaculum.

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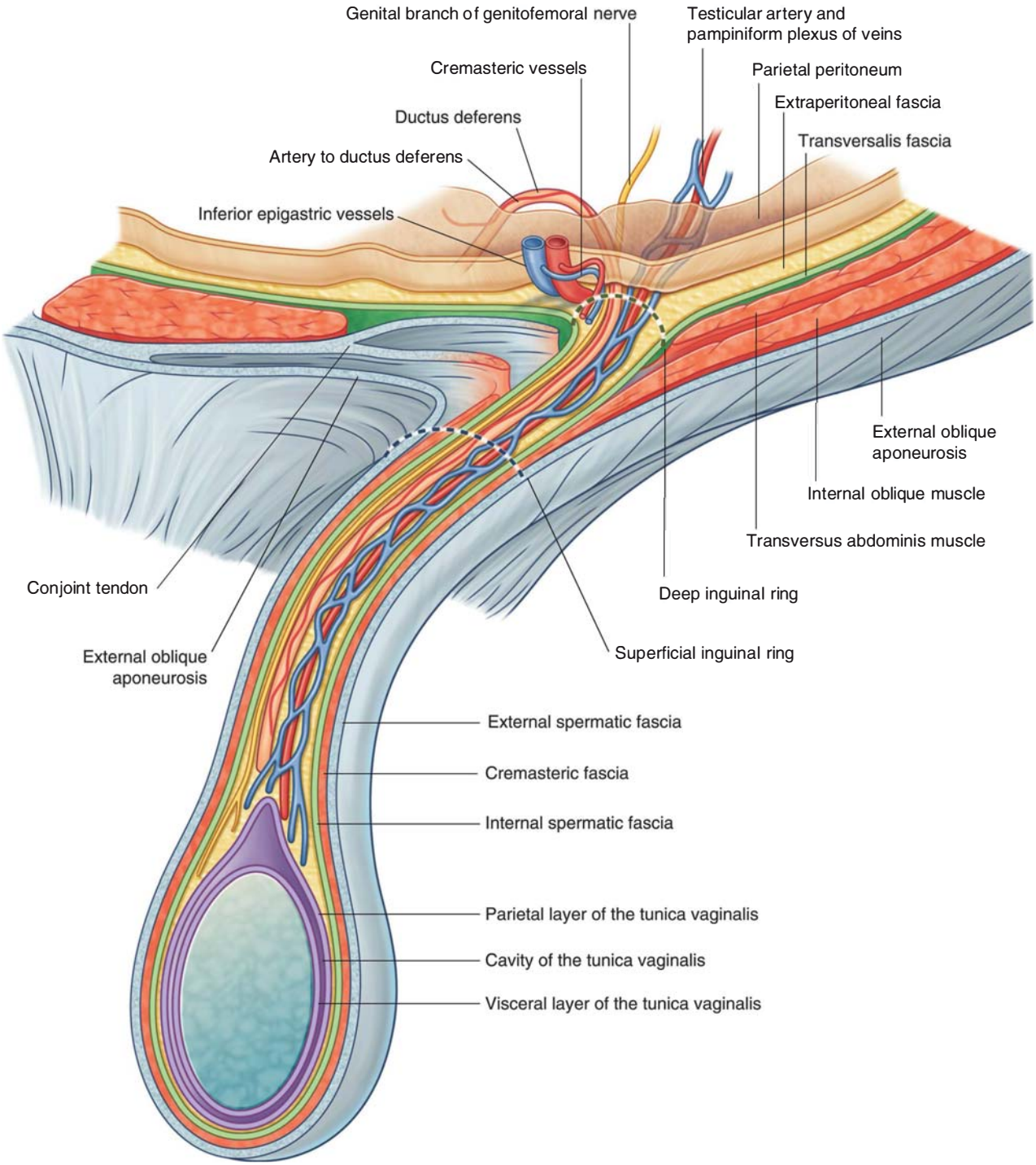


Fig. 4.47 Spermatic cord.

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In the clinic

Cremasteric reflex

In men, the cremaster muscle and cremasteric fascia form the middle or second covering of the spermatic cord. This muscle and its associated fascia are supplied by the genital branch of the genitofemoral nerve (L1/L2). Contraction of this muscle and the resulting elevation of the testis can be stimulated by a reflex arc. Gently touching the skin at and around the anterior aspect of the superior part of the thigh stimulates the sensory fibers in the ilio-inguinal nerve. These sensory fibers enter the spinal cord at level L1. At this level, the sensory fibers stimulate the motor fibers carried in the genital branch of the genitofemoral nerve, which results in contraction of the cremaster muscle and elevation of the testis.

The cremasteric reflex is more active in children, tending to diminish with age. As with many reflexes, it may be absent in certain neurological disorders. Although it can be used for testing spinal cord function at level L1 in men, its clinical use is limited.

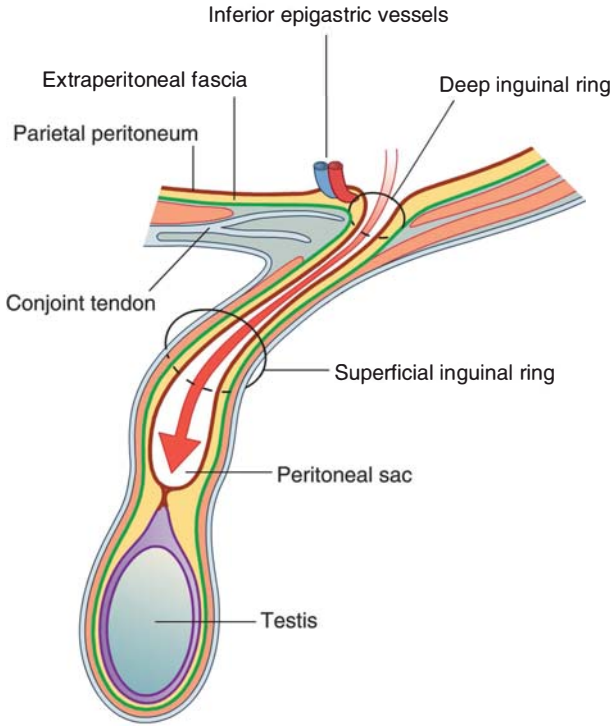


Fig. 4.48 Indirect inguinal hernia.

Inguinal hernias

An inguinal hernia is the protrusion or passage of a peritoneal sac, with or without abdominal contents, through a weakened part of the abdominal wall in the groin. It occurs because the peritoneal sac enters the inguinal canal either:

- indirectly, through the deep inguinal ring, or
- directly, through the posterior wall of the inguinal canal.

Inguinal hernias are therefore classified as either indirect or direct.

Indirect inguinal hernias

The indirect inguinal hernia is the most common of the two types of inguinal hernia and is much more common in men than in women (Fig. 4.48). It occurs because some part, or all, of the embryonic processus vaginalis remains open or patent. It is therefore referred to as being congenital in origin.

The protruding peritoneal sac enters the inguinal canal by passing through the deep inguinal ring, just lateral to the inferior epigastric vessels. The extent of its excursion down the inguinal canal depends on the amount of processus vaginalis that remains patent. If the entire processus vaginalis remains patent, the peritoneal sac may traverse the length of the canal, exit the superficial inguinal ring, and continue into the scrotum in men or the labia majus in women. In this case, the protruding peritoneal sac acquires the same three coverings as those associated with the spermatic cord in men or the round ligament of the uterus in women.

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Direct inguinal hernias

A peritoneal sac that enters the medial end of the inguinal canal directly through a weakened posterior wall is a direct inguinal hernia (Fig. 4.49). It is usually described as acquired because it develops when abdominal musculature has been weakened, and is commonly seen in mature men. The bulging occurs medial to the inferior epigastric vessels in the inguinal triangle (Hesselbach's triangle), which is bounded:

- laterally by the inferior epigastric artery,
- medially by the rectus abdominis muscle, and
- inferiorly by the inguinal ligament (Fig. 4.50).

Internally, a thickening of the transversalis fascia (the iliopubic tract) follows the course of the inguinal ligament (Fig. 4.50).

This type of inguinal hernia does not traverse the entire length of the inguinal canal but may exit through the superficial inguinal ring. When this occurs, the peritoneal sac acquires a layer of external spermatic fascia and can extend, like an indirect hernia, into the scrotum.

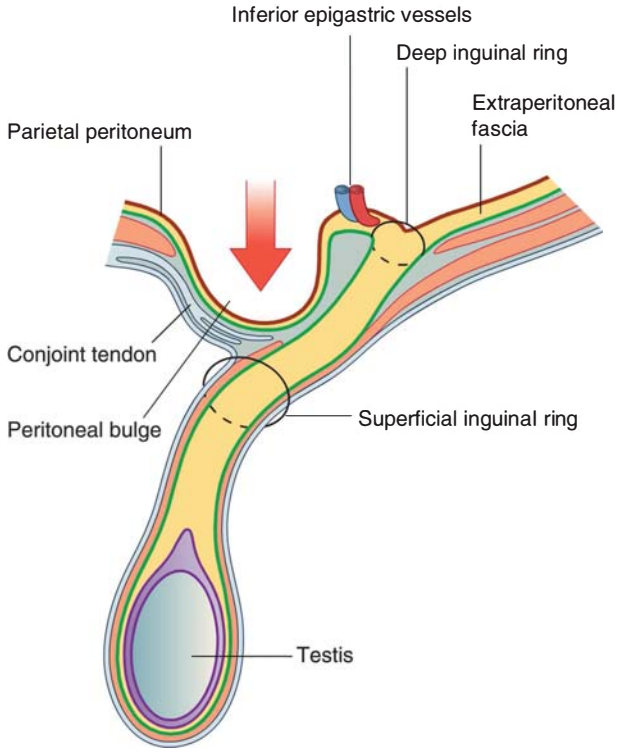


Fig. 4.49 Direct inguinal hernia.

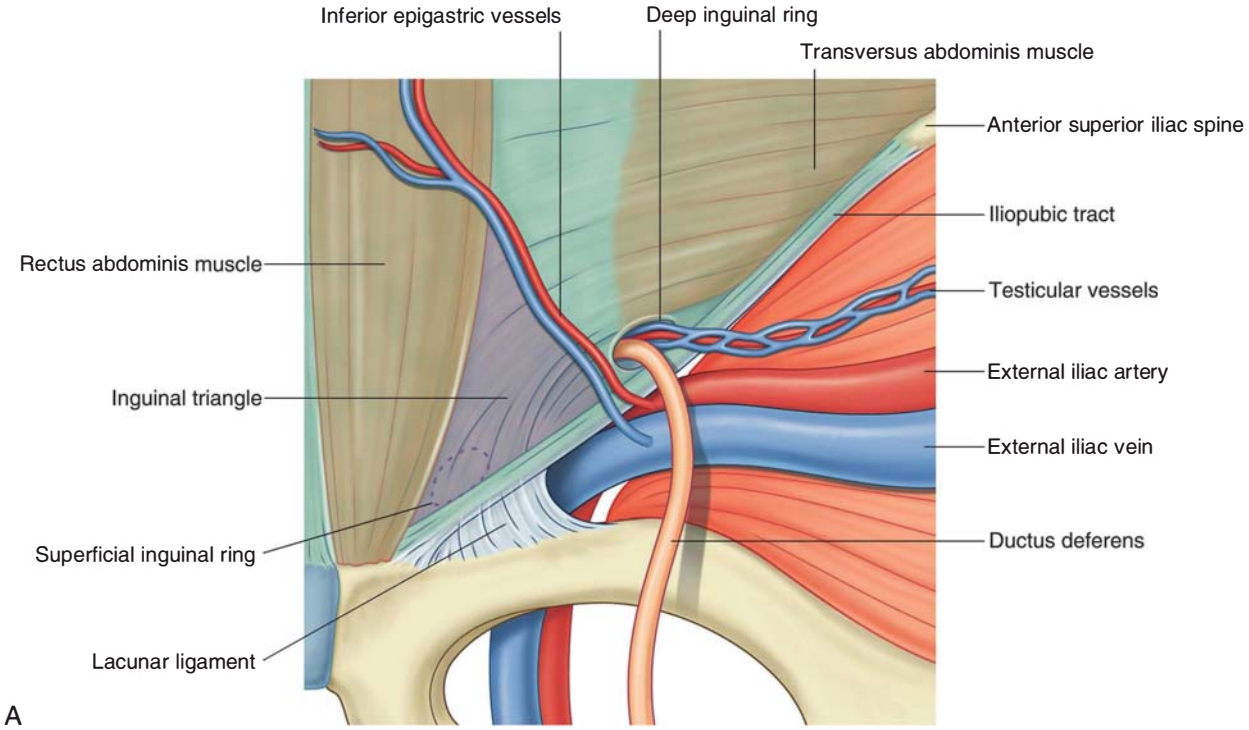


Fig. 4.50 Right inguinal triangle. **A.** Internal view.

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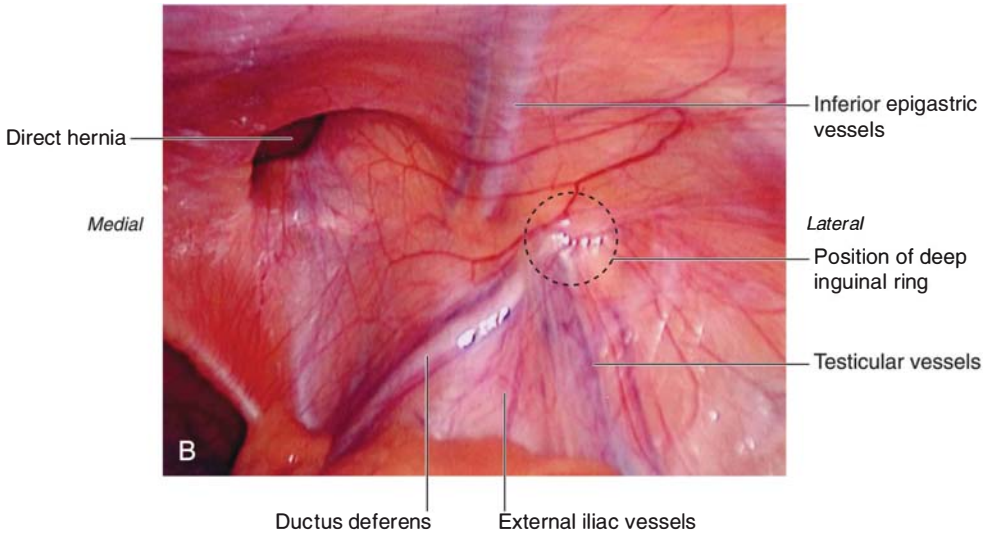


Fig. 4.50, cont'd B. Laparoscopic view showing the parietal peritoneum still covering the area.

In the clinic

Masses around the groin

Around the groin there is a complex confluence of anatomical structures. Careful examination and good anatomical knowledge allows determination of the correct anatomical structure from which the mass arises and therefore the diagnosis. The most common masses in the groin are hernias.

The key to groin examination is determining the position of the inguinal ligament. The inguinal ligament passes between the anterior superior iliac spine laterally and the pubic tubercle medially. Inguinal hernias are above the inguinal ligament and are usually more apparent on standing. A visual assessment of the lump is necessary, bearing in mind the anatomical landmarks of the inguinal ligament.

In men, it is wise to examine the scrotum to check for a lump. If an abnormal mass is present, an inability to feel its upper edge suggests that it may originate from the inguinal canal and might be a hernia. By placing the hand over the lump and asking the patient to cough, the lump bulges outward.

An attempt should be made to reduce the swelling by applying gentle, firm pressure over the lump. If the lump is reducible, the hand should be withdrawn and careful observation will reveal recurrence of the mass.

The position of an abnormal mass in the groin relative to the pubic tubercle is very important, as are the presence of increased temperature and pain, which may represent early signs of strangulation or infection.

As a general rule:

- An inguinal hernia appears through the superficial inguinal ring above the pubic tubercle and crest.
- A femoral hernia (see below) appears through the femoral canal below and lateral to the pubic tubercle.

A hernia is the protrusion of a viscus, in part or in whole, through a normal or abnormal opening. The viscus usually carries a covering of parietal peritoneum, which forms the lining of the hernial sac.

Inguinal hernias

Hernias occur in a variety of regions. The commonest site is the groin of the lower anterior abdominal wall. In some patients, inguinal hernias are present from birth (congenital) and are caused by the persistence of the processus vaginalis and the passage of viscera through the inguinal canal. Acquired hernias occur in older patients and causes include raised intraabdominal pressure (e.g., from repeated coughing associated with lung disease), damage to nerves of the anterior abdominal wall (e.g., from surgical abdominal incisions), and weakening of the walls of the inguinal canal.

One of the potential problems with hernias is that bowel and fat may become stuck within the hernial sac. This can cause appreciable pain and bowel obstruction, necessitating urgent surgery. Another potential risk is **strangulation** of the hernia, in which the blood supply to the bowel is cut off at the neck of the hernial sac, rendering the bowel ischemic and susceptible to perforation.

The hernial sac of an **indirect inguinal hernia** enters the deep inguinal ring and passes through the inguinal canal. If the hernia is large enough, the hernial sac may

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In the clinic—cont'd

emerge through the superficial inguinal ring. In men, such a hernia may extend into the scrotum (Fig. 4.51).

The hernial sac of a **direct inguinal hernia** pushes forward through the posterior wall of the inguinal canal immediately posterior to the superficial inguinal ring. The hernia protrudes directly forward medial to the inferior epigastric vessels and through the superficial inguinal ring.

The differentiation between an indirect and a direct inguinal hernia is made during surgery when the inferior epigastric vessels are identified at the medial edge of the deep internal ring:

- An indirect hernial sac passes lateral to the inferior epigastric vessels.
- A direct hernia is medial to the inferior epigastric vessels.

Inguinal hernias occur more commonly in men than in women possibly because men have a much larger inguinal canal than women.

Femoral hernias

A **femoral hernia** passes through the femoral canal and into the medial aspect of the anterior thigh. The femoral canal lies at the medial edge of the femoral sheath, which contains the femoral artery, femoral vein, and lymphatics. The neck of the femoral canal is extremely narrow and is prone to trapping bowel within the sac, so making this type of hernia irreducible and susceptible to bowel strangulation. Femoral hernias are usually acquired, are not congenital, and most commonly occur in middle-aged and elderly populations. In addition, because women generally have wider pelvises than men, they tend to occur more commonly in women.

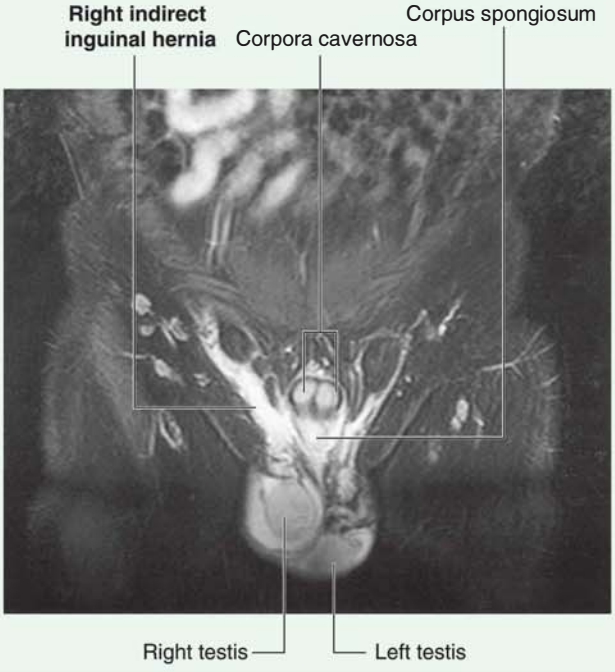


Fig. 4.51 Right indirect inguinal hernia. T2, fat saturated, weighted magnetic resonance image in the coronal plane of a male groin.

Sportsmen's groin/sportsmen's hernia

The groin can loosely be defined as the area where the leg meets the trunk near the midline. Here the abdominal muscles of the trunk blend in with the adductor muscles of the thigh, the medial end of the inguinal ligament attaches to the pubic tubercle, the pubic symphysis attaches the two pubic bones together, and the superficial (external) inguinal ring occurs. It also is in and around this region where there is considerable translation of force during most athletic and sporting activities. Pain in the groin or pubic region can be due to numerous causes, which include inflammatory changes at the pubic symphysis, insertional problems of the rectus abdominis/adductor longus, and hernias.

Umbilical hernias

Umbilical hernias are rare. Occasionally, they are congenital and result from failure of the small bowel to return to the abdominal cavity from the umbilical cord during development. After birth, umbilical hernias may result from incomplete closure of the umbilicus (navel). Overall, most of these hernias close in the first year of life, and surgical repair is not generally attempted until later.

Para-umbilical hernias may occur in adults at and around the umbilicus and often have small necks, so requiring surgical treatment.

Incisional hernias

Incisional hernias occur through a defect in a scar of a previous abdominal operation. Usually, the necks of these hernias are wide and do not therefore strangulate the viscera they contain.

Other hernias

A **spigelian hernia** passes upward through the arcuate line into the lateral border at the lower part of the posterior rectus sheath. It may appear as a tender mass on one side of the lower anterior abdominal wall.

Abdominopelvic cavity hernias can also develop in association with the pelvic walls, and sites include the obturator canal, the greater sciatic foramen and above and below the piriformis muscle.

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ABDOMINAL VISCERA

Peritoneum

A thin membrane (the peritoneum) lines the walls of the abdominal cavity and covers much of the viscera. The parietal peritoneum lines the walls of the cavity and the visceral peritoneum covers the viscera. Between the parietal and visceral layers of peritoneum is a potential space (the peritoneal cavity). Abdominal viscera either are suspended in the peritoneal cavity by folds of peritoneum (**mesenteries**) or are outside the peritoneal cavity. Organs suspended in the cavity are referred to as intraperitoneal (Fig. 4.52); organs outside the peritoneal cavity, with only one surface or part of one surface covered by peritoneum, are retroperitoneal.

Innervation of the peritoneum

The parietal peritoneum associated with the abdominal wall is innervated by somatic afferents carried in branches of the associated spinal nerves and is therefore sensitive to well-localized pain. The visceral peritoneum is innervated by visceral afferents that accompany autonomic nerves (sympathetic and parasympathetic) back to the central nervous system. Activation of these fibers can lead to referred and poorly localized sensations of discomfort, and to reflex visceral motor activity.

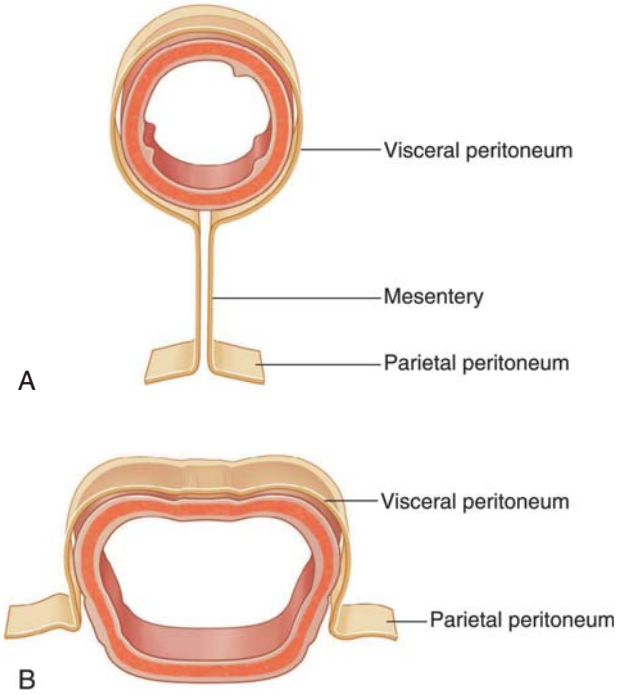


Fig. 4.52 **A.** Intraperitoneal. **B.** Retroperitoneal.

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Peritoneal cavity

The peritoneal cavity is subdivided into the greater sac and the omental bursa (lesser sac; Fig. 4.53).

- The greater sac accounts for most of the space in the peritoneal cavity, beginning superiorly at the diaphragm and continuing inferiorly into the pelvic cavity. It is entered once the parietal peritoneum has been penetrated.
- The omental bursa is a smaller subdivision of the peritoneal cavity posterior to the stomach and liver and is continuous with the greater sac through an opening, the omental (epiploic) foramen (Fig. 4.54).

Surrounding the omental (epiploic) foramen are numerous structures covered with peritoneum. They include the portal vein, hepatic artery proper, and bile duct anteriorly; the inferior vena cava posteriorly; the caudate lobe of the liver superiorly; and the first part of the duodenum inferiorly.

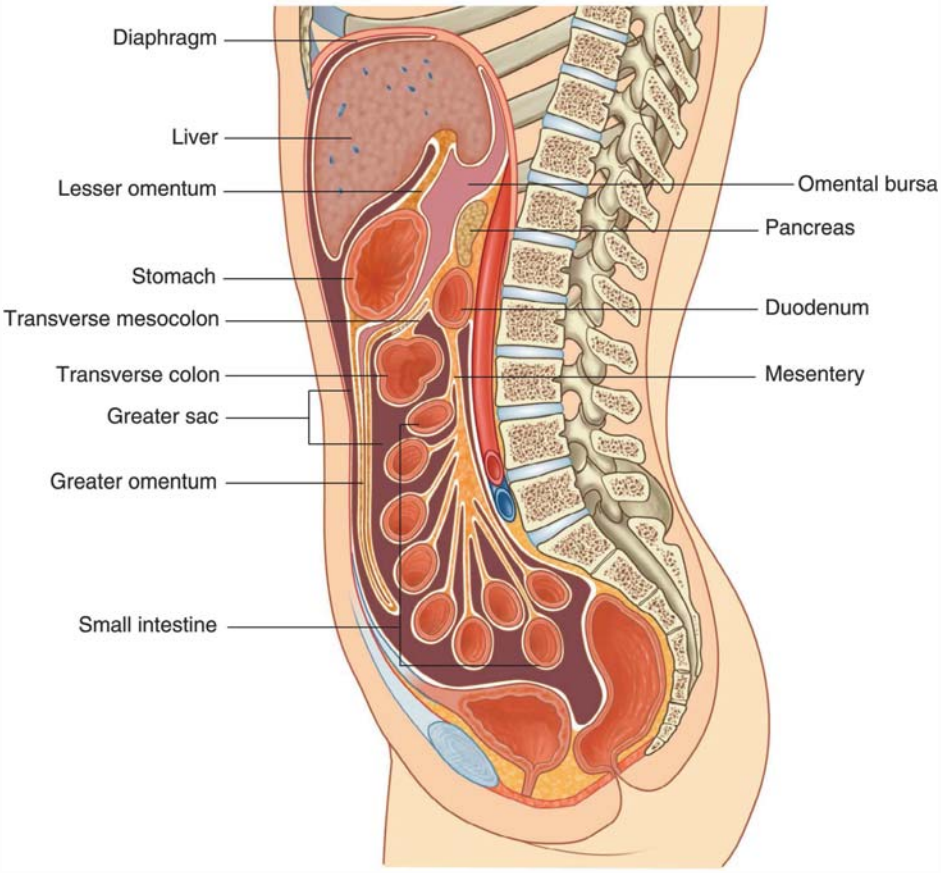


Fig. 4.53 Greater and lesser sacs of the peritoneal cavity.

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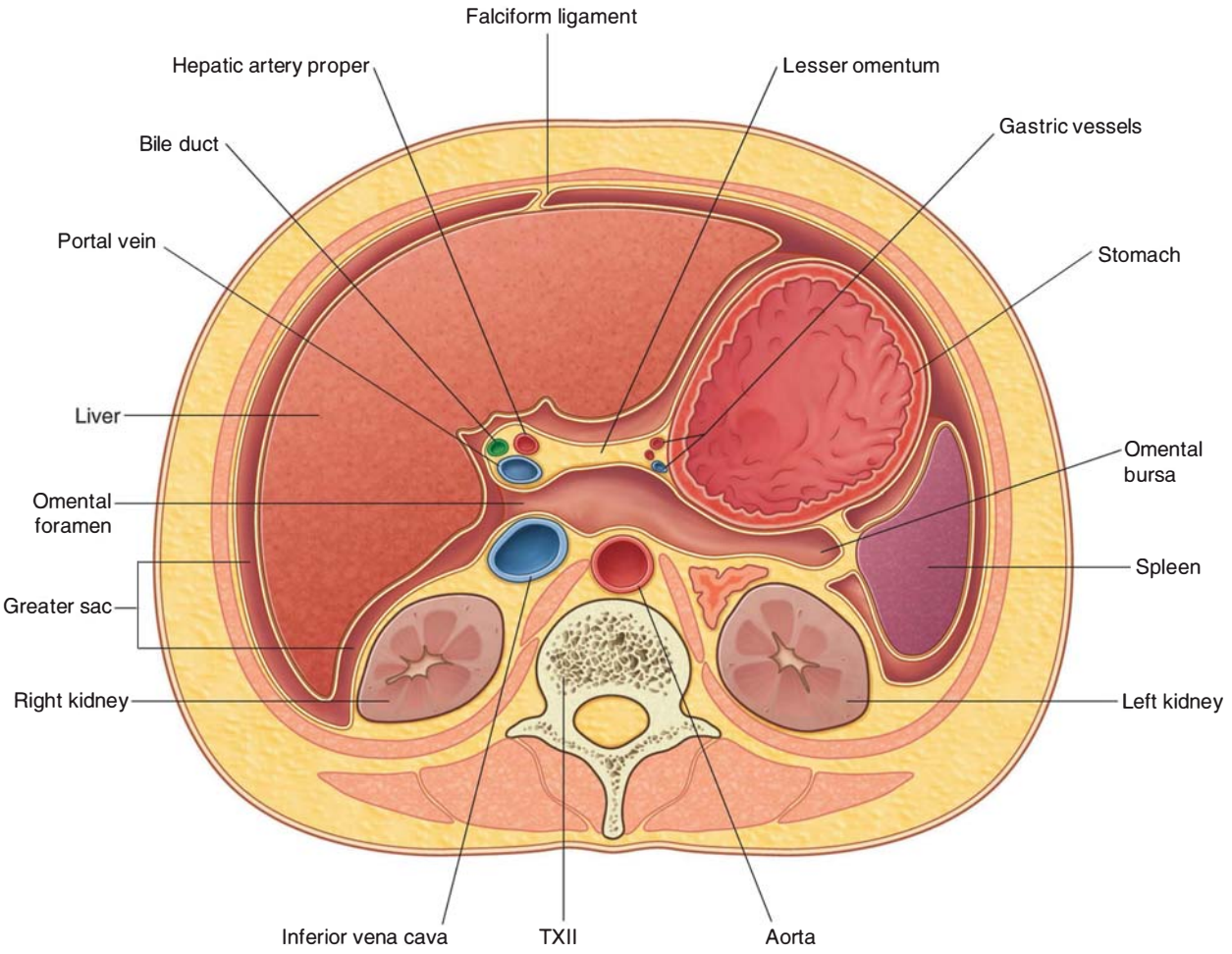


Fig. 4.54 Transverse section illustrating the continuity between the greater and lesser sacs through the omental (epiploic) foramen.

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In the clinic

Peritoneum

A small volume of peritoneal fluid within the peritoneal cavity lubricates movement of the viscera suspended in the abdominal cavity.

The peritoneal space has a large surface area, which facilitates the spread of disease through the peritoneal cavity and over the bowel and visceral surfaces. Conversely, this large surface area can be used for administering certain types of treatment and a number of procedures.

Ventriculoperitoneal shunts

Patients with obstructive hydrocephalus (an excessive accumulation of cerebrospinal fluid within the cerebral ventricular system) require continuous drainage of this fluid. This is achieved by placing a fine-bore catheter through the skull into the cerebral ventricles and placing the extracranial part of the tube beneath the scalp and skin of the chest wall and then passing it through the abdominal wall into the peritoneal cavity. Cerebrospinal fluid drains through the tube into the peritoneal cavity, where it is absorbed.

Dialysis and peritoneal dialysis

People who develop renal failure require dialysis to live. There are two methods.

In the first method (**hemodialysis**), blood is taken from the circulation, dialyzed through a complex artificial membrane, and returned to the body. A high rate of blood flow is required to remove excess body fluid, exchange electrolytes, and remove noxious metabolites. To accomplish this, either an arteriovenous fistula is established surgically (by connecting an artery to a vein, usually in the upper limb, and requiring approximately six weeks to "mature") and is cannulated each time the patient returns for dialysis, or a large-bore cannula is

(continues)

In the clinic—cont'd

placed into the right atrium, through which blood can be aspirated and returned.

In the second method of dialysis, the peritoneum is used as the dialysis membrane. The large surface area of the peritoneal cavity is an ideal dialysis membrane for fluid and electrolyte exchange. To accomplish dialysis, a small tube is inserted through the abdominal wall and dialysis fluid is injected into the peritoneal cavity. Electrolytes and molecules are exchanged across the peritoneum between the fluid and blood. Once dialysis is completed, the fluid is drained.

Peritoneal spread of disease

The large surface area of the peritoneal cavity allows infection and malignant disease to spread easily throughout the abdomen (Fig. 4.55). If malignant cells enter the peritoneal cavity by direct invasion (e.g., from colon or ovarian cancer), spread may be rapid. Similarly, a surgeon excising a malignant tumor and releasing malignant cells into the peritoneal cavity may cause an appreciable worsening of the patient's prognosis. Infection can also spread across the large surface area.

The peritoneal cavity can also act as a barrier to, and container of, disease. Intraabdominal infection therefore tends to remain below the diaphragm rather than spread into other body cavities.

Perforated bowel

A perforated bowel (e.g., caused by a perforated duodenal ulcer) often leads to the release of gas into the peritoneal cavity. This peritoneal gas can be easily visualized on an erect chest radiograph—gas can be demonstrated in extremely small amounts beneath the diaphragm. A patient with severe abdominal pain and subdiaphragmatic gas needs a laparotomy.

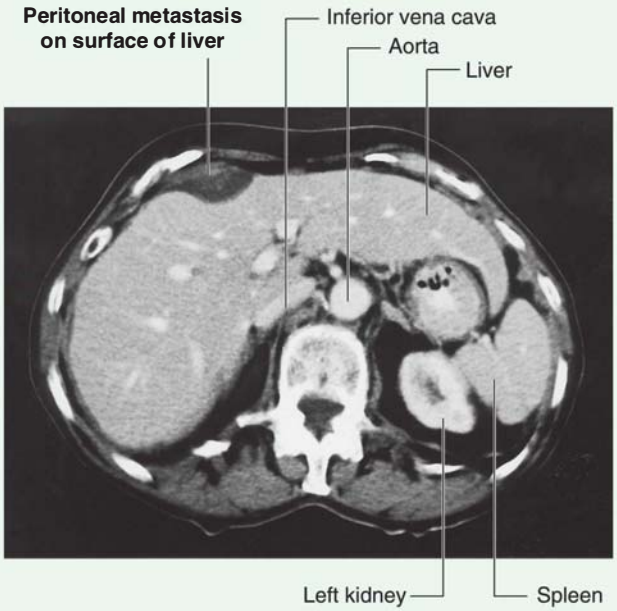


Fig. 4.55 Peritoneal metastasis on the surface of the liver. Computed tomogram in the axial plane of the upper abdomen.

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Omenta, mesenteries, and ligaments

Throughout the peritoneal cavity numerous peritoneal folds connect organs to each other or to the abdominal wall. These folds (omenta, mesenteries, and ligaments) develop from the original dorsal and ventral mesenteries, which suspend the developing gastrointestinal tract in the embryonic coelomic cavity. Some contain vessels and nerves supplying the viscera, while others help maintain the proper positioning of the viscera.

Omenta

The omenta consist of two layers of peritoneum, which pass from the stomach and the first part of the duodenum to other viscera. There are two:

- the greater omentum, derived from the dorsal mesentery, and
- the lesser omentum, derived from the ventral mesentery.

Greater omentum

The **greater omentum** is a large, apron-like, peritoneal fold that attaches to the greater curvature of the stomach and the first part of the duodenum (Fig. 4.56). It drapes inferiorly over the transverse colon and the coils of the jejunum and ileum (see Fig. 4.53). Turning posteriorly, it ascends to associate with, and become adherent to, the peritoneum on the superior surface of the transverse colon and the anterior layer of the transverse mesocolon before arriving at the posterior abdominal wall.

Usually a thin membrane, the greater omentum always contains an accumulation of fat, which may become substantial in some individuals. Additionally, there are two arteries and accompanying veins, the **right** and **left gastro-omental vessels**, between this double-layered peritoneal apron just inferior to the greater curvature of the stomach.

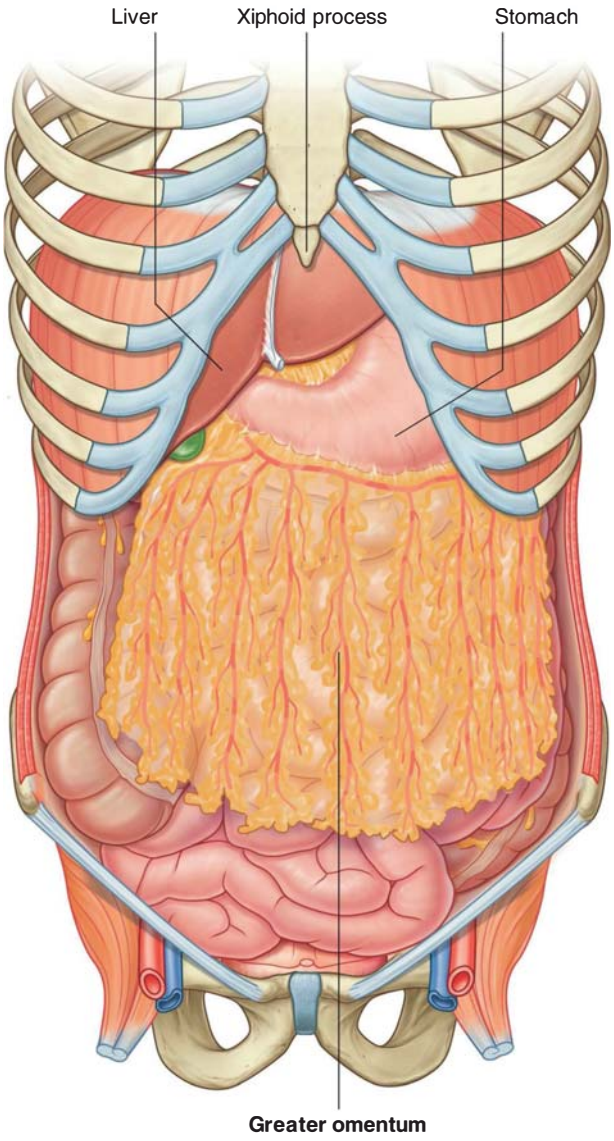


Fig. 4.56 Greater omentum.

Lesser omentum

The other two-layered peritoneal omentum is the **lesser omentum** (Fig. 4.57). It extends from the lesser curvature of the stomach and the first part of the duodenum to the inferior surface of the liver (Figs. 4.53 and 4.57).

A thin membrane continuous with the peritoneal coverings of the anterior and posterior surfaces of the stomach and the first part of the duodenum, the lesser omentum is divided into:

- a medial hepatogastric ligament, which passes between the stomach and liver, and
- a lateral hepatoduodenal ligament, which passes between the duodenum and liver.

The hepatoduodenal ligament ends laterally as a free margin and serves as the anterior border of the omental foramen (Fig. 4.54). Enclosed in this free edge are the hepatic artery proper, the bile duct, and the portal vein. Additionally, the right and left gastric vessels are between the layers of the lesser omentum near the lesser curvature of the stomach.

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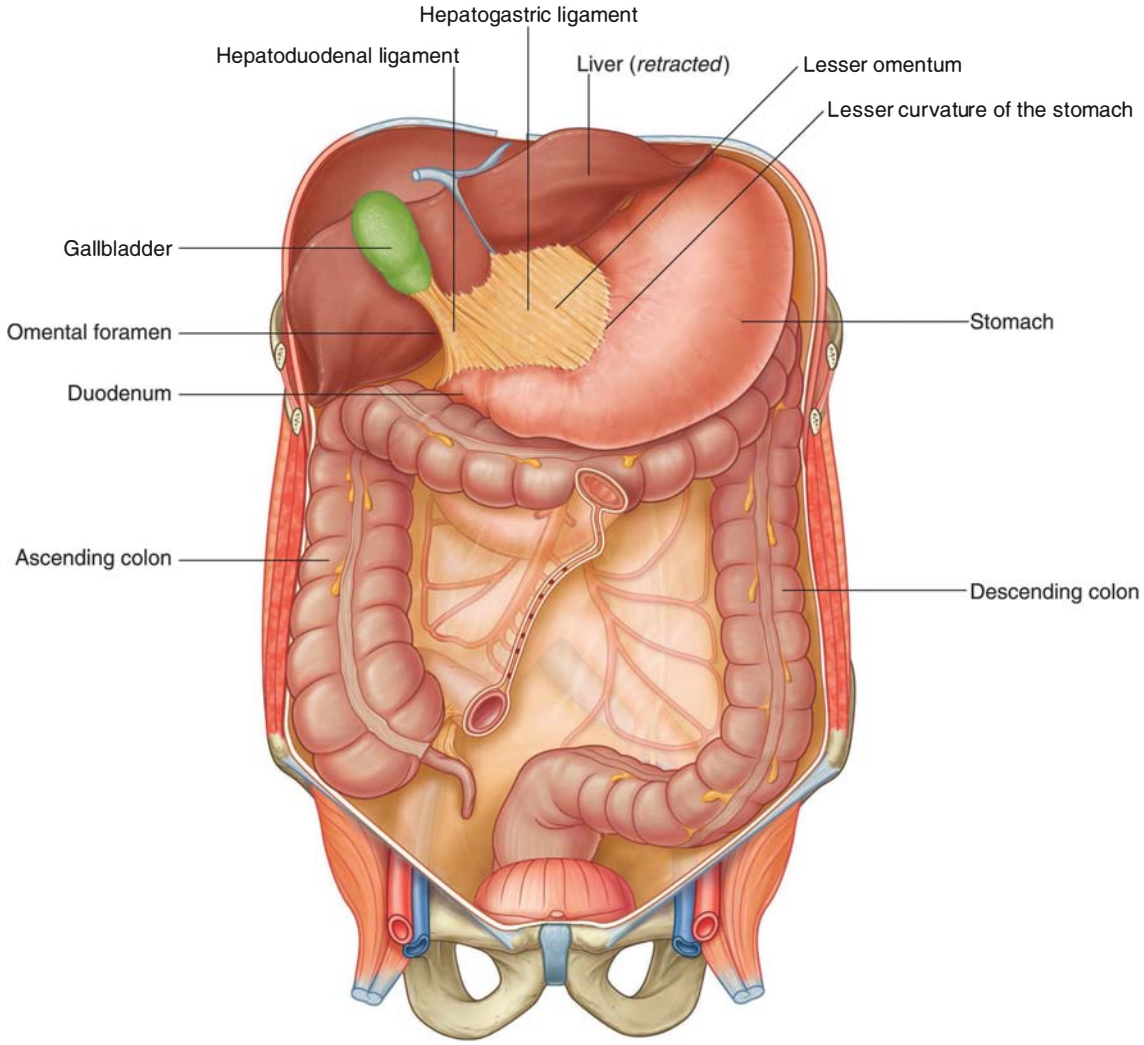


Fig. 4.57 Lesser omentum.

In the clinic

The greater omentum

When a laparotomy is performed and the peritoneal cavity is opened, the first structure usually encountered is the greater omentum. This fatty double-layered vascular membrane hangs like an apron from the greater curvature of the stomach, drapes over the transverse colon, and lies freely suspended within the abdominal cavity. It is often referred to as the "policeman of the abdomen" because of its apparent ability to migrate to any inflamed area and wrap itself around the organ to wall off inflammation. When a part of bowel becomes inflamed, it ceases peristalsis. This aperistaltic area is referred to as a local paralytic ileus. The remaining noninflamed part of the bowel continues to move and "massages" the greater omentum to the region where there is no peristalsis. The localized inflammatory reaction spreads to the greater omentum, which then adheres to the diseased area of bowel.

The greater omentum is also an important site for metastatic tumor spread. Direct omental spread by a transcoelomic route is common for carcinoma of the ovary. As the metastases develop within the greater omentum, it becomes significantly thickened.

In computed tomography imaging and during laparotomy, the thickened omentum is referred to as an "omental cake."

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Mesenteries

Mesenteries are peritoneal folds that attach viscera to the posterior abdominal wall. They allow some movement and provide a conduit for vessels, nerves, and lymphatics to reach the viscera and include:

- the mesentery—associated with parts of the small intestine,
- the transverse mesocolon—associated with the transverse colon, and
- the sigmoid mesocolon—associated with the sigmoid colon.

All of these are derivatives of the dorsal mesentery.

Mesentery

The **mesentery** is a large, fan-shaped, double-layered fold of peritoneum that connects the jejunum and ileum to the posterior abdominal wall (Fig. 4.58). Its superior attachment is at the duodenojejunal junction, just to the left of the upper lumbar part of the vertebral column. It passes obliquely downward and to the right, ending at the ileocecal junction near the upper border of the right sacro-iliac joint. In the fat between the two peritoneal layers of the mesentery are the arteries, veins, nerves, and lymphatics that supply the jejunum and ileum.

Transverse mesocolon

The **transverse mesocolon** is a fold of peritoneum that connects the transverse colon to the posterior abdominal wall (Fig. 4.58). Its two layers of peritoneum leave the posterior abdominal wall across the anterior surface of the head and body of the pancreas and pass outward to surround the transverse colon. Between its layers are the arteries, veins, nerves, and lymphatics related to the transverse colon. The anterior layer of the transverse mesocolon is adherent to the posterior layer of the greater omentum.

Sigmoid mesocolon

The **sigmoid mesocolon** is an inverted, V-shaped peritoneal fold that attaches the sigmoid colon to the abdominal wall (Fig. 4.58). The apex of the V is near the division of the left common iliac artery into its internal and external branches, with the left limb of the descending V along the medial border of the left psoas major muscle and the right limb descending into the pelvis to end at the level of vertebra SIII. The sigmoid and superior rectal vessels, along with the nerves and lymphatics associated with the sigmoid colon, pass through this peritoneal fold.

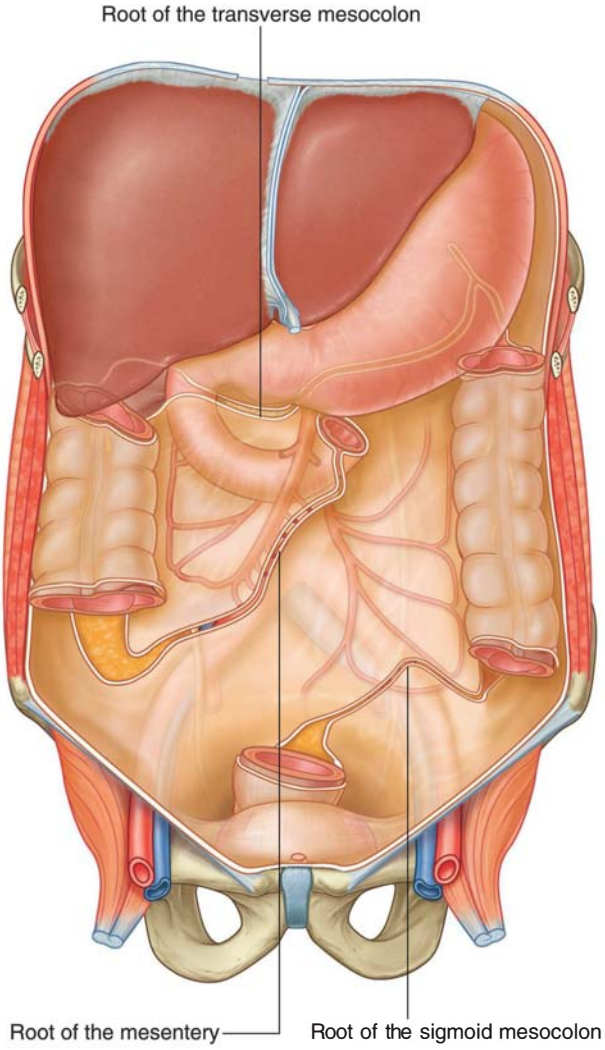


Fig. 4.58 Peritoneal reflections, forming mesenteries, outlined on the posterior abdominal wall.

Ligaments

Peritoneal ligaments consist of two layers of peritoneum that connect two organs to each other or attach an organ to the body wall, and may form part of an omentum. They are usually named after the structures being connected. For example, the splenorenal ligament connects the left kidney to the spleen and the gastrophrenic ligament connects the stomach to the diaphragm.

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Organs

Abdominal esophagus

The abdominal esophagus represents the short distal part of the esophagus located in the abdominal cavity. Emerging through the right crus of the diaphragm, usually at the level of vertebra TX, it passes from the esophageal hiatus to the cardial orifice of the stomach just left of the midline (Fig. 4.59).

Associated with the esophagus, as it enters the abdominal cavity, are the anterior and posterior vagal trunks:

- The **anterior vagal trunk** consists of several smaller trunks whose fibers mostly come from the left vagus nerve; rotation of the gut during development moves these trunks to the anterior surface of the esophagus.
- Similarly, the **posterior vagal trunk** consists of a single trunk whose fibers mostly come from the right vagus nerve, and rotational changes during development move this trunk to the posterior surface of the esophagus.

The arterial supply to the abdominal esophagus (Fig. 4.60) includes:

- esophageal branches from the left gastric artery (from the celiac trunk), and
- esophageal branches from the left inferior phrenic artery (from the abdominal aorta).

Stomach

The stomach is the most dilated part of the gastrointestinal tract and has a J-like shape (Figs. 4.61 and 4.62). Positioned between the abdominal esophagus and the small intestine, the stomach is in the epigastric, umbilical, and left hypochondrium regions of the abdomen.

The stomach is divided into four regions:

- the cardia, which surrounds the opening of the esophagus into the stomach;
- the fundus of the stomach, which is the area above the level of the cardial orifice;
- the body of the stomach, which is the largest region of the stomach; and
- the pyloric part, which is divided into the pyloric antrum and pyloric canal and is the distal end of the stomach.

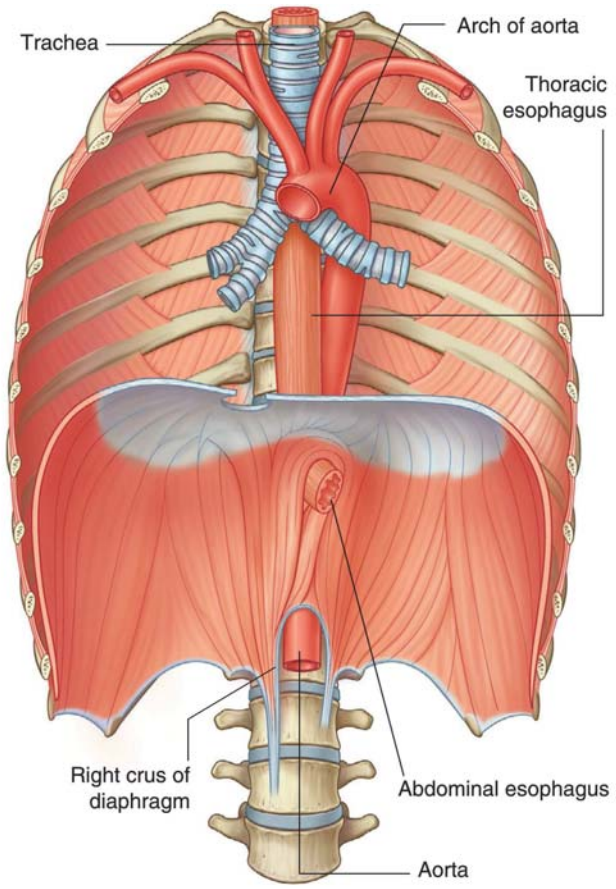


Fig. 4.59 Abdominal esophagus.

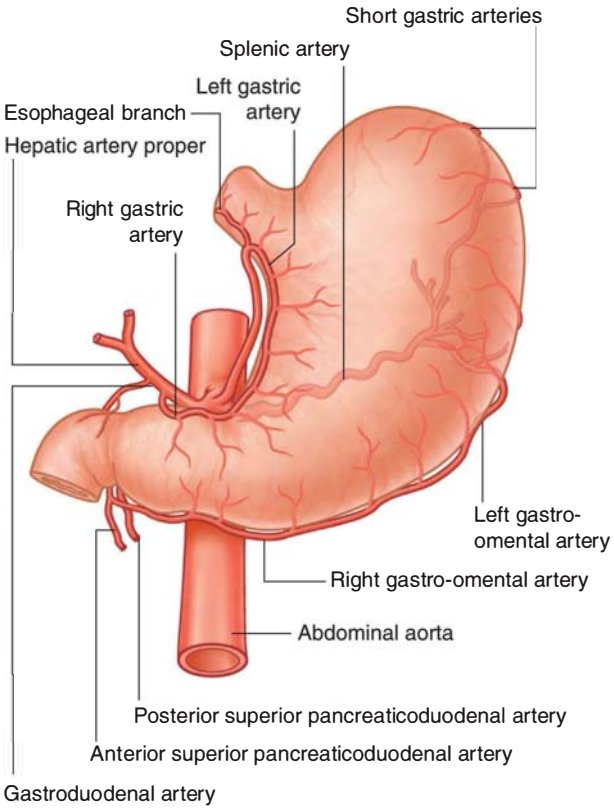


Fig. 4.60 Arterial supply to the abdominal esophagus and stomach.

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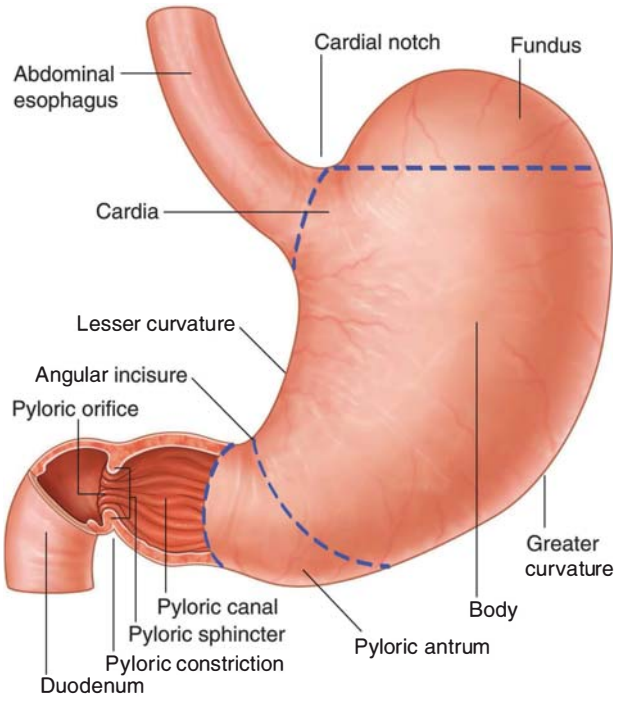


Fig. 4.61 Stomach.

The most distal portion of the pyloric part of the stomach is the **pylorus** (Fig. 4.61). It is marked on the surface of the organ by the **pyloric constriction** and contains a thickened ring of gastric circular muscle, the **pyloric sphincter**, that surrounds the distal opening of the stomach, the **pyloric orifice** (Figs. 4.61 and 4.62B). The pyloric orifice is just to the right of midline in a plane that passes through the lower border of vertebra LI (the **transpyloric plane**).

Other features of the stomach include:

- the **greater curvature**, which is a point of attachment for the gastrosplenic ligament and the greater omentum;
- the **lesser curvature**, which is a point of attachment for the lesser omentum;
- the **cardial notch**, which is the superior angle created when the esophagus enters the stomach; and
- the **angular incisure**, which is a bend on the lesser curvature.

The arterial supply to the stomach (Fig. 4.60) includes:

- the left gastric artery from the celiac trunk,
- the right gastric artery from the hepatic artery proper,

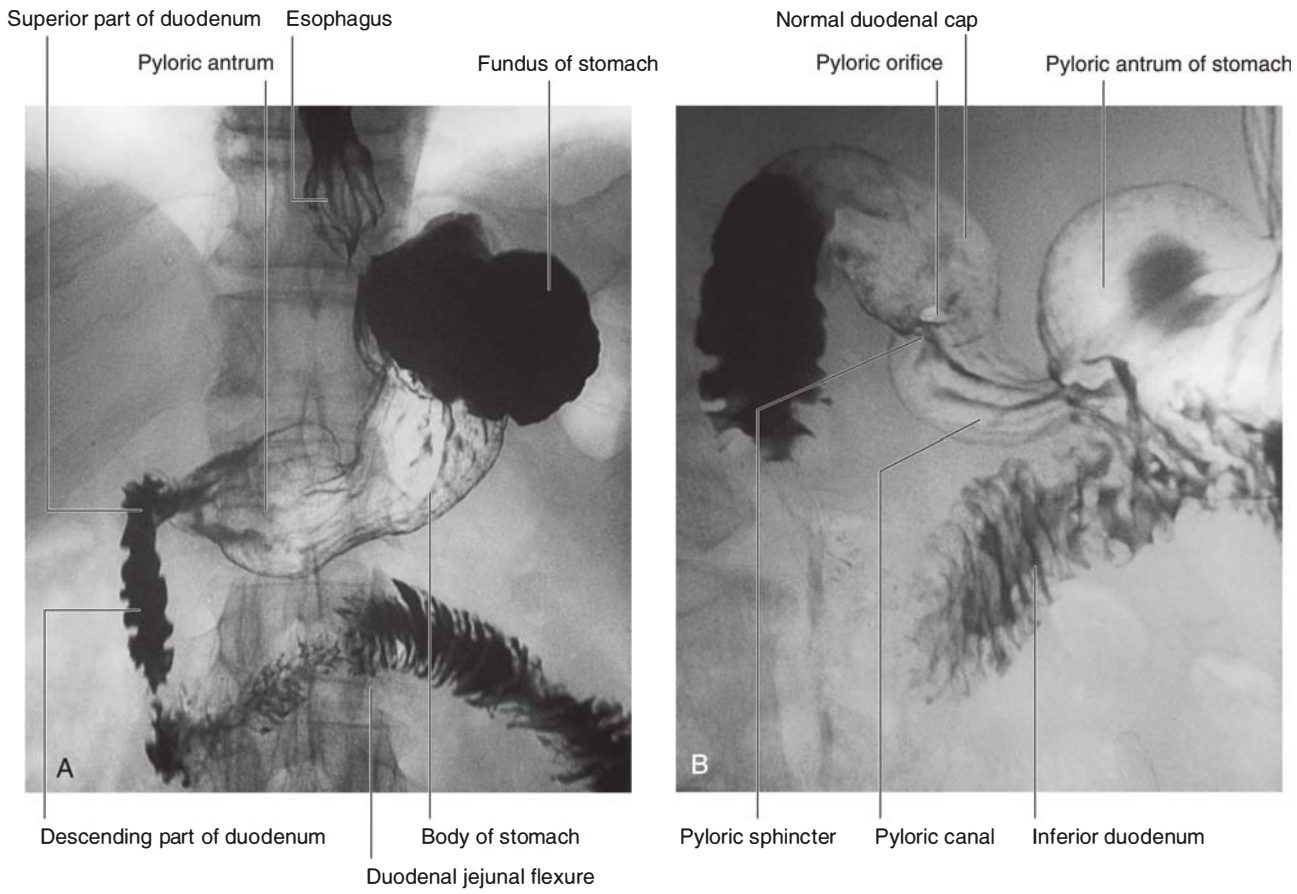


Fig. 4.62 Radiograph, using barium, showing the stomach and duodenum. **A.** Double-contrast radiograph of the stomach. **B.** Double-contrast radiograph showing the duodenal cap.

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- the right gastro-omental artery from the gastroduodenal artery,
- the left gastro-omental artery from the splenic artery, and
- the posterior gastric artery from the splenic artery (variant and not always present).

Small intestine

The small intestine is the longest part of the gastrointestinal tract and extends from the pyloric orifice of the stomach to the ileocecal fold. This hollow tube, which is approximately 6 to 7 m long with a narrowing diameter from beginning to end, consists of the duodenum, the jejunum, and the ileum.

Duodenum

The first part of the small intestine is the duodenum. This C-shaped structure, adjacent to the head of the pancreas, is 20 to 25 cm long and is above the level of the umbilicus; its lumen is the widest of the small intestine (Fig. 4.63). It is retroperitoneal except for its beginning, which is connected to the liver by the hepatoduodenal ligament, a part of the lesser omentum.

The duodenum is divided into four parts (Fig. 4.63).

- The **superior part** (first part) extends from the pyloric orifice of the stomach to the neck of the gallbladder, is just to the right of the body of vertebra LI, and passes anteriorly to the bile duct, gastroduodenal artery, portal vein, and inferior vena cava. Clinically, the beginning of this part of the duodenum is referred to as the ampulla or duodenal cap, and most duodenal ulcers occur in this part of the duodenum.
- The **descending part** (second part) of the duodenum is just to the right of midline and extends from the neck of the gallbladder to the lower border of vertebra LIII. Its anterior surface is crossed by the transverse colon, posterior to it is the right kidney, and medial to it is the head of the pancreas. This part of the duodenum contains the **major duodenal papilla**, which is the

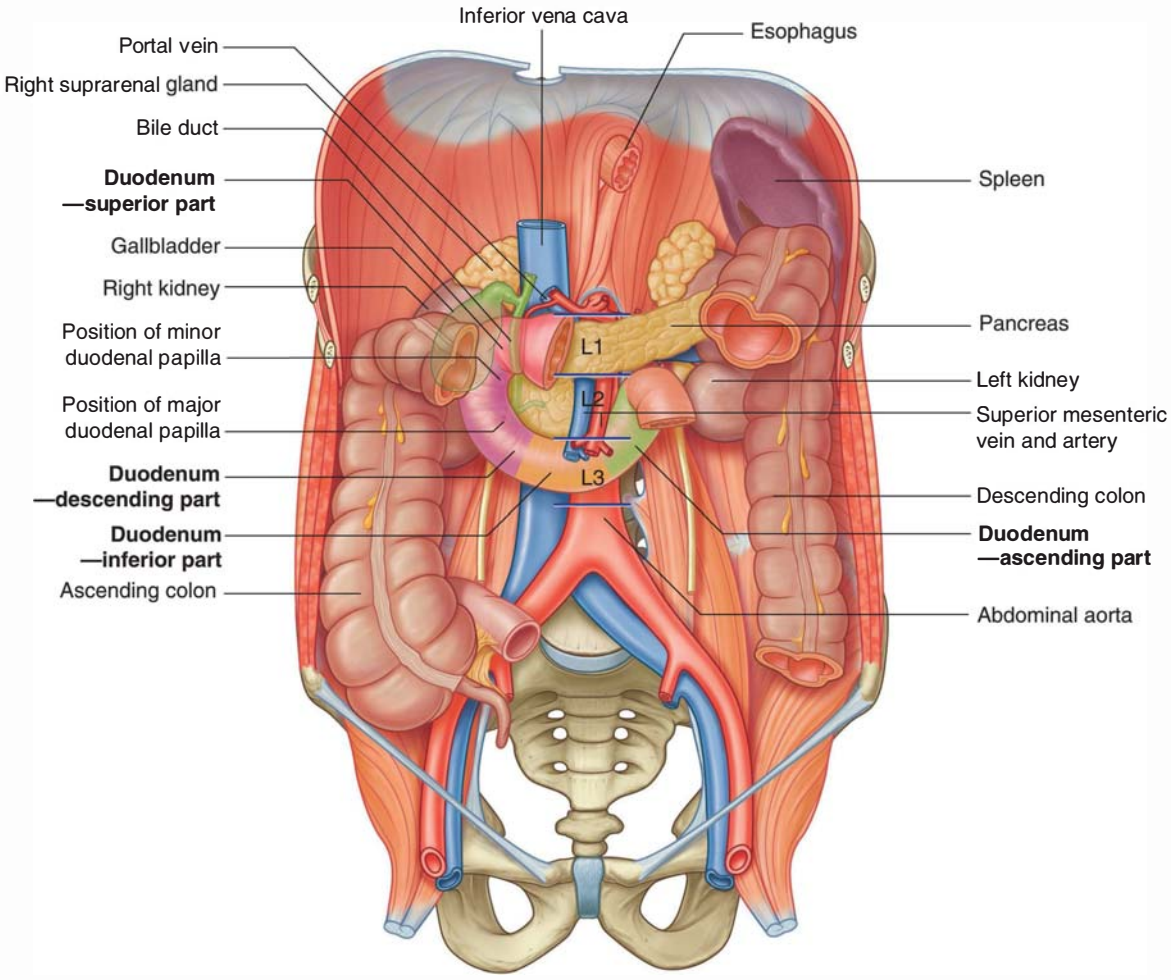


Fig. 4.63 Duodenum.

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common entrance for the bile and pancreatic ducts, and the **minor duodenal papilla**, which is the entrance for the accessory pancreatic duct, and the junction of the foregut and the midgut just below the major duodenal papilla.

- The **inferior part** (third part) of the duodenum is the longest section, crossing the inferior vena cava, the aorta, and the vertebral column (Figs. 4.62B and 4.63). It is crossed anteriorly by the superior mesenteric artery and vein.
- The **ascending part** (fourth part) of the duodenum passes upward on, or to the left of, the aorta to approximately the upper border of vertebra LII and terminates at the **duodenojejunal flexure**.

This duodenojejunal flexure is surrounded by a fold of peritoneum containing muscle fibers called the **suspensory muscle (ligament) of duodenum (ligament of Treitz).**

The arterial supply to the duodenum (Fig. 4.64) includes:

- branches from the gastroduodenal artery,
- the supraduodenal artery from the gastroduodenal artery,
- duodenal branches from the anterior superior pancreaticoduodenal artery (from the gastroduodenal artery),
- duodenal branches from the posterior superior pancreaticoduodenal artery (from the gastroduodenal artery),
- duodenal branches from the anterior inferior pancreaticoduodenal artery (from the inferior pancreaticoduodenal artery—a branch of the superior mesenteric artery),
- duodenal branches from the posterior inferior pancreaticoduodenal artery (from the inferior pancreaticoduodenal artery—a branch of the superior mesenteric artery), and
- the first jejunal branch from the superior mesenteric artery.

Jejunum

The jejunum and ileum make up the last two sections of the small intestine (Fig. 4.65). The jejunum represents the proximal two-fifths. It is mostly in the left upper quadrant

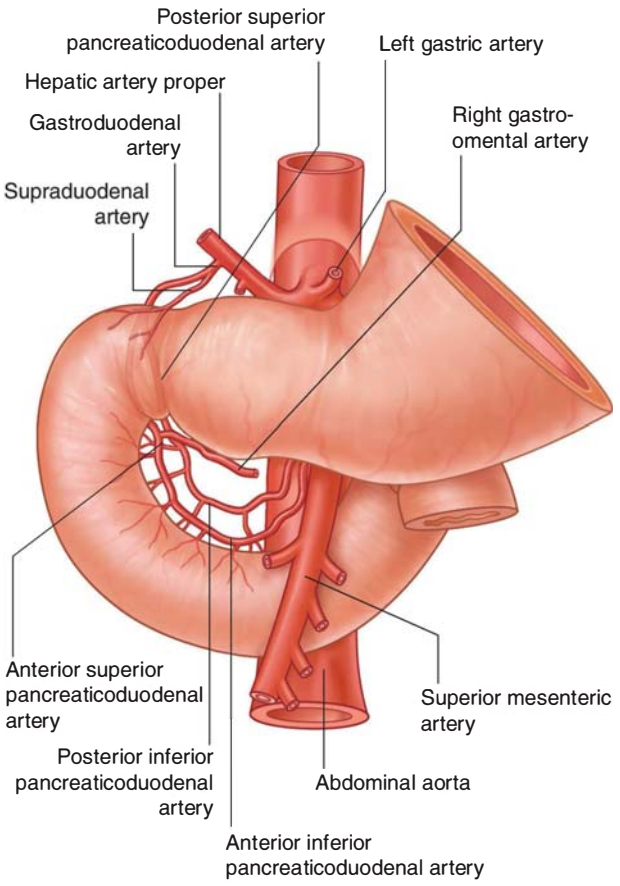


Fig. 4.64 Arterial supply to the duodenum.

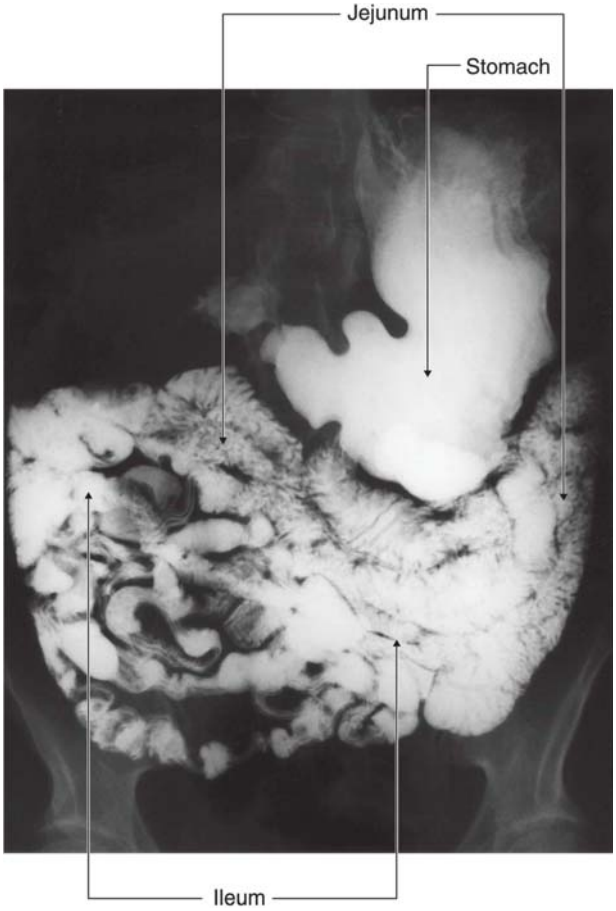


Fig. 4.65 Radiograph, using barium, showing the jejunum and ileum.

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of the abdomen and is larger in diameter and has a thicker wall than the ileum. Additionally, the inner mucosal lining of the jejunum is characterized by numerous prominent folds that circle the lumen (plicae circulares). The less prominent arterial arcades and longer vasa recta (straight arteries) compared to those of the ileum are a unique characteristic of the jejunum (Fig. 4.66).

The arterial supply to the jejunum includes jejunal arteries from the superior mesenteric artery.

Ileum

The ileum makes up the distal three-fifths of the small intestine and is mostly in the right lower quadrant. Compared to the jejunum, the ileum has thinner walls, fewer and less prominent mucosal folds (plicae circulares), shorter vasa recta, more mesenteric fat, and more arterial arcades (Fig. 4.66).

The ileum opens into the large intestine, where the cecum and ascending colon join together. Two flaps projecting into the lumen of the large intestine (the **ileocecal fold**) surround the opening (Fig. 4.67). The flaps of the

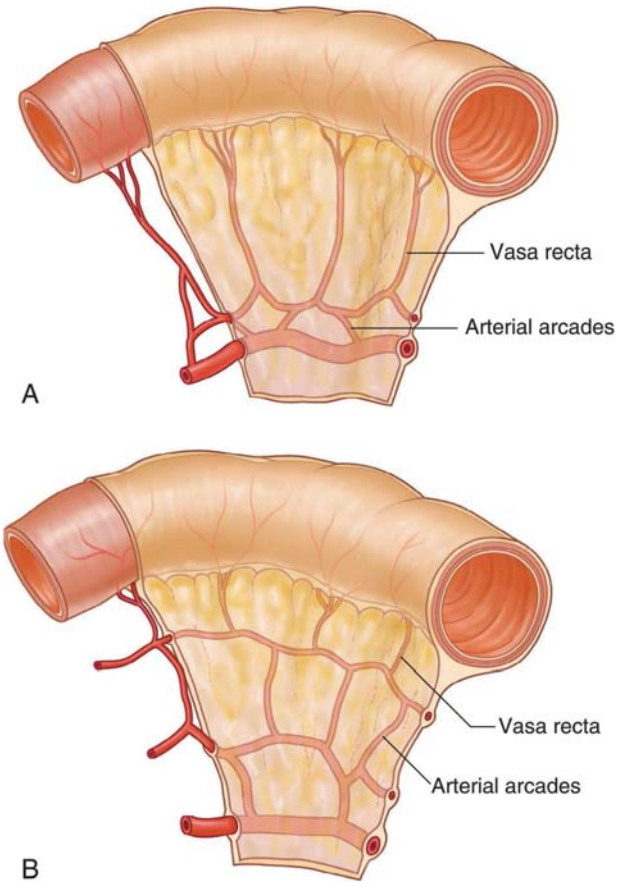


Fig. 4.66 Differences in the arterial supply to the small intestine. **A.** Jejunum. **B.** Ileum.

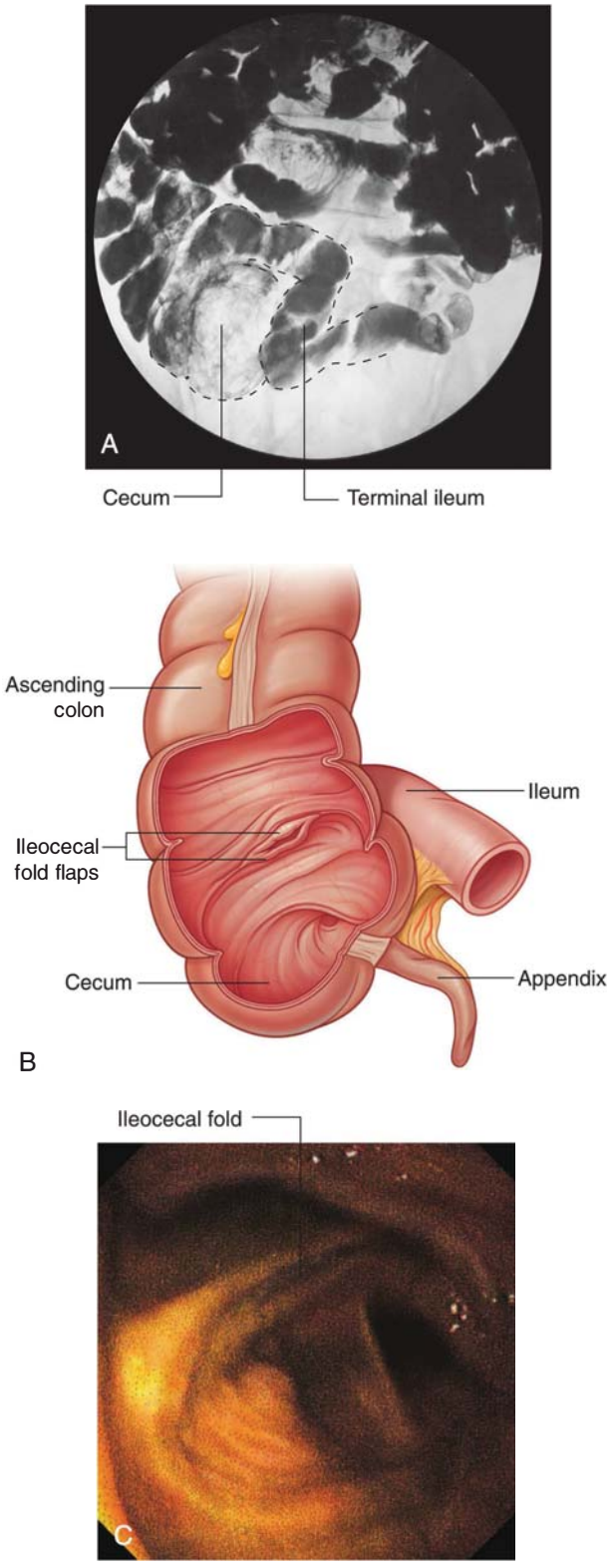


Fig. 4.67 Ileocecal junction. **A.** Radiograph showing ileocecal junction. **B.** Illustration showing ileocecal junction and the ileocecal fold. **C.** Endoscopic image of the ileocecal fold.

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ileocecal fold come together at their end, forming ridges. Musculature from the ileum continues into each flap, forming a sphincter. Possible functions of the ileocecal fold include preventing reflux from the cecum to the ileum, and regulating the passage of contents from the ileum to the cecum.

The arterial supply to the ileum (Fig. 4.68) includes:

- ileal arteries from the superior mesenteric artery, and
- an ileal branch from the ileocolic artery (from the superior mesenteric artery).

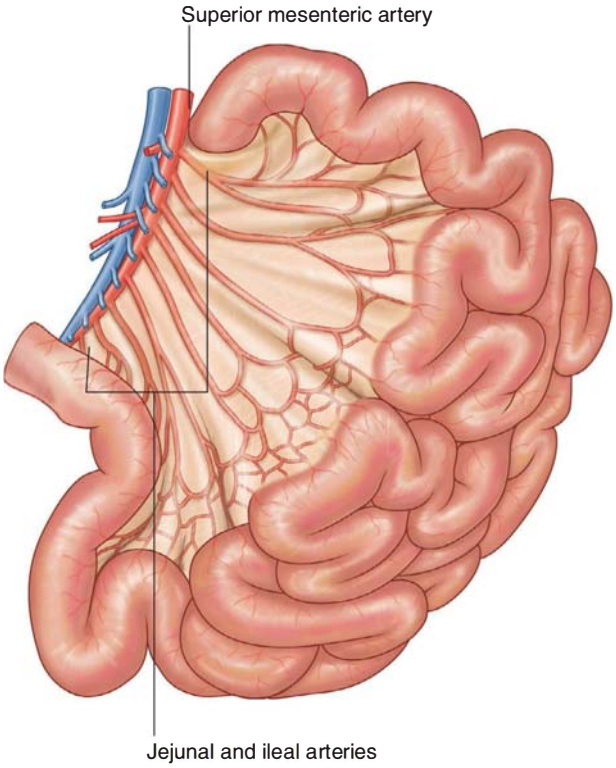


Fig. 4.68 Arterial supply to the ileum.

In the clinic

Epithelial transition between the abdominal esophagus and stomach

The gastroesophageal junction is demarcated by a transition from one epithelial type to another epithelial type. In some people, the histological junction does not lie at the physiological gastroesophageal junction but is in the lower one-third of the esophagus. This may predispose these people to esophageal ulceration and is also associated with an increased risk of adenocarcinoma.

In the clinic

Duodenal ulceration

Duodenal ulcers usually occur in the superior part of the duodenum and are much less common than they were 50 years ago. At first, there was no treatment and patients died from hemorrhage or peritonitis. As surgical techniques developed, patients with duodenal ulcers were subjected to extensive upper gastrointestinal surgery to prevent ulcer recurrence and for some patients the treatment was dangerous. As knowledge and understanding of the mechanisms for acid secretion in the stomach increased, drugs were developed to block acid stimulation and secretion indirectly (histamine H_2-receptor antagonists) and these have significantly reduced the morbidity and mortality rates of this disease. Pharmacological therapy can now directly inhibit the cells of the stomach that produce acid with, for example, proton pump inhibitors. Patients are also screened for the bacteria *Helicobacter pylori*, which when eradicated (by antibiotic treatment) significantly reduces duodenal ulcer formation.

Anatomically, duodenal ulcers tend to occur either anteriorly or posteriorly.

Posterior duodenal ulcers erode either directly onto the gastroduodenal artery or, more commonly, onto the posterior superior pancreaticoduodenal artery, which can produce torrential hemorrhage, which may be fatal in some patients. Treatment may involve extensive upper abdominal surgery with ligation of the vessels or by endovascular means whereby the radiologist may place a very fine catheter retrogradely from the femoral artery into the celiac artery. The common hepatic artery and the gastroduodenal artery are cannulated and the bleeding area may be blocked using small coils, which stem the flow of blood.

Anterior duodenal ulcers erode into the peritoneal cavity, causing peritonitis. This intense inflammatory reaction and the local ileus promote adhesion of the greater omentum, which attempts to seal off the perforation. The stomach and duodenum usually contain considerable amounts of gas, which enters the peritoneal cavity and can be observed on a chest radiograph of an erect patient as subdiaphragmatic gas. In most instances, treatment for the ulcer is surgical.

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In the clinic

Examination of the upper gastrointestinal tract

It is often necessary to examine the esophagus, stomach, duodenum, and proximal jejunum for disease. After taking an appropriate history and examining the patient, most physicians arrange a series of simple blood tests to look for bleeding, inflammation, and tumors. The next steps in the investigation assess the three components of any loop of bowel, namely, the lumen, the wall, and masses extrinsic to the bowel, which may compress or erode into it.

Examination of the bowel lumen

Barium sulfate solutions may be swallowed by the patient and can be visualized using an X-ray fluoroscopy unit. The lumen can be examined for masses (e.g., polyps and tumors) and peristaltic waves can be assessed. Patients may also be given carbon dioxide–releasing granules to fill the stomach so that the barium thinly coats the mucosa, resulting in images displaying fine mucosal detail. These tests are relatively simple and can be used to image the esophagus, stomach, duodenum, and small bowel.

Examination of the bowel wall and extrinsic masses

Endoscopy is a minimally invasive diagnostic medical procedure that can be used to assess the interior surfaces of an organ by inserting a tube into the body. The instrument is typically made of a flexible plastic material through which a light source and eyepiece are attached at one end. Some systems allow passage of small instruments through the main bore of the endoscope to obtain biopsies and to also undertake small procedures (e.g., the removal of polyps).

In gastrointestinal and abdominal medicine an endoscope is used to assess the esophagus, stomach, duodenum, and proximal small bowel (Figs. 4.69 to 4.72). The tube is swallowed by the patient under light sedation and is extremely well tolerated.

Assessment of the colon is performed by passage of the tube through the anus and into the rectum. The whole of the colon can be readily assessed; biopsies and the placement of stents can also be performed using this device.



Fig. 4.69 The endoscope is a flexible plastic tube that can be controlled from the proximal end. Through a side portal various devices can be inserted, which run through the endoscope and can be used to obtain biopsies and to perform minor endoluminal surgical procedures (e.g., excision of polyps).

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In the clinic—cont'd

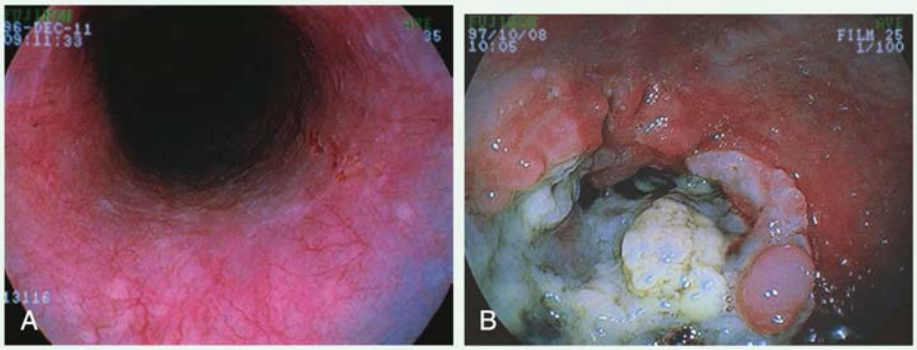


Fig. 4.70 Endoscopic images of the gastroesophageal junction. **A.** Normal. **B.** Esophageal cancer at esophageal junction.



Fig. 4.71 Endoscopic image of the pyloric antrum of the stomach looking toward the pylorus.

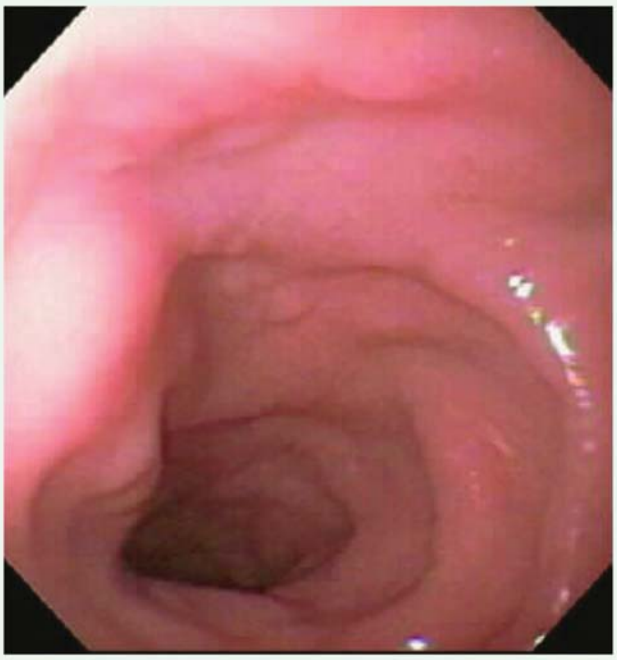


Fig. 4.72 Endoscopic image showing normal appearance of the second part of the duodenum.

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In the clinic

Meckel's diverticulum

A Meckel's diverticulum (Fig. 4.73) is the remnant of the proximal part of the yolk stalk (vitelline duct) that extends into the umbilical cord in the embryo and lies on the antimesenteric border of the ileum. It appears as a blind-ended tubular outgrowth of bowel. Although it is an uncommon finding (occurring in approximately 2% of the population), it is always important to consider the diagnosis of Meckel's diverticulum because it does produce symptoms in a small number of patients. Typical findings include hemorrhage, intussusception, diverticulitis, ulceration, and obstruction.

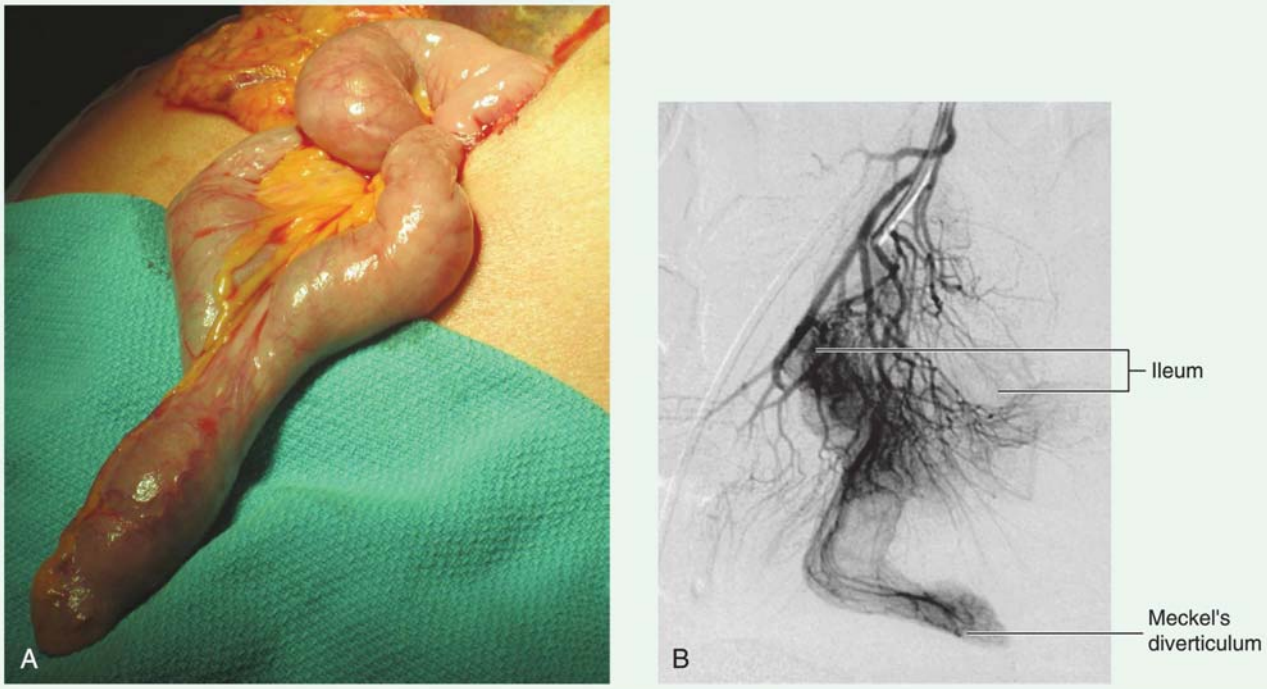


Fig. 4.73 Vasculature associated with a Meckel's diverticulum. **A.** Surgical image of Meckel's diverticulum. **B.** Digital subtraction angiography.

In the clinic

Computed tomography (CT) scanning and magnetic resonance imaging (MRI)

These imaging techniques can provide important information about the wall of the bowel that may not be obtained from barium or endoscopic studies. Thickening of the wall may indicate inflammatory change or tumor and is always regarded with suspicion. If a tumor is demonstrated, the locoregional spread can be assessed, along with lymphadenopathy and metastatic spread.

Advanced imaging methods

A small ultrasound device placed on the end of the endoscope can produce extremely high-powered views of the mucosa and submucosa of the upper gastrointestinal tract. These views can show whether a tumor is resectable and guide the clinician in taking a biopsy.

In the clinic

Carcinoma of the stomach

Carcinoma of the stomach is a common gastrointestinal malignancy. Chronic gastric inflammation (gastritis), pernicious anemia, and polyps predispose to the development of this aggressive cancer, which is usually not diagnosed until late in the course of the disease. Symptoms include vague epigastric pain, early fullness with eating, bleeding leading to chronic anemia, and obstruction.

The diagnosis may be made using barium and conventional radiology or endoscopy, which allows a biopsy to be obtained at the same time. Ultrasound scanning is used to check the liver for metastatic spread, and, if negative, computed tomography is carried out to assess for surgical resectability. If carcinoma of the stomach is diagnosed early, a curative surgical resection is possible. However, because most patients do not seek treatment until late in the disease, the overall 5-year survival rate is between 5% and 20%, with a mean survival time of between 5 and 8 months.

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Large intestine

The large intestine extends from the distal end of the ileum to the anus, a distance of approximately 1.5 m in adults. It absorbs fluids and salts from the gut contents, thus forming feces, and consists of the cecum, appendix, colon, rectum, and anal canal (Figs. 4.74 and 4.75).

Beginning in the right groin as the cecum, with its associated appendix, the large intestine continues upward as the **ascending colon** through the right flank and into the right hypochondrium (Fig. 4.76). Just below the liver, it bends to the left, forming the **right colic flexure** (**hepatic flexure**), and crosses the abdomen as the **transverse colon** to the left hypochondrium. At this position, just below the spleen, the large intestine bends downward, forming the **left colic flexure** (**splenic flexure**), and continues as the **descending colon** through the left flank and into the left groin.

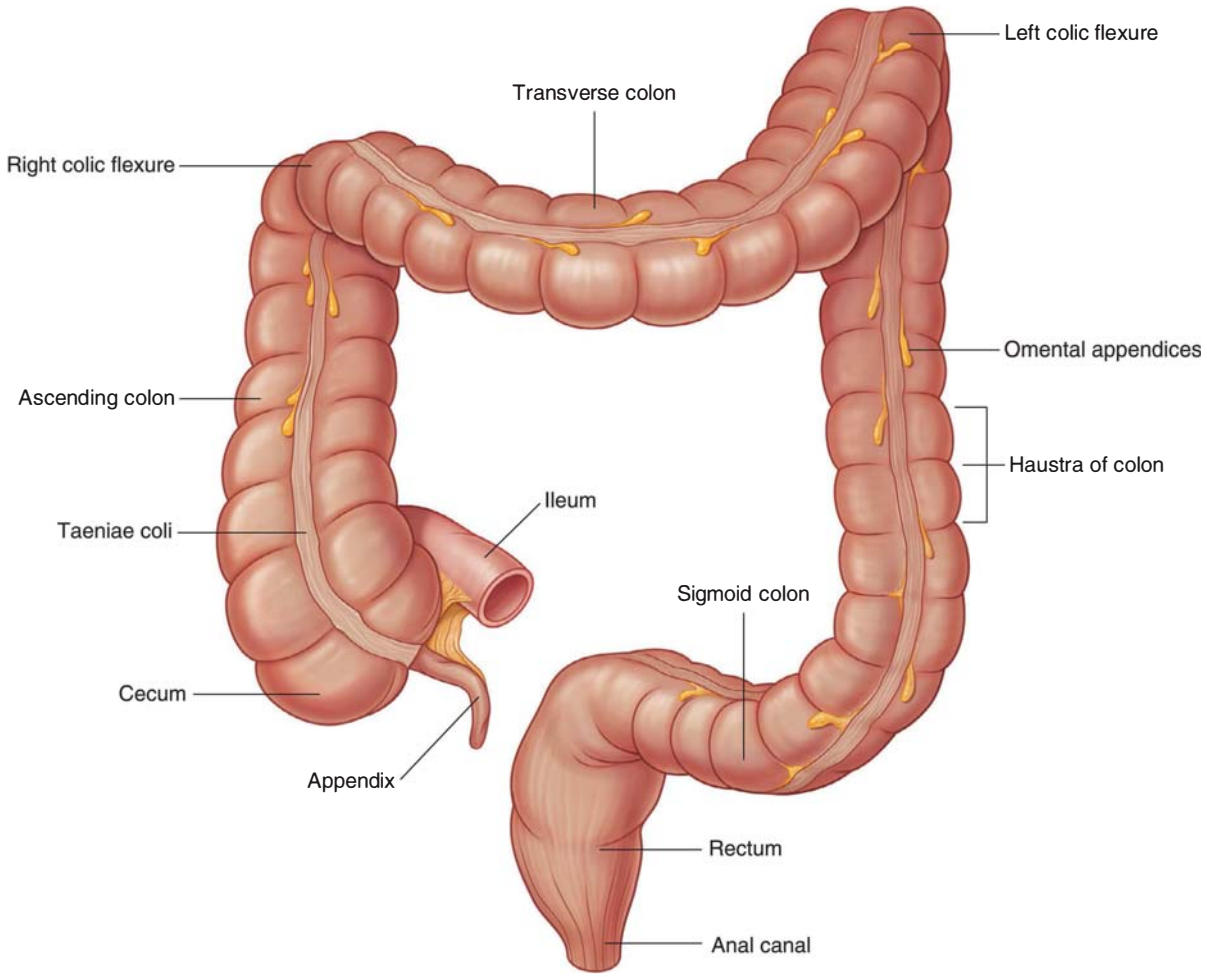


Fig. 4.74 Large intestine.

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Fig. 4.75 Radiograph, using barium, showing the large intestine.

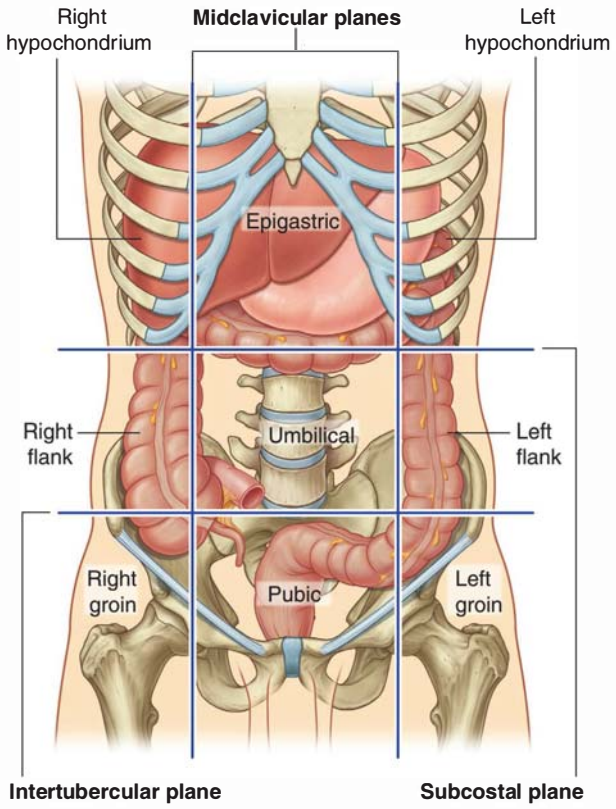


Fig. 4.76 Position of the large intestine in the nine-region organizational pattern.

It enters the upper part of the pelvic cavity as the sigmoid colon, continues on the posterior wall of the pelvic cavity as the rectum, and terminates as the anal canal.

The general characteristics of most of the large intestine (Fig. 4.74) are:

- its large internal diameter compared to that of the small intestine;
- peritoneal-covered accumulations of fat (the **omental appendices**) are associated with the colon;
- the segregation of longitudinal muscle in its walls into three narrow bands (the **taeniae coli**), which are primarily observed in the cecum and colon and less visible in the rectum; and
- the sacculations of the colon (the **haustra of the colon**).

Cecum and appendix

The **cecum** is the first part of the large intestine (Fig. 4.77). It is inferior to the ileocecal opening and in the right iliac fossa. It is an intraperitoneal structure because of its mobility, not because of its suspension by a mesentery.

The cecum is continuous with the ascending colon at the entrance of the ileum and is usually in contact with

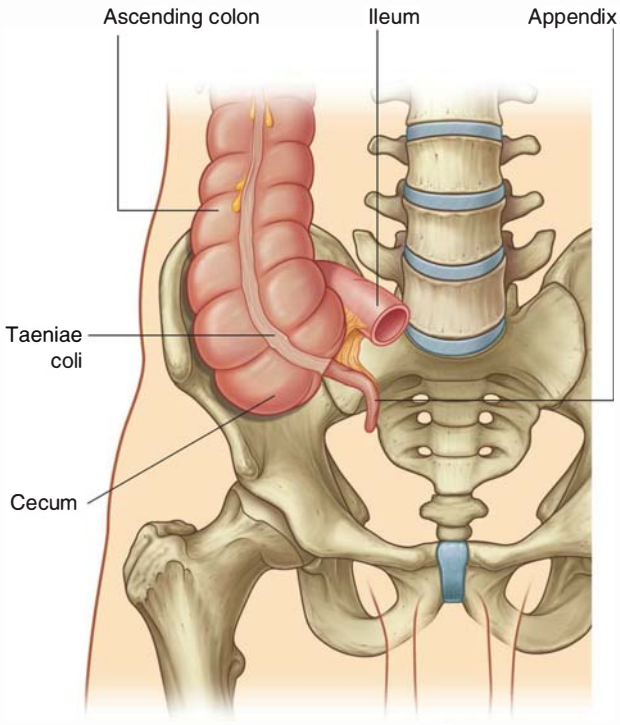


Fig. 4.77 Cecum and appendix.

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the anterior abdominal wall. It may cross the pelvic brim to lie in the true pelvis. The appendix is attached to the posteromedial wall of the cecum, just inferior to the end of the ileum (Fig. 4.77).

The **appendix** is a narrow, hollow, blind-ended tube connected to the cecum. It has large aggregations of lymphoid tissue in its walls and is suspended from the terminal ileum by the **mesoappendix** (Fig. 4.78), which contains the **appendicular vessels**. Its point of attachment to the cecum is consistent with the highly visible free taeniae leading directly to the base of the appendix, but the location of the rest of the appendix varies considerably (Fig. 4.79). It may be:

- posterior to the cecum or the lower ascending colon, or both, in a retrocecal or retrocolic position;
- suspended over the pelvic brim in a pelvic or descending position;
- below the cecum in a subcecal location; or
- anterior to the terminal ileum, possibly contacting the body wall, in a pre-ileal position or posterior to the terminal ileum in a postileal position.

The surface projection of the base of the appendix is at the junction of the lateral and middle one-third of a line from the anterior superior iliac spine to the umbilicus (McBurney's point). People with appendicular problems may describe pain near this location.

The arterial supply to the cecum and appendix (Fig. 4.80) includes:

- the anterior cecal artery from the ileocolic artery (from the superior mesenteric artery),
- the posterior cecal artery from the ileocolic artery (from the superior mesenteric artery), and
- the appendicular artery from the ileocolic artery (from the superior mesenteric artery).

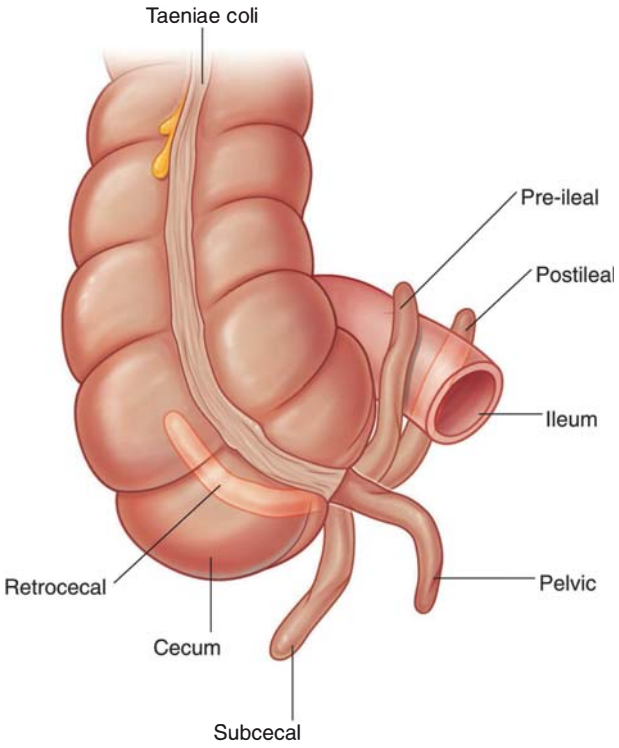


Fig. 4.79 Positions of the appendix.

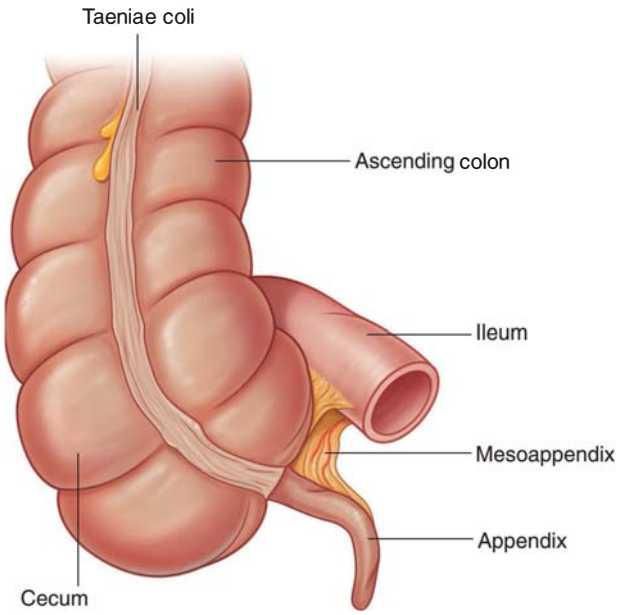


Fig. 4.78 Mesoappendix and appendicular vessels.

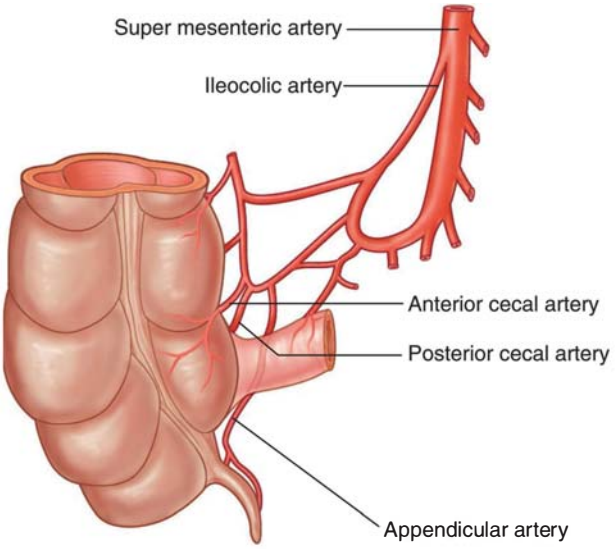


Fig. 4.80 Arterial supply to the cecum and appendix.

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In the clinic

Appendicitis

Acute appendicitis is an abdominal emergency. It usually occurs when the appendix is obstructed by either a fecalith or enlargement of the lymphoid nodules. Within the obstructed appendix, bacteria proliferate and invade the appendix wall, which becomes damaged by pressure necrosis. In some instances, this may resolve spontaneously; in other cases, inflammatory change (Fig. 4.81) continues and perforation ensues, which may lead to localized or generalized peritonitis.

Most patients with acute appendicitis have localized tenderness in the right groin. Initially, the pain begins as a central, periumbilical, colicky type of pain, which tends to come and go. After 6 to 10 hours, the pain tends to localize in the right iliac fossa and becomes constant. Patients may develop a fever, nausea, and vomiting. The etiology of the pain for appendicitis is described in Case 1 of Chapter 1 on p. 50.

The treatment for appendicitis is appendectomy.

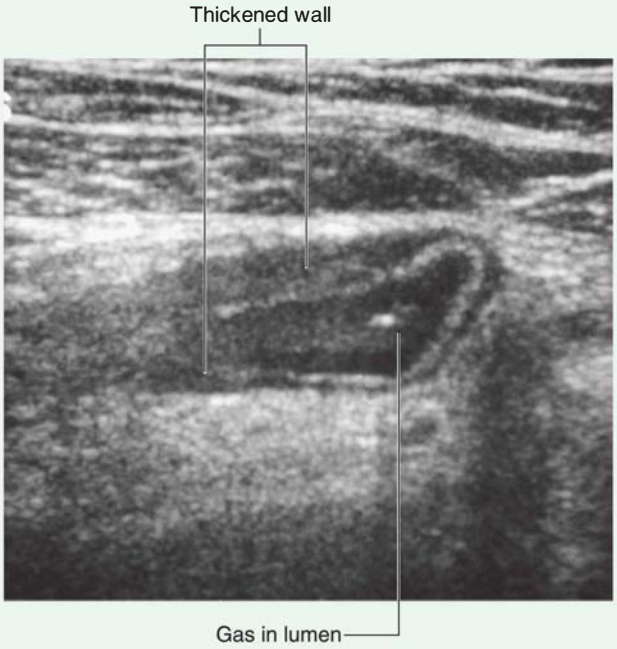


Fig. 4.81 Inflamed appendix. Ultrasound scan.

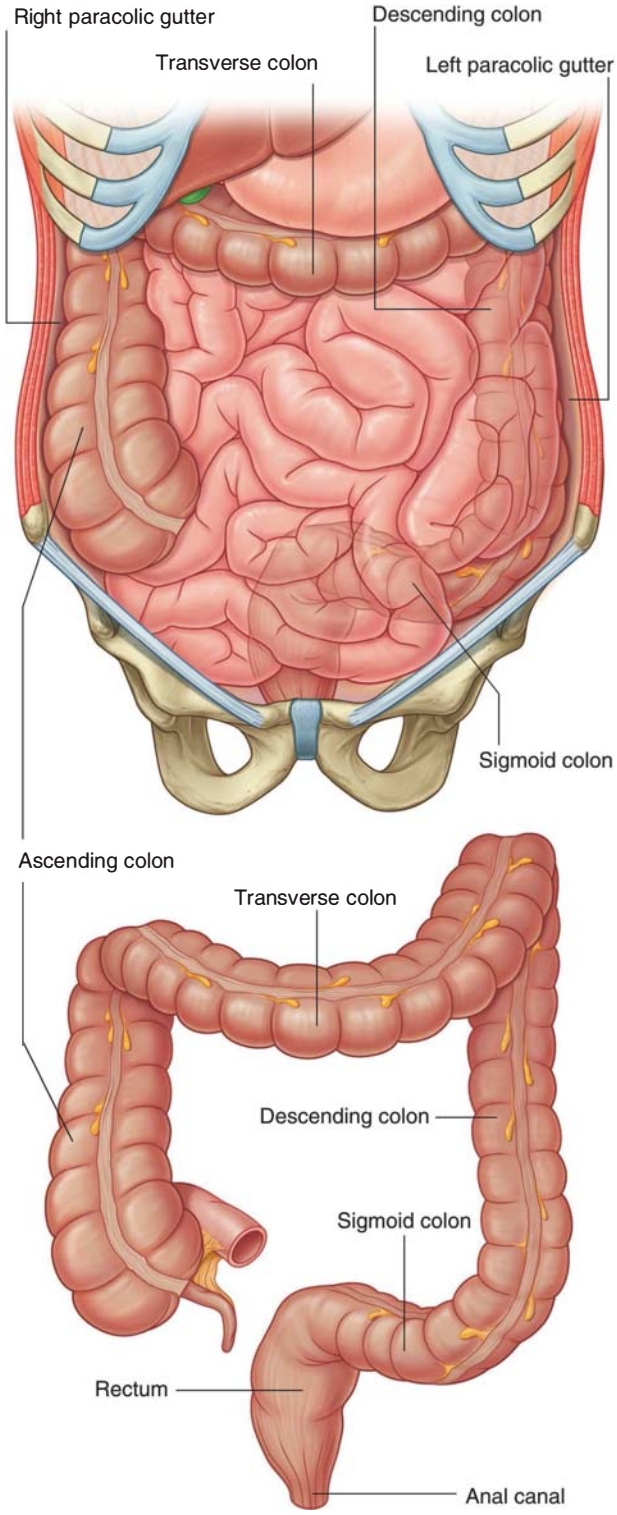


Fig. 4.82 Colon.

Colon

The colon extends superiorly from the cecum and consists of the ascending, transverse, descending, and sigmoid colon (Fig. 4.82). Its ascending and descending segments are (secondarily) retroperitoneal and its transverse and sigmoid segments are intraperitoneal.

At the junction of the ascending and transverse colon is the right colic flexure, which is just inferior to the right lobe of the liver (Fig. 4.83). A similar, but more acute bend (the left colic flexure) occurs at the junction of the transverse and descending colon. This bend is just inferior to the

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spleen, is higher and more posterior than the right colic flexure, and is attached to the diaphragm by the phrenicocolic ligament.

Immediately lateral to the ascending and descending colon are the **right** and **left paracolic gutters** (Fig. 4.82). These depressions are formed between the lateral margins of the ascending and descending colon and the posterolateral abdominal wall and are gutters through which material can pass from one region of the peritoneal cavity to another. Because major vessels and lymphatics are on the medial or posteromedial sides of the ascending and descending colon, a relatively blood-free mobilization of the ascending and descending colon is possible by cutting the peritoneum along these lateral paracolic gutters.

The final segment of the colon (the sigmoid colon) begins above the pelvic inlet and extends to the level of vertebra SIII, where it is continuous with the rectum (Fig. 4.82). This S-shaped structure is quite mobile except at its beginning, where it continues from the descending colon, and at its end, where it continues as the rectum. Between these points, it is suspended by the sigmoid mesocolon.

The arterial supply to the ascending colon (Fig. 4.84) includes:

- the colic branch from the ileocolic artery (from the superior mesenteric artery),
- the anterior cecal artery from the ileocolic artery (from the superior mesenteric artery),
- the posterior cecal artery from the ileocolic artery (from the superior mesenteric artery), and
- the right colic artery from the superior mesenteric artery.

The arterial supply to the transverse colon (Fig. 4.84) includes:

- the right colic artery from the superior mesenteric artery,
- the middle colic artery from the superior mesenteric artery, and
- the left colic artery from the inferior mesenteric artery.

The arterial supply to the descending colon (Fig. 4.84) includes the left colic artery from the inferior mesenteric artery.

The arterial supply to the sigmoid colon (Fig. 4.84) includes sigmoidal arteries from the inferior mesenteric artery.

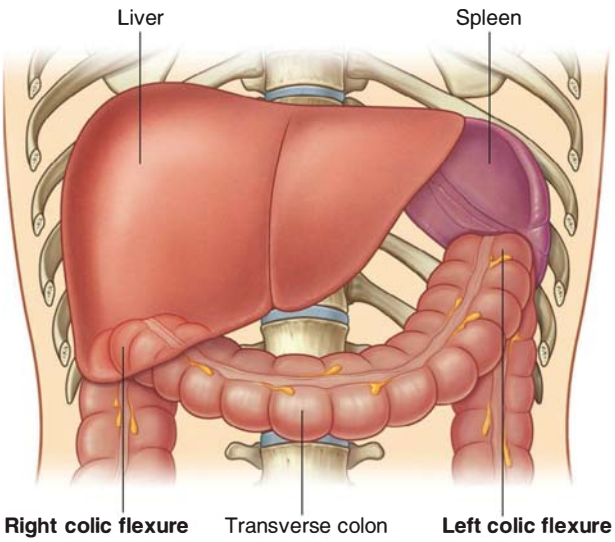


Fig. 4.83 Right and left colic flexures.

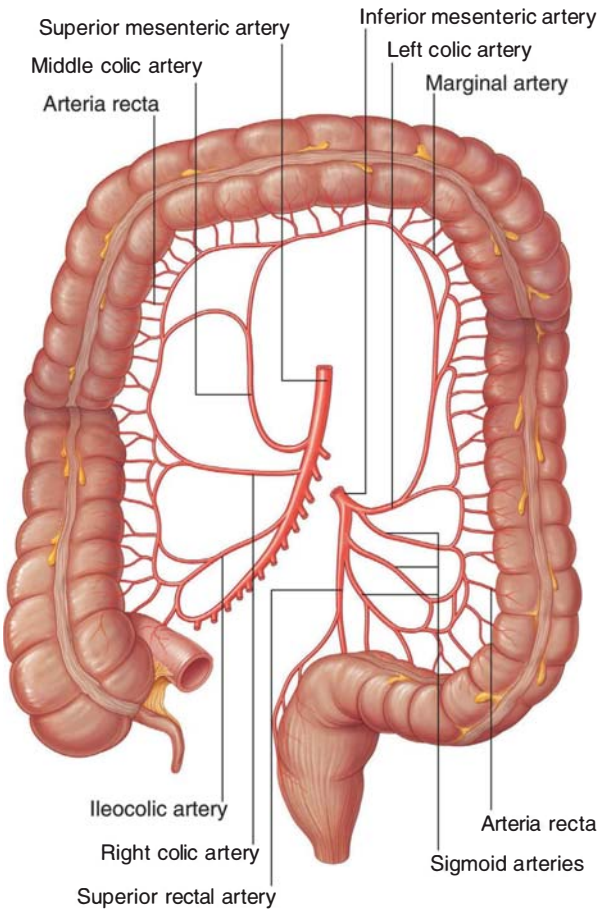


Fig. 4.84 Arterial supply to the colon.

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Rectum and anal canal

Extending from the sigmoid colon is the rectum (Fig. 4.85). The rectosigmoid junction is usually described as being at the level of vertebra SIII or at the end of the sigmoid mesocolon because the rectum is a retroperitoneal structure.

The anal canal is the continuation of the large intestine inferior to the rectum.

The arterial supply to the rectum and anal canal (Fig. 4.86) includes:

- the superior rectal artery from the inferior mesenteric artery,
- the middle rectal artery from the internal iliac artery, and
- the inferior rectal artery from the internal pudendal artery (from the internal iliac artery).

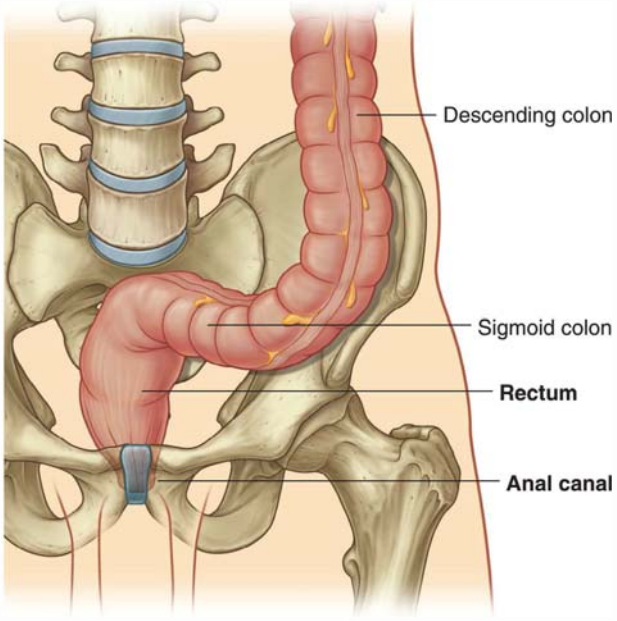


Fig. 4.85 Rectum and anal canal.

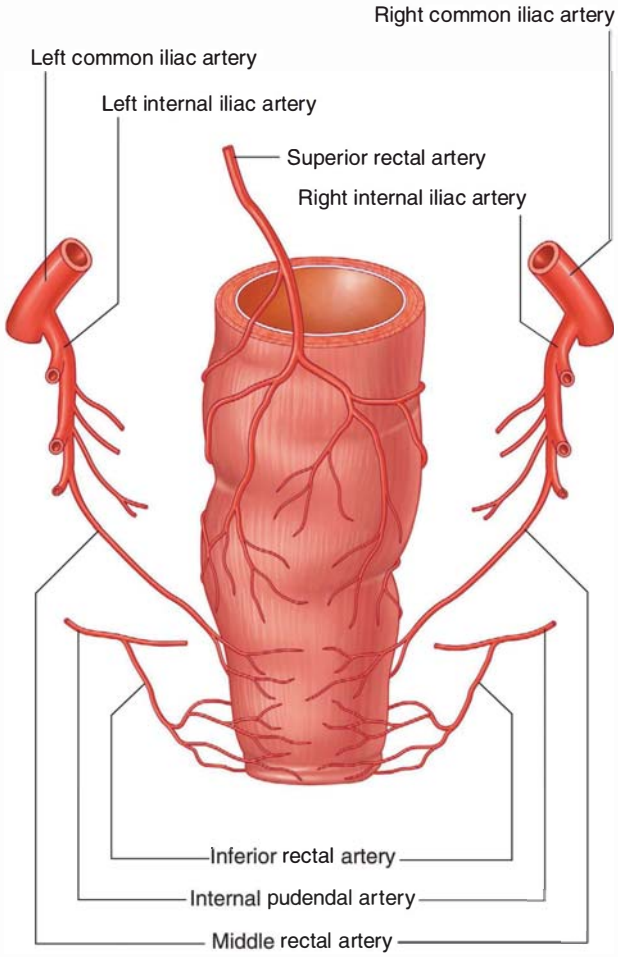


Fig. 4.86 Arterial supply to the rectum and anal canal. Posterior view.

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In the clinic

Congenital disorders of the gastrointestinal tract

The normal positions of the abdominal viscera result from a complex series of rotations that the gut tube undergoes and from the growth of the abdominal cavity to accommodate changes in the size of the developing organs (see pp. 265-268). A number of developmental anomalies can occur during gut development, many of which appear in the neonate or infant, and some of which are surgical emergencies. Occasionally, such disorders are diagnosed only in adults.

Malrotation and midgut volvulus

Malrotation is incomplete rotation and fixation of the midgut after it has passed from the umbilical sac and returned to the abdominal coelom (Figs. 4.87 and 4.88).

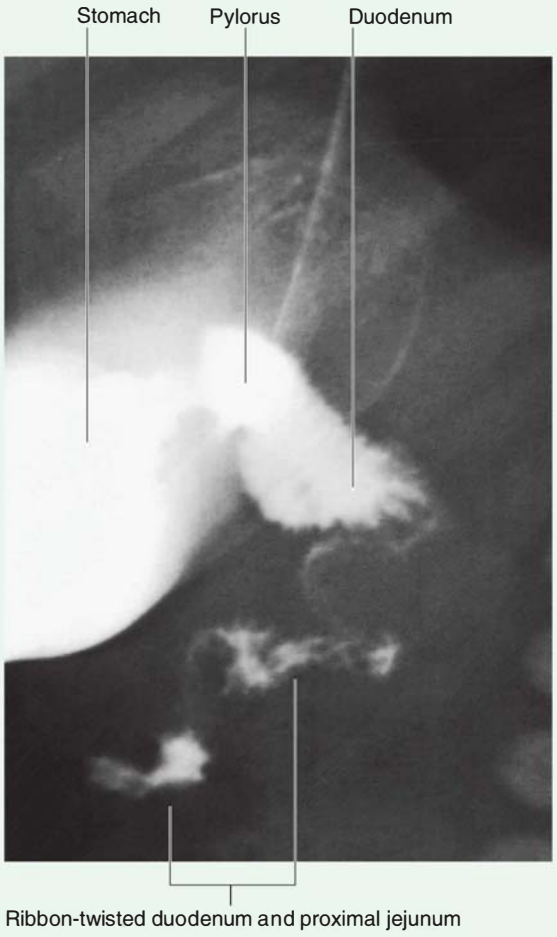


Fig. 4.87 Small bowel malrotation and volvulus. Radiograph of stomach, duodenum, and upper jejunum using barium.

The proximal attachment of the small bowel mesentery begins at the **suspensory muscle of duodenum (ligament of Treitz)**, which determines the position of the duodenojejunal junction. The mesentery of the small bowel ends at the level of the ileocecal junction in the right lower quadrant. This long line of fixation of the mesentery prevents accidental twists of the gut.

If the duodenojejunal flexure or the cecum does not end up in its usual site, the origin of the small bowel mesentery shortens, which permits twisting of the small bowel around the axis of the superior mesenteric artery. Twisting of the bowel, in general, is termed **volvulus**. Volvulus of the small bowel may lead to a reduction of blood flow and infarction.

In some patients, the cecum ends up in the midabdomen. From the cecum and the right side of the colon a series of peritoneal folds (**Ladd's bands**) develop that extend to the right undersurface of the liver and compress the duodenum. A small bowel volvulus may then occur as well as duodenal obstruction. Emergency surgery may be necessary to divide the bands.

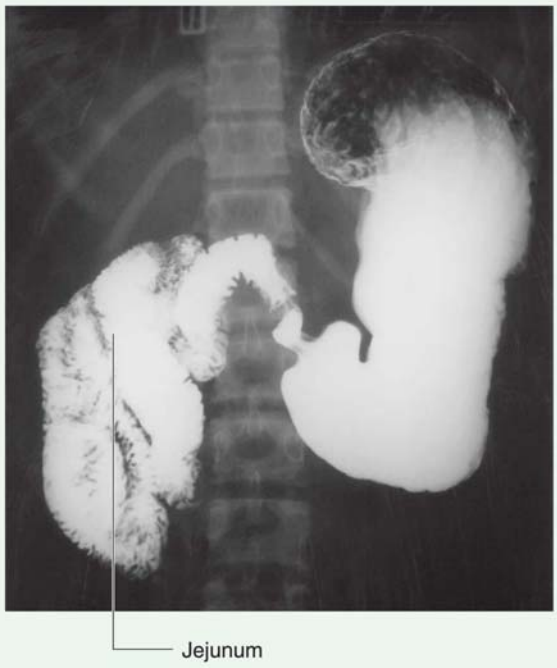


Fig. 4.88 Small bowel malrotation. Radiograph of stomach, duodenum, and jejunum using barium.

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In the clinic

Bowel obstruction

A bowel obstruction can be either functional or due to a true obstruction. Mechanical obstruction is caused by an intraluminal, mural, or extrinsic mass which can be secondary to a foreign body, obstructing tumor in the wall, or extrinsic compression from an adhesion, or embryological band (Fig. 4.89).

A functional obstruction is usually due to an inability of the bowel to peristalse, which again has a number of causes, and most frequently is a postsurgical state due to excessive intraoperative bowel handling. Other causes may well include abnormality of electrolytes (e.g., sodium and potassium) rendering the bowel paralyzed until correction has occurred.

The signs and symptoms of obstruction depend on the level at which the obstruction has occurred. The primary symptom is central abdominal, intermittent, colicky pain as the peristaltic waves try to overcome the obstruction. Abdominal distention will occur if it is a low obstruction (distal), allowing more proximal loops of bowel to fill with fluid. A high obstruction (in the proximal small bowel) may not produce abdominal distention.

Vomiting and absolute constipation, including the inability to pass flatus, will ensue.

Early diagnosis is important because considerable fluid and electrolytes enter the bowel lumen and fail to be reabsorbed, which produces dehydration and electrolyte abnormalities. Furthermore, the bowel continues to distend, compromising the blood supply within the bowel wall, which may lead to ischemia and perforation. The symptoms and signs are variable and depend on the level of obstruction.

Small bowel obstruction is typically caused by adhesions following previous surgery, and history should always be sought for any operations or abdominal interventions (e.g., previous appendectomy). Other causes include bowel passing into hernias (e.g., inguinal) and bowel twisting on its own mesentery (volvulus). Examination of hernial orifices is mandatory in patients with bowel obstruction.

Large bowel obstruction is commonly caused by a tumor. Other potential causes include hernias and inflammatory diverticular disease of the sigmoid colon.

The treatment is intravenous replacement of fluid and electrolytes, analgesia, and relief of obstruction. The passage of a nasogastric tube allows aspiration of fluid from the stomach. In many instances, small bowel obstruction, typically secondary to adhesions, will settle with nonoperative management. Large bowel obstruction may require an urgent operation to remove the obstructing lesion, or a temporary bypass procedure (e.g., defunctioning colostomy) (Fig. 4.90).



Fig. 4.89 This radiograph of the abdomen, anteroposterior view, demonstrates a number of dilated loops of small bowel. Small bowel can be identified by the plicae circulares that pass from wall to wall as indicated. The large bowel is not dilated. The cause of the small bowel dilatation is an adhesion after pelvic surgery.

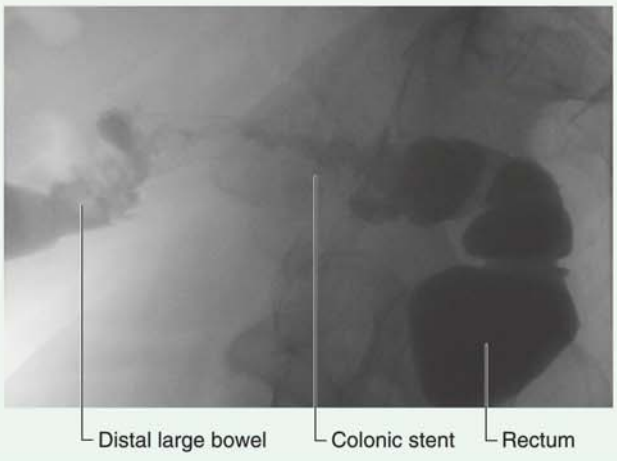


Fig. 4.90 This oblique radiograph demonstrates contrast passing through a colonic stent that has been placed to relieve bowel obstruction prior to surgery.

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In the clinic

Diverticular disease

Diverticular disease is the development of multiple colonic diverticula, predominantly throughout the sigmoid colon, though the whole colon may be affected (Fig. 4.91). The sigmoid colon has the smallest diameter of any portion of the colon and is therefore the site where intraluminal pressure is potentially the highest. Poor dietary fiber intake and obesity are also linked to diverticular disease.

The presence of multiple diverticula does not necessarily mean the patient requires any treatment. Moreover, many patients have no other symptoms or signs.

Patients tend to develop symptoms and signs when the neck of the diverticulum becomes obstructed by feces and becomes infected. Inflammation may spread along the wall, causing abdominal pain. When the sigmoid colon becomes inflamed (diverticulitis), abdominal pain and fever ensue.

Because of the anatomical position of the sigmoid colon there are a number of complications that may occur. The diverticula can perforate to form an abscess in the pelvis. The inflammation may produce an inflammatory mass, obstructing the left ureter. Inflammation may also spread to the bladder, producing a fistula between the sigmoid colon and the bladder. In these circumstances patients may develop a urinary tract infection and rarely have fecal material and gas passing per urethra.

The diagnosis is based upon clinical examination and often CT scanning. In the first instance, patients will be treated with antibiotic therapy; however, a surgical resection may be necessary if symptoms persist.



Fig. 4.91 This double-contrast barium enema demonstrates numerous small outpouchings throughout the distal large bowel predominantly within the descending colon and the sigmoid colon. These small outpouchings are diverticula and in most instances remain quiescent.

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In the clinic

Ostomies

It is occasionally necessary to surgically externalize bowel to the anterior abdominal wall. Externalization of bowel plays an important role in patient management. These extraanatomical bypass procedures use our anatomical knowledge and in many instances are life saving.

Gastrostomy

Gastrostomy is performed when the stomach is attached to the anterior abdominal wall and a tube is placed through the skin into the stomach. Typically this is performed to feed the patient when it is impossible to take food and fluid orally (e.g., complex head and neck cancer). The procedure can be performed either surgically or through a direct needlestick puncture under sedation in the anterior abdominal wall.

Jejunostomy

Similarly the jejunum is brought to the anterior abdominal wall and fixed. The jejunostomy is used as a site where a feeding tube is placed through the anterior abdominal wall into the proximal efferent small bowel.

Ileostomy

An ileostomy is performed when small bowel contents need to be diverted from the distal bowel. An ileostomy is often performed to protect a distal surgical anastomosis, such as in the colon to allow healing after surgery.

Colostomy

There are a number of instances when a colostomy may be necessary. In many circumstances it is performed to protect the distal large bowel after surgery. A further

(continues)

In the clinic—cont'd

indication would include large bowel obstruction with imminent perforation wherein a colostomy allows decompression of the bowel and its contents. This is a safe and temporizing procedure performed when the patient is too unwell for extensive bowel surgery. It is relatively straightforward and carries reduced risk, preventing significant morbidity and mortality.

An end colostomy is necessary when the patient has undergone a surgical resection of the rectum and anus (typically for cancer).

Ileal conduit

An ileal conduit is an extraanatomical procedure and is performed after resection of the bladder for tumor. In this situation a short segment of small bowel is identified. The bowel is divided twice to produce a 20-cm segment of small bowel on its own mesentery. This isolated segment of bowel is used as a conduit. The remaining bowel is joined together. The proximal end is anastomosed to the ureters, and the distal end is anastomosed to the anterior abdominal wall. Hence, urine passes from the kidneys into the ureters and through the short segment of small bowel to the anterior abdominal wall.

When patients have either an ileostomy, colostomy, or ileal conduit it is necessary for them to fix a collecting bag onto the anterior abdominal wall. Contrary to one's initial thoughts these bags are tolerated extremely well by most patients and allow patients to live a nearly normal and healthy life.

Liver

The liver is the largest visceral organ in the body and is primarily in the right hypochondrium and epigastric region, extending into the left hypochondrium (or in the right upper quadrant, extending into the left upper quadrant) (Fig. 4.92).

Surfaces of the liver include:

- a **diaphragmatic surface** in the anterior, superior, and posterior directions; and
- a **visceral surface** in the inferior direction (Fig. 4.93).

Diaphragmatic surface

The diaphragmatic surface of the liver, which is smooth and domed, lies against the inferior surface of the diaphragm (Fig. 4.94). Associated with it are the subphrenic and hepatorenal recesses (Fig. 4.93):

- The **subphrenic recess** separates the diaphragmatic surface of the liver from the diaphragm and is divided into right and left areas by the **falciform ligament**, a structure derived from the ventral mesentery in the embryo.
- The hepatorenal recess is a part of the peritoneal cavity on the right side between the liver and the right kidney and right suprarenal gland.

The subphrenic and hepatorenal recesses are continuous anteriorly.

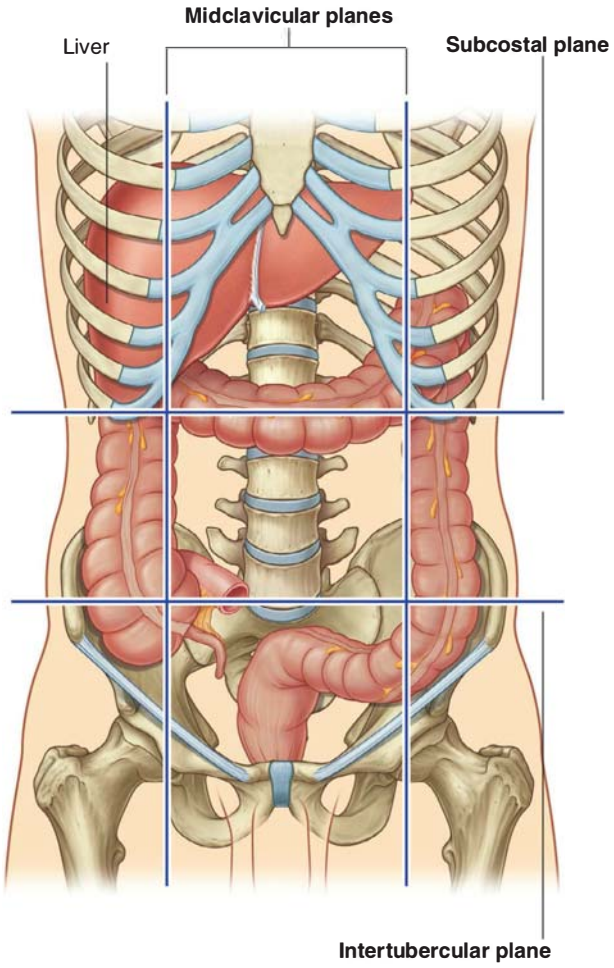


Fig. 4.92 Position of the liver in the abdomen.

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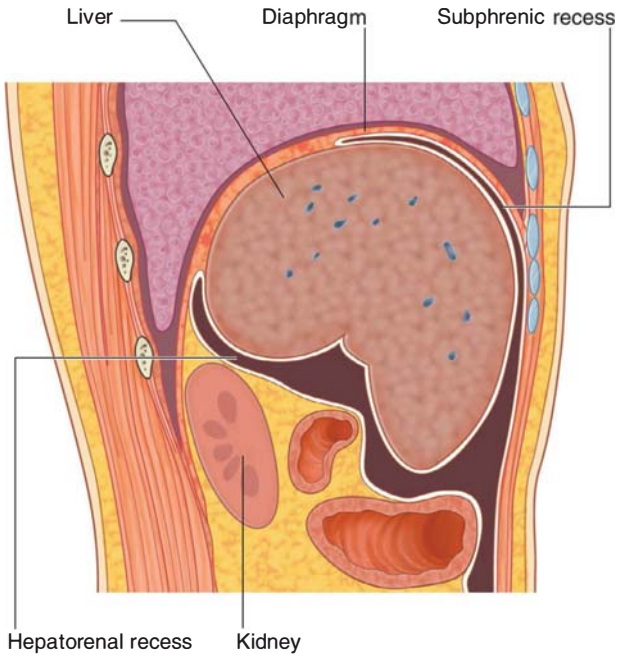


Fig. 4.93 Surfaces of the liver and recesses associated with the liver.

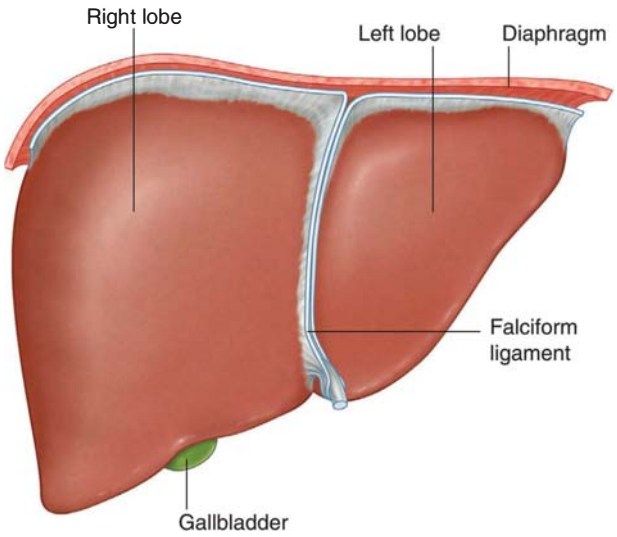


Fig. 4.94 Diaphragmatic surface of the liver.

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Visceral surface

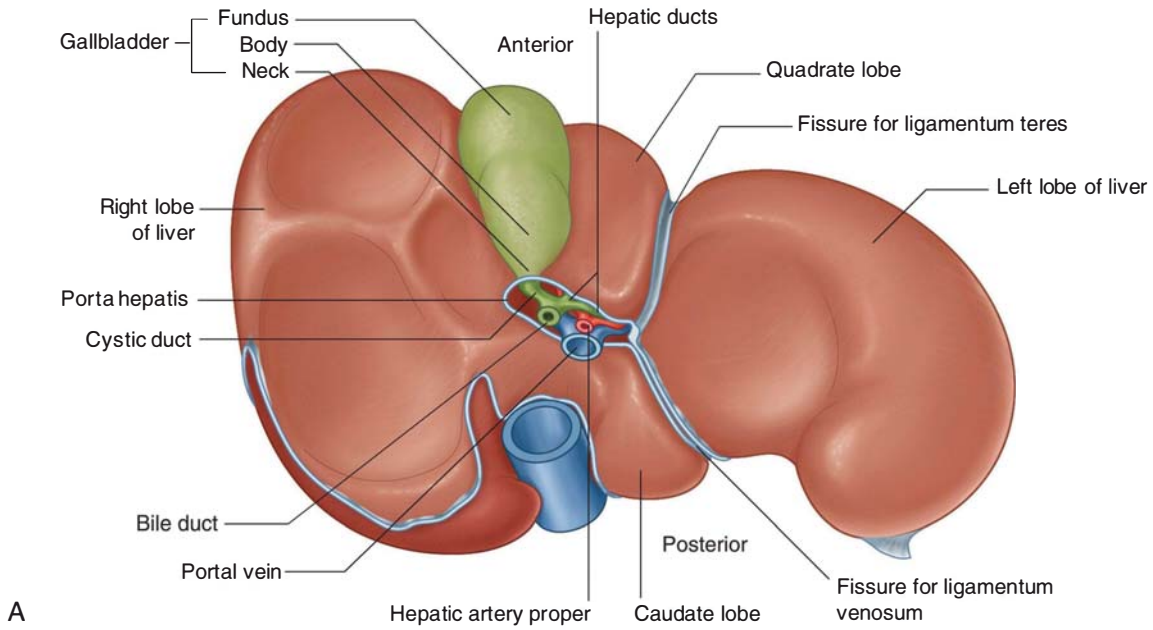
The visceral surface of the liver is covered with visceral peritoneum except in the **fossa for the gallbladder** and at the **porta hepatis** (gateway to the liver; Fig. 4.95), and structures related to it include the following (Fig. 4.96):

- esophagus,
- right anterior part of the stomach,
- superior part of the duodenum,
- lesser omentum,
- gallbladder,
- right colic flexure,
- right transverse colon,
- right kidney, and
- right suprarenal gland.

The **porta hepatis** serves as the point of entry into the liver for the hepatic arteries and the portal vein, and the exit point for the hepatic ducts (Fig. 4.95).

Associated ligaments

The liver is attached to the anterior abdominal wall by the **falciform ligament** and, except for a small area of the



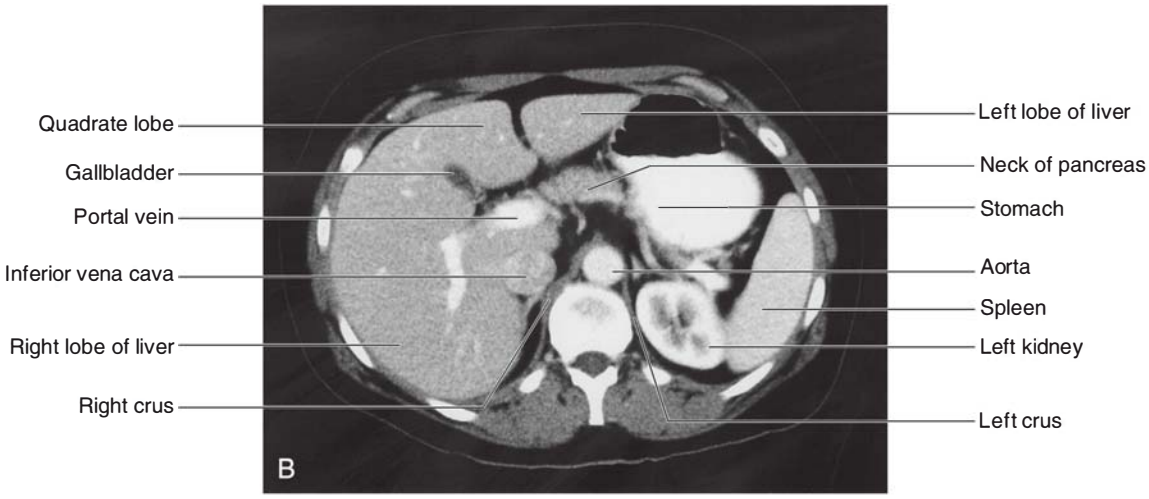


Fig. 4.95 Visceral surface of the liver. **A.** Illustration. **B.** Abdominal computed tomogram, with contrast, in the axial plane.

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liver against the diaphragm (the **bare area**), the liver is almost completely surrounded by visceral peritoneum (Fig. 4.96). Additional folds of peritoneum connect the liver to the stomach (**hepatogastric ligament**), the duodenum (**hepatoduodenal ligament**), and the diaphragm (**right** and **left triangular ligaments** and **anterior** and **posterior coronary ligaments**).

The bare area of the liver is a part of the liver on the diaphragmatic surface where there is no intervening peritoneum between the liver and the diaphragm (Fig. 4.96):

- The anterior boundary of the bare area is indicated by a reflection of peritoneum—the anterior coronary ligament.
- The posterior boundary of the bare area is indicated by a reflection of peritoneum—the posterior coronary ligament.
- Where the coronary ligaments come together laterally, they form the right and left triangular ligaments.

Lobes

The liver is divided into right and left lobes by fossae for the gallbladder and the inferior vena cava (Fig. 4.95). The **right lobe of the liver** is the largest lobe, whereas the **left lobe of the liver** is smaller. The quadrate and caudate lobes are described as arising from the right lobe of the liver but functionally are distinct.

- The **quadrate lobe** is visible on the anterior part of the visceral surface of the liver and is bounded on the left by the fissure for the ligamentum teres and on the right by the fossa for the gallbladder. Functionally, it is related to the left lobe of the liver.
- The **caudate lobe** is visible on the posterior part of the visceral surface of the liver. It is bounded on the left by the fissure for the ligamentum venosum and on the right by the groove for the inferior vena cava. Functionally, it is separate from the right and the left lobes of the liver.

The arterial supply to the liver includes:

- the right hepatic artery from the hepatic artery proper (a branch of the common hepatic artery from the celiac trunk), and
- the left hepatic artery from the hepatic artery proper (a branch of the common hepatic artery from the celiac trunk).

Gallbladder

The **gallbladder** is a pear-shaped sac lying on the visceral surface of the right lobe of the liver in a fossa between the right and quadrate lobes (Fig. 4.95). It has:

- a rounded end (**fundus of the gallbladder**), which may project from the inferior border of the liver;

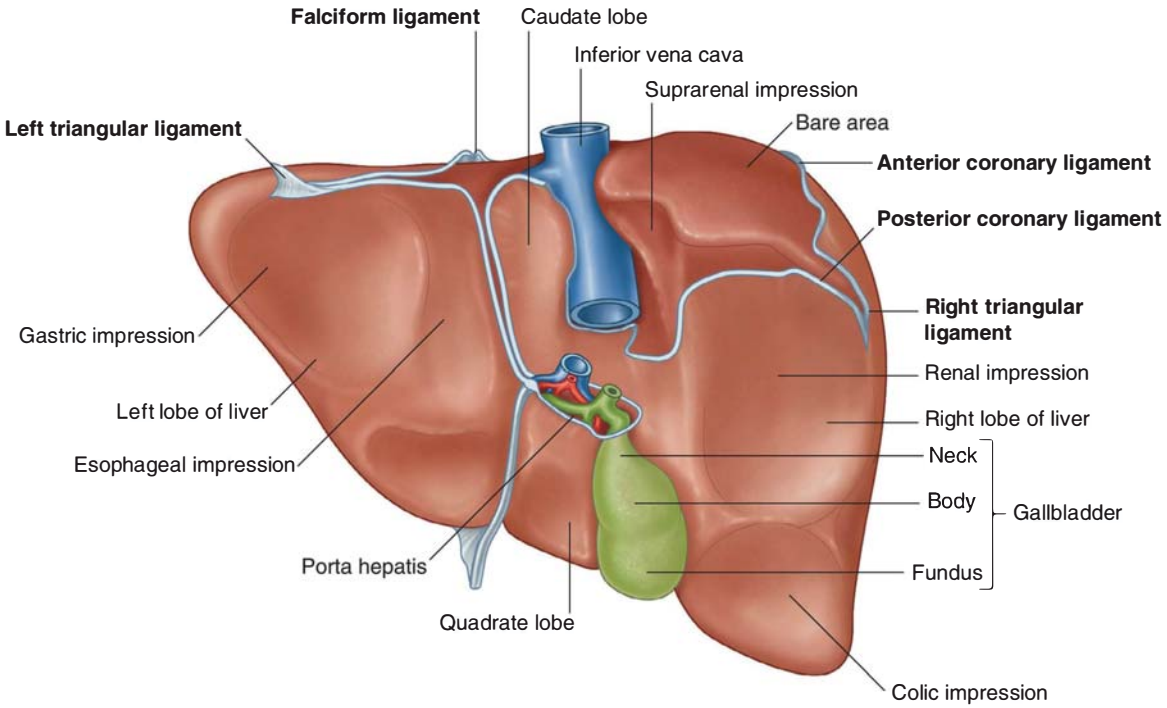


Fig. 4.96 Posterior view of the bare area of the liver and associated ligaments.

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- a major part in the fossa (**body of the gallbladder**), which may be against the transverse colon and the superior part of the duodenum; and
- a narrow part (**neck of the gallbladder**) with mucosal folds forming the spiral fold.

The arterial supply to the gallbladder (Fig. 4.97) is the cystic artery from the right hepatic artery (a branch of the hepatic artery proper).

The gallbladder receives, concentrates, and stores bile from the liver.

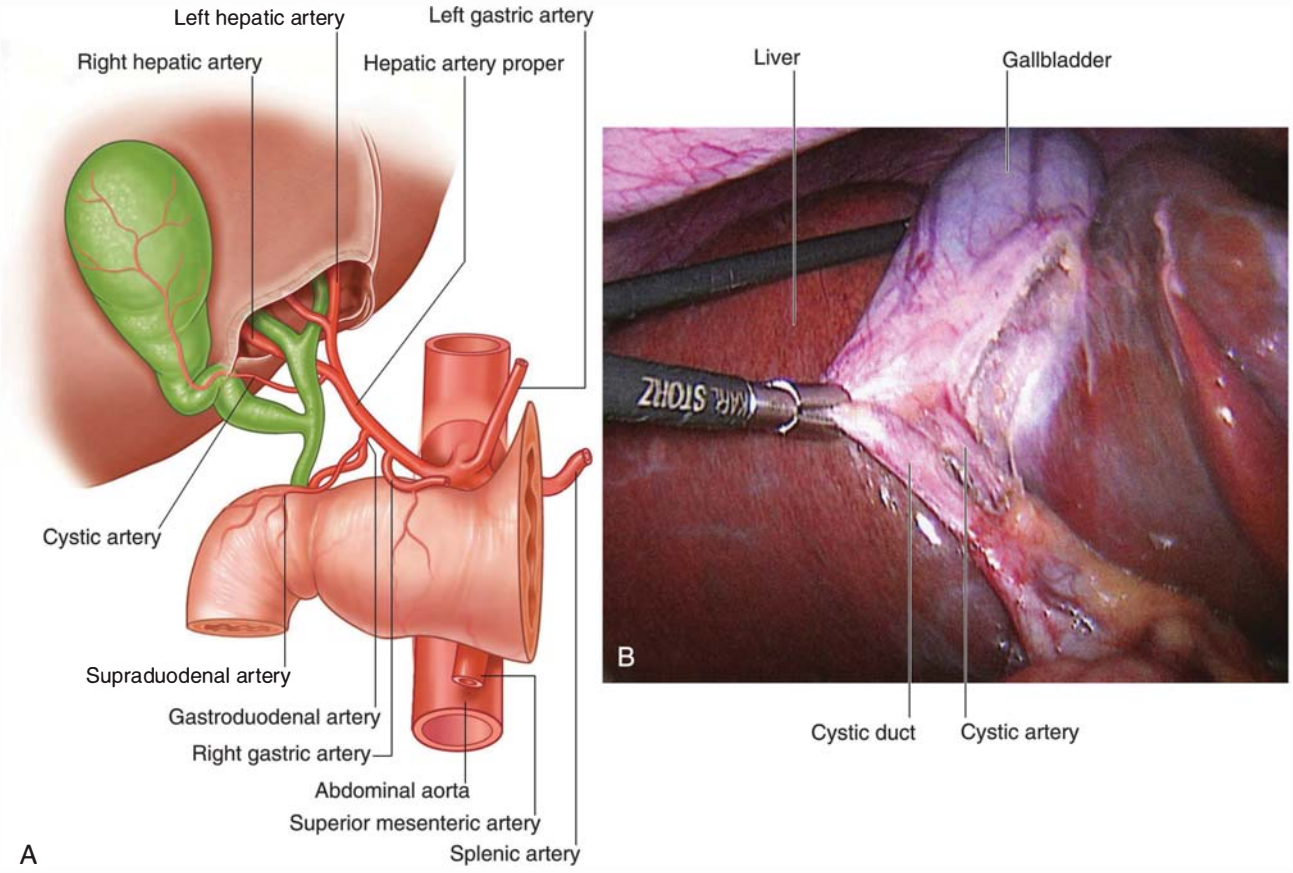


Fig. 4.97 Arterial supply to the liver and gallbladder. **A.** Schematic. **B.** Laparoscopic surgical view of cystic duct and cystic artery.

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Pancreas

The pancreas lies mostly posterior to the stomach (Figs. 4.98 and 4.99). It extends across the posterior abdominal wall from the duodenum, on the right, to the spleen, on the left.

The pancreas is (secondarily) retroperitoneal except for a small part of its tail and consists of a head, uncinate process, neck, body, and tail.

- The **head of the pancreas** lies within the C-shaped concavity of the duodenum.

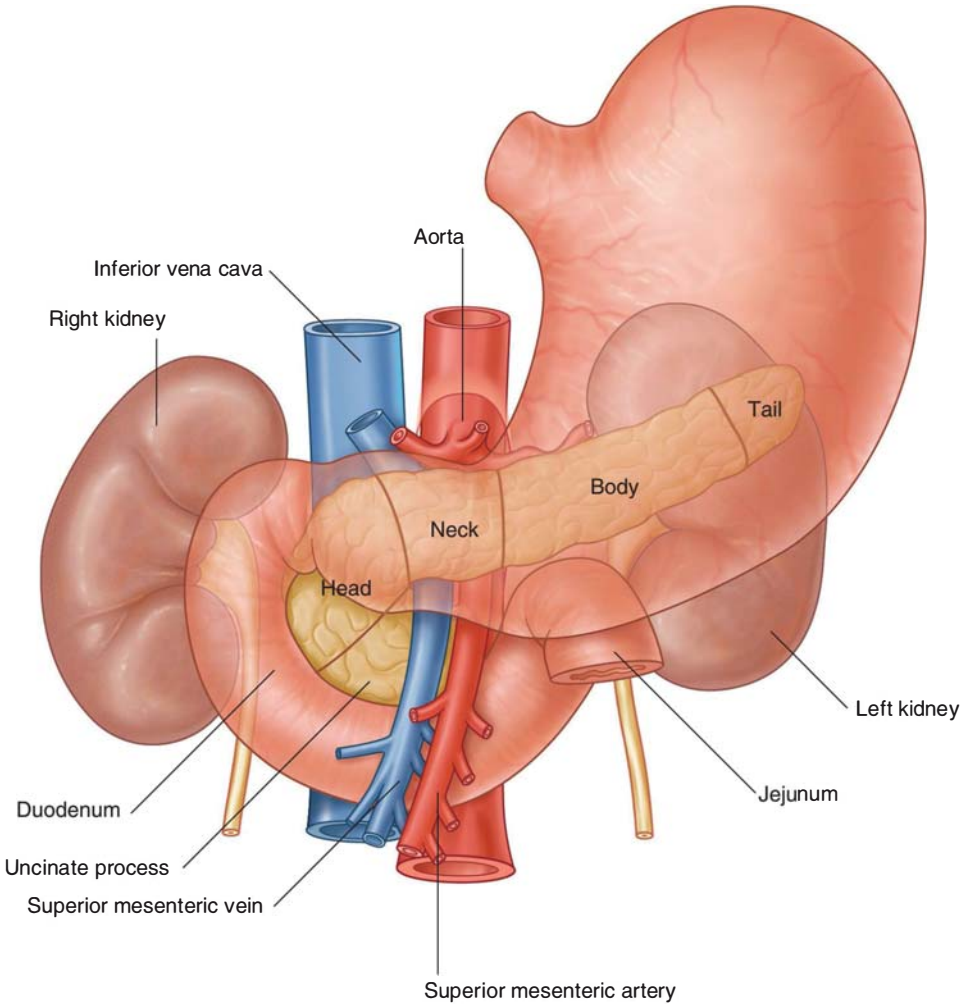


Fig. 4.98 Pancreas.

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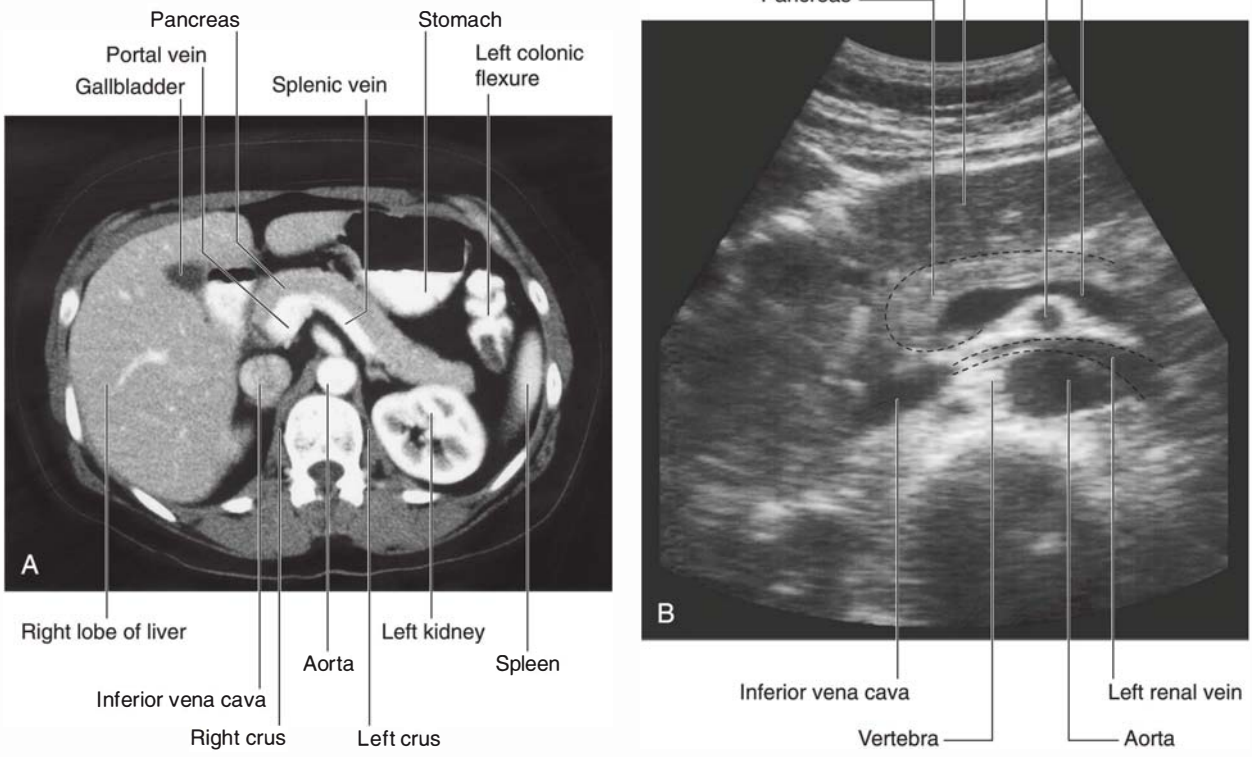


Fig. 4.99 Abdominal images. **A.** Abdominal computed tomogram, with contrast, in the axial plane. **B.** Abdominal ultrasound scan.

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- Projecting from the lower part of the head is the **uncinate process**, which passes posterior to the superior mesenteric vessels.
- The **neck of the pancreas** is anterior to the superior mesenteric vessels. Posterior to the neck of the pancreas, the superior mesenteric and splenic veins join to form the portal vein.
- The **body of the pancreas** is elongate and extends from the neck to the tail of the pancreas.
- The **tail of the pancreas** passes between layers of the splenorenal ligament.

The **pancreatic duct** begins in the tail of the pancreas (Fig. 4.100). It passes to the right through the body of the pancreas and, after entering the head of the pancreas, turns inferiorly. In the lower part of the head of the pancreas, the pancreatic duct joins the bile duct. The joining of these two structures forms the **hepatopancreatic ampulla** (ampulla of Vater), which enters the descending (second) part of the duodenum at the **major duodenal papilla**. Surrounding the ampulla is the **sphincter of ampulla** (sphincter of Oddi), which is a collection of smooth muscles.

The **accessory pancreatic duct** empties into the duodenum just above the major duodenal papilla at the **minor duodenal papilla** (Fig. 4.100). If the accessory duct is followed from the minor papilla into the head of the pancreas, a branch point is discovered:

- One branch continues to the left, through the head of the pancreas, and may connect with the pancreatic duct at the point where it turns inferiorly.
- A second branch descends into the lower part of the head of the pancreas, anterior to the pancreatic duct, and ends in the uncinate process.

The main and accessory pancreatic ducts usually communicate with each other. The presence of these two ducts reflects the embryological origin of the pancreas from dorsal and ventral buds from the foregut.

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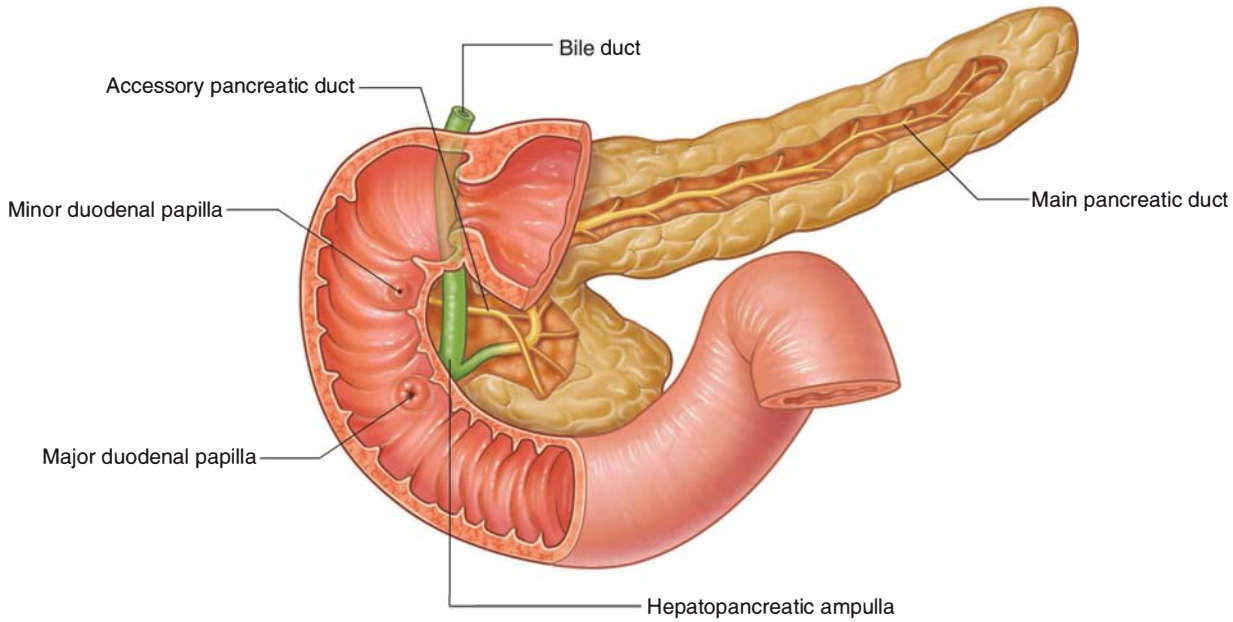


Fig. 4.100 Pancreatic duct system.

The arterial supply to the pancreas (Fig. 4.101) includes the:

- gastroduodenal artery from the common hepatic artery (a branch of the celiac trunk),
- anterior superior pancreaticoduodenal artery from the gastroduodenal artery,
- posterior superior pancreaticoduodenal artery from the gastroduodenal artery,
- dorsal pancreatic artery from the inferior pancreatic artery (a branch of the splenic artery),
- great pancreatic artery from the inferior pancreatic artery (a branch of the splenic artery),
- anterior inferior pancreaticoduodenal artery from the inferior pancreaticoduodenal artery (a branch of the superior mesenteric artery), and
- posterior inferior pancreaticoduodenal artery from the inferior pancreaticoduodenal artery (a branch of the superior mesenteric artery).

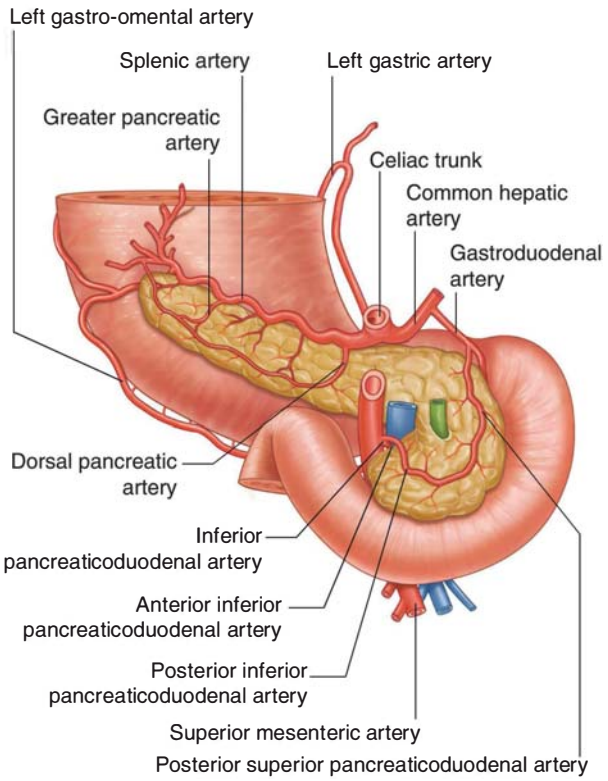


Fig. 4.101 Arterial supply to the pancreas. Posterior view.

In the clinic

Annular pancreas

The pancreas develops from ventral and dorsal buds from the foregut. The dorsal bud forms most of the head, neck, and body of the pancreas. The ventral bud rotates around the bile duct to form part of the head and the uncinate process. If the ventral bud splits (becomes bifid), the two segments may encircle the duodenum. The duodenum is therefore constricted and may even undergo atresia, and be absent at birth because of developmental problems. After birth, the child may fail to thrive and may vomit due to poor gastric emptying.

Sometimes an annular pancreas is diagnosed in utero by ultrasound scanning. The obstruction of the duodenum may prevent the fetus from swallowing enough amniotic fluid, which may increase the overall volume of amniotic fluid in the amniotic sac surrounding the fetus (**polyhydramnios**).

In the clinic

Pancreatic cancer

Pancreatic cancer accounts for a significant number of deaths and is often referred to as the "silent killer." Malignant tumors of the pancreas may occur anywhere within the pancreas but are most frequent within the head and the neck. There are a number of nonspecific findings in patients with pancreatic cancer, including upper abdominal pain, loss of appetite, and weight loss. Depending on the exact site of the cancer, obstruction of the bile duct may occur, which can produce obstructive jaundice. Although surgery is indicated in patients where there is a possibility of cure, most detected cancers have typically spread locally, invading the portal vein and superior mesenteric vessels, and may extend into the porta hepatis. Lymph node spread also is common and these factors would preclude curative surgery.

Given the position of the pancreas, a surgical resection is a complex procedure involving resection of the region of pancreatic tumor usually with part of the duodenum, necessitating a complex bypass procedure.

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Duct system for bile

The duct system for the passage of bile extends from the liver, connects with the gallbladder, and empties into the descending part of the duodenum (Fig. 4.102). The coalescence of ducts begins in the liver parenchyma and continues until the **right** and **left hepatic ducts** are formed. These drain the respective lobes of the liver.

The two hepatic ducts combine to form the **common hepatic duct**, which runs near the liver, with the hepatic artery proper and portal vein in the free margin of the lesser omentum.

As the common hepatic duct continues to descend, it is joined by the **cystic duct** from the gallbladder. This completes the formation of the **bile duct**. At this point, the bile duct lies to the right of the hepatic artery proper and usually to the right of, and anterior to, the portal vein in the free margin of the lesser omentum. The **omental foramen** is posterior to these structures at this point.

The bile duct continues to descend, passing posteriorly to the superior part of the duodenum before joining with the pancreatic duct to enter the descending part of the duodenum at the major duodenal papilla (Fig. 4.102).

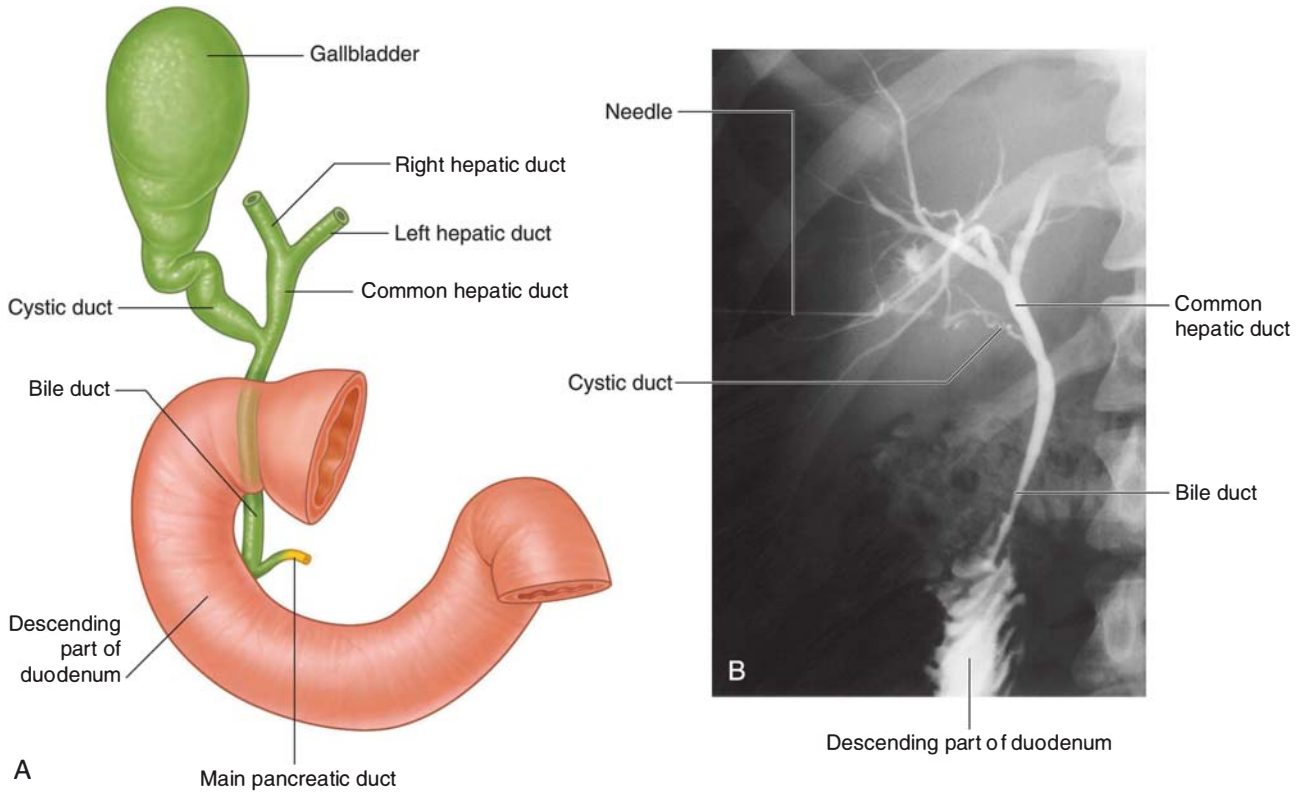


Fig. 4.102 Bile drainage. **A.** Duct system for passage of bile. **B.** Percutaneous transhepatic cholangiogram demonstrating the bile duct system.

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Spleen

The spleen develops as part of the vascular system in the part of the dorsal mesentery that suspends the developing stomach from the body wall. In the adult, the spleen lies against the diaphragm, in the area of rib IX to rib X (Fig. 4.103). It is therefore in the left upper quadrant, or left hypochondrium, of the abdomen.

The spleen is connected to the:

- greater curvature of the stomach by the gastrosplenic ligament, which contains the short gastric and gastro-omental vessels; and
- left kidney by the splenorenal ligament (Fig. 4.104), which contains the splenic vessels.

Both these ligaments are parts of the greater omentum.

The spleen is surrounded by visceral peritoneum except in the area of the hilum on the medial surface of the spleen (Fig. 4.105). The **splenic hilum** is the entry point for the splenic vessels, and occasionally the tail of the pancreas reaches this area.

The arterial supply to the spleen (Fig. 4.106) is the splenic artery from the celiac trunk.

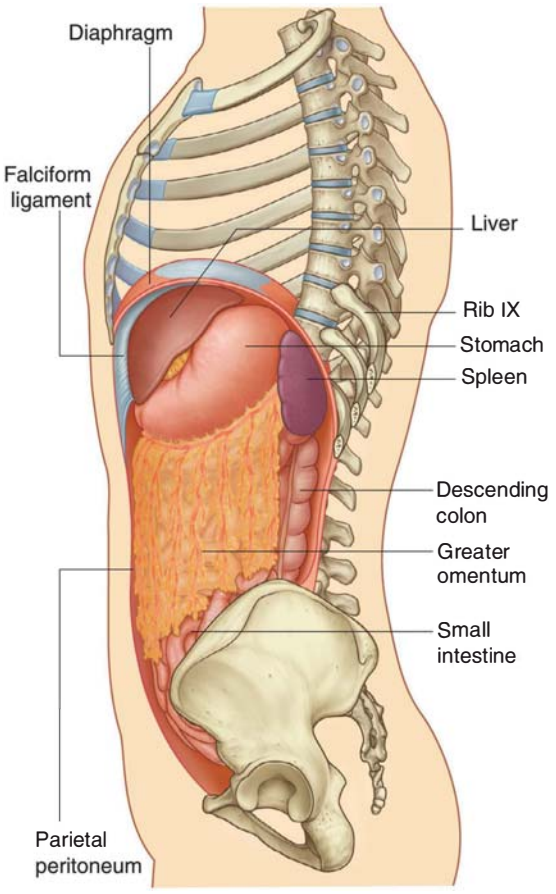


Fig. 4.103 Spleen.

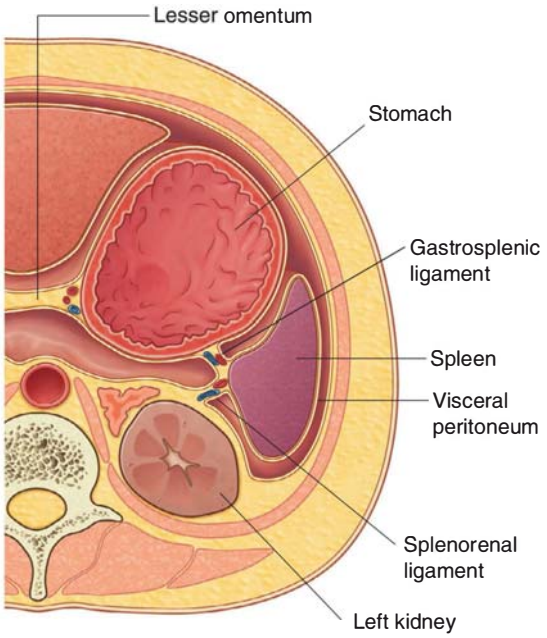


Fig. 4.104 Splenic ligaments and related vasculature.

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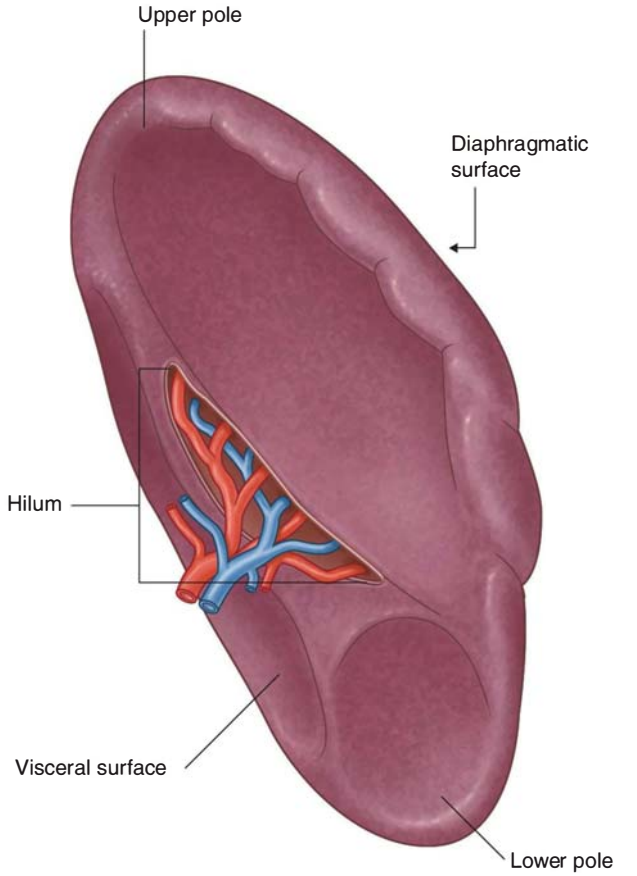


Fig. 4.105 Surfaces and hilum of the spleen.

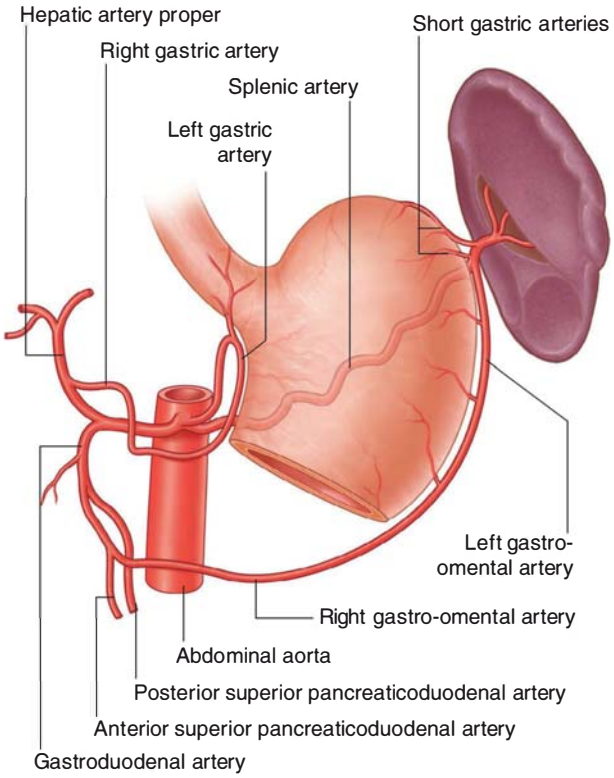


Fig. 4.106 Arterial supply to the spleen.

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In the clinic

Segmental anatomy of the liver

For many years the segmental anatomy of the liver was of little importance. However, since the development of liver resection surgery, the size, shape, and segmental anatomy of the liver have become clinically important, especially with regard to liver resection for metastatic disease. Indeed, with detailed knowledge of the segments, curative surgery can be performed in patients with tumor metastases.

The liver is divided by the **principal plane**, which divides the organ into halves of approximately equal size. This imaginary line is defined by a parasagittal line that passes through the gallbladder fossa to the inferior vena cava. It is in this plane that the middle hepatic vein is found. Importantly, the principal plane divides the left half of the liver from the right half. The lobes of the liver are unequal in size and bear only little relevance to operative anatomy.

(continues)

In the clinic—cont'd

The traditional eight-segment anatomy of the liver relates to the hepatic arterial, portal, and biliary drainage of these segments (Fig. 4.107).

The caudate lobe is defined as segment I, and the remaining segments are numbered in a clockwise fashion up to segment VIII. The features are extremely consistent between individuals.

From a surgical perspective, a right hepatectomy would involve division of the liver in the principal plane in which segments V, VI, VII, and VIII would be removed, leaving segments I, II, III, and IV.

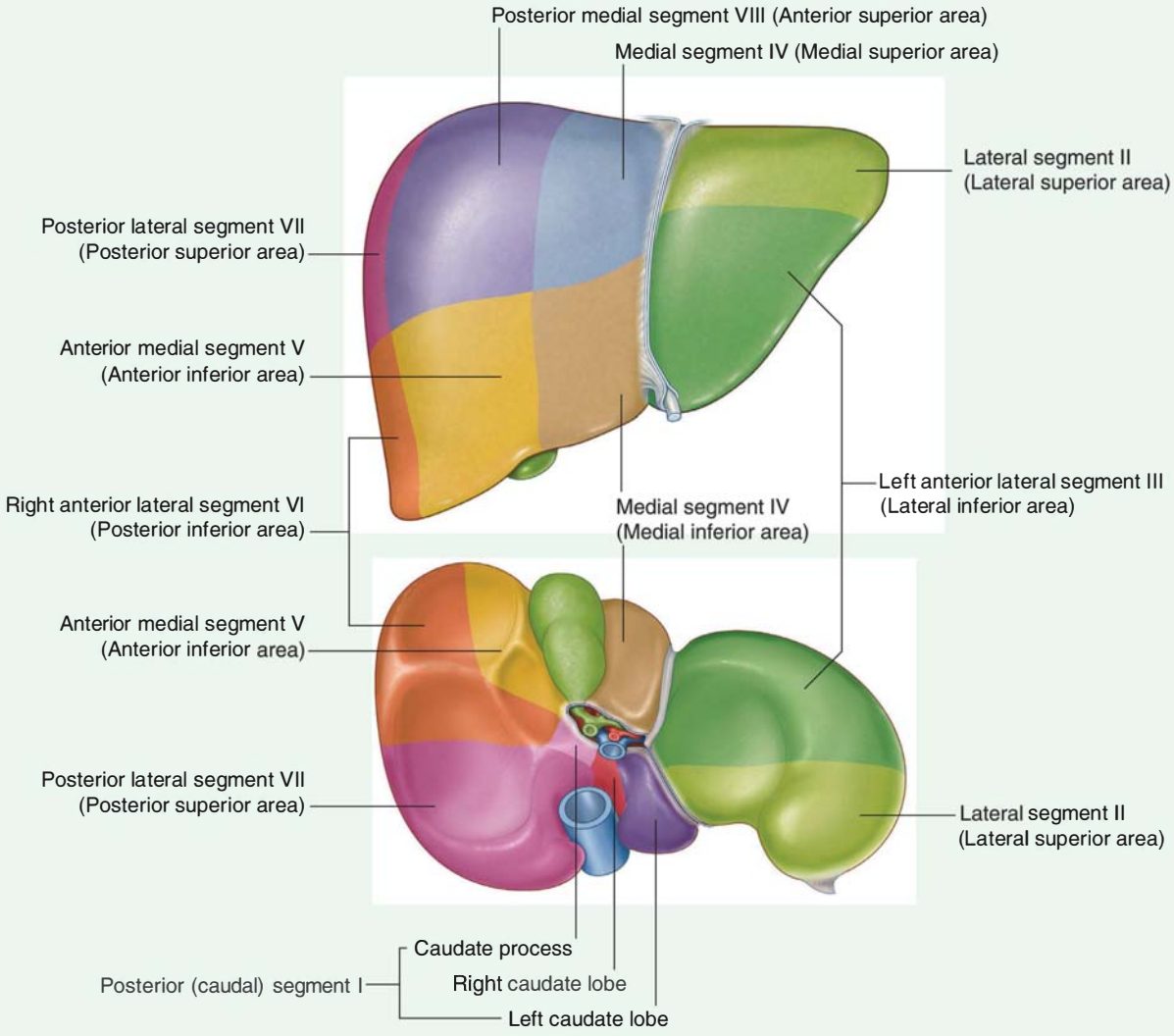


Fig. 4.107 Division of the liver into segments based upon the distributions of the bile ducts and hepatic vessels (Couinaud's segments).

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In the clinic

Gallstones

Gallstones are present in approximately 10% of people over the age of 40 and are more common in women. They consist of a variety of components but are predominantly a mixture of cholesterol and bile pigment. They may undergo calcification, which can be demonstrated on plain radiographs. Gallstones may be visualized incidentally as part of a routine abdominal ultrasound scan (Fig. 4.108) or on a plain radiograph.

From time to time, gallstones impact in the region of **Hartmann's pouch**, which is a bulbous region of the neck of the gallbladder. When the gallstone lodges in this area, the gallbladder cannot empty normally and contractions of the gallbladder wall produce severe pain. If this persists, a **cholecystectomy** (removal of the gallbladder) may be necessary.

Sometimes the gallbladder may become inflamed (**cholecystitis**). If the inflammation involves the related parietal peritoneum of the diaphragm, pain may not only occur in the right upper quadrant of the abdomen but may also be referred to the shoulder on the right side. This referred pain is due to the innervation of the visceral peritoneum of the diaphragm by spinal cord levels (C3 to C5) that also innervate skin over the shoulder. In this case, one somatic sensory region of low sensory output (diaphragm) is referred to another somatic sensory region of high sensory output (dermatomes).



Fig. 4.108 Gallbladder containing multiple stones. Ultrasound scan.

From time to time, small gallstones pass into the bile duct and are trapped in the region of the sphincter of the ampulla, which obstructs the flow of bile into the duodenum. This, in turn, produces jaundice.

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In the clinic

Jaundice

Jaundice is a yellow discoloration of the skin caused by excess bile pigment (bilirubin) within the plasma. The yellow color is best appreciated by looking at the normally white sclerae of the eyes, which turn yellow.

The extent of the elevation of the bile pigments and the duration for which they have been elevated account for the severity of jaundice.

Simplified explanation to understanding the types of jaundice and their anatomical causes

When red blood cells are destroyed by the reticuloendothelial system, the iron from the hemoglobin molecule is recycled, whereas the porphyrin ring (globin) compounds are broken down to form fat-soluble bilirubin. On reaching the liver via the bloodstream, the fat-soluble bilirubin is converted to a water-soluble form of bilirubin. This water-soluble bilirubin is secreted into the biliary tree and then in turn into the bowel, where it forms the dark color of the stool.

Prehepatic jaundice

This type of jaundice is usually produced by conditions where there is an excessive breakdown of red blood cells (e.g., in incompatible blood transfusion and hemolytic anemia).

Hepatic jaundice

The complex biochemical reactions for converting fat-soluble into water-soluble bilirubin may be affected by inflammatory change within the liver (e.g., from hepatitis or chronic liver disease, such as liver cirrhosis) and poisons (e.g., paracetamol overdose).

Posthepatic jaundice

Any obstruction of the biliary tree can produce jaundice, but the two most common causes are gallstones within the bile duct and an obstructing tumor at the head of the pancreas.

In the clinic

Spleen disorders

From a clinical point of view, there are two main categories of spleen disorders: rupture and enlargement.

Splenic rupture

This tends to occur when there is localized trauma to the left upper quadrant. It may be associated with left lower rib fractures. Because the spleen has such an extremely thin capsule, it is susceptible to injury even when there is no damage to surrounding structures, and because the spleen is highly vascular, when ruptured, it bleeds profusely into the peritoneal cavity. Splenic rupture should always be suspected with blunt abdominal injury. Current treatments preserve as much of the spleen as possible, but some patients require splenectomy.

Splenic enlargement

The spleen is an organ of the reticuloendothelial system. Diseases that affect the reticuloendothelial system (e.g., leukemia, lymphoma, and certain infections) may produce generalized lymphadenopathy and enlargement of the spleen (**splenomegaly**) (Fig. 4.109).



Fig. 4.109 Coronal CT of the abdomen containing a massively enlarged spleen (splenomegaly).

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Arterial supply

The **abdominal aorta** begins at the aortic hiatus of the diaphragm, anterior to the lower border of vertebra TXII (Fig. 4.110). It descends through the abdomen, anterior to the vertebral bodies, and by the time it ends at the level of vertebra LIV it is slightly to the left of midline. The terminal branches of the abdominal aorta are the two **common iliac arteries**.

Anterior branches of the abdominal aorta

The abdominal aorta has anterior, lateral, and posterior branches as it passes through the abdominal cavity. The three anterior branches supply the gastrointestinal viscera: the **celiac trunk** and the **superior mesenteric** and **inferior mesenteric arteries** (Fig. 4.110).

The primitive gut tube can be divided into foregut, midgut, and hindgut regions. The boundaries of these

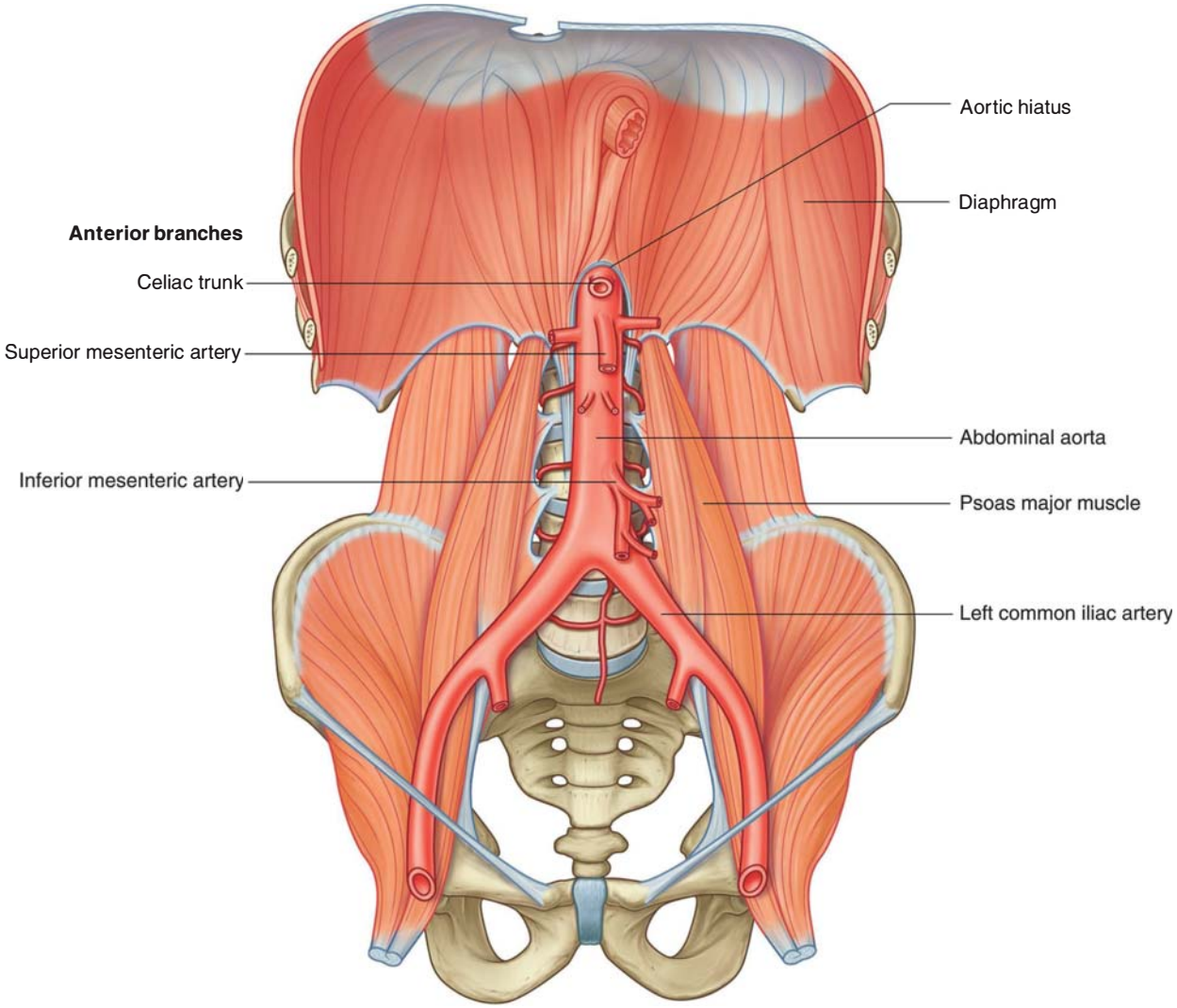


Fig. 4.110 Anterior branches of the abdominal aorta.

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regions are directly related to the areas of distribution of the three anterior branches of the abdominal aorta (Fig. 4.111).

- The **foregut** begins with the abdominal esophagus and ends just inferior to the major duodenal papilla, midway along the descending part of the duodenum. It includes the abdominal esophagus, stomach, duodenum (superior to the major papilla), liver, pancreas, and gallbladder. The spleen also develops in relation to the foregut region. The foregut is supplied by the celiac trunk.
- The **midgut** begins just inferior to the major duodenal papilla, in the descending part of the duodenum, and ends at the junction between the proximal two-thirds and distal one-third of the transverse colon. It includes the duodenum (inferior to the major duodenal papilla), jejunum, ileum, cecum, appendix, ascending colon, and right two-thirds of the transverse colon. The midgut is supplied by the superior mesenteric artery (Fig. 4.111).
- The **hindgut** begins just before the left colic flexure (the junction between the proximal two-thirds and distal one-third of the transverse colon) and ends midway through the anal canal. It includes the left one-third of the transverse colon, descending colon, sigmoid colon, rectum, and upper part of the anal canal. The hindgut is supplied by the inferior mesenteric artery (Fig. 4.111).

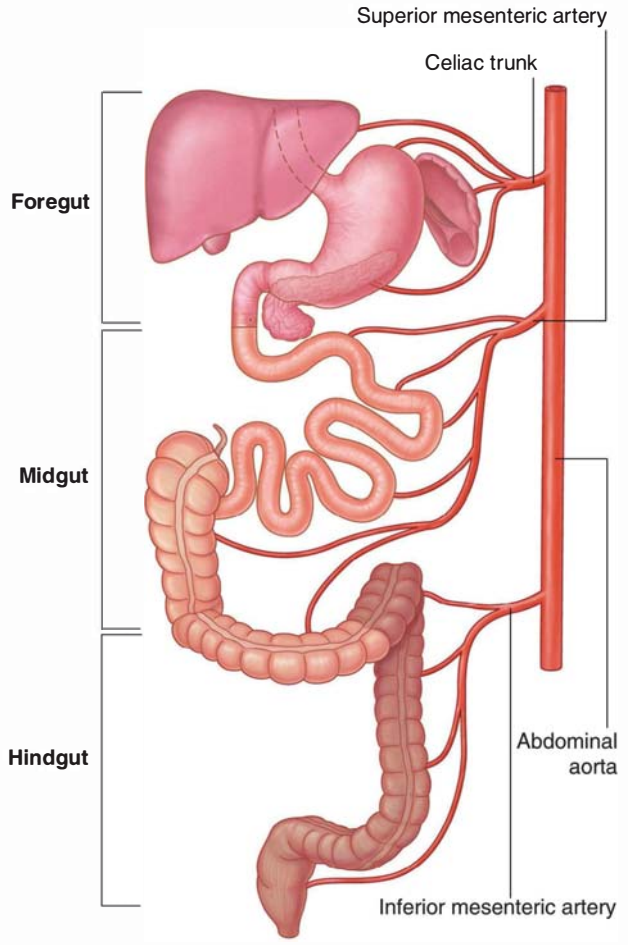


Fig. 4.111 Divisions of the gastrointestinal tract into foregut, midgut, and hindgut, summarizing the primary arterial supply to each segment.

Celiac trunk

The celiac trunk is the anterior branch of the abdominal aorta supplying the foregut. It arises from the abdominal aorta immediately below the aortic hiatus of the diaphragm (Fig. 4.112), anterior to the upper part of vertebra LI. It immediately divides into the left gastric, splenic, and common hepatic arteries.

Left gastric artery

The **left gastric artery** is the smallest branch of the celiac trunk. It ascends to the cardioesophageal junction and sends **esophageal branches** upward to the abdominal part of the esophagus (Fig. 4.112). Some of these branches continue through the esophageal hiatus of the diaphragm and anastomose with esophageal branches from the thoracic aorta. The left gastric artery itself turns to the right and descends along the lesser curvature of the stomach in the lesser omentum. It supplies both surfaces of the stomach in this area and anastomoses with the right gastric artery.

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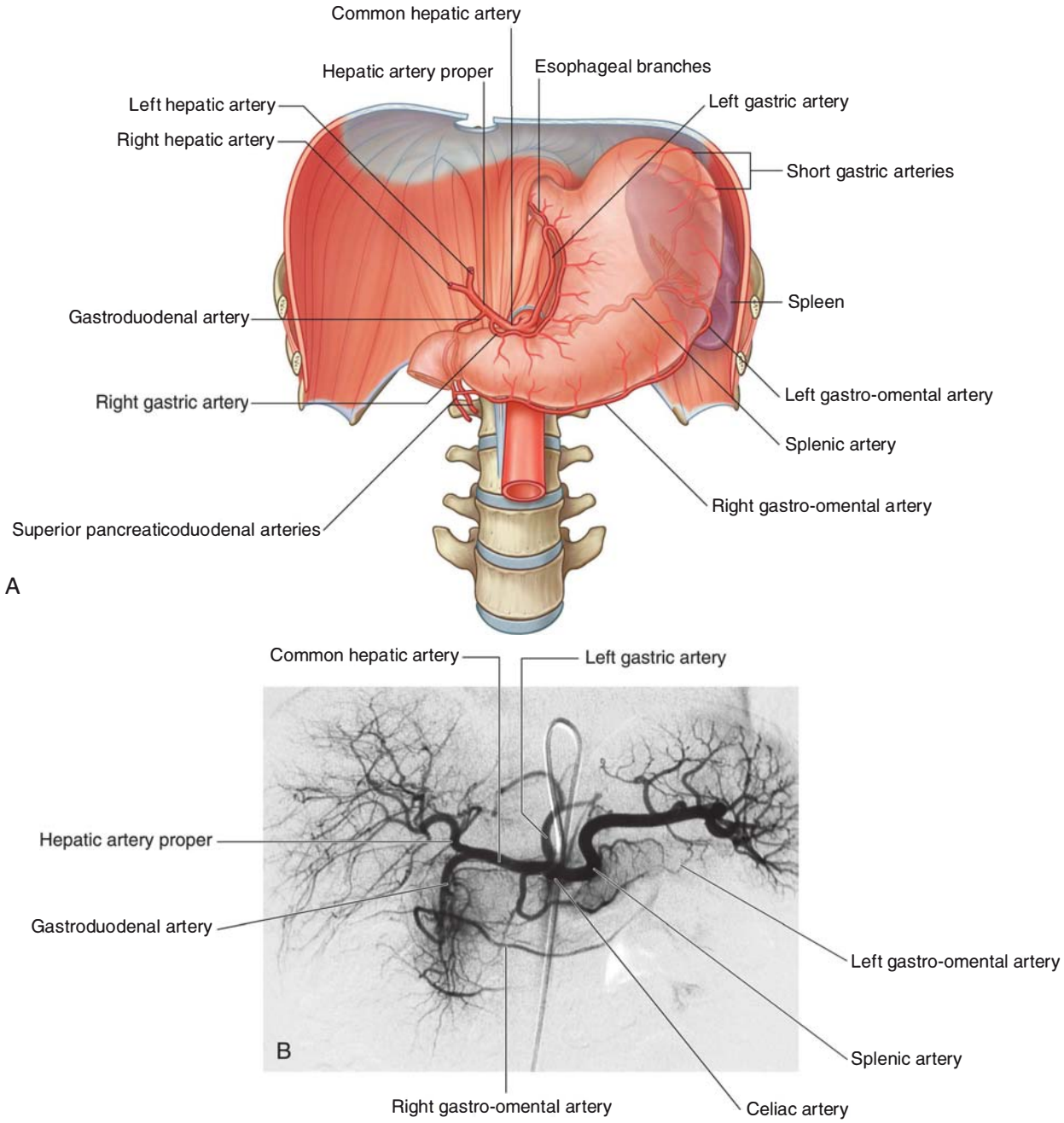


Fig. 4.112 Celiac trunk. **A.** Distribution of the celiac trunk. **B.** Digital subtraction angiography of the celiac trunk and its branches.

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Splenic artery

The **splenic artery**, the largest branch of the celiac trunk, takes a tortuous course to the left along the superior border of the pancreas (Fig. 4.112). It travels in the splenorenal ligament and divides into numerous branches, which enter the hilum of the spleen. As the splenic artery passes along the superior border of the pancreas, it gives off numerous small branches to supply the neck, body, and tail of the pancreas (Fig. 4.113).

Approaching the spleen, the splenic artery gives off **short gastric arteries**, which pass through the gastrosplenic ligament to supply the fundus of the stomach. It also gives off the **left gastro-omental artery**, which runs to the right along the greater curvature of the stomach, and anastomoses with the right gastro-omental artery.

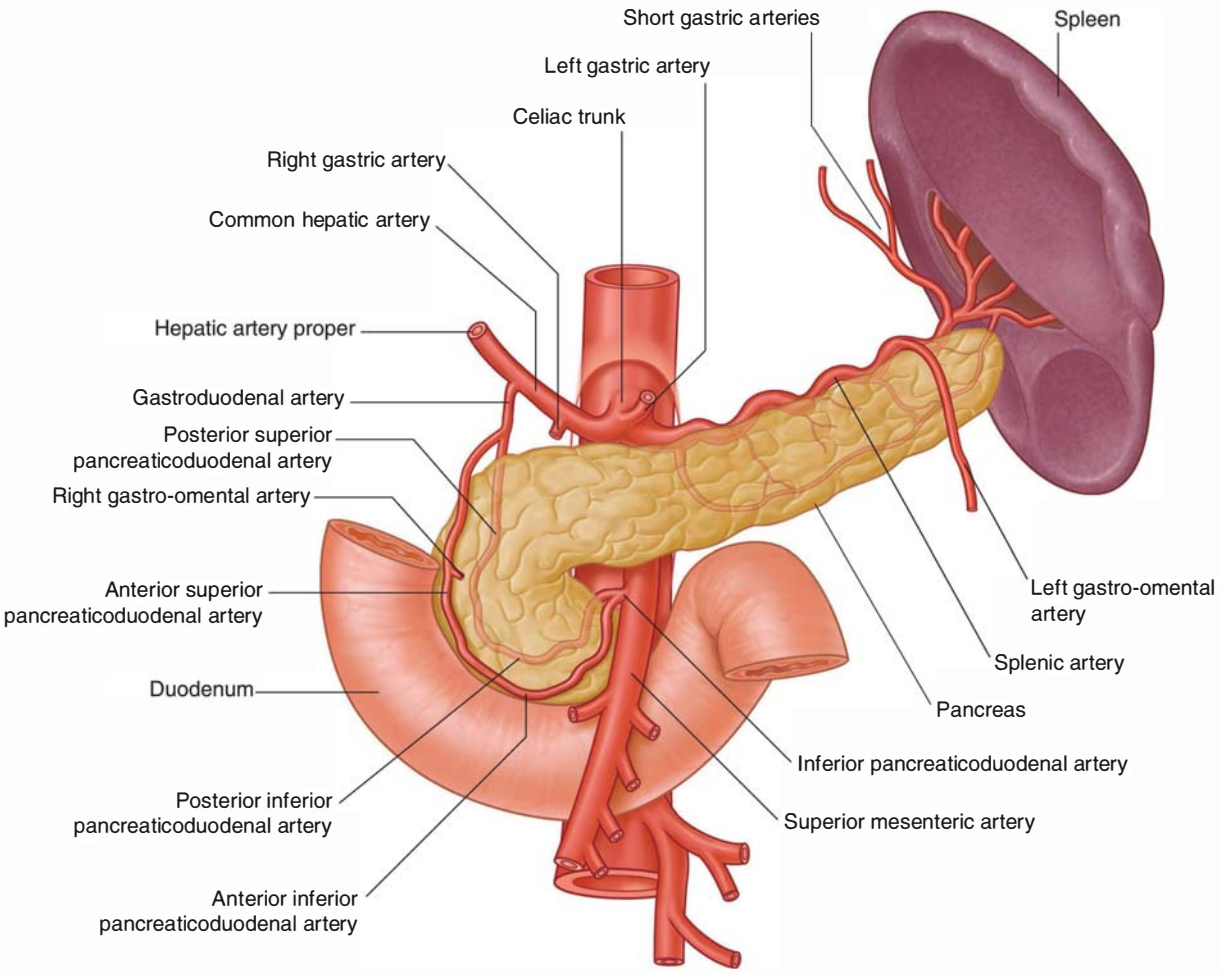


Fig. 4.113 Arterial supply to the pancreas.

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Common hepatic artery

The **common hepatic artery** is a medium-sized branch of the celiac trunk that runs to the right and divides into its two terminal branches, the **hepatic artery proper** and the **gastroduodenal artery** (Figs. 4.112 and 4.113).

The hepatic artery proper ascends toward the liver in the free edge of the lesser omentum. It runs to the left of the bile duct and anterior to the portal vein, and divides into the **right** and **left hepatic arteries** near the porta hepatis (Fig. 4.114).

As the right hepatic artery nears the liver, it gives off the cystic artery to the gallbladder.

The gastroduodenal artery may give off the **supraduodenal artery** and does give off the posterior superior pancreaticoduodenal artery near the upper border of the superior part of the duodenum. After these branches the gastroduodenal artery continues descending posterior to the superior part of the duodenum. Reaching the lower border of the superior part of the duodenum, the gastroduodenal artery divides into its terminal branches, the **right gastro-omental artery** and the **anterior superior pancreaticoduodenal artery** (Fig. 4.113).

The right gastro-omental artery passes to the left, along the greater curvature of the stomach, eventually anastomosing with the left gastro-omental artery from the splenic artery. The right gastro-omental artery sends branches to both surfaces of the stomach and additional branches descend into the greater omentum.

The anterior superior pancreaticoduodenal artery descends and, along with the posterior superior pancreaticoduodenal artery, supplies the head of the pancreas and the duodenum (Fig. 4.113). These vessels eventually anastomose with the anterior and posterior branches of the inferior pancreaticoduodenal artery.

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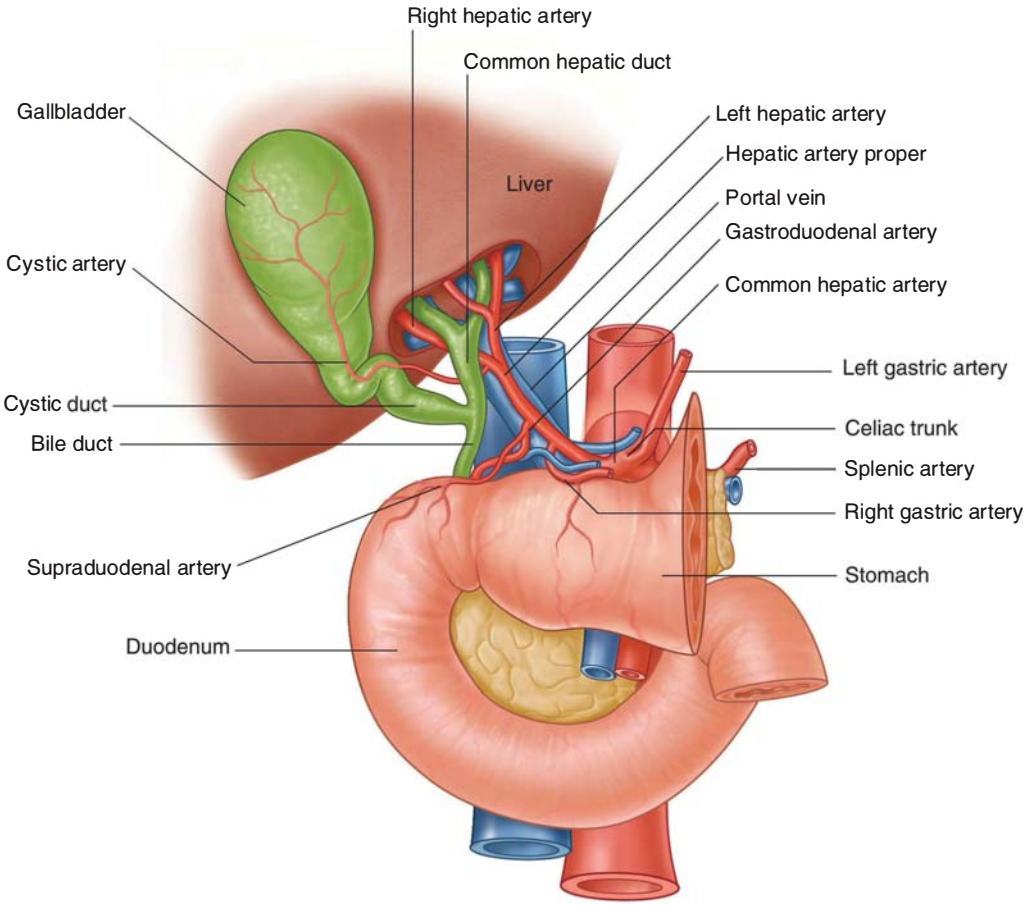


Fig. 4.114 Distribution of the common hepatic artery.

Superior mesenteric artery

The superior mesenteric artery is the anterior branch of the abdominal aorta supplying the midgut. It arises from the abdominal aorta immediately below the celiac artery (Fig. 4.115), anterior to the lower part of vertebra LI.

The superior mesenteric artery is crossed anteriorly by the splenic vein and the neck of the pancreas. Posterior to the artery are the left renal vein, the uncinate process of the pancreas, and the inferior part of the duodenum. After giving off its first branch (the **inferior pancreaticoduodenal artery**), the superior mesenteric artery gives off **jejunal** and **ileal arteries** on its left (Fig. 4.115). Branching from the right side of the main trunk of the superior mesenteric artery are three vessels—the **middle colic, right colic**, and **ileocolic arteries**—which supply the terminal ileum, cecum, ascending colon, and two-thirds of the transverse colon.

Inferior pancreaticoduodenal artery

The inferior pancreaticoduodenal artery is the first branch of the superior mesenteric artery. It divides immediately into anterior and posterior branches, which ascend on the corresponding sides of the head of the pancreas. Superiorly, these arteries anastomose with anterior and posterior superior pancreaticoduodenal arteries (see Figs. 4.114 and 4.115). This arterial network supplies the head and uncinate process of the pancreas and the duodenum.

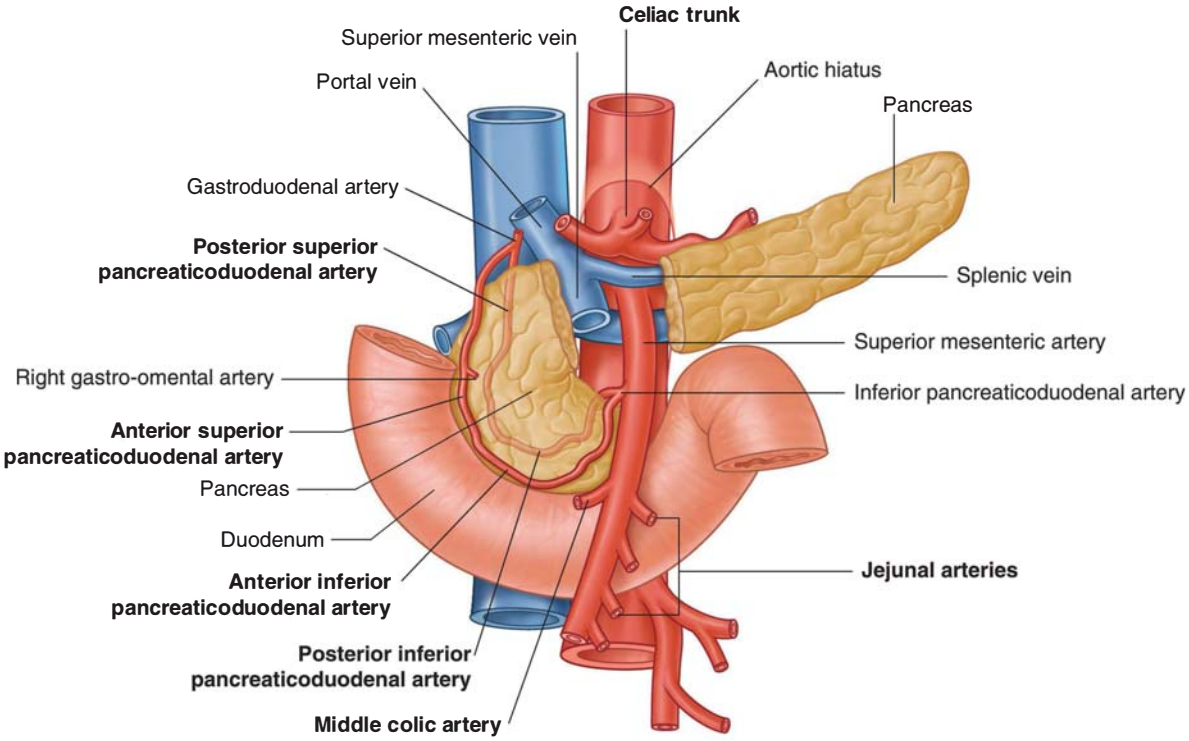


Fig. 4.115 Initial branching and relationships of the superior mesenteric artery.

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Jejunal and ileal arteries

Distal to the inferior pancreaticoduodenal artery, the superior mesenteric artery gives off numerous branches. Arising on the left is a large number of jejunal and ileal arteries supplying the jejunum and most of the ileum (Fig. 4.116). These branches leave the main trunk of the artery, pass between two layers of the mesentery, and form anastomosing arches or arcades as they pass outward to supply the small intestine. The number of arterial arcades increases distally along the gut.

There may be single and then double arcades in the area of the jejunum, with a continued increase in the number

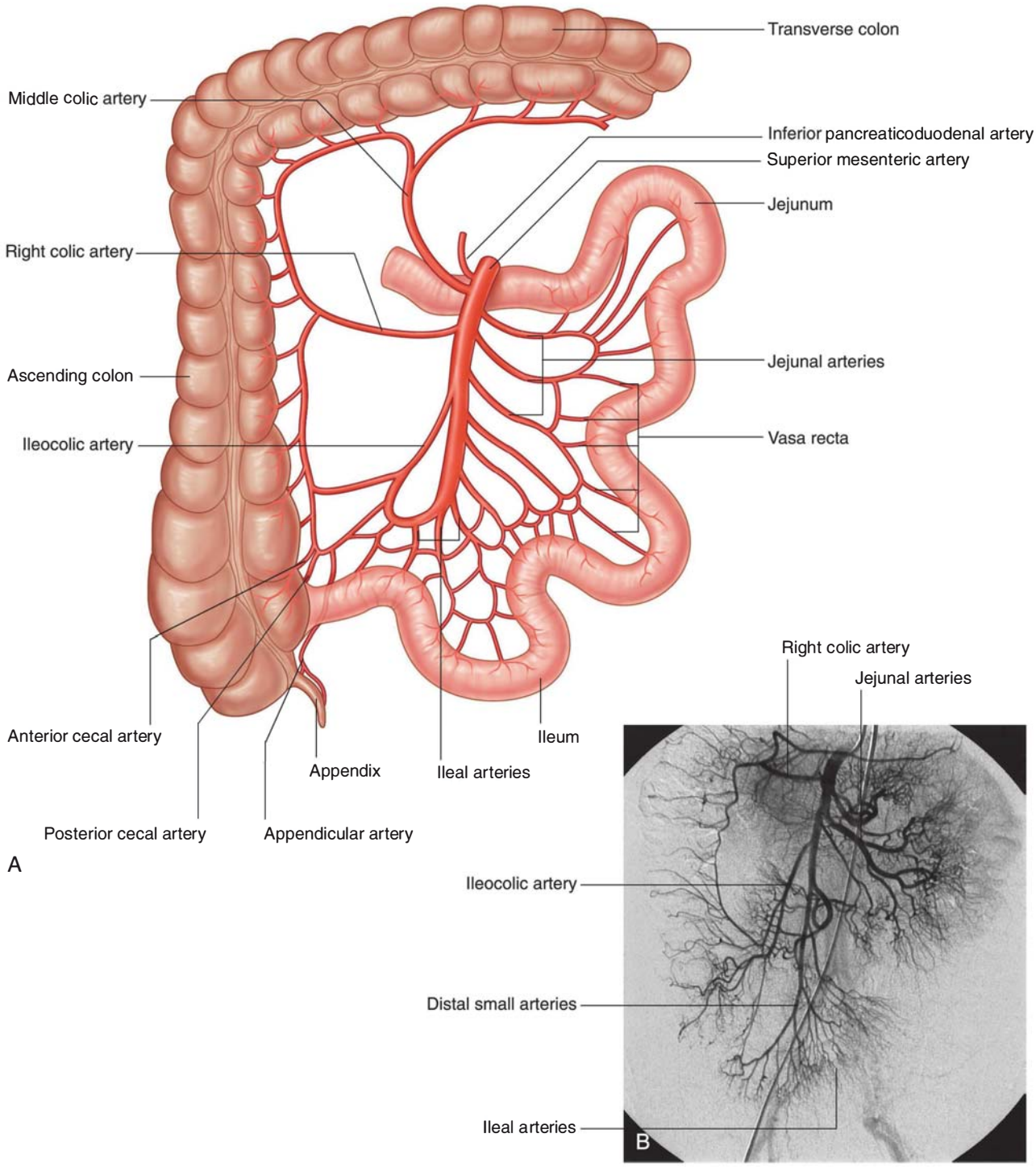


Fig. 4.116 Superior mesenteric artery. **A.** Distribution of the superior mesenteric artery. **B.** Digital subtraction angiography of the superior mesenteric artery and its branches.

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of arcades moving into and through the area of the ileum. Extending from the terminal arcade are vasa recta (straight arteries), which provide the final direct vascular supply to the walls of the small intestine. The **vasa recta** supplying the jejunum are usually long and close together, forming narrow windows visible in the mesentery. The vasa recta supplying the ileum are generally short and far apart, forming low broad windows.

Middle colic artery

The middle colic artery is the first of the three branches from the right side of the main trunk of the superior mesenteric artery (Fig. 4.116). Arising as the superior mesenteric artery emerges from beneath the pancreas, the middle colic artery enters the transverse mesocolon and divides into right and left branches. The right branch anastomoses with the right colic artery while the left branch anastomoses with the left colic artery, which is a branch of the inferior mesenteric artery.

Right colic artery

Continuing distally along the main trunk of the superior mesenteric artery, the right colic artery is the second of the three branches from the right side of the main trunk of the superior mesenteric artery (Fig. 4.115). It is an inconsistent branch, and passes to the right in a retroperitoneal position to supply the ascending colon. Nearing the colon, it divides into a descending branch, which anastomoses with the ileocolic artery, and an ascending branch, which anastomoses with the middle colic artery.

Ileocolic artery

The final branch arising from the right side of the superior mesenteric artery is the ileocolic artery (Fig. 4.116). This passes downward and to the right toward the right iliac fossa where it divides into superior and inferior branches:

- The superior branch passes upward along the ascending colon to anastomose with the right colic artery.
- The inferior branch continues toward the ileocolic junction, dividing into **colic, cecal, appendicular**, and **ileal branches** (Fig. 4.116).

The specific pattern of distribution and origin of these branches is variable:

- The colic branch crosses to the ascending colon and passes upward to supply the first part of the ascending colon.
- Anterior and posterior cecal branches, arising either as a common trunk or as separate branches, supply corresponding sides of the cecum.
- The appendicular branch enters the free margin of and supplies the mesoappendix and the appendix.
- The ileal branch passes to the left and ascends to supply the final part of the ileum before anastomosing with the superior mesenteric artery.

Inferior mesenteric artery

The inferior mesenteric artery is the anterior branch of the abdominal aorta that supplies the hindgut. It is the smallest of the three anterior branches of the abdominal aorta and arises anterior to the body of vertebra LIII. Initially, the inferior mesenteric artery descends anteriorly to the aorta and then passes to the left as it continues inferiorly (Fig. 4.117). Its branches include the **left colic artery, several sigmoid arteries**, and the **superior rectal artery**.

Left colic artery

The left colic artery is the first branch of the inferior mesenteric artery (Fig. 4.117). It ascends retroperitoneally, dividing into ascending and descending branches:

- The ascending branch passes anteriorly to the left kidney, then enters the transverse mesocolon, and passes superiorly to supply the upper part of the descending colon and the distal part of the transverse colon; it anastomoses with branches of the middle colic artery.
- The descending branch passes inferiorly, supplying the lower part of the descending colon, and anastomoses with the first sigmoid artery.

Sigmoid arteries

The sigmoid arteries consist of two to four branches, which descend to the left, in the sigmoid mesocolon, to supply the lowest part of the descending colon and the sigmoid colon (Fig. 4.117). These branches anastomose superiorly with branches from the left colic artery and inferiorly with branches from the superior rectal artery.

Superior rectal artery

The terminal branch of the inferior mesenteric artery is the superior rectal artery (Fig. 4.117). This vessel descends into the pelvic cavity in the sigmoid mesocolon, crossing the left common iliac vessels. Opposite vertebra SIII, the superior rectal artery divides. The two terminal branches descend on each side of the rectum, dividing into smaller branches in the wall of the rectum. These smaller branches continue inferiorly to the level of the internal anal sphincter, anastomosing along the way with branches from the middle rectal arteries (from the internal iliac artery) and the inferior rectal arteries (from the internal pudendal artery).

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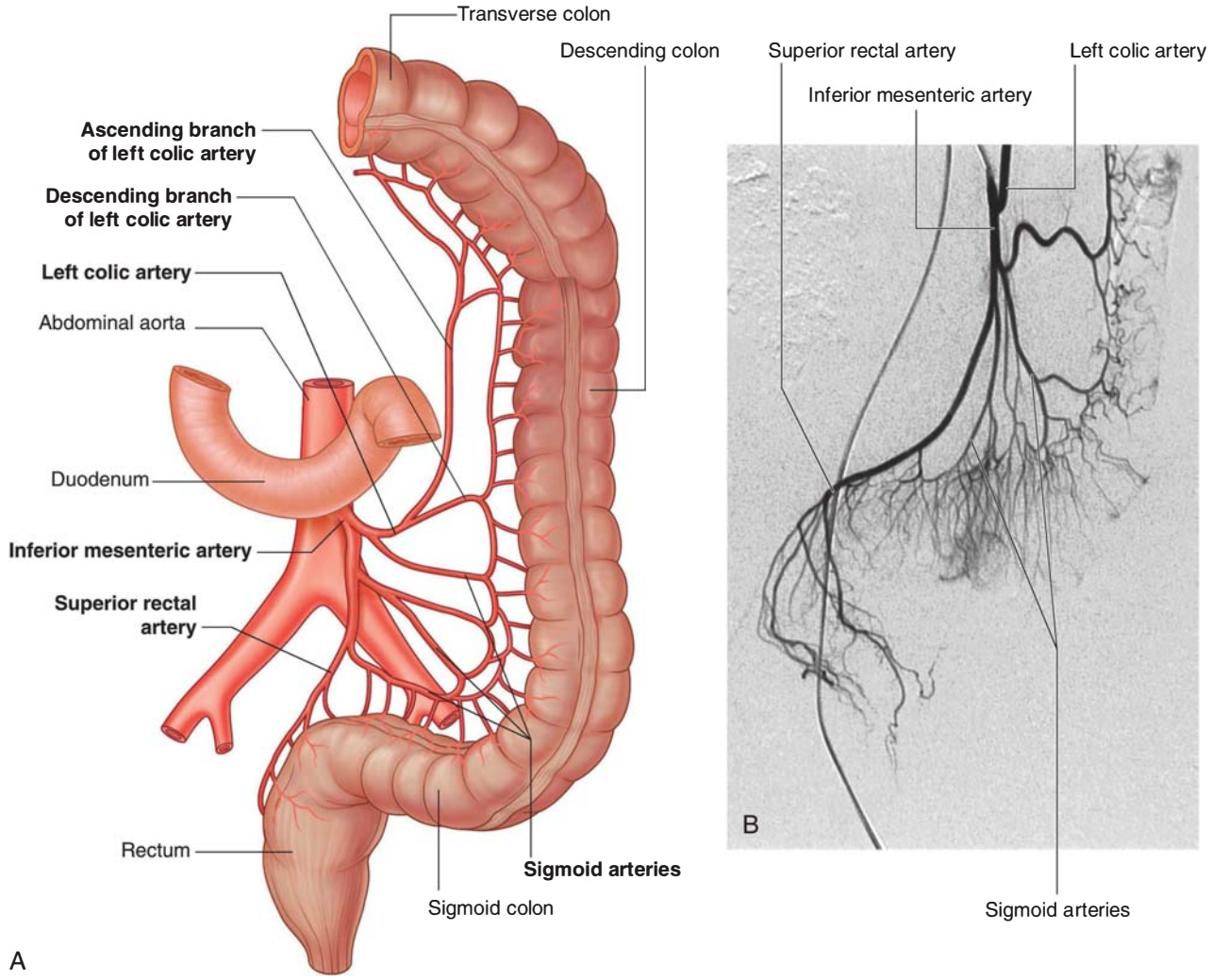


Fig. 4.117 Inferior mesenteric artery. **A.** Distribution of the inferior mesenteric artery. **B.** Digital subtraction angiography of the inferior mesenteric artery and its branches.

In the clinic

Vascular supply to the gastrointestinal system

The abdominal parts of the gastrointestinal system are supplied mainly by the celiac trunk and the superior mesenteric and inferior mesenteric arteries (Fig. 4.118):

- The celiac trunk supplies the lower esophagus, stomach, superior part of the duodenum, and proximal half of the descending part of the duodenum.
- The superior mesenteric artery supplies the rest of the duodenum, the jejunum, the ileum, the ascending colon, and the proximal two-thirds of the transverse colon.
- The inferior mesenteric artery supplies the rest of the transverse colon, the descending colon, the sigmoid colon, and most of the rectum.

Along the descending part of the duodenum there is a potential watershed area between the celiac trunk blood supply and the superior mesenteric arterial blood supply. It is unusual for this area to become ischemic, whereas the watershed area between the superior mesenteric artery and the inferior mesenteric artery, at the splenic flexure, is extremely vulnerable to ischemia.

In certain disease states, the region of the splenic flexure of the colon can become ischemic. When this occurs, the mucosa sloughs off, rendering the patient susceptible to infection and perforation of the large bowel, which then requires urgent surgical attention.

Arteriosclerosis may occur throughout the abdominal aorta and at the openings of the celiac trunk and the superior mesenteric and inferior mesenteric arteries. Not

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In the clinic—cont'd

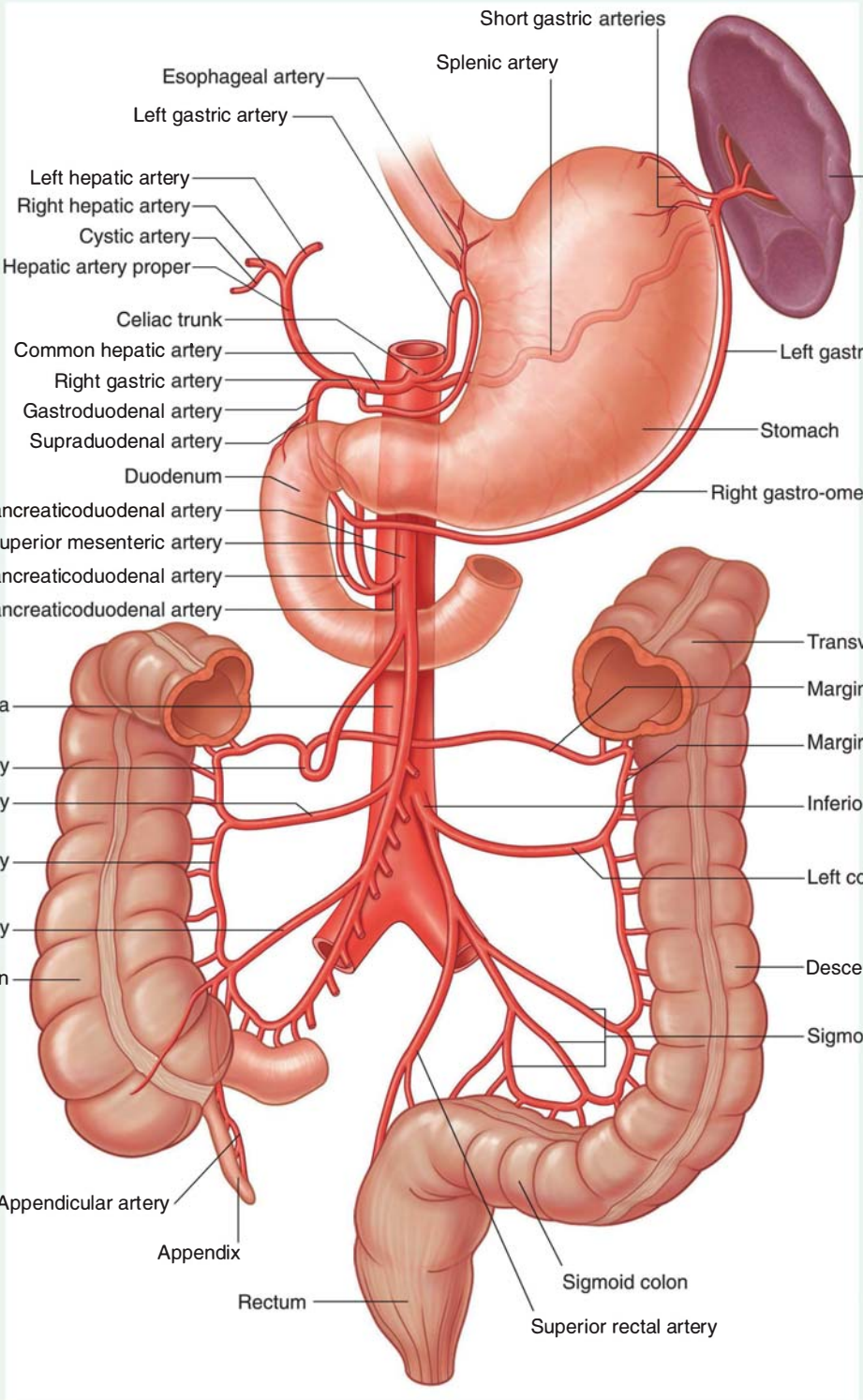


Fig. 4.118 Arterial supply to the abdominal parts of the gastrointestinal system and to the spleen.

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In the clinic—cont'd

infrequently, the inferior mesenteric artery becomes occluded. Interestingly, many of these patients do not suffer any complications, because anastomoses between the right, middle, and left colic arteries gradually enlarge, forming a continuous **marginal artery**. The distal large bowel therefore becomes supplied by this enlarged marginal artery (marginal artery of Drummond), which replaces the blood supply of the inferior mesenteric artery (Fig. 4.119).

If the openings of the celiac trunk and superior mesenteric artery become narrowed, the blood supply to the gut is diminished. After a heavy meal, the oxygen demand of the bowel therefore outstrips the limited supply of blood through the stenosed vessels, resulting in severe pain and discomfort (**mesenteric angina**). Patients with this condition tend not to eat because of the pain and rapidly lose weight. The diagnosis is determined by aortic angiography, and the stenoses of the celiac trunk and superior mesenteric artery are best appreciated in the lateral view.



Fig. 4.119 Enlarged marginal artery connecting the superior and inferior mesenteric arteries. Digital subtraction angiogram.

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Venous drainage

Venous drainage of the spleen, pancreas, gallbladder, and abdominal part of the gastrointestinal tract, except for the inferior part of the rectum, is through the portal system of veins, which deliver blood from these structures to the liver. Once blood passes through the hepatic sinusoids, it passes through progressively larger veins until it enters the hepatic veins, which return the venous blood to the inferior vena cava just inferior to the diaphragm.

Portal vein

The **portal vein** is the final common pathway for the transport of venous blood from the spleen, pancreas, gallbladder, and abdominal part of the gastrointestinal tract. It is formed by the union of the **splenic vein** and the **superior mesenteric vein** posterior to the neck of the pancreas at the level of vertebra LII (Fig. 4.120).

Ascending toward the liver, the portal vein passes posterior to the superior part of the duodenum and enters the right margin of the lesser omentum. As it passes through this part of the lesser omentum, it is anterior to the omental foramen and posterior to both the bile duct, which is slightly to its right, and the hepatic artery proper, which is slightly to its left (see Fig. 4.114, p. 347).

On approaching the liver, the portal vein divides into **right** and **left branches**, which enter the liver parenchyma. Tributaries to the portal vein include:

- **right** and **left gastric veins** draining the lesser curvature of the stomach and abdominal esophagus,
- **cystic veins** from the gallbladder, and
- the **para-umbilical veins**, which are associated with the obliterated umbilical vein and connect to veins on the anterior abdominal wall (Fig. 4.122 on p. 357).

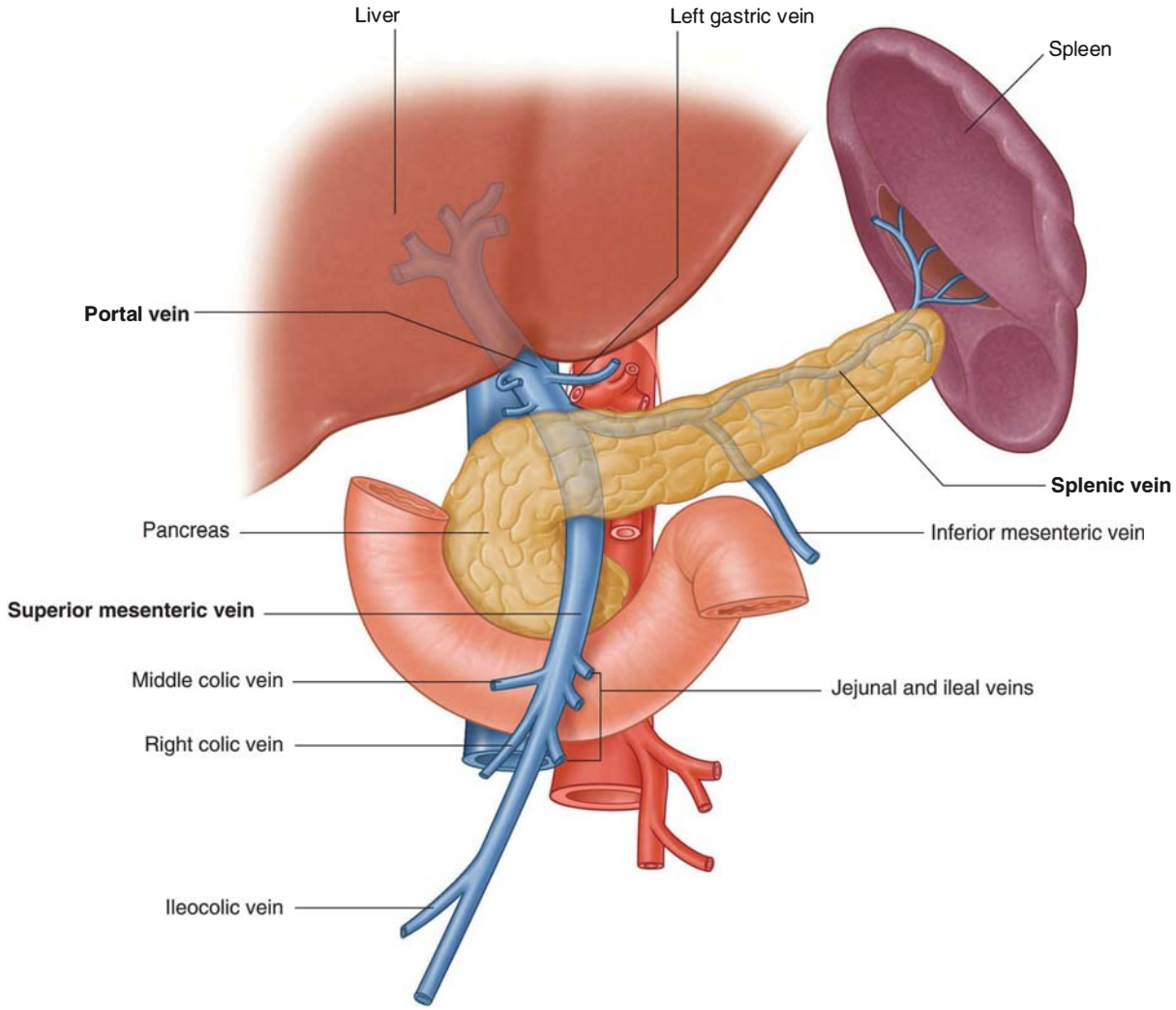


Fig. 4.120 Portal vein.

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Splenic vein

The splenic vein forms from numerous smaller vessels leaving the hilum of the spleen (Fig. 4.121). It passes to the right, passing through the splenorenal ligament with the splenic artery and the tail of the pancreas. Continuing to the right, the large, straight splenic vein is in contact with the body of the pancreas as it crosses the posterior abdominal wall. Posterior to the neck of the pancreas, the splenic vein joins the superior mesenteric vein to form the portal vein.

Tributaries to the splenic vein include:

- **short gastric veins** from the fundus and left part of the greater curvature of the stomach,
- the **left gastro-omental vein** from the greater curvature of the stomach,
- **pancreatic veins** draining the body and tail of the pancreas, and
- usually the **inferior mesenteric vein**.

Superior mesenteric vein

The superior mesenteric vein drains blood from the small intestine, cecum, ascending colon, and transverse colon (Fig. 4.121). It begins in the right iliac fossa as veins draining the terminal ileum, cecum, and appendix join, and ascends in the mesentery to the right of the superior mesenteric artery.

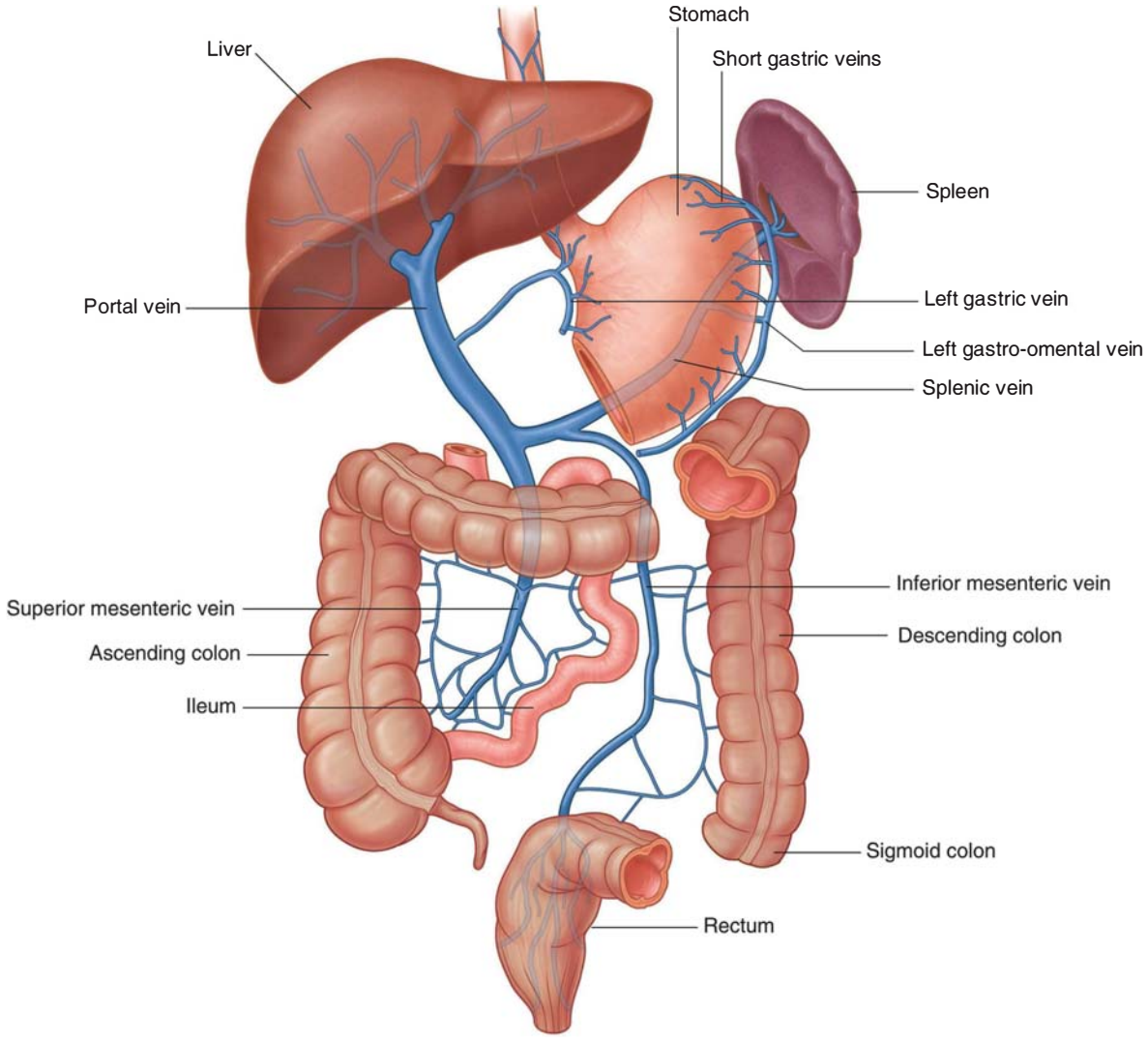


Fig. 4.121 Venous drainage of the abdominal portion of the gastrointestinal tract.

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Posterior to the neck of the pancreas, the superior mesenteric vein joins the splenic vein to form the portal vein.

As a corresponding vein accompanies each branch of the superior mesenteric artery, tributaries to the superior mesenteric vein include jejunal, ileal, ileocolic, right colic, and middle colic veins. Additional tributaries include:

- the **right gastro-omental vein**, draining the right part of the greater curvature of the stomach, and
- the **anterior** and **posterior inferior pancreaticoduodenal veins**, which pass alongside the arteries of the same name; the anterior superior pancreaticoduodenal vein usually empties into the right gastro-omental vein, and the posterior superior pancreaticoduodenal vein usually empties directly into the portal vein.

Inferior mesenteric vein

The **inferior mesenteric vein** drains blood from the rectum, sigmoid colon, descending colon, and **splenic flexure** (Fig. 4.121). It begins as the **superior rectal vein** and ascends, receiving tributaries from the sigmoid veins and the **left colic vein**. All these veins accompany arteries of the same name. Continuing to ascend, the inferior mesenteric vein passes posterior to the body of the pancreas and usually joins the splenic vein. Occasionally, it ends at the junction of the splenic and superior mesenteric veins or joins the superior mesenteric vein.

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In the clinic

Hepatic cirrhosis

Cirrhosis is a complex disorder of the liver, the diagnosis of which is confirmed histologically. When a diagnosis is suspected, a liver biopsy is necessary.

Cirrhosis is characterized by widespread hepatic fibrosis interspersed with areas of nodular regeneration and abnormal reconstruction of preexisting lobular architecture. The presence of cirrhosis implies previous or continuing liver cell damage.

The etiology of cirrhosis is complex and includes toxins (alcohol), viral inflammation, biliary obstruction, vascular outlet obstruction, nutritional (malnutrition) causes, and inherited anatomical and metabolic disorders.

As the cirrhosis progresses, the intrahepatic vasculature is distorted, which in turn leads to increased pressure in the portal vein and its draining tributaries (portal hypertension). Portal hypertension produces increased pressure in the splenic venules, leading to splenic enlargement. At the sites of portosystemic anastomosis (see below), large dilated veins (varices) develop. These veins are susceptible to bleeding and may produce marked blood loss, which in some instances can be fatal.

The liver is responsible for the production of numerous proteins, including those of the clotting cascade. Any disorder of the liver (including infection and cirrhosis) may decrease the production of these proteins and so prevent adequate blood clotting. Patients with severe cirrhosis of the liver have a significant risk of serious bleeding, even from small cuts; in addition, when varices rupture, there is a danger of rapid exsanguination.

As the liver progressively fails, the patient develops salt and water retention, which produces skin and subcutaneous edema. Fluid (ascites) is also retained in the peritoneal cavity, which can hold many liters.

The poorly functioning liver cells (hepatocytes) are unable to break down blood and blood products, leading to an increase in the serum bilirubin level, which manifests as jaundice.

With the failure of normal liver metabolism, toxic metabolic by-products do not convert to nontoxic metabolites. This buildup of noxious compounds is made worse by the numerous portosystemic shunts, which allow the toxic metabolites to bypass the liver. Patients may develop severe neurological features, which may lead to epileptic fits, dementia, and irreversible neurological damage.

Portosystemic anastomosis

The hepatic portal system drains blood from the visceral organs of the abdomen to the liver. In normal individuals, 100% of the portal venous blood flow can be recovered from the hepatic veins, whereas in patients with elevated portal vein pressure (e.g., from cirrhosis), there is significantly less blood flow to the liver. The rest of the blood enters collateral channels, which drain into the

In the clinic—cont'd

systemic circulation at specific points (Fig. 4.122). The largest of these collaterals occur at:

- the gastroesophageal junction around the cardia of the stomach—where the left gastric vein and its tributaries form a portosystemic anastomosis with tributaries to the azygos system of veins of the caval system;
- the anus—the superior rectal vein of the portal system anastomoses with the middle and inferior rectal veins of the systemic venous system; and
- the anterior abdominal wall around the umbilicus—the para-umbilical veins anastomose with veins on the anterior abdominal wall.

When the pressure in the portal vein is elevated, venous enlargement (varices) tend to occur at and around the sites of portosystemic anastomoses and these enlarged veins are called:

- varices at the anorectal junction,
- esophageal varices at the gastroesophageal junction, and
- caput medusae at the umbilicus.

Esophageal varices are susceptible to trauma and, once damaged, may bleed profusely, requiring urgent surgical intervention.

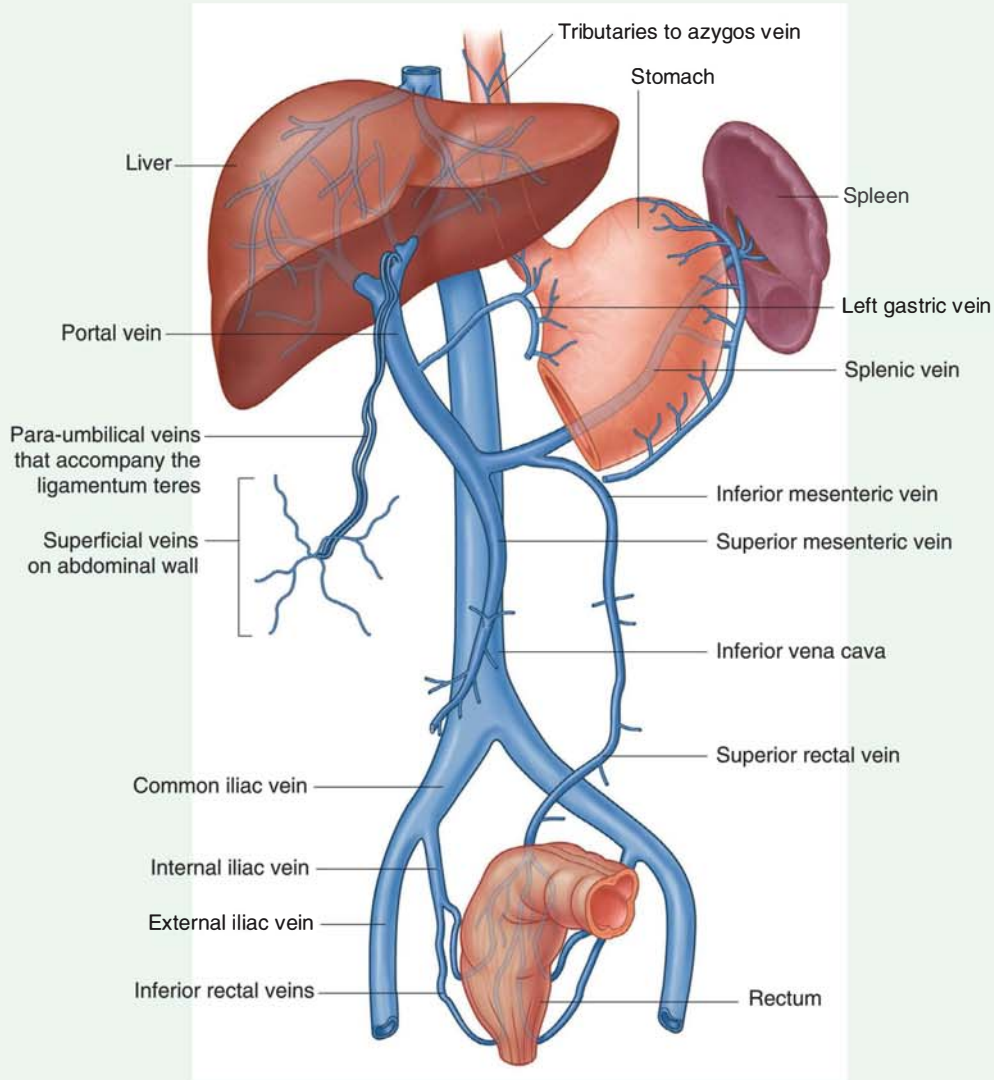


Fig. 4.122 Portosystemic anastomoses.

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Lymphatics

Lymphatic drainage of the abdominal part of the gastrointestinal tract, as low as the inferior part of the rectum, as well as the spleen, pancreas, gallbladder, and liver, is through vessels and nodes that eventually end in large collections of **pre-aortic lymph nodes** at the origins of the three anterior branches of the abdominal aorta, which supply these structures. These collections are therefore referred to as the **celiac, superior mesenteric**, and **inferior mesenteric** groups of pre-aortic lymph nodes. Lymph from viscera is supplied by three routes:

- The celiac trunk (i.e., structures that are part of the abdominal foregut) drains to pre-aortic nodes near the origin of the celiac trunk (Fig. 4.123)—these celiac nodes also receive lymph from the superior mesenteric and inferior mesenteric groups of pre-aortic nodes, and lymph from the celiac nodes enters the **cisterna chyli**.
- The superior mesenteric artery (i.e., structures that are part of the abdominal midgut) drains to pre-aortic nodes near the origin of the superior mesenteric artery (Fig. 4.123)—these superior mesenteric nodes also receive lymph from the inferior mesenteric groups of pre-aortic nodes, and lymph from the superior mesenteric nodes drains to the celiac nodes.
- The inferior mesenteric artery (i.e., structures that are part of the abdominal hindgut) drains to pre-aortic nodes near the origin of the inferior mesenteric artery (Fig. 4.123), and lymph from the inferior mesenteric nodes drains to the superior mesenteric nodes.

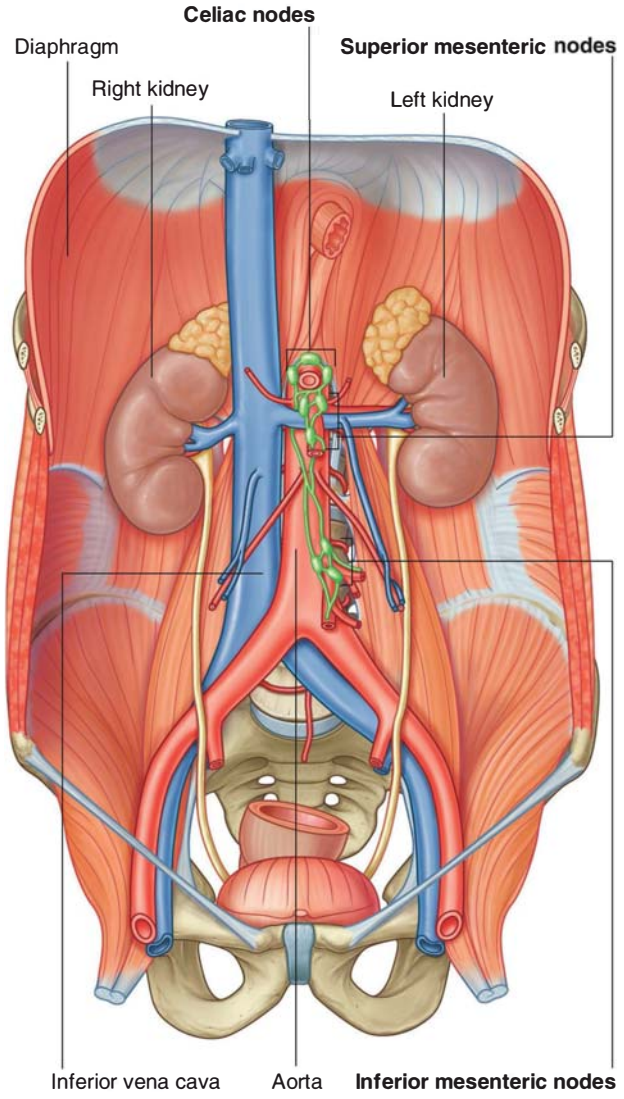


Fig. 4.123 Lymphatic drainage of the abdominal portion of the gastrointestinal tract.

Innervation

Abdominal viscera are innervated by both extrinsic and intrinsic components of the nervous system:

- Extrinsic innervation involves receiving motor impulses from, and sending sensory information to, the central nervous system.
- Intrinsic innervation involves the regulation of digestive tract activities by a generally self-sufficient network of sensory and motor neurons (the enteric nervous system).

Abdominal viscera receiving extrinsic innervation include the abdominal part of the gastrointestinal tract, the spleen, the pancreas, the gallbladder, and the liver. These viscera send sensory information back to the central nervous system through visceral afferent fibers and receive motor impulses from the central nervous system through visceral efferent fibers.

The visceral efferent fibers are part of the sympathetic and parasympathetic parts of the autonomic division of the peripheral nervous system.

Structural components serving as conduits for these afferent and efferent fibers include posterior and anterior roots of the spinal cord, respectively, spinal nerves, anterior rami, white and gray rami communicantes, the sympathetic trunks, splanchnic nerves carrying sympathetic fibers (thoracic, lumbar, and sacral), parasympathetic fibers (pelvic), the prevertebral plexus and related ganglia, and the vagus nerves [X].

The enteric nervous system consists of motor and sensory neurons in two interconnected plexuses in the walls of the gastrointestinal tract. These neurons control the coordinated contraction and relaxation of intestinal smooth muscle and regulate gastric secretion and blood flow.

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Sympathetic trunks

The sympathetic trunks are two parallel nerve cords extending on either side of the vertebral column from the base of the skull to the coccyx (Fig. 4.124). As they pass through the neck, they lie posterior to the carotid sheath. In the upper thorax, they are anterior to the necks of the ribs, while in the lower thorax they are on the lateral aspect of the vertebral bodies. In the abdomen, they are anterolateral to the lumbar vertebral bodies and, continuing into the pelvis, they are anterior to the sacrum. The two sympathetic trunks come together anterior to the coccyx to form the **ganglion impar**.

Throughout the extent of the sympathetic trunks, small raised areas are visible. These collections of neuronal cell bodies outside the CNS are the paravertebral sympathetic ganglia. There are usually:

- three ganglia in the cervical region,
- eleven or twelve ganglia in the thoracic region,
- four ganglia in the lumbar region,
- four or five ganglia in the sacral region, and
- the ganglion impar anterior to the coccyx (Fig. 4.124).

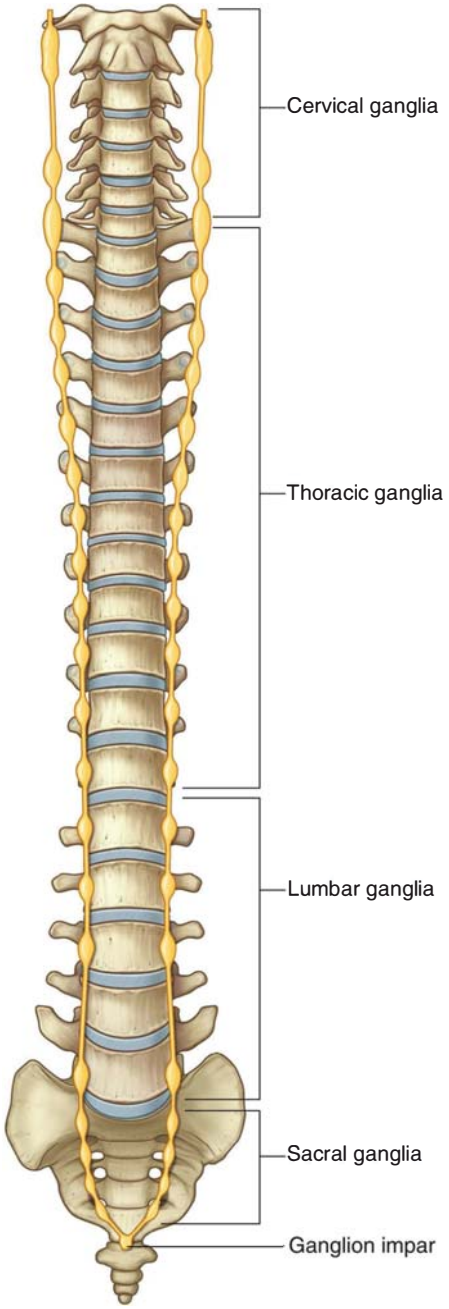


Fig. 4.124 Sympathetic trunks.

The ganglia and trunks are connected to adjacent spinal nerves by gray rami communicantes throughout the length of the sympathetic trunk and by white rami communicantes in the thoracic and upper lumbar parts of the trunk (T1 to L2). Neuronal fibers found in the sympathetic trunks include **preganglionic** and **postganglionic sympathetic fibers** and **visceral afferent fibers**.

Splanchnic nerves

The splanchnic nerves are important components in the innervation of the abdominal viscera. They pass from the sympathetic trunk or sympathetic ganglia associated with the trunk, to the prevertebral plexus and ganglia anterior to the abdominal aorta.

There are two different types of splanchnic nerves, depending on the type of visceral efferent fiber they are carrying:

- The thoracic, lumbar, and sacral splanchnic nerves carry preganglionic sympathetic fibers from the sympathetic trunk to ganglia in the prevertebral plexus, and also visceral afferent fibers.
- The pelvic splanchnic nerves (parasympathetic root) carry preganglionic parasympathetic fibers from anterior rami of S2, S3, and S4 spinal nerves to an extension of the prevertebral plexus in the pelvis (the **inferior hypogastric plexus** or **pelvic plexus**).

Thoracic splanchnic nerves

Three **thoracic splanchnic nerves** pass from sympathetic ganglia along the sympathetic trunk in the thorax to the prevertebral plexus and ganglia associated with the abdominal aorta in the abdomen (Fig. 4.125):

- The greater splanchnic nerve arises from the fifth to the ninth (or tenth) thoracic ganglia and travels to the

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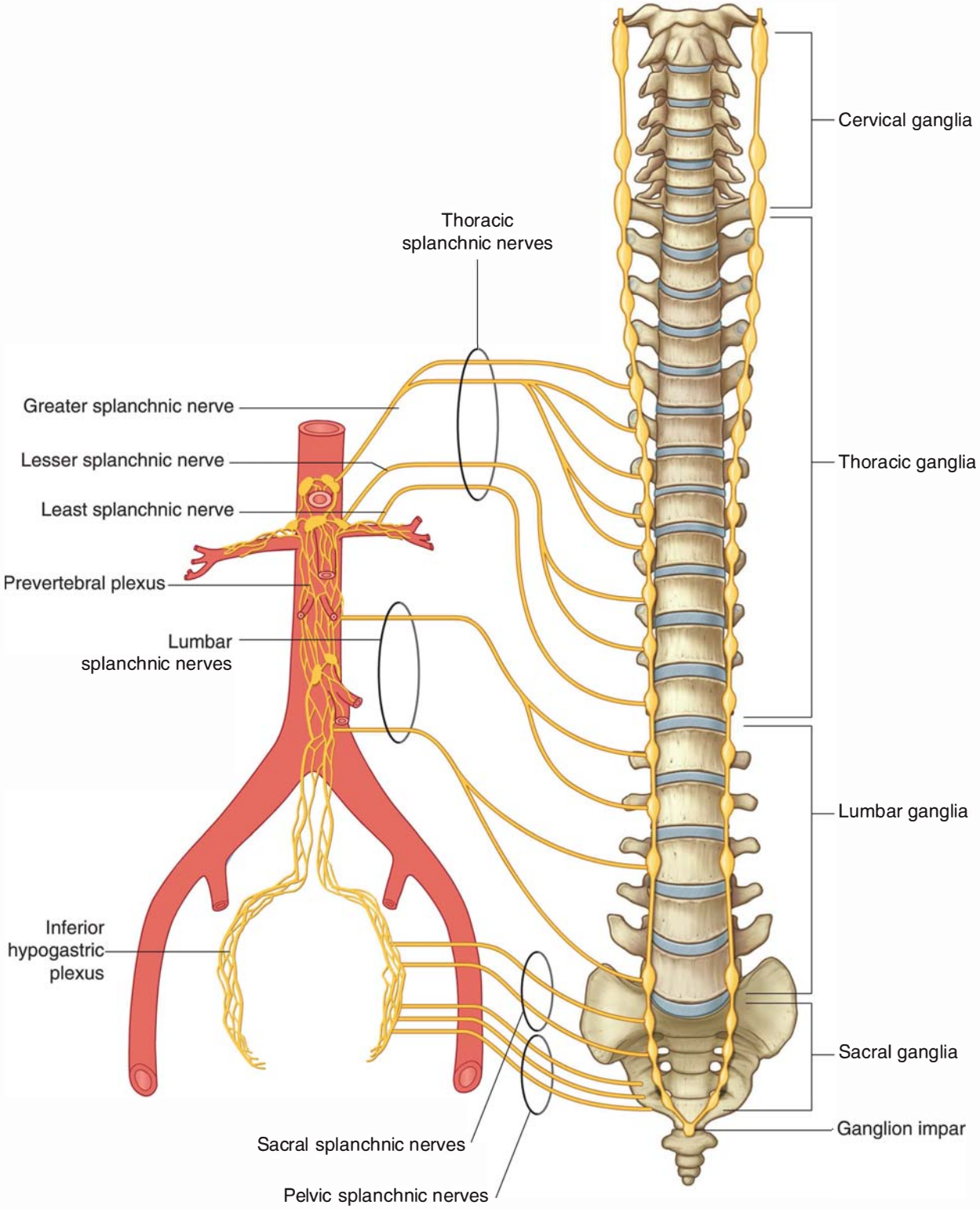


Fig. 4.125 Splanchnic nerves.

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celiac ganglion in the abdomen (a prevertebral ganglion associated with the celiac trunk).
- The lesser splanchnic nerve arises from the ninth and tenth (or tenth and eleventh) thoracic ganglia and travels to the aorticorenal ganglion.
- The least splanchnic nerve, when present, arises from the twelfth thoracic ganglion and travels to the renal plexus.

Lumbar and sacral splanchnic nerves

There are usually two to four **lumbar splanchnic nerves**, which pass from the lumbar part of the sympathetic trunk or associated ganglia and enter the prevertebral plexus (Fig. 4.125).

Similarly, the **sacral splanchnic nerves** pass from the sacral part of the sympathetic trunk or associated ganglia and enter the inferior hypogastric plexus, which is an extension of the prevertebral plexus into the pelvis.

Pelvic splanchnic nerves

The **pelvic splanchnic nerves** (**parasympathetic root**) are unique. They are the only splanchnic nerves that carry parasympathetic fibers. In other words, they do not originate from the sympathetic trunks. Rather, they originate directly from the anterior rami of S2 to S4. Preganglionic parasympathetic fibers originating in the sacral spinal cord pass from the S2 to S4 spinal nerves to the inferior hypogastric plexus (Fig. 4.125). Once in this plexus, some of these fibers pass upward, enter the abdominal prevertebral plexus, and distribute with the arteries supplying the hindgut. This provides the pathway for innervation of the distal one-third of the transverse colon, the descending colon, and the sigmoid colon by preganglionic parasympathetic fibers.

Abdominal prevertebral plexus and ganglia

The abdominal prevertebral plexus is a collection of nerve fibers that surrounds the abdominal aorta and is continuous onto its major branches. Scattered throughout the length of the abdominal prevertebral plexus are cell bodies of postganglionic sympathetic fibers. Some of these cell bodies are organized into distinct ganglia, while others are more random in their distribution. The ganglia are usually associated with specific branches of the abdominal aorta and named after these branches.

The three major divisions of the abdominal prevertebral plexus and associated ganglia are the celiac, aortic, and superior hypogastric plexuses (Fig. 4.126).

- The celiac plexus is the large accumulation of nerve fibers and ganglia associated with the roots of the celiac trunk and superior mesenteric artery immediately below the aortic hiatus of the diaphragm. Ganglia associated with the celiac plexus include two celiac ganglia, a single superior mesenteric ganglion, and two aorticorenal ganglia.
- The aortic plexus consists of nerve fibers and associated ganglia on the anterior and lateral surfaces of the abdominal aorta extending from just below the origin of the superior mesenteric artery to the bifurcation of the aorta into the two common iliac arteries. The major ganglion in this plexus is the inferior mesenteric ganglion at the root of the inferior mesenteric artery.
- The superior hypogastric plexus contains numerous small ganglia and is the final part of the abdominal prevertebral plexus before the prevertebral plexus continues into the pelvic cavity.

Each of these major plexuses gives origin to a number of secondary plexuses, which may also contain small ganglia. These plexuses are usually named after the vessels with which they are associated. For example, the celiac plexus is usually described as giving origin to the superior mesenteric plexus and the renal plexus, as well as other plexuses that extend out along the various branches of the celiac trunk. Similarly, the aortic plexus has secondary plexuses consisting of the inferior mesenteric plexus, the spermatic plexus, and the external iliac plexus.

Inferiorly, the superior hypogastric plexus divides into the **hypogastric nerves**, which descend into the pelvis and contribute to the formation of the inferior hypogastric or pelvic plexus (Fig. 4.126).

The abdominal prevertebral plexus receives:

- preganglionic parasympathetic and visceral afferent fibers from the vagus nerves [X],
- preganglionic sympathetic and visceral afferent fibers from the thoracic and lumbar splanchnic nerves, and
- preganglionic parasympathetic fibers from the pelvic splanchnic nerves.

Parasympathetic innervation

Parasympathetic innervation of the abdominal part of the gastrointestinal tract and of the spleen, pancreas, gallbladder, and liver is from two sources—the vagus nerves [X] and the pelvic splanchnic nerves.

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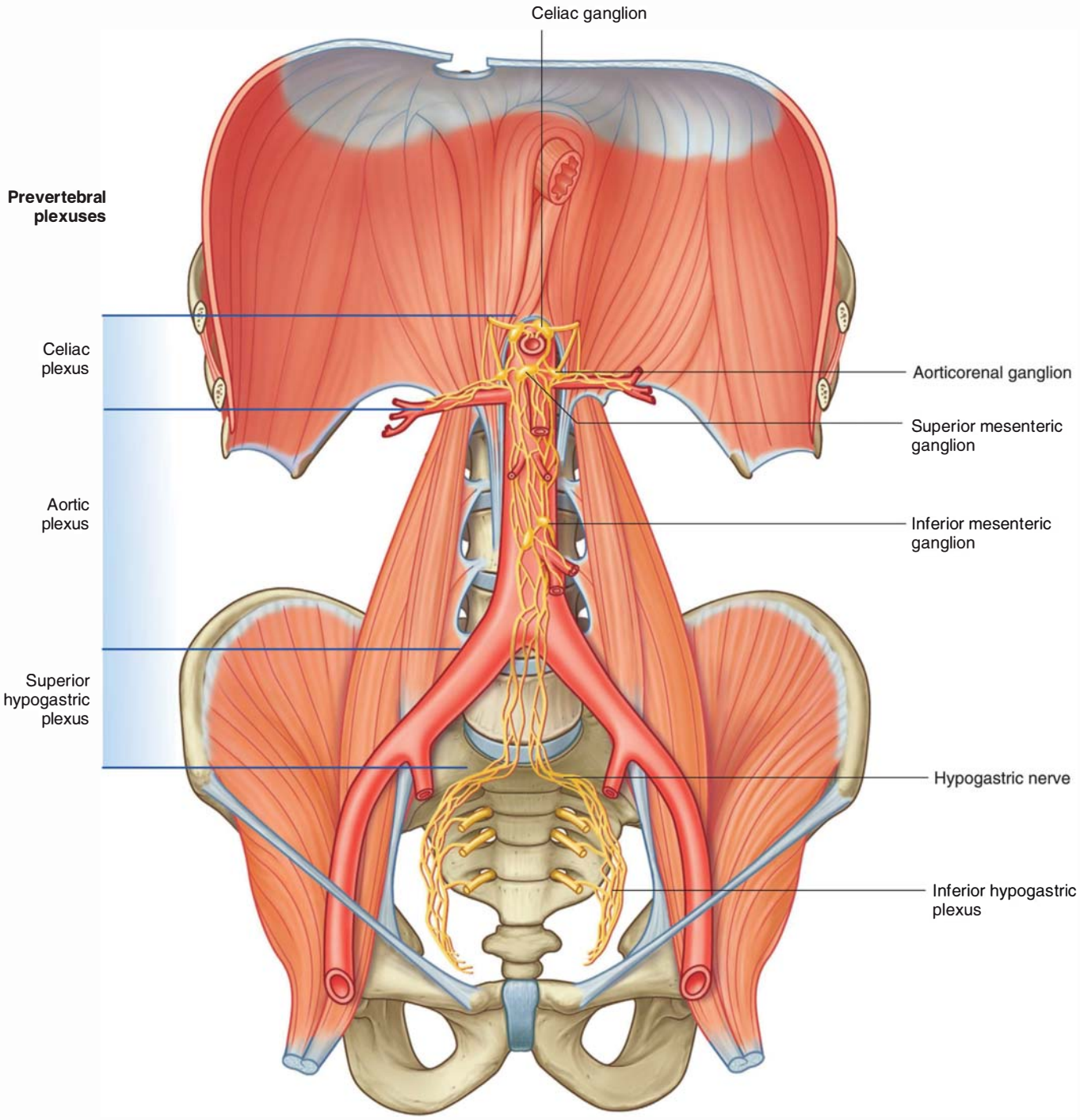


Fig. 4.126 Abdominal prevertebral plexus and ganglia.

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Vagus nerves

The **vagus nerves** [X] enter the abdomen associated with the esophagus as the esophagus passes through the diaphragm (Fig. 4.127) and provide parasympathetic innervation to the foregut and midgut.

After entering the abdomen as the **anterior** and **posterior vagal trunks**, they send branches to the abdominal prevertebral plexus. These branches contain preganglionic parasympathetic fibers and visceral afferent fibers, which are distributed with the other components of the prevertebral plexus along the branches of the abdominal aorta.

Pelvic splanchnic nerves

The **pelvic splanchnic nerves**, carrying preganglionic parasympathetic fibers from S2 to S4 spinal cord levels, enter the inferior hypogastric plexus in the pelvis. Some of these fibers move upward into the inferior mesenteric part of the prevertebral plexus in the abdomen (Fig. 4.127). Once there, these fibers are distributed with branches of

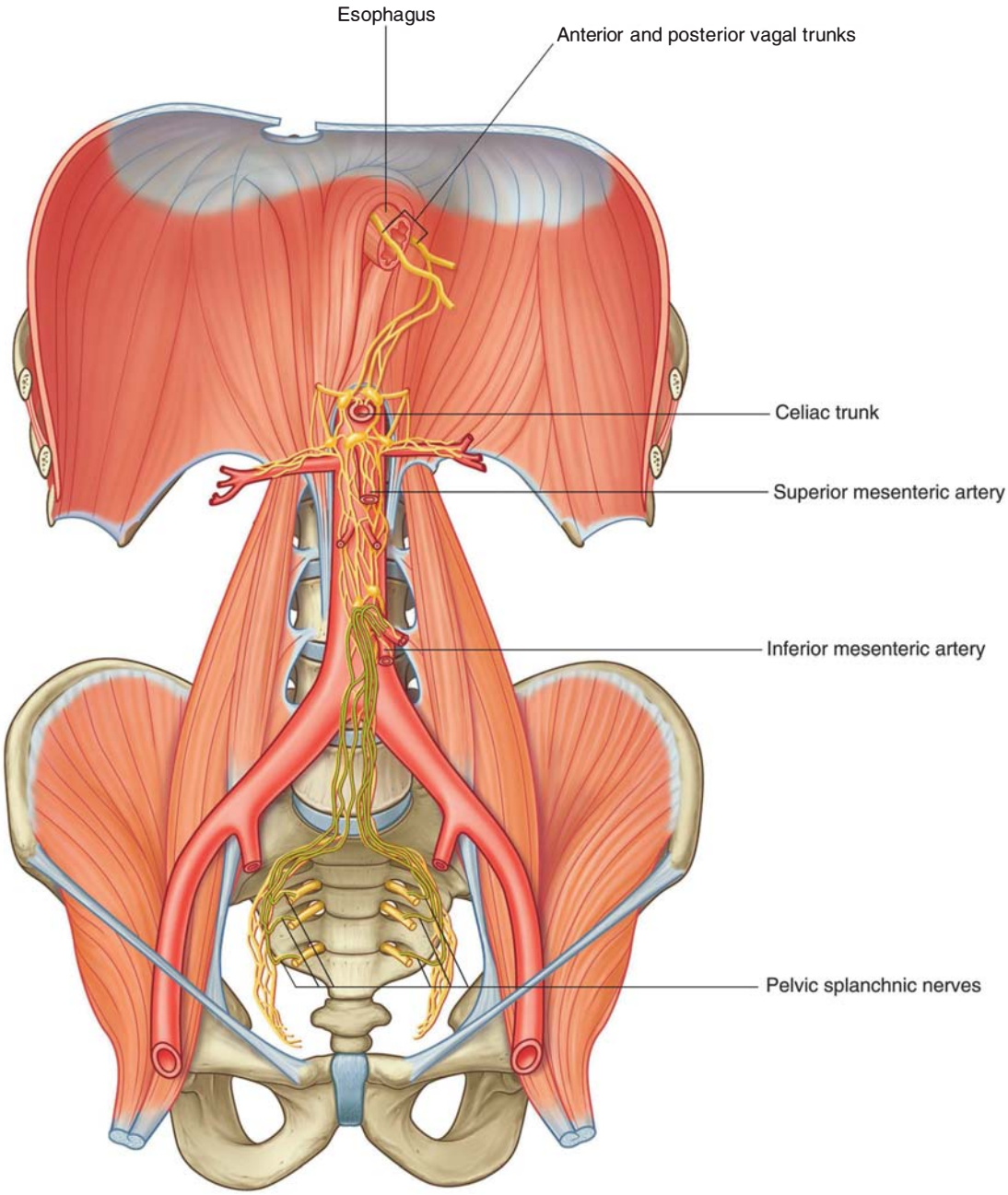


Fig. 4.127 Parasympathetic innervation of the abdominal portion of the gastrointestinal tract.

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the inferior mesenteric artery and provide parasympathetic innervation to the hindgut.

Enteric system

The enteric system is a division of the visceral part of the nervous system and is a local neuronal circuit in the wall of the gastrointestinal tract. It consists of motor and sensory neurons organized into two interconnected plexuses (the **myenteric** and **submucosal plexuses**) between the layers of the gastrointestinal wall, and the associated nerve fibers that pass between the plexuses and from the plexuses to the adjacent tissue (Fig. 4.128).

The enteric system regulates and coordinates numerous gastrointestinal tract activities, including gastric secretory activity, gastrointestinal blood flow, and the contraction and relaxation cycles of smooth muscle (**peristalsis**).

Although the enteric system is generally independent of the central nervous system, it does receive input from postganglionic sympathetic and preganglionic parasympathetic neurons that modifies its activities.

Sympathetic innervation of the stomach

The pathway of sympathetic innervation of the stomach is as follows:

- A preganglionic sympathetic fiber originating at the T6 level of the spinal cord enters an anterior root to leave the spinal cord.
- At the level of the intervertebral foramen, the anterior root (which contains the preganglionic fiber) and a posterior root join to form a spinal nerve.
- Outside the vertebral column, the preganglionic fiber leaves the anterior ramus of the spinal nerve through the white ramus communicans.
- The white ramus communicans, containing the preganglionic fiber, connects to the sympathetic trunk.
- Entering the sympathetic trunk, the preganglionic fiber does not synapse but passes through the trunk and enters the greater splanchnic nerve.
- The greater splanchnic nerve passes through the crura of the diaphragm and enters the celiac ganglion.
- In the celiac ganglion, the preganglionic fiber synapses with a postganglionic neuron.
- The postganglionic fiber joins the plexus of nerve fibers surrounding the celiac trunk and continues along its branches.
- The postganglionic fiber travels through the plexus of nerves accompanying the branches of the celiac trunk supplying the stomach and eventually reaches its point of distribution.
- This input from the sympathetic system may modify the activities of the gastrointestinal tract controlled by the enteric nervous system.

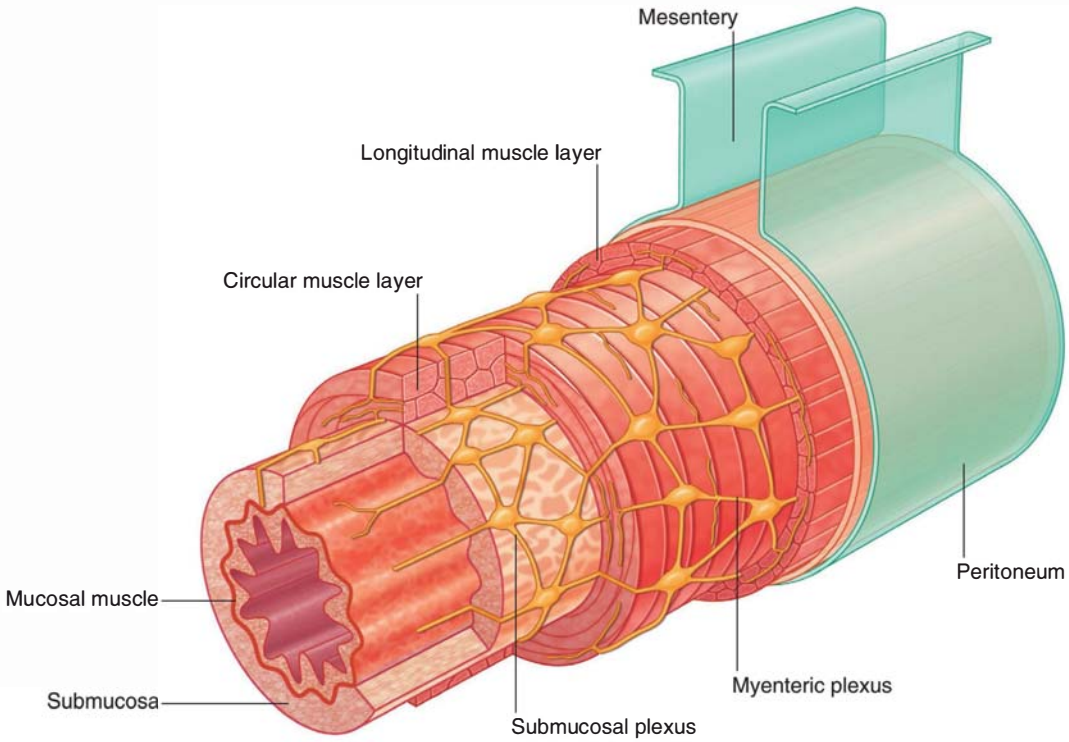


Fig. 4.128 The enteric system.

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In the clinic

Surgery for obesity

Surgery for obesity is also known as weight loss surgery and bariatric surgery. This type of surgery has become increasingly popular over the last few years for patients who are unable to achieve significant weight loss through appropriate diet modification and exercise programs. It is often regarded as a last resort. Importantly, we have to recognize the increasing medical impact that overweight patients pose. With obesity the patient is more likely to develop diabetes and cardiovascular problems and may suffer from increased general health disorders. All of these have a significant impact on health care budgeting and are regarded as serious conditions for the "health of a nation."

There are a number of surgical options to treat obesity. Surgery for patients who are morbidly obese can be categorized into two main groups: malabsorptive procedures and restrictive procedures.

Malabsorptive procedures

There are a variety of bypass procedures that produce a malabsorption state, preventing further weight gain and also producing weight loss. There are complications, which may include anemia, osteoporosis, and diarrhea (e.g., jejunoileal bypass).

Predominantly restrictive procedures

Restrictive procedures involve placing a band or stapling in or around the stomach to decrease the size of the organ. This reduction produces an earlier feeling of satiety and prevents the patient from overeating.

Combination procedure

Probably the most popular procedure currently in the United States, this procedure involves stapling the proximal stomach and joining a loop of small bowel to the small gastric remnant.

Any overweight patient undergoing surgery faces significant risk and increased morbidity, with mortality rates from 1% to 5%.

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POSTERIOR ABDOMINAL REGION

The posterior abdominal region is posterior to the abdominal part of the gastrointestinal tract, the spleen, and the pancreas (Fig. 4.129). This area, bounded by bones and muscles making up the posterior abdominal wall, contains numerous structures that not only are directly involved in the activities of the abdominal contents but also use this area as a conduit between body regions. Examples include the abdominal aorta and its associated nerve plexuses, the inferior vena cava, the sympathetic trunks, and lymphatics. There are also structures originating in this area that are critical to the normal function of other regions of the body (i.e., the lumbar plexus of nerves), and there are organs that associate with this area during development and remain in it in the adult (i.e., the kidneys and suprarenal glands).

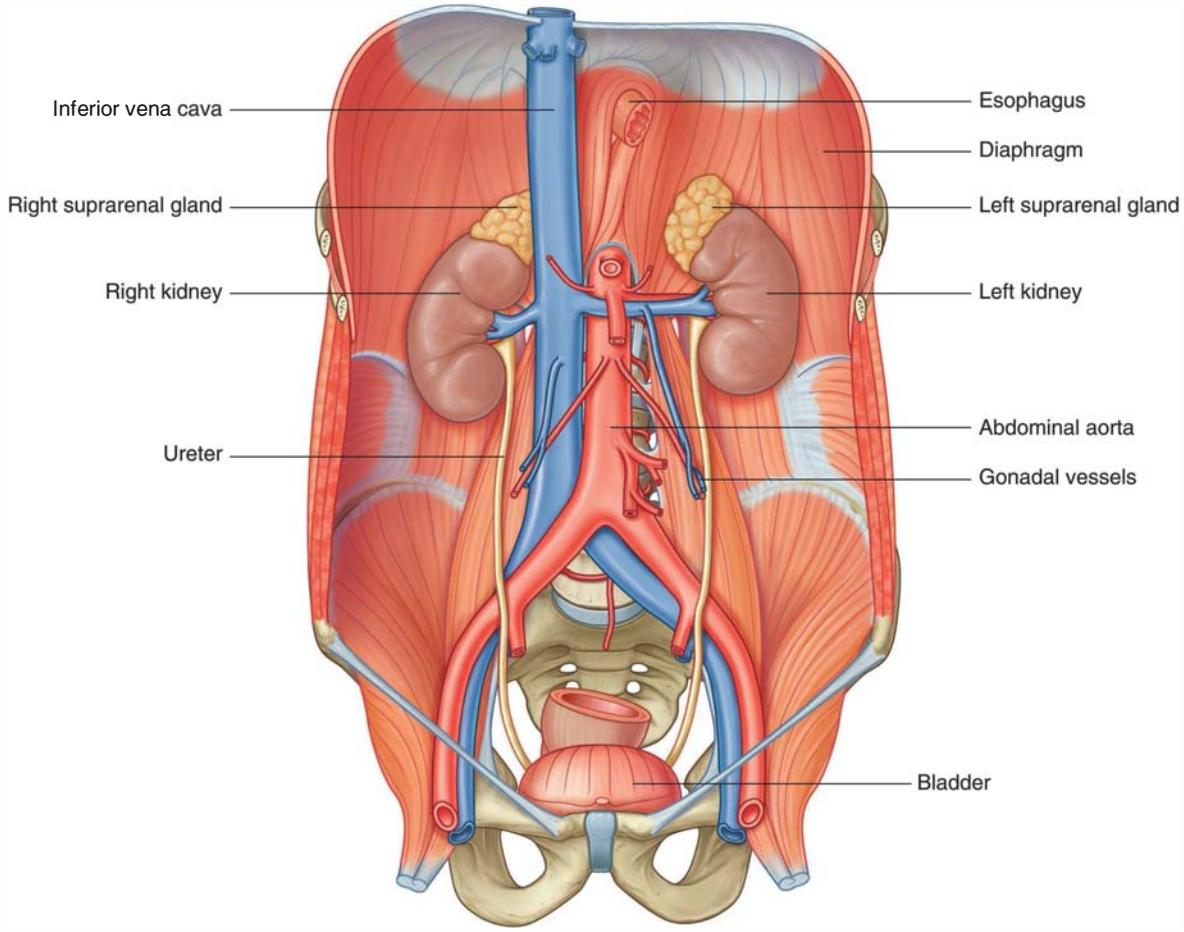


Fig. 4.129 Posterior abdominal region.

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Posterior abdominal wall

Bones

Lumbar vertebrae and the sacrum

Projecting into the midline of the posterior abdominal area are the bodies of the five lumbar vertebrae (Fig. 4.130). The prominence of these structures in this region is due to the secondary curvature (a forward convexity) of the lumbar part of the vertebral column.

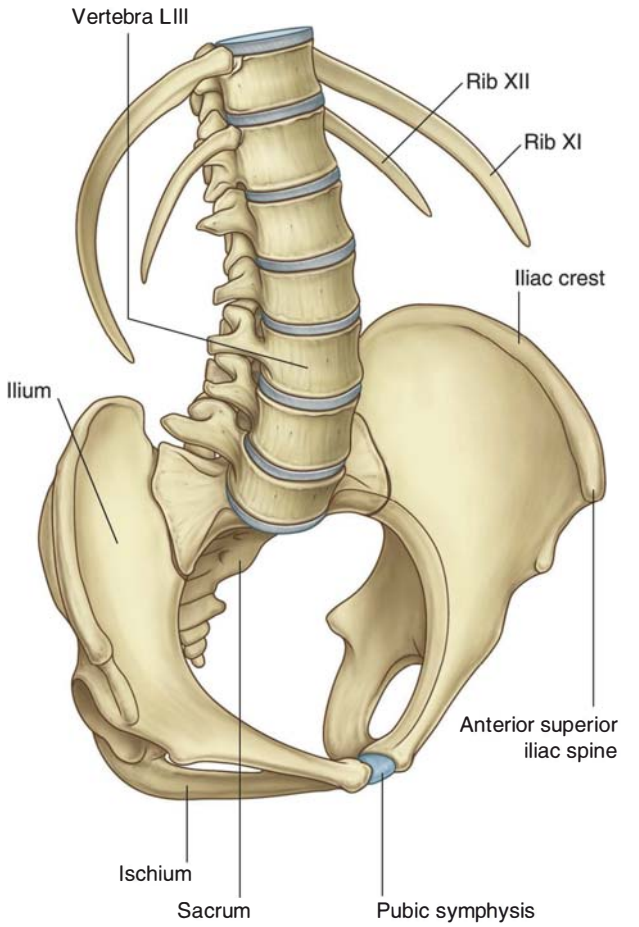


Fig. 4.130 Osteology of the posterior abdominal wall.

The lumbar vertebrae can be distinguished from cervical and thoracic vertebrae because of their size. They are much larger than any other vertebrae in any other region. The vertebral bodies are massive and progressively increase in size from vertebra LI to LV. The pedicles are short and stocky, the transverse processes are long and slender, and the spinous processes are large and stubby. The articular processes are large and oriented medially and laterally, which promotes flexion and extension in this part of the vertebral column.

Between each lumbar vertebra is an intervertebral disc, which completes this part of the midline boundary of the posterior abdominal wall.

The midline boundary of the posterior abdominal wall, inferior to the lumbar vertebrae, consists of the upper margin of the sacrum (Fig. 4.130). The sacrum is formed by the fusion of the five sacral vertebrae into a single, wedge-shaped bony structure that is broad superiorly and narrows inferiorly. Its concave anterior surface and its convex posterior surface contain anterior and posterior sacral foramina for the anterior and posterior rami of spinal nerves to pass through.

Pelvic bones

The **ilia**, which are components of each pelvic bone, attach laterally to the sacrum at the sacro-iliac joints (Fig. 4.130). The upper part of each ilium expands outward into a thin wing-like area (the **iliac fossa**). The medial side of this region of each iliac bone, and the related muscles, are components of the posterior abdominal wall.

Ribs

Superiorly, ribs XI and XII complete the bony framework of the posterior abdominal wall (Fig. 4.130). These ribs are unique in that they do not articulate with the sternum or other ribs, they have a single articular facet on their heads, and they do not have necks or tubercles.

Rib XI is posterior to the superior part of the left kidney, and rib XII is posterior to the superior part of both kidneys. Also, rib XII serves as a point of attachment for numerous muscles and ligaments.

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Muscles

Muscles forming the medial, lateral, inferior, and superior boundaries of the posterior abdominal region fill in the bony framework of the posterior abdominal wall (Table 4.2). Medially are the psoas major and minor muscles, laterally is the quadratus lumborum muscle, inferiorly is the iliacus muscle, and superiorly is the diaphragm (Figs. 4.131 and 4.132).

Psoas major and minor

Medially, the **psoas major** muscles cover the anterolateral surface of the bodies of the lumbar vertebrae, filling in the space between the vertebral bodies and the transverse processes (Fig. 4.131). Each of these muscles arises from the bodies of vertebra TXII and all five lumbar vertebrae, from the intervertebral discs between each vertebra, and from the transverse processes of the lumbar vertebrae. Passing inferiorly along the pelvic brim, each muscle continues into the anterior thigh, under the inguinal ligament, to attach to the lesser trochanter of the femur.

The psoas major muscle flexes the thigh at the hip joint when the trunk is stabilized and flexes the trunk against gravity when the body is supine. It is innervated by anterior rami of nerves L1 to L3.

Associated with the psoas major muscle is the **psoas minor** muscle, which is sometimes absent. Lying on the surface of the psoas major when present, this slender muscle arises from vertebrae TXII and LI and the intervening intervertebral disc; its long tendon inserts into the pectineal line of the pelvic brim and the iliopubic eminence.

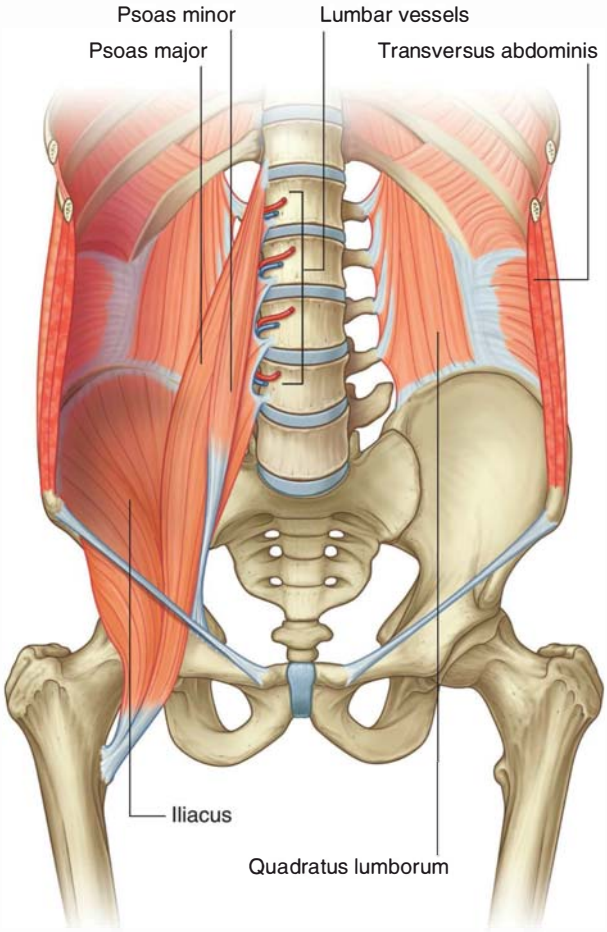


Fig. 4.131 Muscles of the posterior abdominal wall.

Table 4.2 Posterior abdominal wall muscles

Muscle	Origin	Insertion	Innervation	Function
Psoas major	Lateral surface of bodies of TXII and LI to LV vertebrae, transverse processes of the lumbar vertebrae, and the intervertebral discs between TXII and LI to LV vertebrae	Lesser trochanter of the femur	Anterior rami of L1 to L3	Flexion of thigh at hip joint
Psoas minor	Lateral surface of bodies of TXII and LI vertebrae and intervening intervertebral disc	Pectineal line of the pelvic brim and iliopubic eminence	Anterior rami of L1	Weak flexion of lumbar vertebral column
Quadratus lumborum	Transverse process of LV vertebra, iliolumbar ligament, and iliac crest	Transverse processes of LI to LIV vertebrae and inferior border of rib XII	Anterior rami of T12 and L1 to L4	Depress and stabilize rib XII and some lateral bending of trunk
Iliacus	Upper two-thirds of iliac fossa, anterior sacro-iliac and iliolumbar ligaments, and upper lateral surface of sacrum	Lesser trochanter of femur	Femoral nerve (L2 to L4)	Flexion of thigh at hip joint

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The psoas minor is a weak flexor of the lumbar vertebral column and is innervated by the anterior ramus of nerve L1.

Quadratus lumborum

Laterally, the quadratus lumborum muscles fill the space between rib XII and the iliac crest on both sides of the vertebral column (Fig. 4.131). They are overlapped medially by the psoas major muscles; along their lateral borders are the transversus abdominis muscles.

Each quadratus lumborum muscle arises from the transverse process of vertebra LV, the iliolumbar ligament, and the adjoining part of the iliac crest. The muscle attaches superiorly to the transverse process of the first four lumbar vertebrae and the inferior border of rib XII.

The quadratus lumborum muscles depress and stabilize the twelfth ribs and contribute to lateral bending of the trunk. Acting together, the muscles may extend the lumbar part of the vertebral column. They are innervated by anterior rami of T12 and L1 to L4 spinal nerves.

Iliacus

Inferiorly, an **iliacus** muscle fills the iliac fossa on each side (Fig. 4.131). From this expansive origin covering the iliac fossa, the muscle passes inferiorly, joins with the psoas major muscle, and attaches to the lesser trochanter of the femur. As they pass into the thigh, these combined muscles are referred to as the **iliopsoas** muscle.

Like the psoas major muscle, the iliacus flexes the thigh at the hip joint when the trunk is stabilized and flexes the trunk against gravity when the body is supine. It is innervated by branches of the femoral nerve.

Diaphragm

Superiorly, the diaphragm forms the boundary of the posterior abdominal region. This musculotendinous sheet also separates the abdominal cavity from the thoracic cavity.

Structurally, the diaphragm consists of a central tendinous part into which the circumferentially arranged muscle fibers attach (Fig. 4.132). The diaphragm is

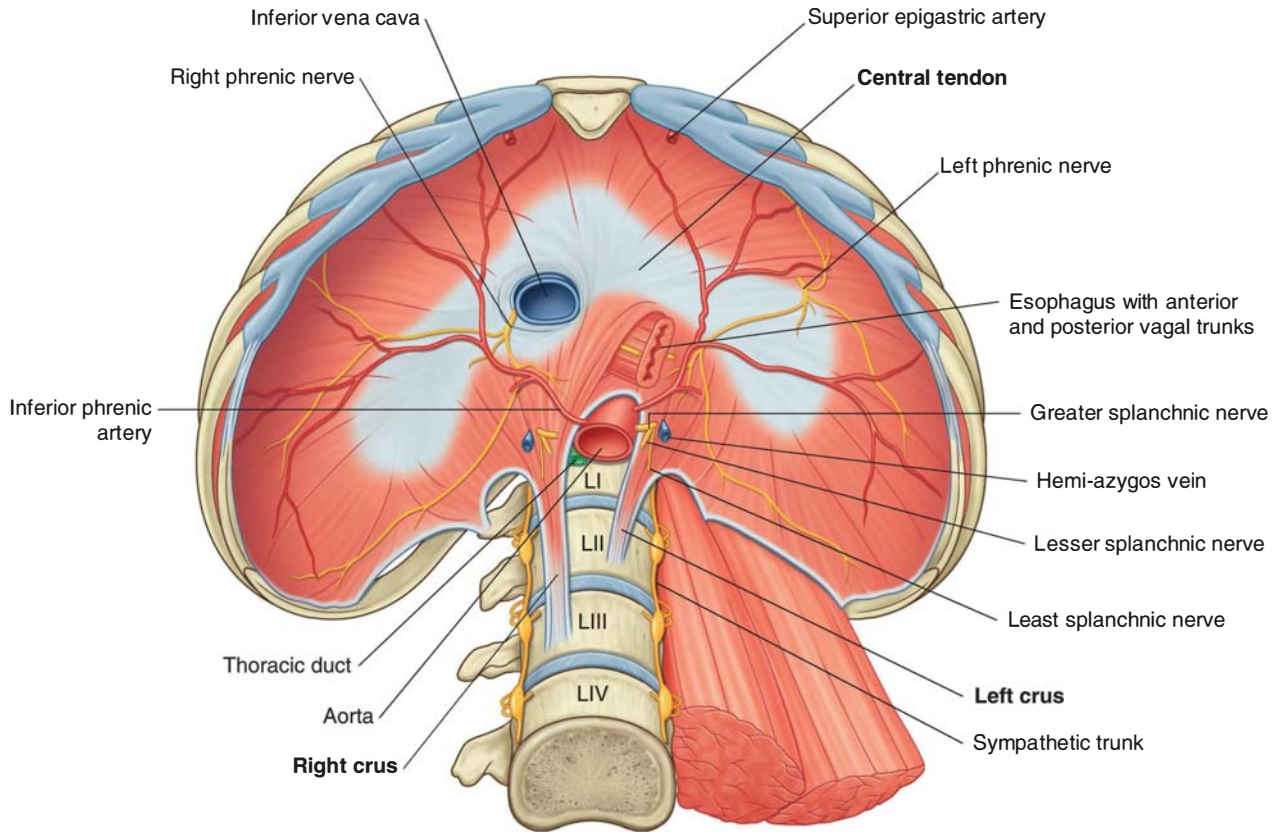


Fig. 4.132 Diaphragm.

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anchored to the lumbar vertebrae by musculotendinous crura, which blend with the anterior longitudinal ligament of the vertebral column:

- The right crus is the longest and broadest of the crura and is attached to the bodies of vertebrae LI to LIII and the intervening intervertebral discs (Fig. 4.133).
- Similarly, the left crus is attached to vertebrae LI and LII and the associated intervertebral disc.

The crura are connected across the midline by a tendinous arch (the **median arcuate ligament**), which passes anterior to the aorta (Fig. 4.133).

Lateral to the crura, a second tendinous arch is formed by the fascia covering the upper part of the psoas major muscle. This is the **medial arcuate ligament**, which is attached medially to the sides of vertebrae LI and LII and laterally to the transverse process of vertebra LI (Fig. 4.133).

A third tendinous arch, the **lateral arcuate ligament**, is formed by a thickening in the fascia that covers the quadratus lumborum. It is attached medially to the transverse process of vertebra LI and laterally to rib XII (Fig. 4.133).

The medial and lateral arcuate ligaments serve as points of origin for some of the muscular components of the diaphragm.

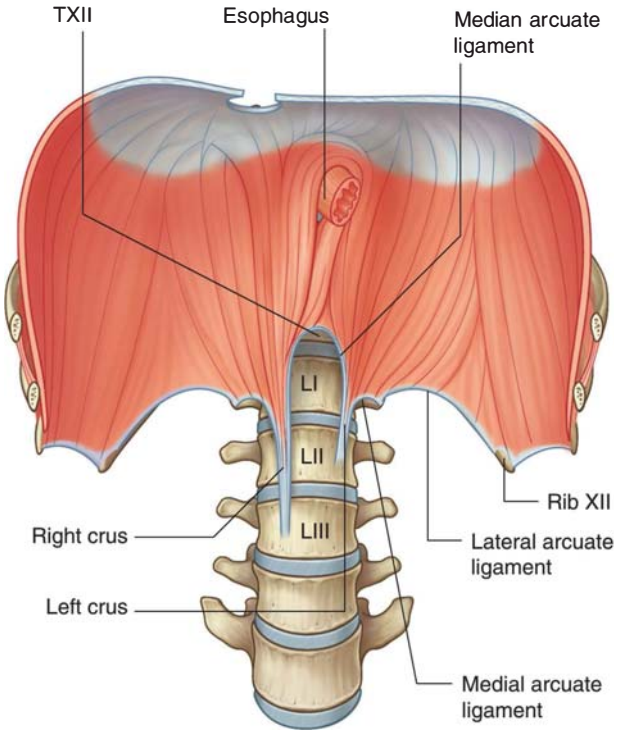


Fig. 4.133 Crura of the diaphragm.

Structures passing through or around the diaphragm

Numerous structures pass through or around the diaphragm (Fig. 4.132):

- The aorta passes posterior to the diaphragm and anterior to the vertebral bodies at the lower level of vertebra TXII; it is between the two crura of the diaphragm and posterior to the median arcuate ligament, just to the left of midline.
- Accompanying the aorta through the aortic hiatus is the thoracic duct and, sometimes, the azygos vein.
- The esophagus passes through the musculature of the right crus of the diaphragm at the level of vertebra TX, just to the left of the aortic hiatus.
- Passing through the esophageal hiatus with the esophagus are the anterior and posterior vagal trunks, the esophageal branches of the left gastric artery and vein, and a few lymphatic vessels.
- The third large opening in the diaphragm is the caval opening, through which the inferior vena cava passes from the abdominal cavity to the thoracic cavity (Fig. 4.132) at approximately vertebra TVIII in the central tendinous part of the diaphragm.
- Accompanying the inferior vena cava through the caval opening is the right phrenic nerve.
- The left phrenic nerve passes through the muscular part of the diaphragm just anterior to the central tendon on the left side.

Additional structures pass through small openings either in or just outside the diaphragm as they pass from the thoracic cavity to the abdominal cavity (Fig. 4.132):

- The greater, lesser, and least (when present) splanchnic nerves pass through the crura, on either side.
- The hemi-azygos vein passes through the left crus.
- Passing posterior to the medial arcuate ligament, on either side, are the sympathetic trunks.
- Passing anterior to the diaphragm, just deep to the ribs, are the superior epigastric vessels.
- Other vessels and nerves (i.e., the musculophrenic vessels and intercostal nerves) also pass through the diaphragm at various points.

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Domes

The classic appearance of the right and left domes of the diaphragm is caused by the underlying abdominal contents pushing these lateral areas upward, and by the fibrous pericardium, which is attached centrally, causing a flattening of the diaphragm in this area (Fig. 4.134).

The domes are produced by:

- the liver on the right, with some contribution from the right kidney and the right suprarenal gland, and
- the fundus of the stomach and spleen on the left, with contributions from the left kidney and the left suprarenal gland.

Although the height of these domes varies during breathing, a reasonable estimate in normal expiration places the left dome at the fifth intercostal space and the right dome at rib V. This is important to remember when percussing the thorax.

During inspiration, the muscular part of the diaphragm contracts, causing the central tendon of the diaphragm to be drawn inferiorly. This results in some flattening of the domes, enlargement of the thoracic cavity, and a reduction in intrathoracic pressure. The physiological effect of these changes is that air enters the lungs and venous return to the heart is enhanced.



Fig. 4.134 Right and left domes of the diaphragm. Chest radiograph.

Blood supply

There is blood supply to the diaphragm on its superior and inferior surfaces:

- Superiorly, the musculophrenic and pericardiacophrenic arteries, both branches of the internal thoracic artery, and the superior phrenic artery, a branch of the thoracic aorta, supply the diaphragm.
- Inferiorly, the inferior phrenic arteries, branches of the abdominal aorta, supply the diaphragm (see Fig. 4.132).

Venous drainage is through companion veins to these arteries.

Innervation

Innervation of the diaphragm is primarily by the **phrenic nerves**. These nerves, from the C3 to C5 spinal cord levels, provide all motor innervation to the diaphragm and sensory fibers to the central part. They pass through the thoracic cavity, between the mediastinal pleura and the pericardium, to the superior surface of the diaphragm. At this point, the right phrenic nerve accompanies the inferior vena cava through the diaphragm and the left phrenic nerve passes through the diaphragm by itself (see Fig. 4.132). Additional sensory fibers are supplied to the peripheral areas of the diaphragm by intercostal nerves.

In the clinic

Psoas muscle abscess

At first glance, it is difficult to appreciate why the psoas muscle sheath is of greater importance than any other muscle sheath. The psoas muscle and its sheath arise not only from the lumbar vertebrae but also from the intervertebral discs between each vertebra. This disc origin is of critical importance. In certain types of infection, the intervertebral disc is preferentially affected (e.g., tuberculosis and salmonella discitis). As the infection of the disc develops, the infection spreads anteriorly and anterolaterally. In the anterolateral position, the infection passes into the psoas muscle sheath, and spreads within the muscle and sheath, and may appear below the inguinal ligament as a mass.

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In the clinic

Diaphragmatic hernias

To understand why a hernia occurs through the diaphragm, it is necessary to consider the embryology of the diaphragm.

The diaphragm is formed from four structures—the septum transversum, the posterior esophageal mesentery, the pleuroperitoneal membrane, and the peripheral rim—which eventually fuse together, so separating the abdominal cavity from the thoracic cavity. The septum transversum forms the central tendon, which develops from a mesodermal origin in front of the embryo's head and then moves to its more adult position during formation of the head fold.

Fusion of the various components of the diaphragm may fail, and hernias may occur through the failed points of fusion (Fig. 4.135). The commonest sites are:

- between the xiphoid process and the costal margins on the right (Morgagni's hernia), and
- through an opening on the left when the pleuroperitoneal membrane fails to close the pericardioperitoneal canal (Bochdalek's hernia).

Hernias may also occur through the central tendon and through a congenitally large esophageal hiatus.

Morgagni's and Bochdalek's hernias tend to appear at or around the time of birth or in early infancy. They allow abdominal bowel to enter the thoracic cavity, which may compress the lungs and reduce respiratory function. Most of these hernias require surgical closure of the diaphragmatic defect.

Occasionally, small defects within the diaphragm fail to permit bowel through, but do allow free movement of fluid. Patients with ascites may develop pleural effusions, while patients with pleural effusions may develop ascites when these defects are present.

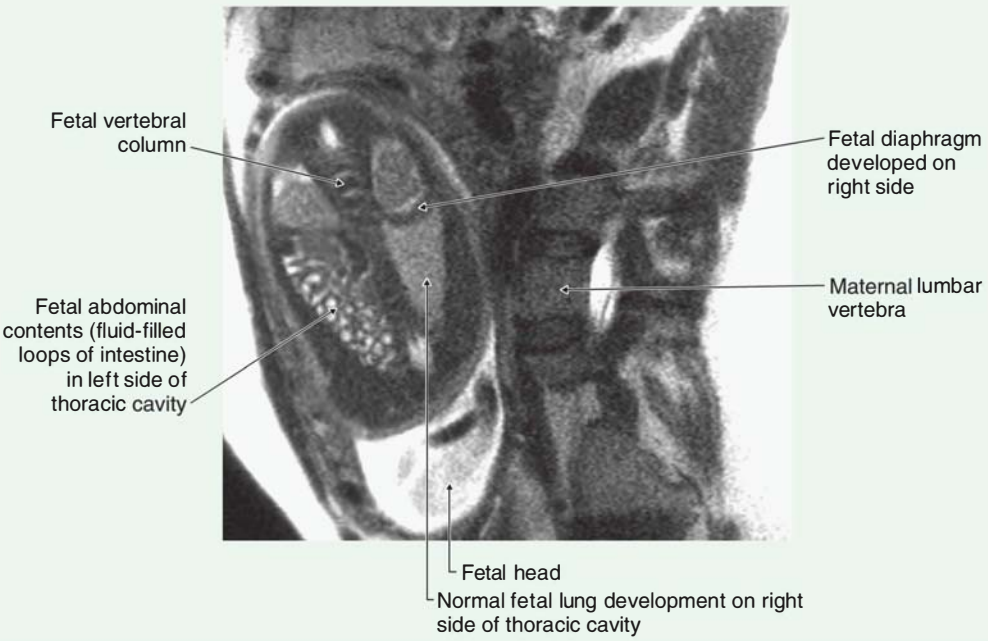


Fig. 4.135 Fetal diaphragmatic hernia in utero. T2-weighted MR image. Fetus in coronal plane, mother in sagittal plane.

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